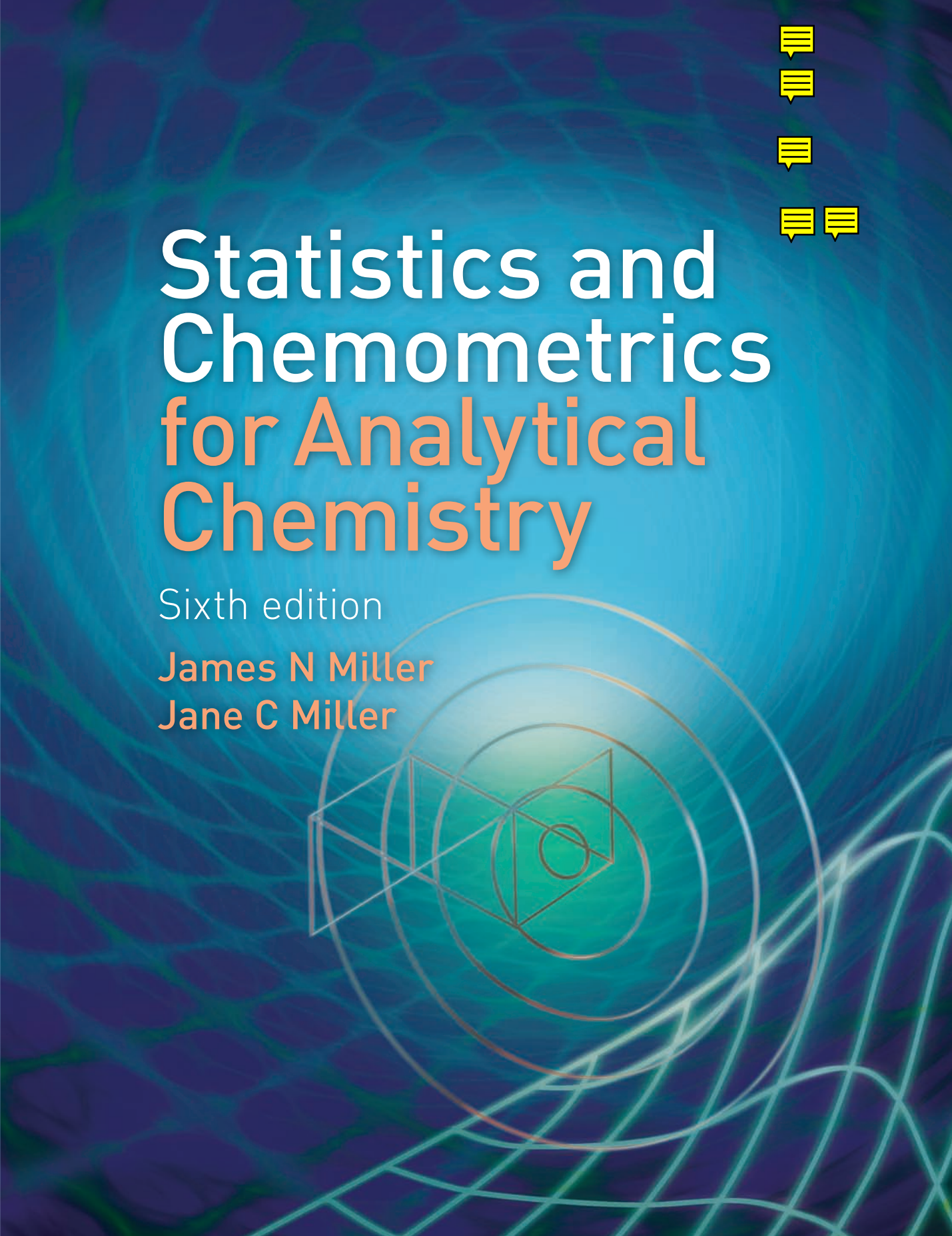


Statistics and Chemometrics for Analytical Chemistry

Sixth edition

James N Miller

Jane C Miller

Statistics and Chemometrics for Analytical Chemistry

Sixth Edition

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James N. Miller
Jane C. Miller

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Preface to the sixth edition

Since the publication of the fifth edition of this book in 2005 the use of elementary and advanced statistical methods in the teaching and the practice of the analytical sciences has continued to increase in extent and quality. This new edition attempts to keep pace with these developments in several chapters, while retaining the basic approach of previous editions by adopting a pragmatic and, as far as possible, non-mathematical approach to statistical calculations.

The results of many analytical experiments are conventionally evaluated using established significance testing methods. In recent years, however, Bayesian methods have become more widely used, especially in areas such as forensic science and clinical chemistry. The basis and methodology of Bayesian statistics have some distinctive features, which are introduced in a new section of Chapter 3. The quality of analytical results obtained when different laboratories study identical sample materials continues, for obvious practical reasons, to be an area of major importance and interest. Such comparative studies form a major part of the process of validating the use of a given method by a particular laboratory. Chapter 4 has therefore been expanded to include a new section on method validation. The most popular form of inter-laboratory comparison, proficiency testing schemes, often yields suspect or unexpected results. The latter are now generally treated using robust statistical methods, and the treatment of several such methods in Chapter 6 has thus been expanded. Uncertainty estimates have become a widely accepted feature of many analyses, and a great deal of recent attention has been focused on the uncertainty contributions that often arise from the all-important sampling process: this topic has also been covered in Chapter 4. Calibration methods lie at the core of most modern analytical experiments. In Chapter 5 we have expanded our treatments of the standard additions approach, of weighted regression, and of regression methods where both x - and y -axes are subject to errors or variations.

A topic that analytical laboratories have not, perhaps, given the attention it deserves has been the proper use of experimental designs. Such designs have distinctive nomenclature and approaches compared with post-experiment data analysis, and this perhaps accounts for their relative neglect, but many experimental designs are relatively simple, and again excellent software support is available. This has encouraged us to expand significantly the coverage of experimental designs in Chapter 7. New and ever more sophisticated multivariate analysis

methods are now used by many researchers, and also in some everyday applications of analytical methods. They really deserve a separate text to themselves, but for this edition we have modestly expanded Chapter 8, which deals with these methods.

We have continued to include in the text many examples of calculations performed by two established pieces of software, Excel[®] and Minitab[®]. The former is accessible from most personal computers, and is much used in the collection and processing of data from analytical instruments, while the latter is frequently adopted in education as well as by practising scientists. In each program the calculations, at least the simple ones used in this book, are easily accessible and simply displayed, and many texts are available as general introductions to the software. Macros and add-ins that usefully expand the capacities and applications of Excel[®] and Minitab[®] are widely and freely available, and both programs offer graphical displays that provide opportunities for better understanding and further data interpretation. These extra facilities are utilised in some examples provided in the Instructors' Manual, which again accompanies this edition of our book. The Manual also contains ideas for classroom and laboratory work, a complete set of figures for use as OHP masters, and fully worked solutions to the exercises in this volume: this text now contains only outline solutions.

We are very grateful to many correspondents and staff and student colleagues who continue to provide us with constructive comments and suggestions, and to point out minor errors and omissions. We also thank the Royal Society of Chemistry for permission to use data from papers published in *The Analyst*. Finally we thank Rufus Curnow and his editorial colleagues at Pearson Education, Nicola Chilvers and Ros Woodward, for their perfect mixture of expertise, patience and enthusiasm; any errors that remain despite their best efforts are ours alone.

James N. Miller
Jane C. Miller
December 2009

Preface to the first edition

To add yet another volume to the already numerous texts on statistics might seem to be an unwarranted exercise, yet the fact remains that many highly competent scientists are woefully ignorant of even the most elementary statistical methods. It is even more astonishing that analytical chemists, who practise one of the most quantitative of all sciences, are no more immune than others to this dangerous, but entirely curable, affliction. It is hoped, therefore, that this book will benefit analytical scientists who wish to design and conduct their experiments properly, and extract as much information from the results as they legitimately can. It is intended to be of value to the rapidly growing number of students specialising in analytical chemistry, and to those who use analytical methods routinely in everyday laboratory work.

There are two further and related reasons that have encouraged us to write this book. One is the enormous impact of microelectronics, in the form of microcomputers and handheld calculators, on statistics: these devices have brought lengthy or difficult statistical procedures within the reach of all practising scientists. The second is the rapid development of new 'chemometric' procedures, including pattern recognition, optimisation, numerical filter techniques, simulations and so on, all of them made practicable by improved computing facilities. The last chapter of this book attempts to give the reader at least a flavour of the potential of some of these newer statistical methods. We have not, however, included any computer programs in the book – partly because of the difficulties of presenting programs that would run on all the popular types of microcomputer, and partly because there is a substantial range of suitable and commercially available books and software.

The availability of this tremendous computing power naturally makes it all the more important that the scientist applies statistical methods rationally and correctly. To limit the length of the book, and to emphasise its practical bias, we have made no attempt to describe in detail the theoretical background of the statistical tests described. But we have tried to make it clear to the practising analyst which tests are appropriate to the types of problem likely to be encountered in the laboratory. There are worked examples in the text, and exercises for the reader at the end of each chapter. Many of these are based on the data provided by research papers published in *The Analyst*. We are deeply grateful to Mr. Phil Weston, the

Editor, for allowing us thus to make use of his distinguished journal. We also thank our colleagues, friends and family for their forbearance during the preparation of the book; the sources of the statistical tables, individually acknowledged in the appendices; the Series Editor, Dr. Bob Chalmers; and our publishers for their efficient cooperation and advice.

J. C. Miller

J. N. Miller

April 1984

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Glossary of symbols

a	– intercept of regression line
b	– gradient of regression line
c	– number of columns in two-way ANOVA
C	– correction term in two-way ANOVA
C	– used in Cochran's test for homogeneity of variance
d	– difference between estimated and standard concentrations in Shewhart control charts
F	– the ratio of two variances
G	– used in Grubbs' test for outliers
h	– number of samples in one-way ANOVA
k	– coverage factor in uncertainty estimates
μ	– arithmetic mean of a population
M	– number of minus signs in Wald–Wolfowitz runs test
n	– sample size
N	– number of plus signs in Wald–Wolfowitz runs test
N	– total number of measurements in two-way ANOVA
ν	– number of degrees of freedom
$P(r)$	– probability of r
Q	– Dixon's Q , used to test for outliers
r	– product-moment correlation coefficient
r	– number of rows in two-way ANOVA
r	– number of smallest and largest observations omitted in trimmed mean calculations
R^2	– coefficient of determination
R'^2	– adjusted coefficient of determination
r_s	– Spearman rank correlation coefficient
s	– standard deviation of a sample
$s_{y/x}$	– standard deviation of y -residuals
s_b	– standard deviation of slope of regression line
s_a	– standard deviation of intercept of regression line
$s_{(y/x)w}$	– standard deviation of y -residuals of weighted regression line
s_{x_0}	– standard deviation of x -value estimated using regression line
s_B	– standard deviation of blank
s_{x_E}	– standard deviation of extrapolated x -value

$s_{x_{0w}}$	– standard deviation of x -value estimated by using weighted regression line
σ	– standard deviation of a population
σ_0^2	– measurement variance
σ_1^2	– sampling variance
t	– quantity used in the calculation of confidence limits and in significance testing of mean (see Section 2.4)
T	– grand total in ANOVA
T_1 and T_2	– test statistics used in the Wilcoxon rank sum test
u	– standard uncertainty
U	– expanded uncertainty
w	– range
w_i	– weight given to point on regression line
$\bar{x}$	– arithmetic mean of a sample
x_0	– x -value estimated by using regression line
x_0	– outlier value of x
$\tilde{x}_i$	– pseudo-value in robust statistics
x_E	– extrapolated x -value
$\bar{x}_w$	– arithmetic mean of weighted x -values
X^2	– quantity used to test for goodness-of-fit
$\hat{y}$	– y -values predicted by regression line
y_0	– signal from test material in calibration experiments
$\bar{y}_w$	– arithmetic mean of weighted y -values
y_B	– signal from blank
z	– standard normal variable

1

Introduction

Major topics covered in this chapter

- Errors in analytical measurements
- Gross, random and systematic errors
- Precision, repeatability, reproducibility, bias, accuracy
- Planning experiments
- Using calculators and personal computers

1.1

Analytical problems

Analytical chemists face both qualitative and quantitative problems. For example, the presence of boron in distilled water is very damaging in the manufacture of electronic components, so we might be asked the qualitative question ‘Does this distilled water sample contain any boron?’ The comparison of soil samples in forensic science provides another qualitative problem: ‘Could these two soil samples have come from the same site?’ Other problems are quantitative ones: ‘How much albumin is there in this sample of blood serum?’ ‘What is the level of lead in this sample of tap-water?’ ‘This steel sample contains small amounts of chromium, tungsten and manganese – how much of each?’ These are typical examples of single- and multiple-component quantitative analyses.

Modern analytical chemistry is overwhelmingly a **quantitative** science, as a quantitative result will generally be much more valuable than a qualitative one. It may be useful to have detected boron in a water sample, but it is much more useful to be able to say *how much* boron is present. Only then can we judge whether the boron level is worrying, or consider how it might be reduced. Sometimes it is *only* a quantitative result that has any value at all: almost all samples of blood serum contain albumin, so the only question is, how much?

Even when only a qualitative answer is required, quantitative methods are often used to obtain it. In reality, an analyst would never simply report ‘I can/cannot detect boron in this water sample’. A quantitative method capable of detecting

boron at, say, $1\text{ }\mu\text{g ml}^{-1}$ levels would be used. If it gave a negative result, the outcome would be described in the form, 'This sample contains less than $1\text{ }\mu\text{g ml}^{-1}$ boron'. If the method gave a positive result, the sample will be reported to contain at least $1\text{ }\mu\text{g ml}^{-1}$ boron (with other information too – see below). More complex approaches can be used to compare two soil samples. The soils might be subjected to a particle size analysis, in which the proportions of the soil particles falling within a number, say 10, of particle-size ranges are determined. Each sample would then be characterised by these 10 pieces of data, which can be used (see Chapter 8) to provide a quantitative rather than just a qualitative assessment of their similarity.

1.2 Errors in quantitative analysis

Once we accept that quantitative methods will be the norm in an analytical laboratory, we must also accept that the errors that occur in such methods are of crucial importance. Our guiding principle will be that no *quantitative results are of any value unless they are accompanied by some estimate of the errors inherent in them*. (This principle naturally applies not only to analytical chemistry but to any field of study in which numerical experimental results are obtained.) Several examples illustrate this idea, and they also introduce some types of statistical problem that we shall meet and solve in later chapters.

Suppose we synthesise an analytical reagent which we believe to be entirely new. We study it using a spectrometric method and it gives a value of 104 (normally our results will be given in proper units, but in this hypothetical example we use purely arbitrary units). On checking the reference books, we find that no compound previously discovered has given a value above 100 when studied by the same method in the same experimental conditions. So have we really discovered a new compound? The answer clearly lies in the reliance that we can place on that experimental value of 104. What errors are associated with it? If further work suggests that the result is correct to within 2 (arbitrary) units, i.e. the true value probably lies in the range 104 ± 2 , then a new compound has probably been discovered. But if investigations show that the error may amount to 10 units (i.e. 104 ± 10), then it is quite likely that the true value is actually less than 100, in which case a new discovery is far from certain. So our knowledge of the experimental errors is crucial (in this and every other case) to the proper interpretation of the results. Statistically this example involves the comparison of our experimental result with an assumed or reference value: this topic is studied in detail in Chapter 3.

Analysts commonly perform several replicate determinations in the course of a single experiment. (The value and significance of such replicates is discussed in detail in the next chapter.) Suppose we perform a titration four times and obtain values of 24.69, 24.73, 24.77 and 25.39 ml. (Note that titration values are reported to the nearest 0.01 ml: this point is also discussed in Chapter 2.) All four values are different, because of the errors inherent in the measurements, and the fourth value (25.39 ml) is substantially different from the other three. So can this fourth value be safely rejected, so that (for example) the mean result is reported as 24.73 ml, the average of the other three readings? In statistical terms, is the value 25.39 ml an *outlier*? The major topic of outlier rejection is discussed in detail in Chapters 3 and 6.

Another frequent problem involves the comparison of two (or more) sets of results. Suppose we measure the vanadium content of a steel sample by two separate methods. With the first method the average value obtained is 1.04%, with an estimated error of 0.07%, and with the second method the average value is 0.95%, with an error of 0.04%. Several questions then arise. Are the two average values significantly different, or are they indistinguishable within the limits of the experimental errors? Is one method significantly less error-prone than the other? Which of the mean values is actually closer to the truth? Again, Chapter 3 discusses these and related questions.

Many instrumental analyses are based on graphical methods. Instead of making repeated measurements on the same sample, we perform a series of measurements on a small group of *standards* containing known analyte concentrations covering a considerable range. The results yield a calibration graph that is used to estimate by interpolation the concentrations of *test samples* ('unknowns') studied by the same procedure. All the measurements on the standards and on the test samples will be subject to errors. We shall need to assess the errors involved in drawing the calibration graph, and the error in the concentration of a single sample determined using the graph. We can also estimate the limit of detection of the method, i.e. the smallest quantity of analyte that can be detected with a given degree of confidence. These and related methods are described in Chapter 5.

These examples represent only a small fraction of the possible problems arising from the occurrence of experimental errors in quantitative analysis. All such problems have to be solved if the quantitative data are to have any real meaning, so clearly we must study the various types of error in more detail.

1.3 Types of error

Experimental scientists make a fundamental distinction between three types of error. These are known as **gross**, **random** and **systematic** errors. Gross errors are readily described: they are so serious that there is no alternative to abandoning the experiment and making a completely fresh start. Examples include a complete instrument breakdown, accidentally dropping or discarding a crucial sample, or discovering during the course of the experiment that a supposedly pure reagent was in fact badly contaminated. Such errors (which occur even in the best laboratories!) are normally easily recognised. But we still have to distinguish carefully between **random** and **systematic** errors.

We can make this distinction by careful study of a real experimental situation. Four students (A–D) each perform an analysis in which *exactly* 10.00 ml of *exactly* 0.1 M sodium hydroxide is titrated with *exactly* 0.1 M hydrochloric acid. Each student performs five replicate titrations, with the results shown in Table 1.1.

The results obtained by student A have two characteristics. First, they are all very close to each other; all the results lie between 10.08 and 10.12 ml. In everyday terms we would say that the results are highly *repeatable*. The second feature is that they are *all too high*: in this experiment (somewhat unusually) we know the correct answer: the result should be exactly 10.00 ml. Evidently two entirely separate types of error have occurred. First, there are **random errors** – *these cause replicate results to*

Table 1.1 Data demonstrating random and systematic errors

Student	Results (ml)					Comment
A	10.08	10.11	10.09	10.10	10.12	Precise, biased
B	9.88	10.14	10.02	9.80	10.21	Imprecise, unbiased
C	10.19	9.79	9.69	10.05	9.78	Imprecise, biased
D	10.04	9.98	10.02	9.97	10.04	Precise, unbiased

differ from one another, so that the individual results fall on both sides of the average value (10.10 ml in this case). Random errors affect the **precision**, or **repeatability**, of an experiment. In the case of student A it is clear that the random errors are small, so we say that the results are **precise**. In addition, however, there are **systematic errors** – *these cause all the results to be in error in the same sense* (in this case they are all too high). The total systematic error (in a given experiment there may be several sources of systematic error, some positive and others negative; see Chapter 2) is called the **bias** of the measurement. (The opposite of bias, or lack of bias, is sometimes referred to as **trueness** of a method: see Section 4.15.) The random and systematic errors here are readily distinguishable by inspection of the results, and may also have quite distinct causes in terms of experimental technique and equipment (see Section 1.4). We can extend these principles to the data obtained by student B, which are in direct contrast to those of student A. The average of B's five results (10.01 ml) is very close to the true value, so there is no evidence of bias, but the spread of the results is very large, indicating poor precision, i.e. substantial random errors. Comparison of these results with those obtained by student A shows clearly that random and systematic errors can occur independently of one another. This conclusion is reinforced by the data of students C and D. Student C's work has poor precision (range 9.69–10.19 ml) and the average result (9.90 ml) is (negatively) biased. Student D has achieved both precise (range 9.97–10.04 ml) and unbiased (average 10.01 ml) results. The distinction between random and systematic errors is summarised in Table 1.2, and in Fig. 1.1 as a series of *dot-plots*. This simple graphical method of displaying data, in which individual results are plotted as dots on a linear scale, is frequently used in exploratory data analysis (EDA, also called initial data analysis, IDA: see Chapters 3 and 6).

Table 1.2 Random and systematic errors

Random errors	Systematic errors
Affect precision – repeatability or reproducibility	Produce bias – an overall deviation of a result from the true value even when random errors are very small
Cause replicate results to fall on either side of a mean value	Cause all results to be affected in one sense only, all too high or all too low
Can be estimated using replicate measurements	Cannot be detected simply by using replicate measurements
Can be minimised by good technique but not eliminated	Can be corrected, e.g. by using standard methods and materials
Caused by both humans and equipment	Caused by both humans and equipment

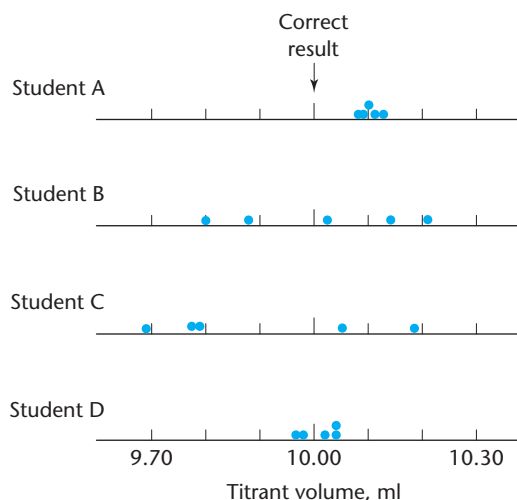


Figure 1.1 Bias and precision: dot-plots of the data in Table 1.1.

In most analytical experiments the most important question is, how far is the result from the true value of the concentration or amount that we are trying to measure? This is expressed as the **accuracy** of the experiment. Accuracy is defined by the International Organization for Standardization (ISO) as ‘the closeness of agreement between a test result and the accepted reference value’ of the analyte. Under this definition the accuracy of a single result may be affected by *both* random and systematic errors. The accuracy of an average result also has contributions from both error sources: even if systematic errors are absent, the average result will probably not equal the reference value exactly, because of the occurrence of random errors (see Chapters 2 and 3). The results obtained by student B demonstrate this. Four of B’s five measurements show significant inaccuracy, i.e. are well removed from the true value of 10.00. But the average of the results (10.01) is very accurate, so it seems that the inaccuracy of the individual results is due largely to random errors and not to systematic ones. By contrast, all of student A’s individual results, and the resulting average, are inaccurate: given the good precision of A’s work, it seems certain that these inaccuracies are due to systematic errors. Note that, contrary to the implications of many dictionaries, accuracy and precision have entirely different meanings in the study of experimental errors.

In summary, precision describes random error, bias describes systematic error and the accuracy, i.e. closeness to the true value of a single measurement or a mean value, incorporates both types of error.

Another important area of terminology is the difference between **reproducibility** and **repeatability**. We can illustrate this using the students’ results again. In the normal way each student would do the five replicate titrations in rapid succession, taking only an hour or so. The same set of solutions and the same glassware would be used throughout, the same preparation of indicator would be added to each titration flask, and the temperature, humidity and other laboratory conditions would remain much the same. In such cases the precision measured would be the

within-run precision: this is called the **repeatability**. Suppose, however, that for some reason the titrations were performed by different staff on five different occasions in different laboratories, using different pieces of glassware and different batches of indicator. It would not be surprising to find a greater spread of the results in this case. The resulting data would reflect the *between-run* precision of the method, i.e. its **reproducibility**.

- **Repeatability** describes the precision of within-run replicates.
- **Reproducibility** describes the precision of between-run replicates.
- The reproducibility of a method is normally expected to be poorer (i.e. with larger random errors) than its repeatability.

One further lesson may be learned from the titration experiments. Clearly the data obtained by student C are unacceptable, and those of student D are the best. Sometimes, however, two methods may be available for a particular analysis, one of which is believed to be precise but biased, and the other imprecise but without bias. In other words we may have to choose between the types of results obtained by students A and B respectively. Which type of result is preferable? It is impossible to give a dogmatic answer to this question, because in practice the choice of analytical method will often be based on the cost, ease of automation, speed of analysis, and so on. But it is important to realise that a method which is substantially free from systematic errors may still, if it is very imprecise, give an average value that is (by chance) a long way from the correct value. On the other hand a method that is precise but biased (e.g. student A) can be converted into one that is both precise *and* unbiased (e.g. student D) *if the systematic errors can be discovered and hence removed*. Random errors can never be eliminated, though by careful technique we can minimise them, and by making repeated measurements we can measure them and evaluate their significance. Systematic errors can in many cases be removed by careful checks on our experimental technique and equipment. This crucial distinction between the two major types of error is further explored in the next section.

When an analytical laboratory is supplied with a sample and requested to determine the concentrations of one of its constituents, it will estimate, or perhaps know from previous experience, the extent of the major random and systematic errors occurring. The customer supplying the sample may well want this information incorporated in a single statement, giving the *range within which the true concentration is reasonably likely to lie*. This range, which should be given with a probability (e.g. 'it is 95% probable that the concentration lies between . . . and . . .'), is called the **uncertainty** of the measurement. Uncertainty estimates are now very widely used in analytical chemistry and are discussed in more detail in Chapter 4.

1.4

Random and systematic errors in titrimetric analysis

The students' titrimetric experiments showed clearly that random and systematic errors can occur independently of one another, and thus presumably arise at different stages of an experiment. A complete titrimetric analysis can be summarised by the following steps:

- 1 Making up a standard solution of one of the reactants. This involves (a) weighing a weighing bottle or similar vessel containing some solid material, (b) transferring the solid material to a standard flask and weighing the bottle again to obtain by subtraction the weight of solid transferred (weighing *by difference*), and (c) filling the flask up to the mark with water (assuming that an aqueous titration is to be used).
- 2 Transferring an aliquot of the standard material to a titration flask by filling and draining a pipette properly.
- 3 Titrating the liquid in the flask with a solution of the other reactant, added from a burette. This involves (a) filling the burette and allowing the liquid in it to drain until the meniscus is at a constant level, (b) adding a few drops of indicator solution to the titration flask, (c) reading the initial burette volume, (d) adding liquid to the titration flask from the burette until the end point is adjudged to have been reached, and (e) measuring the final level of liquid in the burette.

So the titration involves some ten separate steps, the last seven of which are normally repeated several times, giving replicate results. In principle, we should examine each step to evaluate the random and systematic errors that might occur. In practice, it is simpler to examine separately those stages which utilise weighings (steps 1(a) and 1(b)), and the remaining stages involving the use of volumetric equipment. (It is not intended to give detailed descriptions of the experimental techniques used in the various stages. Similarly, methods for calibrating weights, glassware, etc. will not be given.) The tolerances of weights used in the gravimetric steps, and of the volumetric glassware, may contribute significantly to the experimental errors. Specifications for these tolerances are issued by such bodies as the British Standards Institute (BSI) and the American Society for Testing and Materials (ASTM). The tolerance of a top-quality 100 g weight can be as low as ± 0.25 mg, although for a weight used in routine work the tolerance would be up to four times as large. Similarly the tolerance for a grade A 250 ml standard flask is ± 0.12 ml: grade B glassware generally has tolerances twice as large as grade A glassware. If a weight or a piece of glassware is within the tolerance limits, but not of exactly the correct weight or volume, a systematic error will arise. Thus, if the standard flask actually has a volume of 249.95 ml, this error will be reflected in the results of all the experiments based on the use of that flask. Repetition of the experiment will not reveal the error: in each replicate the volume will be assumed to be 250.00 ml when in fact it is less than this. If, however, the results of an experiment using this flask are compared with the results of several other experiments (e.g. in other laboratories) done with other flasks, then if all the flasks have slightly different volumes they will contribute to the random variation, i.e. the reproducibility, of the results.

Weighing procedures are normally associated with very small *random* errors. In routine laboratory work a 'four-place' balance is commonly used, and the random error involved should not be greater than ca. 0.0002 g (the next chapter describes in detail the statistical terms used to express random errors). Since the quantity being weighed is normally of the order of 1 g or more, the random error, expressed as a percentage of the weight involved, is not more than 0.02%. A good standard material for volumetric analysis should (amongst other properties) have as high a formula weight as possible, to minimise these random weighing errors when a solution of a specified molarity is being made up.

Systematic errors in weighings can be appreciable, arising from adsorption of moisture on the surface of the weighing vessel; corroded or dust-contaminated weights;

and the buoyancy effect of the atmosphere, acting to different extents on objects of different density. For the best work, weights must be calibrated against standards provided by statutory bodies and authorities (see above). This calibration can be very accurate indeed, e.g. to ± 0.01 mg for weights in the range 1–10 g. Some simple experimental precautions can be taken to minimise these systematic weighing errors. Weighing by difference (see above) cancels systematic errors arising from (for example) the moisture and other contaminants on the surface of the bottle. (See also Section 2.12.) *If such precautions are taken*, the errors in the weighing steps will be small, and in most volumetric experiments weighing errors will probably be negligible compared with the volumetric ones. Indeed, gravimetric methods are usually used for the calibration of items of volumetric glassware, by weighing (in standard conditions) the water that they contain or deliver, and standards for top-quality calibration experiments (Chapter 5) are made up by weighing rather than volume measurements.

Most of the *random* errors in volumetric procedures arise in the use of volumetric glassware. In filling a 250 ml standard flask to the mark, the error (i.e. the distance between the meniscus and the mark) might be about ± 0.03 cm in a flask neck of diameter ca. 1.5 cm. This corresponds to a volume error of about 0.05 ml – only 0.02% of the total volume of the flask. The error in reading a burette (the conventional type graduated in 0.1 ml divisions) is perhaps 0.01–0.02 ml. Each titration involves two such readings (the errors of which are *not* simply additive – see Chapter 2); if the titration volume is ca. 25 ml, the percentage error is again very small. The experiment should be arranged so that the volume of titrant is not too small (say not less than 10 ml), otherwise such errors may become appreciable. (This precaution is analogous to choosing a standard compound of high formula weight to minimise the weighing error.) Even though a volumetric analysis involves several steps, each involving a piece of volumetric glassware, the random errors should evidently be small if the experiments are performed with care. In practice a good volumetric analysis should have a relative standard deviation (see Chapter 2) of not more than about 0.1%. Until fairly recently such precision was not normally attainable in instrumental analysis methods, and it is still not very common.

Volumetric procedures incorporate several important sources of systematic error: the drainage errors in the use of volumetric glassware, calibration errors in the glassware and ‘indicator errors’. Perhaps the commonest error in routine volumetric analysis is to fail to allow enough time for a pipette to drain properly, or a meniscus level in a burette to stabilise. The temperature at which an experiment is performed has two effects. Volumetric equipment is conventionally calibrated at 20 °C, but the temperature in an analytical laboratory may easily be several degrees different from this, and many experiments, for example in biochemical analysis, are carried out in ‘cold rooms’ at ca. 4 °C. The temperature affects both the volume of the glassware and the density of liquids.

Indicator errors can be quite substantial, perhaps larger than the random errors in a typical titrimetric analysis. For example, in the titration of 0.1 M hydrochloric acid with 0.1 M sodium hydroxide, we expect the end point to correspond to a pH of 7. In practice, however, we estimate this end point using an indicator such as methyl orange. Separate experiments show that this substance changes colour over the pH range ca. 3–4. If, therefore, the titration is performed by adding alkali to acid, the indicator will yield an apparent end point when the pH is ca. 3.5, i.e. just before the true end point. The error can be evaluated and corrected by doing a *blank* experiment, i.e. by determining how much alkali is required to produce the indicator colour change in the *absence* of the acid.

It should be possible to consider and estimate the sources of random and systematic error arising at each distinct stage of an analytical experiment. It is very desirable to do this, so as to avoid major sources of error by careful experimental design (Sections 1.5 and 1.6). In many analyses (though not normally in titrimetry) the overall error is in practice dominated by the error in a single step: this point is further discussed in the next chapter.

1.5 Handling systematic errors

Much of the rest of this book will deal with the handling of random errors, using a wide range of statistical methods. In most cases we shall assume that systematic errors are absent (though methods which test for the occurrence of systematic errors will be described). So at this stage we must discuss systematic errors in more detail – how they arise, and how they may be countered. The example of the titrimetric analysis given above shows that systematic errors cause the mean value of a set of replicate measurements to deviate from the true value. It follows that (a) in contrast to random errors, systematic errors cannot be revealed merely by making repeated measurements, and that (b) unless the true result of the analysis is known in advance – an unlikely situation! – very large systematic errors might occur but go entirely undetected unless suitable precautions are taken. That is, it is all too easy totally to overlook substantial sources of systematic error. A few examples will clarify both the possible problems and their solutions.

The levels of transition metals in biological samples such as blood serum are important in many biomedical studies. For many years determinations were made of the levels of (for example) chromium in serum – with some startling results. Different workers, all studying pooled serum samples from healthy subjects, published chromium concentrations varying from <1 to ca. 200 ng ml^{-1} . In general the lower results were obtained later than the higher ones, and it gradually became apparent that the earlier values were due at least in part to contamination of the samples by chromium from stainless-steel syringes, tube caps, and so on. The determination of traces of chromium, e.g. by atomic-absorption spectrometry, is in principle relatively straightforward, and no doubt each group of workers achieved results which seemed satisfactory in terms of precision; but in a number of cases the large systematic error introduced by the contamination was entirely overlooked. Similarly the normal levels of iron in seawater are now known to be in the parts per billion (ng ml^{-1}) range, but until fairly recently the concentration was thought to be much higher, perhaps tens of $\mu\text{g ml}^{-1}$. This misconception arose from the practice of sampling and analysing seawater in ship-borne environments containing high ambient iron levels. Methodological systematic errors of this kind are extremely common.

Another class of systematic error occurs widely when false assumptions are made about the accuracy of an analytical instrument. A monochromator in a spectrometer may gradually go out of adjustment, so that errors of several nanometres in wavelength settings arise, yet many photometric analyses are undertaken without appropriate checks being made. Very simple devices such as volumetric glassware, stopwatches, pH meters and thermometers can all show substantial systematic errors, but many laboratory workers use them as though they are without bias. Most

instrumental analysis systems are now wholly controlled by computers, minimising the number of steps and the skill levels required in many experiments. It is very tempting to regard results from such instruments as beyond reproach, but (unless the devices are 'intelligent' enough to be self-calibrating – see Section 1.7) they are still subject to systematic errors.

Systematic errors arise not only from procedures or apparatus; they can also arise from human bias. Some chemists suffer from astigmatism or colour-blindness (the latter is more common among men than women) which might introduce errors into their readings of instruments and other observations. A number of authors have reported various types of number bias, for example a tendency to favour even over odd numbers, or 0 and 5 over other digits, in the reporting of results. In short, systematic errors of several kinds are a constant, and often hidden, risk for the analyst, so very careful steps to minimise them must be taken.

Several approaches to this problem are available, and any or all of them should be considered in each analytical procedure. The first precautions should be taken before any experimental work is begun. The analyst should consider carefully each stage of the experiment to be performed, the apparatus to be used and the sampling and analytical procedures to be adopted. At this early stage the likely sources of systematic error, such as the instrument functions that need calibrating, and the steps of the analytical procedure where errors are most likely to occur, and the checks that can be made during the analysis, must be identified. Foresight of this kind can be very valuable (the next section shows that similar advance attention should be given to the sources of random error) and is normally well worth the time invested. For example, a little thinking of this kind might well have revealed the possibility of contamination in the serum chromium determinations described above.

The second line of defence against systematic errors lies in the design of the experiment at every stage. We have already seen (Section 1.4) that weighing by difference can remove some systematic gravimetric errors: these can be assumed to occur to the same extent in both weighings, so the subtraction process eliminates them. Another example of careful experimental planning is provided by the spectrometer wavelength error described above. If the concentration of a sample of a single material is to be determined by absorption spectrometry, two procedures are possible. In the first, the sample is studied in a 1 cm pathlength spectrometer cell at a single wavelength, say 400 nm, and the concentration of the test component is determined from the well-known equation $A = \epsilon bc$ (where A , ϵ , c and b are the measured absorbance, a published value of the molar absorptivity (with units $\text{l mole}^{-1} \text{ cm}^{-1}$) of the test component, the molar concentration of this analyte, and the pathlength (cm) of the spectrometer cell) respectively. Several systematic errors can arise here. The wavelength might, as already discussed, be (say) 405 nm rather than 400 nm, thus rendering the published value of ϵ inappropriate; this published value might in any case be wrong; the absorbance scale of the spectrometer might exhibit a systematic error; and the pathlength of the cell might not be exactly 1 cm. Alternatively, the analyst might use the calibration graph approach outlined in Section 1.2 and discussed in detail in Chapter 5. In this case the value of ϵ is not required, and the errors due to wavelength shifts, absorbance errors and pathlength inaccuracies should cancel out, as they occur equally in the calibration and test experiments. If the conditions are truly equivalent for the test and calibration samples (e.g. the same cell is used and the wavelength and absorbance scales do not alter *during the experiment*) all the major sources of systematic error are in principle eliminated.

The final and perhaps most formidable protection against systematic errors is the use of **standard reference materials** and methods. Before the experiment is started, each piece of apparatus is calibrated by an appropriate procedure. We have seen that volumetric equipment can be calibrated by the use of gravimetric methods. Similarly, spectrometer wavelength scales can be calibrated with the aid of standard light sources which have narrow emission lines at well-established wavelengths, and spectrometer absorbance scales can be calibrated with standard solid or liquid filters. Most pieces of equipment can be calibrated so that their systematic errors are known in advance. The importance of this area of chemistry (and other experimental sciences) is reflected in the extensive work of bodies such as the National Physical Laboratory and LGC (in the UK), the National Institute for Science and Technology (NIST) (in the USA) and similar organisations elsewhere. Whole volumes have been written on the standardisation of particular types of equipment, and a number of commercial organisations specialise in the sale of certified reference materials (CRMs).

A further check on the occurrence of systematic errors in a method is to compare the results with those obtained from a different method. If two unrelated methods are used to perform one analysis, and if they consistently yield results showing only random differences, it is a reasonable presumption that no significant systematic errors are present. For this approach to be valid, *each step* of the two analyses has to be independent. Thus in the case of serum chromium determinations, it would not be sufficient to replace the atomic-absorption spectrometry method by a colorimetric one or by plasma spectrometry. The systematic errors would only be revealed by altering the sampling methods also, e.g. by minimising or eliminating the use of stainless-steel equipment. Moreover such comparisons must be made over the whole of the concentration range for which an analytical procedure is to be used. For example, the bromocresol green dye-binding method for the determination of albumin in blood serum agrees well with alternative methods (e.g. immunological ones) at normal or high levels of albumin, but when the albumin levels are abnormally low (these are the cases of most clinical interest, inevitably!) the agreement between the two methods is poor, the dye-binding method giving consistently (and erroneously) higher values. The statistical approaches used in method comparisons are described in detail in Chapters 3 and 5.

The prevalence of systematic errors in everyday analytical work is well illustrated by the results of **collaborative trials (method performance studies)**. If an experienced analyst finds 10 ng ml^{-1} of a drug in a urine sample, it is natural to suppose that other analysts would obtain very similar results for the same sample, any differences being due to random errors only. Unfortunately, this is far from true in practice. Many collaborative studies involving different laboratories, when aliquots of a *single* sample are examined by the same experimental procedures and types of instrument, show variations in the results much greater than those expected from random errors. So in many laboratories substantial systematic errors, both positive and negative, must be going undetected or uncorrected. This situation, which has serious implications for all analytical scientists, has encouraged many studies of the methodology of collaborative trials and **proficiency testing schemes**, and of their statistical evaluation. Such schemes have led to dramatic improvements in the quality of analytical results in a range of fields. These topics are discussed in Chapter 4.

Tackling systematic errors:

- Foresight: identifying problem areas before starting experiments.
- Careful experimental design, e.g. use of calibration methods.
- Checking instrument performance.
- Use of standard reference materials and other standards.
- Comparison with other methods for the same analytes.
- Participation in proficiency testing schemes.

1.6 Planning and design of experiments

Many chemists regard statistical methods only as tools to assess the results of completed experiments. This is indeed a crucial area of application of statistics, but we must also be aware of the importance of statistical concepts in the planning and design of experiments. In the previous section the value of trying to predict systematic errors in advance, thereby permitting the analyst to lay plans for countering them, was emphasised. The same considerations apply to random errors. As we shall see in Chapter 2, combining the random errors of the individual parts of an experiment to give an overall random error requires some simple statistical formulae. In practice, the overall error is often dominated by the error in just one stage of the experiment, the other errors having negligible effects when they are all combined correctly. It is obviously desirable to try to find, *before the experiment begins*, where this single dominant error is likely to arise, and then to try to minimise it. Although random errors can never be eliminated, they can certainly be minimised by particular attention to experimental techniques. For both random and systematic errors, therefore, the moral is clear: every effort must be made to identify the serious sources of error before practical work starts, so that experiments can be designed to minimise such errors.

There is another and more subtle aspect of experimental design. In many analyses, one or more of the desirable features of the method (sensitivity, selectivity, sampling rate, low cost, etc.) will depend on a number of experimental *factors*. We should design the analysis so that we can identify the most important of these factors and then use them in the best combination, thereby obtaining the best sensitivity, selectivity, etc. In the interests of conserving resources, samples, reagents, etc., this process of design and optimisation should again be completed before a method is put into routine or widespread use.

Some of the problems of experimental design and optimisation can be illustrated by a simple example. In enzymatic analyses, the concentration of the analyte is determined by measuring the rate of an enzyme-catalysed reaction. (The analyte is often the substrate, i.e. the compound that is changed in the reaction.) Let us assume that we want the maximum reaction rate in a particular analysis, and that we believe that this rate depends on (amongst other factors) the pH of the reaction mixture and the temperature. How do we establish just how important these factors are, and find their best *levels*, i.e. values? It is easy to identify one possible approach. We could perform a series of experiments in which the temperature is kept constant but the pH is varied. In each case the rate of the reaction would be determined and an optimum

pH value would thus be found – suppose it is 7.5. A second series of reaction-rate experiments could then be performed, with the pH maintained at 7.5 but the temperature varied. An optimum temperature would thus be found, say 40 °C. This approach to studying the factors affecting the experiment is clearly tedious, because in more realistic examples many more than two factors might need investigation. Moreover, if the reaction rate at pH 7.5 and 40 °C was only slightly different from that at (e.g.) pH 7.5 and 37 °C, we would need to know whether the difference was a real one, or merely a result of random experimental errors, and we could distinguish these possibilities only by repeating the experiments. A more fundamental problem is that this ‘one at a time’ approach assumes that the factors affect the reaction rate *independently* of each other, i.e. that the best pH is 7.5 whatever the temperature, and the best temperature is 40 °C at all pH values. This may not be true: for example at a pH other than 7.5 the optimum temperature might not be 40 °C, i.e. the factors may affect the reaction rate *interactively*. It follows that the conditions established in the two sets of experiments just described might not actually be the optimum ones: had the first set of experiments been done at a different pH, a different set of ‘optimum’ values might have been obtained. Experimental design and optimisation can clearly present significant problems. These important topics are considered in more detail in Chapter 7.

1.7

Calculators and computers in statistical calculations

The rapid growth of **chemometrics** – the application of mathematical methods to the solution of chemical problems of all types – is due to the ease with which large quantities of data can be handled, and advanced calculations done, with calculators and computers.

These devices are available to the analytical chemist at several levels of complexity and cost. Handheld calculators are extremely cheap, very reliable and capable of performing many of the routine statistical calculations described in this book with a minimal number of keystrokes. Pre-programmed functions allow calculations of mean and standard deviation (see Chapter 2) and correlation and linear regression (see Chapter 5). Other calculators can be programmed by the user to perform additional calculations such as confidence limits (see Chapter 2), significance tests (see Chapter 3) and non-linear regression (see Chapter 5). For those performing analytical research or routine analyses such calculators will be more than adequate. Their main disadvantage is their inability to handle very large quantities of data.

Most modern analytical instruments (some entirely devoid of manual controls) are controlled by personal computers which also handle and report the data obtained. Portable computers facilitate the recording and calculation of data in the field, and are readily linked to their larger cousins on returning to the laboratory. Additional functions can include checking instrument performance, diagnosing and reporting malfunctions, storing large databases (e.g. of digitised spectra), comparing analytical data with the databases, optimising operating conditions (see Chapter 7), and selecting and using a variety of calibration calculations.

A wealth of excellent general statistical software is available. The memory size and speed of computers are now sufficient for work with all but the largest data sets, and word processors greatly aid the production of analytical reports and papers.

Spreadsheet programs, originally designed for financial calculations, are often invaluable for statistical work, having many built-in statistical functions and excellent graphical presentation facilities. The popularity of spreadsheets derives from their speed and simplicity in use, and their ability to perform almost instant ‘what if?’ calculations: for example, what would the mean and standard deviation of a set of results be if one suspect piece of data is omitted? Spreadsheets are designed to facilitate rapid data entry, and data in spreadsheet format can easily be exported to the more specialist suites of statistics software. **Microsoft Excel**[®] is the most widely used spreadsheet, and offers most of the statistical facilities that users of this book may need. Several examples of its application are provided in later chapters, and helpful texts are listed in the bibliography.

More advanced calculation facilities are provided by specialised suites of statistical software. Amongst these, **Minitab**[®] is very widely used in educational establishments and research laboratories. In addition to the expected simple statistical functions it offers many more advanced calculations, including multivariate methods (see Chapter 8), exploratory data analysis (EDA) and non-parametric tests (see Chapter 6), experimental design (see Chapter 7) and many quality control methods (see Chapter 4). More specialised and excellent programs for various types of multivariate analysis are also available: the best known is **The Unscrambler**[®]. New and updated versions of these programs, with extra facilities and/or improved user interfaces, appear at regular intervals. Although help facilities are always built in, such software is really designed for users rather than students, and does not have a strongly tutorial emphasis. But a program specifically designed for tutorial purposes, **VAMSTAT**[®], is a valuable tool, with on-screen tests for students and clear explanations of many important methods.

A group of computers in separate laboratories can be ‘networked’, i.e. linked so that both operating software and data can be freely passed from one to another. A major use of networks is the establishment of Laboratory Information Management Systems (LIMS), which allow large numbers of analytical specimens to be identified and tracked as they move through one or more laboratories. Samples are identified and tracked by bar-coding or similar systems, and the computers attached to a range of instruments send their analytical results to a central computer which (for example) prints a summary report, including a statistical evaluation.

It must be emphasised that the availability of calculators and computers makes it all the more important that their users understand the principles underlying statistical calculations. Such devices will rapidly perform any statistical test or calculation selected by the user, *whether or not that procedure is suitable for the data under study*. For example, a linear least-squares program will determine a straight line to fit *any* set of x - and y -values, even in cases where visual inspection would show that such a program is wholly inappropriate (see Chapter 5). Similarly a simple program for testing the significance of the difference between the means of two data sets may assume that the variances (see Chapter 2) of the two sets are similar: but the program will blindly perform the calculation on request and provide a ‘result’ even if the variances actually differ significantly. Even comprehensive suites of computer programs often fail to provide advice on the right choice of statistical method for a given set of data. The analyst must thus use both statistical know-how and common sense to ensure that the correct calculation is performed.

Bibliography and resources

Books

- Diamond, D. and Hanratty, V.C.A., 1997, *Spreadsheet Applications in Chemistry Using Microsoft Excel®*, Wiley-Interscience, New York. Clear guidance on the use of Excel, with examples from analytical and physical chemistry.
- Ellison, S.L.R., Barwick, V.J. and Farrant, T.J.D., 2009, *Practical Statistics for the Analytical Scientist*, Royal Society of Chemistry, Cambridge. An excellent summary of basic methods, with worked examples.
- Middleton, M.R., 2003, *Data Analysis Using Microsoft Excel® Updated for Windows XP*, Duxbury Press, Belmont, CA. Many examples of scientific calculations, illustrated by screenshots. Earlier editions cover previous versions of Excel.
- Mullins, E., 2003, *Statistics for the Quality Control Laboratory*, Royal Society of Chemistry, Cambridge. The topics covered by this book are wider than its title suggests and it contains many worked examples.
- Neave, H.R., 1981, *Elementary Statistics Tables*, Routledge, London. Good statistical tables are required by all users of statistics: this set is strongly recommended because it is fairly comprehensive, and contains explanatory notes and useful examples with each table.

Software

- VAMSTAT II®** is a CD-ROM-based learning package on statistics for students and workers in analytical chemistry. On-screen examples and tests are provided and the package is very suitable for self-study, with a simple user interface. Multivariate statistical methods are not covered. Single-user and site licences are available, from LGC, Queens Road, Teddington, Middlesex TW11 0LY, UK.
- Teach Me Data Analysis**, by H. Lohninger (Springer, Berlin, 1999) is a CD-ROM-based package with an explanatory booklet (single- and multi-user versions are available). It is designed for the interactive learning of statistical methods, and covers most areas of basic statistics along with some treatment of multivariate methods. Many of the examples are based on analytical chemistry.
- Minitab®** is a long-established and very widely used statistical package, which has now reached version 15. The range of methods covered is immense, and the software comes with extensive instruction manuals, a users' website with regular updates, newsletters and responses to user problems. There are also separate user groups that provide further help, and many independently written books giving further examples. A trial version is available for downloading from www.minitab.com.
- Microsoft Excel®**, the well-known spreadsheet program normally available as part of the **Microsoft Office®** suite, offers many basic statistical functions, and numerous web-based add-ons have been posted by users. The most recent versions of Excel are recommended, as some of the drawbacks of the statistical facilities in earlier versions have been remedied. Many books on the use of Excel in scientific data analysis are also available.

Exercises

- 1 A standard sample of pooled human blood serum contains 42.0 g of albumin per litre. Five laboratories (A–E) each do six determinations (on the same day) of the albumin concentration, with the following results (g l^{-1} throughout):

A	42.5	41.6	42.1	41.9	41.1	42.2
B	39.8	43.6	42.1	40.1	43.9	41.9
C	43.5	42.8	43.8	43.1	42.7	43.3
D	35.0	43.0	37.1	40.5	36.8	42.2
E	42.2	41.6	42.0	41.8	42.6	39.0

Comment on the bias, precision and accuracy of each of these sets of results.

- 2 Using the same sample and method as in question 1, laboratory A makes six further determinations of the albumin concentration, this time on six successive days. The values obtained are 41.5, 40.8, 43.3, 41.9, 42.2 and 41.7 g l^{-1} . Comment on these results.
- 3 The number of binding sites per molecule in a sample of monoclonal antibody is determined four times, with results of 1.95, 1.95, 1.92 and 1.97. Comment on the bias, precision and accuracy of these results.
- 4 Discuss the degrees of bias and precision desirable or acceptable in the following analyses:
- Determination of the lactate concentration of human blood samples.
 - Determination of uranium in an ore sample.
 - Determination of a drug in blood plasma after an overdose.
 - Study of the stability of a colorimetric reagent by determination of its absorbance at a single wavelength over a period of several weeks.
- 5 For each of the following experiments, try to identify the major probable sources of random and systematic errors, and consider how such errors may be minimised:
- The iron content of a large lump of ore is determined by taking a single small sample, dissolving it in acid, and titrating with ceric sulphate after reduction of Fe(III) to Fe(II).
 - The same sampling and dissolution procedure is used as in (a) but the iron is determined colorimetrically after addition of a chelating reagent and extraction of the resulting coloured and uncharged complex into an organic solvent.
 - The sulphate content of an aqueous solution is determined gravimetrically with barium chloride as the precipitant.

2

Statistics of repeated measurements



Major topics covered in this chapter

- Measures of location and spread; mean, standard deviation, variance
- Normal and log-normal distributions; samples and populations
- Sampling distribution of the mean; central limit theorem
- Confidence limits and intervals
- Presentation and rounding of results
- Propagation of errors in multi-stage experiments

2.1

Mean and standard deviation

In Chapter 1 we saw that it is usually necessary to make repeated measurements in many analytical experiments in order to reveal the presence of random errors. This chapter applies some fundamental statistical concepts to such a situation. We will start by looking again at the example in Chapter 1 which considered the results of five replicate titrations done by each of four students. These results are reproduced below.

Student	Results (ml)				
A	10.08	10.11	10.09	10.10	10.12
B	9.88	10.14	10.02	9.80	10.21
C	10.19	9.79	9.69	10.05	9.78
D	10.04	9.98	10.02	9.97	10.04

Two criteria were used to compare these results, the average value (technically known as a measure of location) and the degree of spread (or dispersion). The average value used was the **arithmetic mean**, $\bar{x}$ (usually abbreviated to the **mean**),

which is the sum of all the measurements, $\sum x_i$, divided by the number of measurements, n .

$$\text{The mean, } \bar{x}, \text{ of } n \text{ measurements is given by } \bar{x} = \frac{\sum x_i}{n} \quad (2.1.1)$$

In Chapter 1 the spread was measured by the difference between the highest and lowest values. A more useful measure, which utilises all the values, is the **standard deviation**, s , which is defined as follows:

The standard deviation, s , of n measurements is given by

$$s = \sqrt{\sum_i (x_i - \bar{x})^2 / (n - 1)} \quad (2.1.2)$$

The calculation of these statistics can be illustrated by an example.

Example 2.1.1

Find the mean and standard deviation of A's results.

	x_i	$(x_i - \bar{x})$	$(x_i - \bar{x})^2$
	10.08	-0.02	0.0004
	10.11	0.01	0.0001
	10.09	-0.01	0.0001
	10.10	0.00	0.0000
	10.12	0.02	0.0004
Totals	50.50	0	0.0010

$$\bar{x} = \frac{\sum x_i}{n} = \frac{50.50}{5} = 10.1 \text{ ml}$$

$$s = \sqrt{\sum_i (x_i - \bar{x})^2 / (n - 1)} = \sqrt{0.001/4} = 0.0158 \text{ ml}$$

Note that $\sum (x_i - \bar{x})$ is always equal to 0.

The answers to this example have been arbitrarily given to three significant figures: further discussion of this important aspect of the presentation of results is considered in Section 2.8. The reader can check that the standard deviations of the results of students B, C and D are 0.172, 0.210 and 0.0332 ml respectively, giving quantitative confirmation of the assessments of precision made in Chapter 1.

In practice all calculators will give these results if the values of x_i are keyed in. However, care must be taken that the correct key is pressed to obtain the standard deviation. Some calculators give two different values for the standard deviation, one calculated by using Eq. (2.1.2) and the other calculated with the denominator of this

equation, i.e. $(n - 1)$, replaced by n . (The reason for this is explained below, p. 20.) Obviously, for large values of n the difference is negligible. Alternatively, readily available computer software can be used to perform these calculations (see Chapter 1).

The square of s is a very important statistical quantity known as the **variance**; its value will become apparent later in this chapter when we consider the propagation of errors.

$$\text{Variance} = \text{the square of the standard deviation, } s^2 \quad (2.1.3)$$

Another widely used measure of spread is the **coefficient of variation (CV)**, also known as the **relative standard deviation (RSD)**, which is given by $100 s/\bar{x}$.

$$\text{Coefficient of variation (CV)} = \text{relative standard deviation (RSD)} = 100 s/\bar{x} \quad (2.1.4)$$

The CV or RSD, the units of which are obviously per cent, is an example of a **relative error**, i.e. an error estimate divided by an estimate of the absolute value of the measured quantity. Relative errors are often used to compare the precision of results which have different units or magnitudes, and are again important in calculations of error propagation.

2.2

The distribution of repeated measurements

Although the standard deviation gives a measure of the spread of a set of results about the mean value, it does not indicate the shape of the distribution. To illustrate this we need quite a large number of measurements such as those in Table 2.1. This gives the results (to two significant figures) of 50 replicate determinations of the levels of nitrate ion, a potentially harmful contaminant, in a particular water specimen.

These results can be summarised in a **frequency table** (Table 2.2). This table shows that, in Table 2.1, the value $0.46 \mu\text{g ml}^{-1}$ appears once, the value $0.47 \mu\text{g ml}^{-1}$ appears three times, and so on. The reader can check that the mean of these results is $0.500 \mu\text{g ml}^{-1}$ and the standard deviation is $0.0165 \mu\text{g ml}^{-1}$ (both values being given to three significant figures). The distribution of the results can most easily be appreciated by drawing a **histogram** as in Fig. 2.1. This shows that the distribution of the measurements is roughly symmetrical about the mean, with the measurements clustered towards the centre.

Table 2.1 Results of 50 determinations of nitrate ion concentration, in $\mu\text{g ml}^{-1}$

0.51	0.51	0.51	0.50	0.51	0.49	0.52	0.53	0.50	0.47
0.51	0.52	0.53	0.48	0.49	0.50	0.52	0.49	0.49	0.50
0.49	0.48	0.46	0.49	0.49	0.48	0.49	0.49	0.51	0.47
0.51	0.51	0.51	0.48	0.50	0.47	0.50	0.51	0.49	0.48
0.51	0.50	0.50	0.53	0.52	0.52	0.50	0.50	0.51	0.51

Table 2.2 Frequency table for measurements of nitrate ion concentration

Nitrate ion concentration ($\mu\text{g ml}^{-1}$)	Frequency
0.46	1
0.47	3
0.48	5
0.49	10
0.50	10
0.51	13
0.52	5
0.53	3

This set of 50 measurements is a **sample** from the theoretically infinite number of measurements which we could make of the nitrate ion concentration. The set of all possible measurements is called the **population**. *If there are no systematic errors*, then the mean of this population, given the symbol μ , is the true value of the nitrate ion concentration we are trying to determine. The mean of the sample, $\bar{x}$, gives us an *estimate of μ* . Similarly, the population has a standard deviation, denoted by $\sigma = \sqrt{\sum_i (x_i - \mu)^2 / n}$. The standard deviation, s , of the sample gives us an *estimate of σ* . Use of Eq. (2.1.2) gives us an *unbiased* estimate of σ . If n , rather than $(n - 1)$, is used in the denominator of the equation the value of s obtained tends to underestimate σ (see p. 19 above).

The distinction between populations and samples is fundamental in statistics: the properties of populations have Greek symbols, samples have English symbols.

The nitrate ion concentrations given in Table 2.2 have only certain discrete values, because of the limitations of the method of measurement. In theory a concentration could take any value, so a continuous curve is needed to describe the form of the population from which the sample was taken. The mathematical model usually used is the **normal** or **Gaussian distribution** which is described by the equation:

$$y = \frac{1}{\sigma\sqrt{2\pi}} \exp[-(x - \mu)^2 / 2\sigma^2] \quad (2.2.1)$$

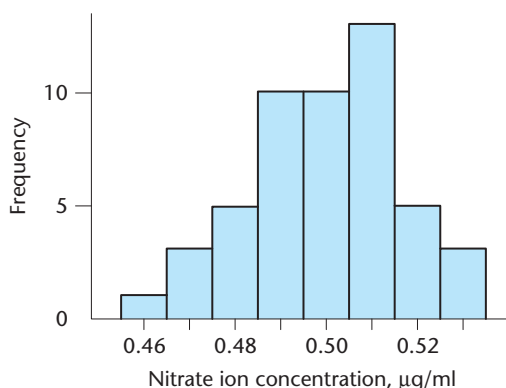


Figure 2.1 Histogram of the nitrate ion concentration data in Table 2.2.

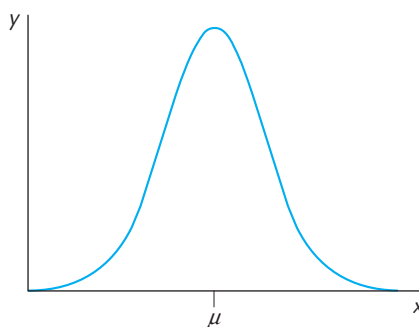


Figure 2.2 The normal distribution, $y = \exp[-(x - \mu)^2/2\sigma^2]/\sigma\sqrt{2\pi}$. The mean is indicated by μ .

where x is the measured value, and y the frequency with which it occurs. The shape of this distribution is shown in Fig. 2.2. There is no need to remember this complicated formula, but some of its general properties are important. The curve is symmetrical about μ and the greater the value of σ the greater the spread of the curve, as shown in Fig. 2.3. More detailed analysis shows that, whatever the values of μ and σ , the normal distribution has the following properties.

For a normal distribution with mean μ and standard deviation σ :

- approximately 68% of the population values lie within $\pm\sigma$ of the mean;
- approximately 95% of population values lie within $\pm2\sigma$ of the mean;
- approximately 99.7% of population values lie within $\pm3\sigma$ of the mean.

These properties are illustrated in Fig. 2.4. This would mean that, if the nitrate ion concentrations (in $\mu\text{g ml}^{-1}$) given in Table 2.2 are normally distributed, then about 68% should lie in the range 0.483–0.517, about 95% in the range 0.467–0.533 and 99.7% in the range 0.450–0.550. In fact 33 out of the 50 results (66%) lie between

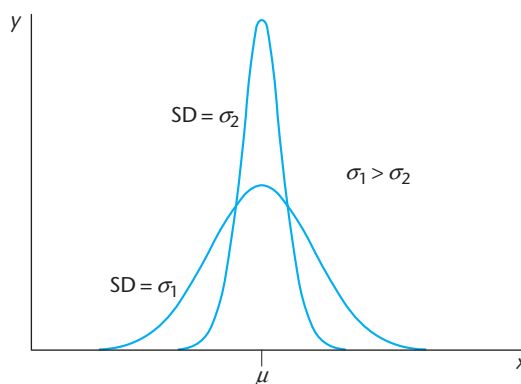


Figure 2.3 Normal distributions with the same mean but different values of the standard deviation.

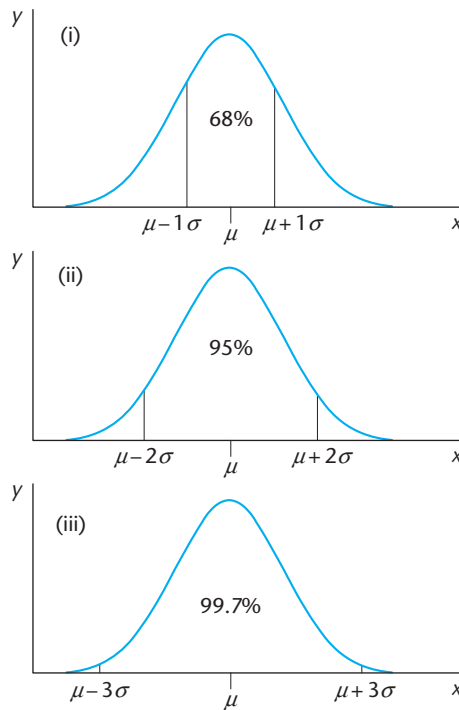


Figure 2.4 Properties of the normal distribution: (i) approximately 68% of values lie within $\pm 1\sigma$ of the mean; (ii) approximately 95% of values lie within $\pm 2\sigma$ of the mean; (iii) approximately 99.7% of values lie within $\pm 3\sigma$ of the mean.

0.483 and 0.517, 49 (98%) between 0.467 and 0.533, and all the results between 0.450 and 0.550, so the agreement with theory is fairly good.

For a normal distribution with known mean, μ , and standard deviation, σ , the exact proportion of values which lie within any interval can be found from tables, provided that the values are first **standardised** so as to give **z-values**. (These are widely used in proficiency testing schemes; see Chapter 4.) This is done by expressing any value of x in terms of its deviation from the mean in units of the standard deviation, σ . That is:

$$\text{Standardised normal variable, } z = \frac{(x - \mu)}{\sigma} \quad (2.2.2)$$

Table A.1 (Appendix 2) gives the proportions of values, $F(z)$, that lie *below* a given value of z . $F(z)$ is called the **standard normal cumulative distribution function**. For example the proportion of values below $z = 2$ is $F(2) = 0.9772$ and the proportion of values below $z = -2$ is $F(-2) = 0.0228$. Thus the *exact* proportion of measurements lying within two standard deviations of the mean is $0.9772 - 0.0228 = 0.9544$.

Example 2.2.1

If repeated values of a titration are normally distributed with mean 10.15 ml and standard deviation 0.02 ml, find the proportion of measurements which lie between 10.12 ml and 10.20 ml.

Standardising the lower limit of the range gives $z = (10.12 - 10.15)/0.02 = -1.5$. From Table A.1, $F(-1.5) = 0.0668$.

Standardising the upper limit of the range gives $z = (10.20 - 10.15)/0.02 = 2.5$. From Table A.1, $F(2.5) = 0.9938$.

Thus the proportion of values between $x = 10.12$ to 10.20 (corresponding to $z = -1.5$ to 2.5) is $0.9938 - 0.0668 = 0.927$.

In practice there is considerable variation in the formatting of tables for calculating proportions from z -values. Some tables give only positive z -values, and the proportions for negative z -values then have to be deduced using considerations of symmetry. $F(z)$ values are also provided by Excel® and Minitab®.

Although it cannot be proved that replicate values of a single analytical quantity are always normally distributed, there is considerable evidence that this assumption is generally at least approximately true. Moreover we shall see when we come to look at sample means that any departure of a population from normality is not usually important in the context of the statistical tests most frequently used.

The normal distribution is not only applicable to repeated measurements made on the same specimen. It also often fits the distribution of results obtained when the same quantity is measured for different materials from similar sources. For example if we measured the concentration of albumin in blood sera taken from healthy adult humans we would find the results were approximately normally distributed.

2.3 Log-normal distribution

In situations where one measurement is made on each of a number of specimens, distributions other than the normal distribution can also occur. In particular the so-called **log-normal distribution** is frequently encountered. For this distribution, frequency plotted against the *logarithm* of the concentration (or other characteristics) gives a normal distribution curve. An example of a variable which has a log-normal distribution is the antibody concentration in human blood sera. When frequency is plotted against concentration for this variable, the asymmetrical histogram shown in Fig. 2.5(a) is obtained. If, however, the frequency is plotted against the logarithm (to the base 10) of the concentration, an approximately normal distribution is obtained, as shown in Fig. 2.5(b). Another example of a variable which may follow a log-normal distribution is the particle size of the droplets formed by the nebulisers used in flame spectroscopy. Particle size distributions in atmospheric aerosols may also take the log-normal form, and the distribution is used to describe equipment failure rates and in gene expression analysis. However, many asymmetric population distributions cannot be converted to normal ones by the logarithmic transformation.

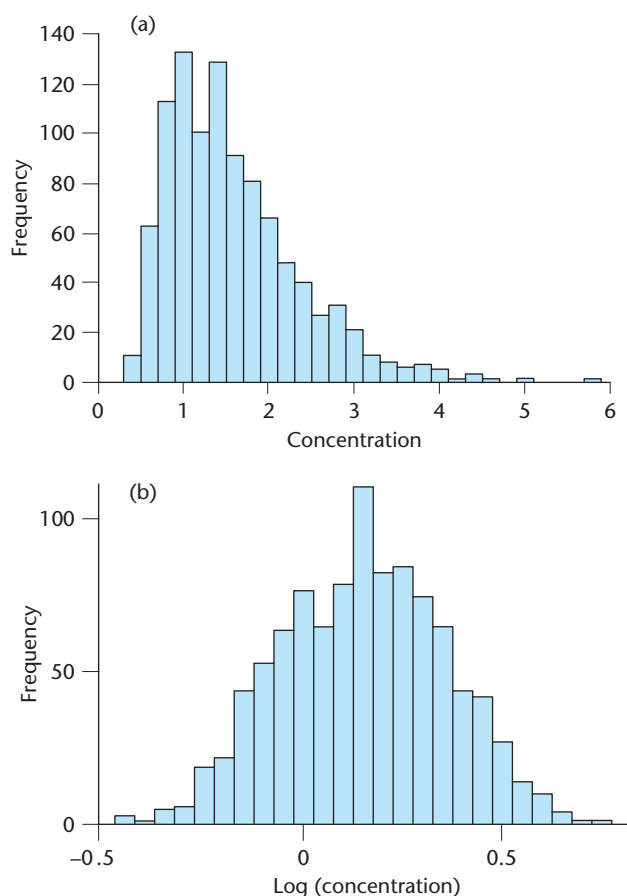


Figure 2.5 (a) An approximately log-normal distribution: concentration of serum immunoglobulin M antibody in male subjects. (b) The results in (a) plotted against the logarithm of the concentration.

The interval containing a given percentage of the measurements for a variable which is log-normally distributed can be found by working with the logarithms of the values. The distribution of the logarithms of the blood serum antibody concentration shown in Fig. 2.5(b) has mean 0.15 and standard deviation 0.20. This means that approximately 95% of the logged values lie in the interval $0.15 - 0.20$ to $0.15 + 0.20$, that is -0.05 to 0.35 . Taking antilogarithms we find that 95% of the original measurements lie in the interval $10^{-0.05}$ to $10^{0.35}$, that is 1.05 to 2.24. The antilogarithm of the mean of the logged values, $10^{0.15} = 1.41$, gives the *geometric* mean of the original distribution where the geometric mean is given by $\sqrt[n]{x_1 x_2 \dots x_n}$. See also Section 2.10.

2.4 Definition of a 'sample'

In this chapter the word 'sample' has been introduced and used in its statistical sense of a group of objects selected from the population of all such objects, for example a sample of 50 measurements of nitrate ion concentration from the (infinite)

population of all such possible measurements, or a sample of healthy human adults chosen from the whole population in order to measure the concentration of serum albumin for each one. The Commission on Analytical Nomenclature of the Analytical Chemistry Division of the International Union of Pure and Applied Chemistry has pointed out that confusion and ambiguity can arise if the term 'sample' is also used in its colloquial sense of the actual material being studied. It recommends that the term 'sample' is confined to its statistical concept. Other words should be used to describe the material on which measurements are being made, in each case preceded by 'test', for example **test solution** or **test extract**. We can then talk unambiguously of a sample of measurements on a test extract, or a sample of tablets from a batch. A test portion from a population which varies with time, such as a river or circulating blood, should be described as a **specimen**. Unfortunately this practice is by no means usual, so the term 'sample' remains in use for two related but distinct purposes.

2.5

The sampling distribution of the mean

We have seen that, in the absence of systematic errors, the mean of a sample of measurements, $\bar{x}$, provides us with an estimate of the true value, μ , of the quantity we are trying to measure. However, even in the absence of systematic errors, the individual measurements vary due to random errors and so it is most unlikely that the mean of the sample will be *exactly* equal to the true value. For this reason it is more useful to give a range of values which is likely to include the true value. The width of this range depends on two factors, the precision of the individual measurements, which in turn depends on the standard deviation of the population; and the number of measurements in the sample. The very fact that we repeat measurements implies that we have more confidence in the mean of several values than in a single value. Intuitively we would expect that the more measurements we make, the more reliable our estimate of μ , the true value, will be.

To pursue this idea, let us return to the nitrate ion determination described in Section 2.2. In practice it would be most unusual to make 50 repeated measurements in such a case: a more likely number would be 5. We can see how the means of samples of this size are spread about μ by treating the results in Table 2.1 as ten samples, each containing five results. Taking each column as one sample, the means are 0.506, 0.504, 0.502, 0.496, 0.502, 0.492, 0.506, 0.504, 0.500 and 0.486. We can see at once that these means are more closely clustered than the original measurements. If we took still more samples of five measurements and calculated their means, those means would have a frequency distribution of their own. The distribution of all possible sample means (in this case, an infinite number) is called the **sampling distribution of the mean**. Its mean is the same as the mean of the original population. Its standard deviation is called the **standard error of the mean** (SEM). There is an exact mathematical relationship between the latter and the standard deviation, σ , of the distribution of the individual measurements:

For a sample of n measurements,

$$\text{standard error of the mean} = \sigma/\sqrt{n} \quad (2.5.1)$$

As expected, the larger n is, the smaller the value of the SEM and consequently the smaller the spread of the sample means about μ .

The term ‘standard error of the mean’ might give the impression that $\sigma/\sqrt{n}$ gives the difference between μ and $\bar{x}$. This is not so: $\sigma/\sqrt{n}$ gives a measure of the variability of $\bar{x}$, as we shall see in the next section.

Another property of the sampling distribution of the mean is that, even if the original population is not normal, the sampling distribution of the mean tends to the normal distribution as n increases. This result is known as the **central limit theorem**. This theorem is of great importance because many statistical tests are performed on the mean and assume that it is normally distributed. Since in practice we can assume that distributions of repeated measurements are at least approximately normally distributed, it is reasonable to assume that the means of quite small samples (say $n > 5$) are also normally distributed.

2.6

Confidence limits of the mean for large samples

Now that we know the form of the sampling distribution of the mean we can return to the problem of using a sample to define a range which we may reasonably assume includes the true value. (Remember that in doing this we are assuming systematic errors to be absent.) Such a range is known as a **confidence interval** and the extreme values of the interval are called the **confidence limits**. The term ‘confidence’ implies that we can assert with a given degree of confidence, i.e. a certain probability, that the confidence interval does include the true value. The size of the confidence interval will obviously depend on how certain we want to be that it includes the true value: the greater the certainty, the *greater* the interval required.

Figure 2.6 shows the sampling distribution of the mean for samples of size n . If we assume that this distribution is normal, then 95% of the sample means will lie in the range given by:

$$\mu - 1.96(\sigma/\sqrt{n}) < \bar{x} < \mu + 1.96(\sigma/\sqrt{n}) \quad (2.6.1)$$

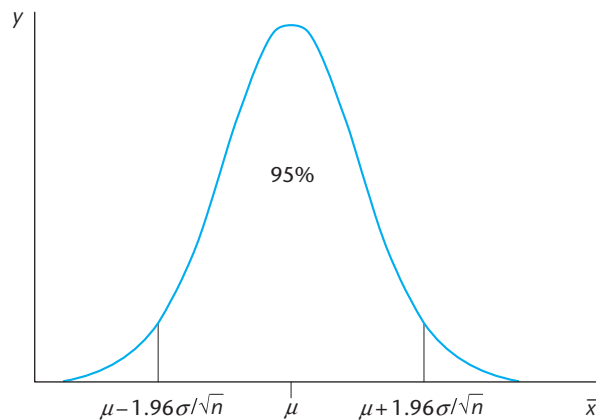


Figure 2.6 The sampling distribution of the mean, showing the range within which 95% of sample means lie.

(The exact value 1.96 has been used in this equation rather than the approximate value, 2, quoted in Section 2.2. The reader can use Table A.1 to check that the proportion of values between $z = -1.96$ and $z = 1.96$ is indeed 0.95.)

In practice, however, we usually have one sample, of known mean, and we require a range for μ , the true value. Equation (2.6.1) can be rearranged to give this:

$$\bar{x} - 1.96(\sigma/\sqrt{n}) < \mu < \bar{x} + 1.96(\sigma/\sqrt{n}) \quad (2.6.2)$$

Equation (2.6.2) gives the **95% confidence interval of the mean**. The **95% confidence limits** are $\bar{x} \pm 1.96\sigma/\sqrt{n}$. In practice we are unlikely to know σ exactly. However, *provided that the sample is large*, σ can be replaced by its estimate, s .

Other confidence limits are sometimes used, in particular the 99% and 99.7% confidence limits.

For large samples, the confidence limits of the mean are given by

$$\bar{x} \pm zs/\sqrt{n} \quad (2.6.3)$$

where the value of z depends on the degree of confidence required.

For 95% confidence limits, $z = 1.96$

For 99% confidence limits, $z = 2.58$

For 99.7% confidence limits, $z = 2.97$

Example 2.6.1

Calculate the 95% and 99% confidence limits of the mean for the nitrate ion concentration measurements in Table 2.1.

From previous examples we have found that $\bar{x} = 0.500$, $s = 0.0165$ and $n = 50$. Using Eq. (2.6.3) gives the 95% confidence limits as:

$$\bar{x} \pm 1.96s/\sqrt{n} = 0.500 \pm 1.96 \times 0.0165/\sqrt{50} = 0.500 \pm 0.005 \mu\text{g ml}^{-1}$$

and the 99% confidence limits as:

$$\bar{x} \pm 2.58s/\sqrt{n} = 0.500 \pm 2.58 \times 0.01651/\sqrt{50} = 0.500 \pm 0.006 \mu\text{g ml}^{-1}$$

In this example it is interesting to note that although the original measurements varied between 0.46 and 0.53, the 99% confidence interval for the mean is from 0.494 to 0.506 – a much narrower range, though admittedly the sample size of 50 is quite large.

2.7

Confidence limits of the mean for small samples

As the sample size gets smaller, s becomes less reliable as an estimate of σ . This can be seen by treating each column of data in Table 2.1 as a sample of size 5. The standard deviations of the ten columns are then 0.009, 0.015, 0.026, 0.021, 0.013, 0.019, 0.013, 0.017, 0.010 and 0.018, i.e. the largest value of s is nearly three times the size

Table 2.3 Values of t for confidence intervals

Degrees of freedom	Values of t for confidence interval of	
	95%	99%
2	4.30	9.92
5	2.57	4.03
10	2.23	3.17
20	2.09	2.85
50	2.01	2.68
100	1.98	2.63

of the smallest. To allow for this effect, Eq. (2.6.3) must be modified using the so-called t -statistic.

For small samples, the confidence limits of the mean are given by

$$\bar{x} \pm t_{n-1}s/\sqrt{n} \quad (2.7.1)$$

The subscript $(n - 1)$ indicates that t depends on this quantity, which is known as the number of **degrees of freedom, d.f.** (usually given the symbol ν). (The term 'degrees of freedom' refers to the number of *independent* deviations $(x_i - \bar{x})$ which are used in calculating s . In this case the number is $(n - 1)$, because when $(n - 1)$ deviations are known the last can be deduced, since $\sum_i (x_i - \bar{x}) = 0$.) The value of t also depends on the degree of confidence required. Some values of t are given in Table 2.3. A more complete version of this table is given in Table A.2 in Appendix 2.

For large n , the values of t_{n-1} for confidence intervals of 95% and 99% respectively are very close to the values 1.96 and 2.58 used in Example 2.6.1. The following example illustrates the use of Eq. (2.7.1).

Example 2.7.1

The sodium ion level in a urine specimen was measured using an ion-selective electrode. The following values were obtained: 102, 97, 99, 98, 101, 106 mM. What are the 95% and 99% confidence limits for the sodium ion concentration?

The mean and standard deviation of these values are 100.5 mM and 3.27 mM respectively. There are six measurements and therefore five degrees of freedom. From Table A.2 the value of t_5 for calculating the 95% confidence limits is 2.57 and from Eq. (2.7.1) the 95% confidence limits of the mean are given by:

$$100.5 \pm 2.57 \times 3.27/\sqrt{6} = 100.5 \pm 3.4 \text{ mM}$$

Similarly the 99% confidence limits are given by:

$$100.5 \pm 4.03 \times 3.27/\sqrt{6} = 100.5 \pm 5.4 \text{ mM}$$

2.8

Presentation of results

Since no quantitative result is of any value unless it comes with an estimate of the errors involved, the presentation of the results and the errors is important. A common practice is to quote the mean as the estimate of the quantity measured and the standard deviation as the estimate of the precision. Less commonly, the standard error of the mean is sometimes quoted instead of the standard deviation, or the result is given in the form of the 95% confidence limits of the mean. There is no universal convention, so it is obviously essential to state clearly the form used. Provided the value of n is given (which it always should be), the three forms are easily inter-converted using Eqs (2.5.1) and (2.7.1). Uncertainty estimates are now often calculated (see Chapter 4).

A related aspect of presenting results is the rounding-off of the answer. The important principle here is that the *number of significant figures given indicates the precision of the experiment*. It would be absurd, for example, to give the result of a titrimetric analysis as 0.107846 M – no analyst could achieve the implied precision of 0.000001 in ca. 0.1, i.e. 0.001%. In practice it is usual to quote as significant figures all the digits which are certain, plus the first uncertain one. For example, the mean of the values 10.09, 10.11, 10.09, 10.10 and 10.12 is 10.102, and their standard deviation is 0.01304. Clearly there is uncertainty in the second decimal place; the results are all 10.1 to one decimal place but disagree in the second decimal place. Using the suggested method the result would be quoted as:

$$\bar{x} \pm s = 10.10 \pm 0.01 \quad (n = 5)$$

If it was felt that this resulted in an unacceptable rounding-off of the standard deviation, then the result could be given as:

$$\bar{x} \pm s = 10.10_2 \pm 0.01_3 \quad (n = 5)$$

where the use of a subscript indicates that the digit is given only to avoid loss of information. The reader could decide whether it was useful or not. Similar principles apply to the presentation of confidence limits.

The number of significant figures quoted is sometimes used *instead of* a specific estimate of the precision of a result. For example the result 0.1046 M is taken to mean that the figures in the first three decimal places are certain but there is doubt about the fourth. Sometimes the uncertainty in the last figure is emphasised by using the formats 0.104(6) M or 0.104₆ M, but it remains preferable to give a specific estimate of precision such as the standard deviation.

One problem is whether a result ending in a 5 should be rounded up or down. For example, if 9.65 is rounded to one decimal place, should it become 9.6 or 9.7? It is evident that the results will be biased if a 5 is always rounded up; this bias can be avoided by rounding the 5 to the nearest *even* number giving, in this case, 9.6. Analogously, 4.75 is rounded to 4.8.

When several measured quantities are to be combined mathematically to calculate a final result (see Section 2.11) these quantities should not be rounded off too much or a needless loss of precision will result. A good rule is to keep one digit beyond the last significant figure for each individual quantity, and leave further rounding until the final result is reached. The same advice applies when

the mean and standard deviation are used to apply a statistical test such as the F - and t -tests (see Chapter 3): un-rounded values of $\bar{x}$ and s should be used in the calculations.

2.9 Other uses of confidence limits

Confidence intervals can be used to test for systematic errors as shown in the following example.

Example 2.9.1

The absorbance scale of a spectrometer is tested at a particular wavelength with a standard solution which has an absorbance given as 0.470. Ten measurements of the absorbance with the spectrometer give $\bar{x} = 0.461$, and $s = 0.003$. Find the 95% confidence interval for the mean absorbance as measured by the spectrometer, and hence decide whether a systematic error is present.

The 95% confidence limits for the absorbance as measured by the spectrometer are (Eq. (2.7.1)):

$$\bar{x} \pm t_{n-1}s/\sqrt{n} = 0.461 \pm 2.26 \times 0.003/\sqrt{10} = 0.461 \pm 0.002$$

(The value of t_9 was obtained from Table A.2.)

Since the confidence interval does *not* include the known absorbance of 0.470, it is likely that a systematic error has occurred.

In practice the type of problem in Example 2.9.1 is usually tackled by a different, but related, approach (see Example 3.2.1).

Confidence limits can also be used in cases where measurements are made on each of a number of specimens. For example if the mean weight of a tablet in a very large batch is required, it would be too time-consuming to weigh each tablet. Similarly, if the mean iron content of the tablets is measured using a destructive method of analysis such as atomic-absorption spectrometry, it is clearly impossible to examine every tablet. In each case, a sample could be taken from the batch (which in such instances forms the population), and from the mean and standard deviation of the sample a confidence interval could be found for the mean value of the quantity measured.

2.10

Confidence limits of the geometric mean for a log-normal distribution

In Section 2.3 we saw that measurements on a number of different specimens may not be normally distributed. If they come from a log-normal distribution, then the confidence limits should be calculated taking this fact into account. Since the log of

the measurements is normally distributed it is more accurate to work with the logarithms of the measurements when calculating a confidence interval. The confidence interval obtained will be the confidence interval for the *geometric* mean.

Example 2.10.1

The following values (expressed as percentages) give the antibody concentration in human blood serum for a sample of eight healthy adults.

2.15, 1.13, 2.04, 1.45, 1.35, 1.09, 0.99, 2.07

Calculate the 95% confidence interval for the geometric mean assuming that antibody concentration is log-normally distributed.

The logarithms (to the base 10) of these values are, from Eq. (2.7.1):

0.332, 0.053, 0.310, 0.161, 0.130, 0.037, -0.004, 0.316

The mean of these logged values is 0.1669, giving $10^{0.1669} = 1.47$ as the geometric mean of the original values. The standard deviation of the logged values is 0.1365.

The 95% confidence limits for the logged values are

$$0.1669 \pm 2.36 \times 0.1365 / \sqrt{8} = 0.1669 \pm 0.1139 = 0.0530 \text{ to } 0.2808$$

The value of t was taken from Table A.2.

Taking antilogarithms of these limits gives the 95% confidence interval of the geometric mean of the antibody concentrations as 1.13 to 1.91.

2.11

Propagation of random errors

In experimental work, the final result is often calculated from a combination of measured quantities. In Chapter 1 it was shown that even a relatively simple operation such as a titration involves several stages, each of which is subject to errors. The final result may be calculated from the sum, difference, product or quotient of two or more measured quantities, or the raising of any quantity to a power.

The procedures used for combining random and systematic errors are completely distinct. This is because random errors to some extent cancel each other out, whereas every systematic error occurs in a definite and known sense. Suppose for example that the final result of an experiment, x , is given by $x = a + b$. If a and b each have a systematic error of +1, it is clear that the systematic error in x is +2. If, however, a and b each have a random error of ± 1 , the random error in x is not ± 2 : this is because there will be occasions when the random error in a is positive while that in b is negative (or vice versa). This section deals only with the propagation of random errors (systematic errors are considered in Section 2.12). If the precision of each observation is known, then simple mathematical rules can be used to estimate the precision of the final result. These rules can be summarised as follows.

2.11.1 Linear combinations

In this case the final value, y , is calculated from a linear combination of measured quantities a , b , c , etc. by:

$$y = k + k_a a + k_b b + k_c c + \dots \quad (2.11.1)$$

where k , k_a , k_b , k_c etc. are constants. Variance (defined as the square of the standard deviation) has the important property that **the variance of a sum or difference of independent quantities is equal to the sum of their variances**. It can be shown that if σ_a , σ_b , σ_c , etc. are the standard deviations of a , b , c , etc., then the standard deviation of y , σ_y , is given by:

$$\sigma_y = \sqrt{(k_a \sigma_a)^2 + (k_b \sigma_b)^2 + (k_c \sigma_c)^2 + \dots} \quad (2.11.2)$$

Example 2.11.1

In a titration the initial reading on the burette is 3.51 ml and the final reading is 15.67 ml, both with a standard deviation of 0.02 ml. What is the volume of titrant used and what is its standard deviation?

$$\text{Volume used} = 15.67 - 3.51 = 12.16 \text{ ml}$$

From Eq. (2.11.2), using two terms on the right-hand side with $k_a = 1$, $\sigma_a = 0.02$, $k_b = -1$, $\sigma_b = 0.02$, we have

$$\text{Standard deviation} = \sqrt{(0.02)^2 + (-0.02)^2} = 0.028 \text{ ml}$$

This example illustrates the important point that the standard deviation for the final result is larger than the standard deviations of the individual burette readings, even though the volume used is calculated from a difference. It is, however, less than the sum of the standard deviations.

2.11.2 Multiplicative expressions

If y is calculated from an expression of the type:

$$y = kab/cd \quad (2.11.3)$$

(where a , b , c and d are independent measured quantities and k is a constant), then there is a relationship between the squares of the *relative* standard deviations:

$$\frac{\sigma_y}{y} = \sqrt{\left(\frac{\sigma_a}{a}\right)^2 + \left(\frac{\sigma_b}{b}\right)^2 + \left(\frac{\sigma_c}{c}\right)^2 + \left(\frac{\sigma_d}{d}\right)^2} \quad (2.11.4)$$

Example 2.11.2

The quantum yield of fluorescence, ϕ , of a material in solution is calculated from the expression:

$$\phi = I_f / kclI_0\varepsilon$$

where the quantities involved are defined below, with an estimate of their relative standard deviations in brackets:

I_0 = incident light intensity (0.5%)

I_f = fluorescence intensity (2%)

ε = molar absorptivity (1%)

c = concentration (0.2%)

l = optical pathlength (0.2%)

k is an instrument constant.

From Eq. (2.11.4), the relative standard deviation (RSD) of ϕ is given by:

$$\text{RSD} = \sqrt{2^2 + 0.2^2 + 0.2^2 + 0.5^2 + 1^2} = 2.3\%$$

It can be seen that the relative standard deviation in the final result is not much larger than the largest relative standard deviation used to calculate it (i.e. 2% for I_f). This is mainly a consequence of the squaring of the relative standard deviations and illustrates an important general point: any efforts to improve the precision of an experiment need to be directed towards improving the precision of the least precise values. By contrast, there is little point in expending effort in increasing the precision of the most precise measurements. Nonetheless small errors are not always unimportant: the combination of small errors at many stages of an experiment, such as the titration discussed in detail in Chapter 1, may produce an appreciable error in the final result.

When a quantity is raised to a power, e.g. b^3 , the error is not calculated as for a multiplication, i.e. $b \times b \times b$, because the quantities involved are not independent. If the relationship is:

$$y = b^n \quad (2.11.5)$$

then the standard deviations of y and b are related by:

$$\left| \frac{\sigma_y}{y} \right| = \left| \frac{n\sigma_b}{b} \right| \quad (2.11.6)$$

(The modulus sign means that the magnitude of the enclosed quantity is taken without respect to sign, e.g. $|-2| = 2$.)

2.11.3 Other functions

If y is a general function of x , i.e. $y = f(x)$, then the standard deviations of x and y are related by:

$$\sigma_y = \left| \sigma_x \frac{dy}{dx} \right| \quad (2.11.7)$$

Example 2.11.3

The absorbance, A , of a solution is given by $A = -\log(T)$ where T is the transmittance. If the measured value of T is 0.501 with a standard deviation of 0.001, calculate A and its standard deviation.

We have:

$$A = -\log 0.501 = 0.300$$

Also:

$$dA/dT = -(\log e)/T = -0.434/T$$

so from Eq. (2.11.7):

$$\sigma_A = |\sigma_T(-\log e/T)| = |0.001 \times (-0.434/0.501)| = 0.00087$$

It is interesting and important that for this widely used experimental method we can also find the conditions for which the relative standard deviation is a minimum. The relative standard deviation of A is given by

$$\text{RSD of } A = 100\sigma_A/A = \left| \frac{-100\sigma_T \log e}{T \log T} \right|$$

Differentiation of this expression with respect to T shows that the RSD of A is a minimum when $T = 1/e = 0.368$.

2.12

Propagation of systematic errors

The rules for the combination of systematic errors can also be divided into three groups.

2.12.1 Linear combinations

If y is calculated from measured quantities by using Eq. (2.11.1), and the systematic errors in a , b , c , etc., are Δa , Δb , Δc , etc., then the systematic error in y , Δy , is calculated from:

$$\Delta y = k_a \Delta a + k_b \Delta b + k_c \Delta c + \dots \quad (2.12.1)$$

Remember that each systematic error is either positive or negative and that these signs must be included in the calculation of Δy .

The total systematic error can sometimes be zero. For example, if a balance with a systematic error of -0.01 g is used for the weighings involved in making a standard solution and the weight of the solute used is found from the difference between two weighings, the systematic errors cancel out. (This applies only to an electronic

balance with a single internal reference weight.) Carefully considered procedures such as this can often minimise the systematic errors (see Chapter 1).

2.12.2 Multiplicative expressions

If y is calculated from the measured quantities by use of Eq. (2.11.3), then *relative* systematic errors are used:

$$(\Delta y/y) = (\Delta a/a) + (\Delta b/b) + (\Delta c/c) + (\Delta d/d) \quad (2.12.2)$$

When a quantity is raised to some power, then Eq. (2.11.6) is used with the modulus sign omitted and the standard deviations replaced by systematic errors.

2.12.3 Other functions

The equation used is identical to Eq. (2.11.7) but with the modulus sign omitted and the standard deviations replaced by systematic errors.

In any real analytical experiment both random and systematic errors will occur. They can be combined to give an **uncertainty** in the final result, which provides a realistic range of values within which the true value of a measured quantity probably lies. This topic is dealt with in detail in Chapter 4.

Bibliography

The following books show the application of the statistical principles from this chapter in important areas of applied analytical chemistry.

Altman, D.G., 1991, *Practical Statistics for Medical Research*, Chapman & Hall, London. Gives a fuller discussion of the log-normal distribution.

Lucy, D., 2005, *Introduction to Statistics for Forensic Scientists*, John Wiley, Chichester. Has a strong emphasis on the use of statistics in presenting evidence in court.

Pentecost, A., 1999, *Analysing Environmental Data*, Pearson Education, Harlow. Contains many examples from the field of environmental analysis.

Exercises

- 1 The reproducibility of a method for the determination of selenium in foods was investigated by taking nine samples from a single batch of brown rice and determining the selenium concentration in each. The following results were obtained:

0.07 0.07 0.08 0.07 0.07 0.08 0.08 0.09 0.08 $\mu\text{g g}^{-1}$

(Moreno-Dominguez, T., Garcia-Moreno, C. and Marine-Font, A., 1983, *Analyst*, **108**: 505)

Calculate the mean, standard deviation and relative standard deviation of these results.

- 2 The morphine levels (%) of seven batches of seized heroin were determined, with the following results:

15.1 21.2 18.5 25.3 19.2 16.0 17.8

Calculate the 95% and 99% confidence limits for these measurements

- 3 Ten replicate analyses of the concentration of mercury in a sample of commercial gas condensate gave the following results:

23.3 22.5 21.9 21.5 19.9 21.3 21.7 23.8 22.6 24.7 ng ml⁻¹

(Shafawi, A., Ebdon, L., Foulkes, M., Stockwell, P. and Corns, W., 1999, *Analyst*, **124**: 185)

Calculate the mean, standard deviation, relative standard deviation and 99% confidence limits of the mean.

Six replicate analyses on another sample gave the following values:

13.8 14.0 13.2 11.9 12.0 12.1 ng ml⁻¹

Repeat the calculations for these values.

- 4 The concentration of lead in the bloodstream was measured for a sample of 50 children from a large school near a busy main road. The sample mean was 10.12 ng ml⁻¹ and the standard deviation was 0.64 ng ml⁻¹. Calculate the 95% confidence interval for the mean lead concentration for all the children in the school.

About how big should the sample have been to reduce the range of the confidence interval to 0.2 ng ml⁻¹ (i.e. ± 0.1 ng ml⁻¹)?

- 5 In an evaluation of a method for the determination of fluorene in seawater, a synthetic sample of seawater was spiked with 50 ng ml⁻¹ of fluorene. Ten replicate determinations of the fluorene concentration in the sample had a mean of 49.5 ng ml⁻¹ with a standard deviation of 1.5 ng ml⁻¹.

(Gonzalez, M.A. and Lopez, M.H., 1998, *Analyst*, **123**: 2217)

Calculate the 95% confidence limits of the mean. Is the spiked value of 50 ng ml⁻¹ within the 95% confidence limits?

- 6 A 0.1 M solution of acid was used to titrate 10 ml of 0.1 M solution of alkali and the following volumes of acid were recorded:

9.88 10.18 10.23 10.39 10.21 ml.

Calculate the 95% confidence limits of the mean and use them to decide whether there is any evidence of systematic error.

- 7 A volume of 250 ml of a 0.05 M solution of a reagent of formula weight (relative molecular mass) 40 was made up, using weighing by difference. The standard deviation of each weighing was 0.0001 g: what were the standard deviation and relative standard deviation of the weight of reagent used? The standard deviation of the volume of solvent used was 0.05 ml. Express this as a relative standard deviation. Hence calculate the relative standard deviation of the molarity of the solution.

Repeat the calculation for a reagent of formula weight 392.

- 8 The solubility product of barium sulphate is 1.3×10^{-10} , with a standard deviation of 0.1×10^{-10} . Calculate the standard deviation of the calculated solubility of barium sulphate in water.

3

Significance tests

Major topics covered in this chapter

- Principles of significance testing: one-sided and two-sided tests
- Applications of the *t*-test for comparing means
- *F*-test for comparing variances
- Testing for outliers
- One-way analysis of variance (ANOVA)
- The χ^2 (chi-squared) test
- Testing for the normal distribution
- Conclusions and errors in significance tests
- Introduction to Bayesian methods

3.1

Introduction

One of the most important properties of an analytical method is that it should be free from bias, so that the value it gives for the amount of the analyte should be the true value. This property may be tested by applying the method to a standard test portion containing a known amount of analyte (see Chapters 1 and 2). However, *even if there are no systematic errors*, random errors make it most unlikely that the measured amount will *exactly* equal the known amount in the standard. To decide whether the difference between the measured and standard amounts can be accounted for by random errors a statistical test known as a **significance test** can be used. As its name implies, this approach tests whether the difference between the two results is significant, or whether it can be accounted for merely by random variations. Significance tests are widely used in the evaluation of experimental results. This chapter considers several tests which are particularly useful to analytical chemists.

3.2 Comparison of an experimental mean with a known value

In making a significance test we are testing the truth of a hypothesis which is known as a **null hypothesis**, often denoted by H_0 . For the example in the previous paragraph we adopt the null hypothesis that the analytical method is *not* subject to systematic error. The term *null* is used to imply that there is no difference between the observed and known values apart from that due to random variation. Assuming that this null hypothesis is true, statistical theory can be used to calculate the *probability* that the observed difference (or a greater one) between the sample mean, $\bar{x}$, and the true value, μ , arises solely as a result of random errors. As the probability that the observed difference occurs by chance falls, it becomes less likely that the null hypothesis is true. Usually the null hypothesis is *rejected* if the probability of such a difference occurring by chance is less than 1 in 20 (i.e. 0.05 or 5%). In such a case the difference is said to be **significant at the $P = 0.05$ (or 5%) level**. Using this level of significance there is, on average, a 1 in 20 chance that we shall reject the null hypothesis *when it is in fact true*. In order to be more certain that we make the correct decision a higher level of significance can be used, usually 0.01 or 0.001 (1% or 0.1%). The significance level is indicated by writing, for example, P (i.e. probability) = 0.05, this number giving the probability of rejecting a true null hypothesis. It is important to appreciate that if the null hypothesis is *retained*, we have not *proved* that it is true, only that it has not been demonstrated to be false. Later in the chapter the probability of retaining a null hypothesis when it is in fact false will be discussed in more detail.

In order to decide whether the difference between $\bar{x}$ and μ is significant, i.e. to test H_0 : mean of the population from which the sample is drawn = μ , the statistic t is calculated from:

$$t = (\bar{x} - \mu)\sqrt{n}/s \quad (3.2.1)$$

where $\bar{x}$ = sample mean, s = sample standard deviation and n = sample size.

If $|t|$ (i.e. the calculated value of t without regard to sign) exceeds a certain **critical value** then the null hypothesis is rejected. The critical value of t for a given significance level can be found from Table A.2. For example for a sample size of ten (i.e. nine degrees of freedom) and a significance level of 0.01, the critical value of t_9 is 3.25, where, as in Chapter 2, the subscript is used to denote the number of degrees of freedom.

Example 3.2.1

In a new method for determining selenourea in water the following values were obtained for tap water samples spiked with 50 ng ml⁻¹ of selenourea:

50.4, 50.7, 49.1, 49.0, 51.1 ng ml⁻¹

(Aller, A.J. and Robles, L.C., 1998, *Analyst*, 123: 919)

Is there any evidence of systematic error?

The mean of these values is 50.06 and the standard deviation is 0.956. Adopting the null hypothesis that there is no systematic error, i.e. that $\mu = 50$, Eq. (3.2.1) gives:

$$t = \frac{(50.06 - 50)\sqrt{5}}{0.956} = 0.14$$

From Table A.2, the critical value is $t_4 = 2.78$ ($P = 0.05$). Since the observed value of $|t|$ is less than the critical value the null hypothesis is retained: there is no evidence of systematic error. Note again that this does not mean that there are no systematic errors, only that they have not been demonstrated.

Critical values from statistical tables were used in significance testing for many years because it was too tedious to calculate the probability of t exceeding the experimental value. Computers have altered this situation, and statistical software usually quotes the results of significance tests in terms of a probability. If the individual data values are entered in Minitab® the result of performing this test is shown below:

t-Test of the mean

Test of $\mu = 50.000$ vs $\mu \text{ not} = 50.000$

Variable	N	Mean	StDev	SE Mean	T	P
selenour	5	50.060	0.956	0.427	0.14	0.90

This gives the additional information that $P(|t| > 0.14) = 0.90$. Since this probability is much greater than 0.05, the result is not significant at $P = 0.05$, in agreement with the previous calculation. Obviously the power to calculate an exact probability is a great advantage, removing the need for statistical tables containing critical values. Examples in this book use critical values, however, as many scientists still perform significance tests using handheld calculators, which do not normally provide P values. Moreover in cases where only means and standard deviations are available, but not the original data values, programs such as Minitab® or Excel® cannot be used. However, where the calculation can be performed using such programs, the P value will also be quoted.

3.3

Comparison of two experimental means

Another way in which the results of a new analytical method may be tested is by comparing them with those obtained by using a second (perhaps a reference) method. In this case the two methods give two sample means, $\bar{x}_1$ and $\bar{x}_2$. The null hypothesis is that the two methods give the same result, i.e. $H_0: \mu_1 = \mu_2$, or $\mu_1 - \mu_2 = 0$, so we need to test whether $(\bar{x}_1 - \bar{x}_2)$ differs significantly from zero. As we saw in the previous section, the t -test can be used to compare an experimental result, $(\bar{x}_1 - \bar{x}_2)$ in this case, with a standard value, obviously zero here. However, we must allow for the fact that the results from the two methods might have different sample sizes, n_1 and n_2 , and that we also have two different standard deviations,

s_1 and s_2 . If these standard deviations are not significantly different (see Section 3.5 for a method of testing this assumption), a pooled estimate, s , of the standard deviation can first be calculated using the equation:

$$s = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{(n_1 + n_2 - 2)} \quad (3.3.1)$$

To decide whether the difference between the two means, $\bar{x}_1$ and $\bar{x}_2$, is significant, i.e. to test the null hypothesis, $H_0: \mu_1 = \mu_2$, the statistic t is then calculated from:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad (3.3.2)$$

where t has $n_1 + n_2 - 2$ degrees of freedom.

Example 3.3.1

In a comparison of two methods for the determination of chromium in rye grass, the following results ($\text{mg kg}^{-1} \text{Cr}$) were obtained:

Method 1: mean = 1.48; standard deviation 0.28

Method 2: mean = 2.33; standard deviation 0.31

For each method five determinations were made.

(Sahuquillo, A., Rubio, R. and Rauret, G., 1999, *Analyst*, 124: 1)

Do these two methods give results having means which differ significantly?

From Eq. (3.3.1), the pooled value of the standard deviation is given by:

$$s^2 = ([4 \times 0.28^2] + [4 \times 0.31^2]) / (5 + 5 - 2) = 0.0872$$

so $s = 0.295$.

From Eq. (3.3.2):

$$t = \frac{2.33 - 1.48}{0.295\sqrt{\frac{1}{5} + \frac{1}{5}}} = 4.56$$

There are eight degrees of freedom, so (Table A.2) the critical value is $t_8 = 2.31$ ($P = 0.05$). Since the experimental value of $|t|$ is greater than this, the difference between the two results is significant at the 5% level and the null hypothesis is rejected. In fact since the critical value of t_8 for $P = 0.01$ is 3.36, the difference is significant at the 1% level. In other words, if the null hypothesis is true the probability of such a large difference arising by chance is less than 1 in 100.

The next example shows another application of this test, where it is used to decide whether a change in the conditions of an experiment affects the result.

Example 3.3.2

In a series of experiments on the determination of tin in foodstuffs, samples were boiled with hydrochloric acid under reflux for different times. Some of the results are shown below:

Refluxing time (min)	Tin found (mg kg ⁻¹)
30	55, 57, 59, 56, 56, 59
75	57, 55, 58, 59, 59, 59

(Analytical Methods Committee, 1983, *Analyst*, **108**: 109).

Does the mean amount of tin found differ significantly for the two boiling times?

The mean and variance (square of the standard deviation) for the two times are:

$$30 \text{ min} \quad \bar{x}_1 = 57.00 \quad s_1^2 = 2.80$$

$$75 \text{ min} \quad \bar{x}_2 = 57.83 \quad s_2^2 = 2.57$$

The null hypothesis is that the refluxing time has no effect on the amount of tin found. From Eq. (3.3.1), the pooled value for the variance is given by:

$$s^2 = ([5 \times 2.80] + [5 \times 2.57])/10 = 2.685$$

$$s = 1.64$$

From Eq. (3.3.2):

$$t = \frac{57.00 - 57.83}{1.64 \sqrt{\frac{1}{6} + \frac{1}{6}}} = -0.88$$

There are 10 degrees of freedom so the critical value is $t_{10} = 2.23$ ($P = 0.05$). The observed value of $|t|$ ($= 0.88$) is less than the critical value so the null hypothesis is retained: there is no evidence that the refluxing time affects the amount of tin found.

The table below shows the result of performing this calculation using Excel®:

t-Test: Two-sample assuming equal variances

	Variable 1	Variable 2
Mean	57	57.833
Variance	2.8	2.567
Observations	6	6
Pooled variance	2.683	
Hypothesized mean difference	0	
df	10	
t Stat	-0.881	
P(T<=t) one-tail	0.199	
t Critical one-tail	1.812	
P(T<=t) two-tail	0.399	
t Critical two-tail	2.228	

The distinction between ‘one-tail’ and ‘two-tail’ will be covered in Section 3.5. For the present, it is sufficient to consider only the two-tail values. These show that $P(|t| > 0.88) = 0.399$. Since this probability is much greater than 0.05, the result is not significant at the 5% level.

If the population standard deviations are unlikely to be equal then it is no longer appropriate to pool sample standard deviations in order to give an overall estimate of standard deviation. An approximate method in these circumstances is given below:

In order to test $H_0: \mu_1 = \mu_2$ when it cannot be assumed that the two samples come from populations with equal standard deviations, the statistic t is calculated where

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \quad (3.3.3)$$

$$\text{with the number of degrees of freedom} = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{s_1^4}{n_1^2(n_1 - 1)} + \frac{s_2^4}{n_2^2(n_2 - 1)}} \quad (3.3.4)$$

the value obtained being truncated to an integer.

Several different equations have been suggested for the number of degrees of freedom for t when s_1 and s_2 differ, reflecting the fact that the method is an approximate one. Equation (3.3.4) is used by both Minitab® and Excel®, but Minitab®, erring on the side of caution in giving a significant result, rounds the value down, while Excel® rounds it to the nearest integer. For example, if the equation gave a value of 4.7, Minitab® would take four degrees of freedom and Excel® would take five.

Example 3.3.3

The data below give the concentration of thiol (mM) in the blood lysate of the blood of two groups of volunteers, the first group being ‘normal’ and the second having rheumatoid arthritis:

Normal: 1.84, 1.92, 1.94, 1.92, 1.85, 1.91, 2.07
 Rheumatoid: 2.81, 4.06, 3.62, 3.27, 3.27, 3.76

(Banford, J.C., Brown, D.H., McConnell, A.A., McNeil, C.J., Smith, W.E., Hazelton, R.A. and Sturrock, R.D., 1983, *Analyst*, **107**: 195)

The null hypothesis adopted is that the mean concentration of thiol is the same for the two groups.

The reader can check that:

$$\begin{aligned} n_1 &= 7 & \bar{x}_1 &= 1.921 & s_1 &= 0.076 \\ n_2 &= 6 & \bar{x}_2 &= 3.465 & s_2 &= 0.440 \end{aligned}$$

Substitution in Eq. (3.3.3) gives $t = -8.48$ and substitution in Eq. (3.3.4) gives 5.3, which is truncated to 5. The critical value is $t_5 = 4.03$ ($P = 0.01$) so the null hypothesis is rejected: there is evidence that the mean concentration of thiol differs between the groups.

The result of performing this calculation using Minitab® (where the non-pooled test is the default option) is shown below.

Two sample *t*-test and confidence interval

Two sample T for Normal vs Rheumatoid

	N	Mean	StDev	SE Mean
Normal	7	1.9214	0.0756	0.029
Rheumato	6	3.465	0.440	0.18

95% CI for mu Normal - mu Rheumato: (-2.012, -1.08)

T-Test mu Normal = mu Rheumato (vs not =): T= -8.48

P = 0.0004 DF = 5

This confirms the values above and also gives the information that $P(|t| \geq 8.48) = 0.0004$. This probability is extremely low: the result is in fact significant at $P = 0.001$.

3.4 Paired *t*-test

It frequently happens that two methods of analysis are compared by applying both of them to the *same* set of test materials, which contain different amounts of analyte. For example, Table 3.1 gives the results of determining the paracetamol concentration (% m/m) in tablets by two different methods. Tablets from ten different batches were analysed to see whether the results obtained by the two methods differed. Each batch is thus characterised by a *pair* of measurements, one value for each method.

In addition to random measurement errors, differences between the tablets and differences between the methods may also contribute to the variation between the measurements. Here we wish to know whether the methods produce significantly different results. The *t*-test for comparing two means (Section 3.3) is not appropriate in this case because it does not separate the variation due to method from that due to variation between tablets: the two effects are said to be *confounded*. This difficulty is overcome by looking at the difference, d , between each pair of results given by the two methods. If there is no difference between the two methods then these differences are drawn from a population with mean $\mu_d = 0$. In order to test this null hypothesis, we test whether $\bar{d}$ differs significantly from 0 using the statistic t .

Table 3.1 Example of paired data

Batch	UV spectrometric assay	Near-infrared reflectance spectroscopy
1	84.63	83.15
2	84.38	83.72
3	84.08	83.84
4	84.41	84.20
5	83.82	83.92
6	83.55	84.16
7	83.92	84.02
8	83.69	83.60
9	84.06	84.13
10	84.03	84.24

(Trafford, A.D., Jee, R.D., Moffat, A.C. and Graham, P., 1999, *Analyst*, **124**: 163)

To test whether n paired results are drawn from the same population, that is $H_0: \mu_d = 0$, we calculate the t -statistic from the equation:

$$t = \bar{d}\sqrt{n}/s_d \quad (3.4.1)$$

where $\bar{d}$ and s_d are the mean and standard deviation respectively of d values, the differences between the paired values. (Eq. (3.4.1) is clearly similar to Eq. (3.2.1).)

The number of degrees of freedom of t is $n - 1$.

Example 3.4.1

Test whether there is a significant difference between the results obtained by the two methods in Table 3.1.

The differences between the pairs of values (subtracting the second value from the first value in each case) are:

$$+1.48, +0.66, +0.24, +0.21, -0.10, -0.61, -0.10, +0.09, -0.07, -0.21$$

These differences have mean, $\bar{d} = 0.159$, and standard deviation, $s_d = 0.570$. Substituting in Eq. (3.4.1), with $n = 10$, gives $t = 0.88$. The critical value is $t_9 = 2.26$ ($P = 0.05$). Since the calculated value of $|t|$ is less than this the null hypothesis is retained: the methods do not give significantly different results for the paracetamol concentration.

Again this calculation can be performed on a computer, giving the result that $P(|t| \geq 0.88) = 0.40$. Since this probability is much greater than 0.05 we reach the same conclusion: the two methods do not differ significantly at $P = 0.05$.

The paired test described above does not require that the precisions of the two methods are equal but it does assume that the differences, d , are normally distributed. In effect this requires that each set of measurements is normally distributed and that the precision and bias (if any) of each method are constant over the range of values for which the measurements were made. The data can consist of single measurements, as in Example 3.4.1, or as the means of replicate measurements. However, it is necessary for the same number of measurements to be made on each sample by the first method and likewise for the second method: that is, l measurements are made on each sample by method 1 and m measurements on each sample by method 2, where l and m do not have to be equal.

There are various circumstances in which it may be necessary or desirable to design an experiment so that each sample is analysed by each of two methods, giving results that are naturally paired. Some examples are:

- the quantity of any one test sample is sufficient for only one determination by each method;
- the test samples may be presented over an extended period so it is necessary to remove the effects of variations in the environmental conditions such as temperature, pressure, etc.;
- the methods are to be compared by using a wide variety of samples from different sources and possibly with very different concentrations (but see the next paragraph).

Analytical methods usually have to be applicable over a wide range of concentrations, so a new method is often compared with a standard method by analysis of samples in which the analyte concentration may vary over several powers of 10. In this case it is inappropriate to use the paired t -test since it is valid only if any errors, either random or systematic, are independent of concentration: over wide ranges of concentration this assumption may no longer be true. An alternative method in such cases is linear regression (see Section 5.9), but this approach also presents some difficulties if it is used uncritically.

3.5 One-sided and two-sided tests

The methods described so far in this chapter have been concerned with testing for a difference between two means *in either direction*. For example, the method described in Section 3.2 tests whether there is a significant difference between the experimental result and the known value for the reference material, regardless of the sign of the difference. In most situations of this kind the analyst has no idea, prior to the experiment, as to whether any difference between the experimental mean and the reference value will be positive or negative. Thus the test used must cover either possibility. Such a test is called **two-sided** (or two-tailed). In a few cases, however, a different kind of test may be appropriate. If, for example, we do an experiment in which we hope to increase the rate of a reaction by addition of a catalyst, it is clear before we begin that the only outcome of interest is whether the new reaction rate is greater than the old, and only an increase need be tested for significance. This kind

of test is called **one-sided** (or one-tailed). For a given value of n and a particular probability level, the critical value for a one-sided test differs from that for a two-sided test. In a one-sided test for an increase, the critical value of t (rather than $|t|$) for $P = 0.05$ is that value which is exceeded with a probability of 5%. Since the sampling distribution of the mean is assumed to be symmetrical, this probability is half the probability that is relevant in the two-sided test. The appropriate value for the one-sided test is thus found in the $P = 0.10$ column of Table A.2. Similarly, for a one-sided test at the $P = 0.01$ level, the $P = 0.02$ column is used. For a one-sided test for a decrease, the critical value of t will be of equal magnitude but with a negative sign. If the test is carried out on a computer, it will be necessary to indicate whether a one- or a two-sided test is required.

Example 3.5.1

It is suspected that an acid–base titrimetric method has a significant indicator error and thus tends to give results with a positive systematic error (i.e. positive bias). To test this an exactly 0.1 M solution of acid is used to titrate 25.00 ml of an exactly 0.1 M solution of alkali, with the following results (ml):

25.06 25.18 24.87 25.51 25.34 25.41

Test for positive bias in these results.

For these data we have:

mean = 25.228 ml, standard deviation = 0.238 ml

Adopting the null hypothesis that there is no bias, $H_0: \mu = 25.00$, and using Eq. (3.2.1) gives:

$$t = (25.228 - 25.00) \times \sqrt{6}/0.238 = 2.35$$

From Table A.2 the critical value is $t_5 = 2.02$ ($P = 0.05$, one-sided test). Since the observed value of t is greater than this, the null hypothesis is rejected and we can conclude that there is evidence for positive bias from the indicator error.

Using a computer gives $P(t \geq 2.35) = 0.033$. Since this is less than 0.05, the result is significant at $P = 0.05$, as before.

It is interesting that if a two-sided test had been made in the example above (for which the critical value for $t_5 = 2.57$), the null hypothesis would not have been rejected! This apparently contradictory result is explained by the fact that the decision on whether to use a one- or two-sided test *depends on the degree of prior knowledge*, in this case a suspicion or expectation of positive bias. It is obviously essential that the decision on using a one- or two-sided test should be made *before* the experiment has been done, and not with hindsight, when the results might prejudice the choice. In general, two-sided tests are much more commonly used than one-sided ones and the relatively rare circumstances in which one-sided tests are necessary are easily identified.

3.6

F-test for the comparison of standard deviations

The significance tests described so far are used for comparing means, and hence for detecting systematic errors. In many cases it is also important to compare the standard deviations, i.e. the random errors of two sets of data. As with tests on means, this comparison can take two forms. Either we may wish to test whether Method A is *more precise* than Method B (i.e. a one-sided test) or we may wish to test whether Methods A and B *differ* in their precision (i.e. a two-sided test). For example, if we wished to test whether a new analytical method is more precise than a standard method we would use a one-sided test; if we wished to test whether two standard deviations differ significantly (e.g. before applying a *t*-test – see Section 3.3 above) a two-sided test would be appropriate.

The *F*-test uses the ratio of the two sample variances, i.e. the ratio of the squares of the standard deviations, s_1^2/s_2^2 .

In order to test whether the difference between two sample variances is significant, that is to test $H_0: \sigma_1^2 = \sigma_2^2$, the statistic *F* is calculated:

$$F = \frac{s_1^2}{s_2^2} \quad (3.6.1)$$

where the subscripts 1 and 2 are allocated in the equation so that *F* is always ≥ 1 .

The number of degrees of freedom of the numerator and denominator are $n_1 - 1$ and $n_2 - 1$ respectively.

The test assumes that the populations from which the samples are taken are normal.

If the null hypothesis is true, then the variance ratio should be close to 1. Differences from 1 can occur because of random variation, but if the difference is too great, it can no longer be attributed to this cause. If the calculated value of *F* exceeds a certain critical value (obtained from tables), then the null hypothesis is rejected. This critical value of *F* depends on the size of both samples, the significance level and the type of test performed. The values for $P = 0.05$ are given in Table A.3 for one-sided tests and in Table A.4 for two-sided tests; the use of these tables is illustrated in the following examples.

Example 3.6.1

A proposed method for the determination of the chemical oxygen demand of wastewater was compared with the standard (mercury salt) method. The following results were obtained for a sewage effluent sample:

	Mean (mg l ⁻¹)	Standard deviation (mg l ⁻¹)
Standard method	72	3.31
Proposed method	72	1.51

For each method eight determinations were made.

(Ballinger, D., Lloyd, A. and Morrish, A., 1982, *Analyst*, 107: 1047)

Is the precision of the proposed method significantly greater than that of the standard method?

We have to decide whether the variance of the standard method is significantly *greater than* that of the proposed method, so this is a case where a *one-sided* test must be used. F is given by the ratio of the variances (Eq. 3.6.1):

$$F = \frac{3.31^2}{1.51^2} = 4.8$$

In Table A.3 the number of degrees of freedom of the denominator is given in the left-hand column and the number of degrees of freedom of the numerator at the top. Both samples contain eight values so the number of degrees of freedom in each case is seven. The critical value is $F_{7,7} = 3.787$ ($P = 0.05$), where the first and second subscripts indicate the number of degrees of freedom of the numerator and denominator respectively. Since the calculated value of F (4.8) exceeds this, the null hypothesis of equal variances is rejected. The variance of the standard method is significantly greater than that of the proposed method at the 5% probability level, i.e. the proposed method is more precise.

Example 3.6.2

In Example 3.3.1 it was assumed that the variances of the two methods for determining chromium in rye grass did not differ significantly. This assumption can now be tested.

The standard deviations were 0.28 and 0.31 (each obtained from five measurements on a specimen of a particular plant). Calculating F using Eq. (3.6.1) so that it is greater than 1, we have:

$$F = \frac{0.31^2}{0.28^2} = 1.23$$

In this case, however, we have no reason to expect in advance that the variance of one method should be greater than the other, so a *two-sided* test is appropriate. The critical values are given in Table A.4. From this table, taking the number of degrees of freedom of both numerator and denominator as four, the critical value is $F_{4,4} = 9.605$. The calculated value is much less than this, so the null hypothesis of equal variances is retained: there is no significant difference between the two variances at the 5% level.

As with the t -test, other significance levels may be used for the F -test and the critical values can be found from the tables listed in the bibliography at the end of Chapter 1. Care must be taken that the correct table is used, depending on whether the test is one- or two-sided: for an $\alpha\%$ significance level the $2\alpha\%$ points of the

F distribution are used for a one-sided test and the $\alpha\%$ points are used for a two-sided test. A computer calculation will provide a P -value. Note that Excel® apparently only carries out a one-sided F -test, and that it is necessary to enter the data with the larger variance as the first sample.

3.7

Outliers

Every experimentalist is familiar with the situation in which one (or possibly more than one) measurement in a set of results appears to differ unexpectedly from the others. In some cases the suspect result may be attributed to a human error. For example, if the following results were given for a titration:

12.12, 12.15, 12.13, 13.14, 12.12 ml

the fourth value is almost certainly due to a slip in writing down the result and should be 12.14. However, even when such clearly erroneous values have been removed or corrected (and other obvious problems such as equipment failure taken into account) suspect data may still occur. (They may also arise in proficiency testing schemes and method performance studies (see Chapter 4) and regression problems (see Chapter 5).) Should such suspect values be *retained*, come what may, or should methods be found to decide whether or not they should be *rejected* as **outliers**? The calculated mean and standard deviation values, which are used to evaluate and compare the precision and accuracy of analytical methods, will obviously depend on whether or not any suspect measurements are rejected, so the occurrence of suspect data, and the ways in which they are treated, must always be described carefully.

The essence of the problem is clear. We saw in Chapter 2 that if the data come from a population with a normal error distribution there is a finite chance that a single value in a set of replicates will be a long way from the mean, even if everything is in order. To reject such a value wrongly might result in the value of the mean being shifted and the standard deviation being too small. On the other hand if a measurement is a genuine outlier it would be wrong to retain it, otherwise the results would be unnecessarily pessimistic. In broad terms, suspect values can be tackled in three ways. Two of these approaches, *median-based* statistics and *robust* statistics, are treated separately in Chapter 6. Here we consider only the application of significance testing, probably still the most common approach to the problem of suspect measurements.

The ISO recommended test for outliers is **Grubbs' test**. This test compares the deviation of the suspect value from the sample mean with the standard deviation of the sample. The suspect value is naturally the value that is furthest away from the mean.

In order to use Grubbs' test for an outlier, i.e. to test the null hypothesis, H_0 , that all measurements come from the same population, the statistic G is calculated:

$$G = |\text{suspect value} - \bar{x}|/s \quad (3.7.1)$$

Note that $\bar{x}$ and s are calculated with the suspect value *included*, as H_0 presumes that there are no outliers.

The test assumes that the population has a normal error distribution.

The critical values for G for $P = 0.05$ are given in Table A.5. If the calculated value of G exceeds the critical value, the suspect value is rejected. The values given are for a two-sided test, which is appropriate when it is not known in advance at which extreme of the data range an outlier may occur.

Example 3.7.1

The following values were obtained for the nitrite concentration (mg l^{-1}) in a sample of river water:

0.403, 0.410, 0.401, 0.380

The last measurement is noticeably lower than the others and is thus suspect: should it be rejected?

The four values have $\bar{x} = 0.3985$ and $s = 0.01292$, giving from Eq. (3.7.1)

$$G = \frac{|0.380 - 0.3985|}{0.01292} = 1.432$$

From Table A.5, for sample size 4, the critical value of G is 1.481, ($P = 0.05$). Since the calculated value of G does not exceed 1.481, the suspect measurement should be retained.

In fact, the suspect value in this data set would have to be considerably lower before it was rejected. Trial and error shows that for the data set

0.403, 0.410, 0.401, b

(where $b < 0.401$), the value of b would have to be as low as 0.356 before it was rejected using the Grubbs' test criterion.

Ideally, further measurements should be made when a suspect value occurs, especially if only a few values have been obtained initially: when the sample size is small, one measurement must be *very* different from the rest before it can safely be rejected. Extra values may make it clearer whether or not the suspect value should be rejected, and will also reduce its effect on the mean and standard deviation if it is retained.

Example 3.7.2

Three further measurements were added to those given in the example above, so that the complete results became:

0.403, 0.410, 0.401, 0.380, 0.400, 0.413, 0.408

Should the value 0.380 still be retained?

The seven values have $\bar{x} = 0.4021$ and $s = 0.01088$.

The calculated value of G is now:

$$G = \frac{|0.380 - 0.4021|}{0.01088} = 2.031$$

The critical value of G ($P = 0.05$) for a sample size 7 is 2.020 so the suspect measurement is now (just) rejected at the 5% significance level.

Dixon's test (sometimes called the **Q-test**) is another test for outliers, popular because the calculation is so simple. For small samples (size 3 to 7) the test assesses a suspect measurement by comparing the difference between it and the measurement nearest to it in size with the range of the measurements. (For larger samples the form of the test is modified slightly. The texts by Barnett and Lewis and by Ellison *et al.* listed in the Bibliography at the end of this chapter give further details.)

In order to use Dixon's test for an outlier, that is to test H_0 : all measurements come from the same population, the statistic Q is calculated:

$$Q = \frac{|\text{suspect value} - \text{nearest value}|}{\text{largest value} - \text{smallest value}} \quad (3.7.2)$$

This test again assumes that the population has a normal error distribution.

The critical values of Q for $P = 0.05$ for a two-sided test are given in Table A.6. If the calculated value of Q exceeds the critical value the suspect value is rejected.

Example 3.7.3

Apply Dixon's test to the data from Example 3.7.2.

Using Eq. (3.7.2) we have

$$Q = \frac{|0.380 - 0.400|}{0.413 - 0.380} = 0.606$$

The critical value of Q ($P = 0.05$) for a sample size 7 is 0.570. The suspect value 0.380 is thus rejected (as it was using Grubbs' test).

Although taking extra measurements can be helpful, as shown above, it is of course possible that the new values may include further suspect measurements! Moreover if two suspect values occur, both of them might be at high end of the measurement range, both at the low end, or one at the high end and one at the low end. This can complicate the use of the tests. Figure 3.1 illustrates in the form of dot-plots two examples of such difficulties in the application of the Dixon test. In Fig. 3.1(a) there are two

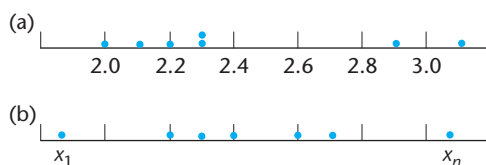


Figure 3.1 Dot-plots illustrating the problem of handling outliers: (a) when there are two suspect results at the high end of the sample data and (b) when there are two suspect results, one at each extreme of the data.

results (2.9, 3.1), both of which are suspiciously high compared with the mean of the data, yet if Q were calculated uncritically using Eq. (3.7.2) we would obtain:

$$Q = \frac{3.1 - 2.9}{3.1 - 2.0} = 0.18$$

a value which is not significant at the $P = 0.05$ level and suggests that the suspect values should be retained. Clearly the possible outlier 3.1 has been **masked** by the other possible outlier, 2.9, giving a low value of Q . A different situation is shown in Fig. 3.1(b), where the two suspect values are at opposite ends of the data set. This gives a large value for the range, so Q is small and again the null hypothesis might be wrongly retained. Handling multiple outliers clearly increases the complexity of outlier testing, though the Grubbs' test has been modified to handle data sets which contain two suspect measurements. These modifications use different formulae for G , depending on whether the suspect measurements occur at the same or opposite ends of the range of values, and two further tables of critical values are thus required. Again, texts by Barnett and Lewis and by Ellison *et al.* listed in the Bibliography give further details.

A more fundamental concern over these outlier tests is that they assume that the sample comes from a population with a normal error distribution. A result that seems to be an outlier on the assumption of such a distribution may well not be an outlier if the sample actually comes from (for example) a log-normal distribution (Section 2.3). Therefore outlier tests should not be used if there is a suspicion that the population may not have a normal error distribution. Moreover there are cases where different outlier tests lead to opposite conclusions. The reader can verify that if the suspect measurement in examples 3.7.2 and 3.7.3 had been 0.382 rather than 0.380, the Dixon test would have recommended its rejection, while Grubbs' test would have recommended its retention. Such conclusions must obviously be regarded with extreme caution. These difficulties, along with the complications arising in cases of multiple outliers, explain the increasing use of the methods described in Chapter 6, especially the robust methods. Such approaches either are insensitive to extreme values or at least give them less weight in calculations, so the problem of whether or not to reject outliers is avoided.

3.8 Analysis of variance

In Section 3.3 a method was described for comparing two means to test whether they differ significantly. In analytical work there are often more than two means to be compared. Some possible situations are: comparing the mean concentration of protein in solution for samples stored under different conditions; comparing the mean results

obtained for the concentration of an analyte by several different methods; comparing the mean titration results obtained by several different experimentalists using the same apparatus. In all these examples there are two possible sources of variation. The first, which is always present, is due to the random error in measurement. This was discussed in detail in the previous chapter: it is this error which causes a different result to be obtained each time a measurement is repeated under the same conditions. The second possible source of variation is due to what is known as a **controlled** or **fixed-effect factor**: for the examples above, the controlled factors are respectively the conditions under which the solution was stored, the method of analysis used, and the experimentalist carrying out the titration. **Analysis of variance** (frequently abbreviated to ANOVA) is an extremely powerful statistical technique which can be used to separate and estimate the different causes of variation. For the examples above, it can be used to separate any variation which is caused by changing the controlled factor from the variation due to random error. It can thus test whether altering the controlled factor leads to a significant difference between the mean values obtained.

ANOVA can also be used in situations where there is more than one source of random variation. Consider, for example, the purity testing of a barrelful of sodium chloride. Samples are taken from different parts of the barrel chosen at random and replicate analyses performed on these samples. In addition to the random error in the measurement of the purity, there may also be variation in the purity of the samples from different parts of the barrel. Since the samples were chosen at random, this variation will be random and is thus sometimes known as a **random-effect factor**. Again, ANOVA can be used to separate and estimate the sources of variation. Both types of statistical analysis described above are known as *one-way* ANOVA as there is one factor, either controlled or random, in addition to the random error in measurements. The arithmetical procedures are similar in the fixed- and random-effect factor cases: examples of the former are given in this chapter and of the latter in the next chapter, where sampling is considered in more detail. More complex situations in which there are two or more factors, possibly interacting with each other, are considered in Chapter 7.

3.9

Comparison of several means

Table 3.2 shows the results obtained in an investigation into the stability of a fluorescent reagent stored under different conditions. The values given are the fluorescence signals (in arbitrary units) from dilute solutions of equal concentration. Three replicate measurements were made on each sample. The table shows that the mean

Table 3.2 Fluorescence from solutions stored under different conditions

Conditions	Replicate measurements	Mean
A Freshly prepared	102, 100, 101	101
B Stored for 1 hour in the dark	101, 101, 104	102
C Stored for 1 hour in subdued light	97, 95, 99	97
D Stored for 1 hour in bright light	90, 92, 94	92
Overall mean		98

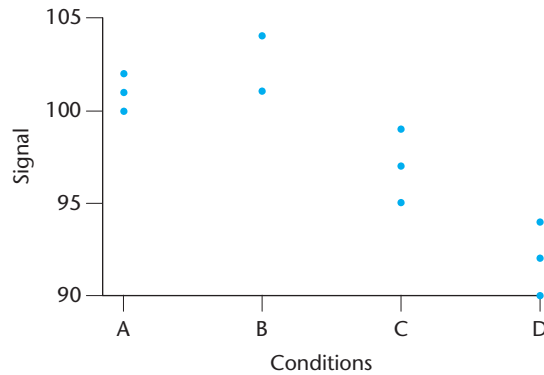


Figure 3.2 Dot-plot of results in Table 3.2.

values for the four samples are different. However, we know that because of random error, even if the true value which we are trying to measure is unchanged, the sample mean may vary from one sample to the next. **ANOVA** tests whether the difference between the sample means is too great to be explained by the random error. Figure 3.2 shows a dot-plot comparing the results obtained in the different conditions. This suggests that there may be little difference between conditions A and B but that conditions C and D differ both from A and B and from each other.

The problem can be generalised to consider h samples each with n members as in Table 3.3 where x_{ij} is the j th measurement of the i th sample. The means of the samples are $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_h$ and the mean of all the values grouped together is $\bar{x}$. The null hypothesis adopted is that all the samples are drawn from a population with mean μ and variance σ_0^2 . On the basis of this hypothesis σ_0^2 can be estimated in two ways, one involving the variation *within* the samples and the other the variation *between* the samples.

1 Within-sample variation

For each sample a variance can be calculated by using the formula

$$\sum (x_i - \bar{x})^2 / (n - 1) \quad (\text{see Eq. (2.1.2)})$$

Using the values in Table 3.2 we have:

$$\text{Variance of sample A} = \frac{(102 - 101)^2 + (100 - 101)^2 + (101 - 101)^2}{(3 - 1)} = 1$$

$$\text{Variance of sample B} = \frac{(101 - 102)^2 + (101 - 102)^2 + (104 - 102)^2}{(3 - 1)} = 3$$

Table 3.3 Generalisation of Table 3.2

				Mean
Sample 1	x_{11}	$x_{12} \cdots x_{1j} \cdots x_{1n}$		$\bar{x}_1$
Sample 2	x_{21}	$x_{22} \cdots x_{2j} \cdots x_{2n}$		$\bar{x}_2$
	$\vdots$	$\vdots \quad \vdots \quad \vdots$		$\vdots$
Sample i	x_{i1}	$x_{i2} \cdots x_{ij} \cdots x_{in}$		$\bar{x}_i$
	$\vdots$	$\vdots \quad \vdots \quad \vdots$		$\vdots$
Sample h	x_{h1}	$x_{h2} \cdots x_{hj} \cdots x_{hn}$		$\bar{x}_h$
Overall mean = $\bar{x}$				

Similarly it can be shown that samples C and D both have variances of 4. Averaging these values gives a *within-sample estimate* of $\sigma_0^2 = (1 + 3 + 4 + 4)/4 = 3$. This estimate has eight degrees of freedom: each sample estimate has two degrees of freedom and there are four samples. Note that this estimate of σ_0^2 does not depend on the means of the samples: for example, if all the measurements for sample A were increased by say, 4, the estimate of σ_0^2 would be unaltered. The general formula for the within-sample estimate of σ_0^2 is:

$$\text{Within-sample estimate of } \sigma_0^2 = \frac{\sum_i \sum_j (x_{ij} - \bar{x}_i)^2}{h(n-1)} \quad (3.9.1)$$

The summation over j and division by $(n-1)$ gives the variance of each sample; the summation over i and division by h averages these sample variances. The expression in Eq. (3.9.1) is known as a **mean square** (MS) since it involves a sum of squared (SS) terms divided by the number of degrees of freedom. In this case the number of degrees of freedom is 8 and the mean square is 3, so the sum of the squared terms is $3 \times 8 = 24$.

2 Between-sample variation

If the samples are all drawn from a population which has a variance σ_0^2 , then their means come from a population with variance σ_0^2/n (cf. the sampling distribution of the mean, Section 2.5). Thus, if the null hypothesis is true, the variance of the means of the samples gives an estimate of σ_0^2/n . From Table 3.2:

$$\begin{aligned} \text{Sample mean variance} &= \frac{(101 - 98)^2 + (102 - 98)^2 + (97 - 98)^2 + (92 - 98)^2}{(4 - 1)} \\ &= 62/3 \end{aligned}$$

So the *between-sample estimate* of σ_0^2 is $(62/3) \times 3 = 62$. This estimate has three degrees of freedom since it is calculated from four sample means. Note that this estimate of σ_0^2 does not depend on the variability *within* each sample, since it is calculated from the sample means. But if, for example, the mean of sample D was changed, then this estimate of σ_0^2 would also be changed.

In general we have:

$$\text{Between-sample estimate of } \sigma_0^2 = n \sum_i (\bar{x}_i - \bar{\bar{x}})^2 / (h - 1) \quad (3.9.2)$$

which again is a 'mean square' involving a sum of squared terms divided by the number of degrees of freedom. In this case the number of degrees of freedom is 3 and the mean square is 62, so the sum of the squared terms is $3 \times 62 = 186$.

Summarising our calculations so far:

Within-sample mean square = 3 with 8 d.f.

Between-sample mean square = 62 with 3 d.f.

If the null hypothesis is correct, then these two estimates of σ_0^2 should not differ significantly. If it is incorrect, the between-sample estimate of σ_0^2 will be *greater than* the within-sample estimate because of between-sample variation. To test whether it is significantly greater, a *one-sided F-test* is used (see Section 3.6):

$$F = 62/3 = 20.7$$

(Remember that each mean *square* is used so no further squaring is necessary.) The numerator has three degrees of freedom and the denominator has eight degrees of freedom, so from Table A.3 the critical value of F is 4.066 ($P = 0.05$). Since the calculated value of F is much greater than this the null hypothesis is rejected: the sample means do differ significantly.

Such a significant difference can arise for several different reasons: for example, one mean may differ from all the others, all the means may differ from each other, the means may fall into two distinct groups, etc. A simple way of deciding the reason for a significant result is to arrange the means in increasing order and compare the difference between adjacent values with a quantity called the **least significant difference**. This is given by $s\sqrt{(2/n) \times t_{h(n-1)}}$ where s is the within-sample estimate of σ_0 and $h(n-1)$ is the number of degrees of freedom of this estimate. For the example above, the sample means arranged in increasing order of size are:

$$\bar{x}_D = 92 \quad \bar{x}_C = 97 \quad \bar{x}_A = 101 \quad \bar{x}_B = 102$$

and the least significant difference is $\sqrt{3} \times \sqrt{2/3} \times 2.31 = 3.27$ ($P = 0.05$). Comparing this value with the differences between the means suggests that conditions D and C give results which differ significantly from each other and from the results obtained in conditions A and B. However, the results obtained in conditions A and B do not differ significantly from each other. This confirms the indications of the dot-plot in Fig. 3.2, and suggests that it is exposure to light which affects the intensity of fluorescence.

The least significant difference method described above is not entirely rigorous: it can be shown that it leads to rather too many significant differences. However, it is a simple follow-up test when ANOVA has indicated that there is a significant difference between the means. Descriptions of other more rigorous tests are given in the books in the Bibliography at the end of this chapter.

3.10 The arithmetic of ANOVA calculations

In the preceding ANOVA calculation σ_0^2 was estimated in two different ways. If the null hypothesis were true, σ_0^2 could also be estimated in a third way by treating the data as one large sample. This would involve summing the squares of the deviations from the overall mean:

$$\begin{aligned} \sum_i \sum_j (x_{ij} - \bar{x})^2 &= 4^2 + 2^2 + 3^2 + 3^2 + 3^2 + 6^2 + 1^2 + 3^2 + 1^2 + 8^2 + 6^2 + 4^2 \\ &= 210 \end{aligned}$$

and dividing by the number of degrees of freedom, $12 - 1 = 11$.

This method of estimating σ_0^2 is not used in the analysis because the estimate depends both on the within- and between-sample variation. However, there is an exact algebraic relationship between this total variation and the sources of variation which contribute to it. This leads to a simplification of the arithmetic involved, especially in more complicated ANOVA calculations. The relationship between the sources of variation is illustrated by Table 3.4, which summarises the sums of squares

Table 3.4 Summary of sums of squares and degrees of freedom

Source of variation	Sum of squares	Degrees of freedom
Between-sample	$n \sum_i (\bar{x}_i - \bar{x})^2 = 186$	$h - 1 = 3$
Within-sample	$\sum_i \sum_j (x_{ij} - \bar{x}_i)^2 = 24$	$h(n - 1) = 8$
Total	$\sum_i \sum_j (x_{ij} - \bar{x})^2 = 210$	$hn - 1 = 11$

and degrees of freedom. Clearly the values for the total variation given in the last row of the table are the sums of the values in the first two rows for both the sum of squares and the degrees of freedom. This additive property holds for all the ANOVA calculations described in this book.

Just as in the calculation of variance, there are formulae which simplify the calculation of the individual sums of squares. These formulae are summarised below:

One-way ANOVA tests for a significant difference between means when there are more than two samples involved. The formulae used are:

Source of variation	Sum of squares	Degrees of freedom
Between-samples	$\sum_i (T_i^2/n) - (T^2/N)$	$h - 1$
Within-samples	by subtraction	by subtraction
Total	$\sum_i \sum_j x_{ij}^2 - (T^2/N)$	$N - 1$

where there are h samples each with n measurements.

$N = nh$ = total number of measurements

T_i = sum of the measurements in the i th sample

T = sum of all the measurements, the grand total

The test statistic is F = between-sample mean square/within-sample mean square and the critical value is $F_{h-1, N-h}$.

These formulae can be illustrated by repeating the ANOVA calculations for the data in Table 3.2. The calculation is given in full below.

Example 3.10.1

Test whether the samples in Table 3.2 are drawn from populations with equal means.

In the calculation of the mean squares all the values in Table 3.2 have had 100 subtracted from them, which simplifies the arithmetic considerably. Note that

this does not affect either the between- or within-sample estimates of variance because the same quantity has been subtracted from every value.

				T_i	T_i^2
A	2	0	1	3	9
B	1	1	4	6	36
C	-3	-5	-1	-9	81
D	-10	-8	-6	-24	576
				$T = -24$	$\sum T_i^2 = 702$

$$n = 3, h = 4, N = 12, \sum_i \sum_j x_{ij}^2 = 258$$

Source of variation	Sum of squares	Degrees of freedom	Mean square
Between-sample	$702/3 - (-24)^2/12 = 186$	3	$186/3 = 62$
Within-sample	by subtraction = 24	8	$24/8 = 3$
Total	$258 - (-24)^2/12 = 210$	11	
	$F = 62/3 = 20.7$		

The critical value $F_{3,8} = 4.066$ ($P = 0.05$). Since the calculated value is greater than this the null hypothesis is rejected: the sample means differ significantly.

The calculations for one-way ANOVA have been given in detail in order to make the principles behind the method clearer. In practice such calculations are normally made on a computer. Both Minitab® and Excel® have an option which performs one-way ANOVA and, as an example, the output given by Excel® is shown below, using the original values.

Anova: Single factor

SUMMARY

Groups	Count	Sum	Average	Variance
A	3	303	101	1
B	3	306	102	3
C	3	291	97	4
D	3	276	92	4

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	186	3	62	20.66667	0.0004	4.06618
Within Groups	24	8	3			
Total	210	11				

Certain assumptions have been made in performing the ANOVA calculations in this chapter. The first is that the variance of the random error is not affected by the treatment used. This assumption is implicit in the pooling of the within-sample variances to calculate an overall estimate of the error variance. In doing this we are assuming what is known as the **homogeneity of variance**. In the particular example given above, where all the measurements are made in the same way, we would expect homogeneity of variance. Methods for testing for this property are given in the Bibliography at the end of this chapter.

A second assumption is that the uncontrolled variation is truly random. This would not be the case if, for example, there was an uncontrolled factor such as a temperature change which produced a trend in the results over a period of time. The effects of such uncontrolled factors can be overcome to a large extent by the techniques of randomisation and blocking which are discussed in Chapter 7.

An important part of ANOVA is clearly the application of the *F*-test. Use of this test (see Section 3.6) simply to compare the variances of two samples depends on the samples being drawn from a normal population. Fortunately, however, the *F*-test as applied in ANOVA is not too sensitive to departures from normality of distribution.

3.11

The chi-squared test

In the significance tests so far described in this chapter the data have taken the form of observations which, apart from any rounding off, have been measured on a continuous scale. In contrast, this section is concerned with *frequency*, i.e. *the number of times* a given event occurs. For example, Table 2.2 gives the frequencies of the different values obtained for the nitrate ion concentration in water when 50 measurements were made. As discussed in Chapter 2, such measurements are usually assumed to be drawn from a population with normally distributed errors. **The chi-squared test can be used to test whether the observed frequencies in a particular case differ significantly from those which would be expected on this null hypothesis.**

To test whether the observed frequencies, O_i , agree with those expected, E_i , according some null hypothesis, the statistic χ^2 is calculated:

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i} \quad (3.11.1)$$

Since the calculation involved in using this statistic to test for normality is relatively complicated, it will not be described here. The principle of the chi-squared test is more easily understood by means of the following example.

Example 3.11.1

The numbers of glassware breakages reported by four laboratory workers over a given period are shown below. Is there any evidence that the workers differ in their reliability?

Numbers of breakages: 24, 17, 11, 9

The null hypothesis is that there is no difference in reliability. Assuming that the workers use the laboratory for an equal length of time, we would thus expect the same number of breakages by each worker. Since the total number of breakages is 61, the expected number of breakages per worker is $61/4 = 15.25$. Obviously it is not possible in practice to have a non-integral number of breakages, but this number, although it is a mathematical concept, can still be used in the test. The nearest practicable 'equal' distribution is 15, 15, 15, 16 in some order. The question to be answered is whether the difference between the observed and expected frequencies is so large that the null hypothesis should be rejected. That there should be *some* difference between the two sets of frequencies can be appreciated by considering a sequence of throws of a die: we should, for example, be most surprised if 30 throws yielded exactly equal frequencies for 1, 2, 3, etc. The calculation of χ^2 is shown below.

Observed frequency, O	Expected frequency, E	$O - E$	$(O - E)^2/E$
24	15.25	8.75	5.020
17	15.25	1.75	0.201
11	15.25	-4.25	1.184
9	15.25	-6.25	2.561
Totals	61	0	$\chi^2 = 8.966$

Note that the total of the $O - E$ column is always zero, thus providing a useful check on the calculation.

If χ^2 exceeds a certain critical value, the null hypothesis is rejected. The critical value depends, as in other significance tests, on the significance level of the test and on the number of degrees of freedom. The number of degrees of freedom is, in an example of this type, one less than the number of classes used, i.e. $4 - 1 = 3$ in this case. The critical values of χ^2 for $P = 0.05$ are given in Table A.7. For three degrees of freedom the critical value is 7.81. Since the calculated value is greater than this, the null hypothesis is rejected at the 5% significance level: there is evidence that the workers *do* differ in their reliability.

The calculation of χ^2 suggests that a significant result is obtained because of the high number of breakages reported by the first worker. To study this further, additional chi-squared tests can be performed. One of them tests whether the second, third and fourth workers differ significantly from each other: in this case each expected frequency is $(17 + 11 + 9)/3$. (Note that the *t*-test cannot be used here as we are dealing with frequencies and not continuous variates.) Alternatively we can test whether the first worker differs significantly from the other three workers taken as a group. In this case there are two classes: the breakages by the first worker with an expected frequency of 15.25 and the total breakages by the other workers with an expected frequency of $15.25 \times 3 = 45.75$. In such cases when there are only two classes and hence one degree of freedom, an adjustment known as **Yates's correction**

should be applied. This involves replacing $O-E$ by $|O-E| - 0.5$. For example, if $O-E = -4.5$, $|O-E| = 4.5$ and $|O-E| - 0.5 = 4$. These further tests are given as an exercise at the end of this chapter.

In general the chi-squared test should be used only if the total number of observations is 50 or more and the individual expected frequencies are no fewer than 5, though this is not a rigid rule. Further information on the chi-squared test and its other applications are provided in the references in the Bibliography at the end of the chapter.

3.12

Testing for normality of distribution

As has been emphasised in this chapter, many statistical tests assume that the data used are drawn from a normal population. Although the chi-squared test can be used to test this assumption it should be used only if there are 50 or more data points, so it is of limited value in analytical work, when we often have only a small set of data. A simple way of seeing whether a set of data is consistent with the assumption of normality is to plot a **cumulative frequency curve on special graph paper known as normal probability paper**. This method is most easily explained by means of an example.

Example 3.12.1

Use normal probability paper to investigate whether the data below could have been drawn from a normal population:

109, 89, 99, 99, 107, 111, 86, 74, 115, 107, 134, 113, 110, 88, 104.

Table 3.5 shows the data arranged in order of increasing size. The second column gives the cumulative frequency for each measurement, i.e. the number

Table 3.5 Data for normal probability paper example

Measurement	Cumulative frequency	% Cumulative frequency
74	1	6.25
86	2	12.50
88	3	18.75
89	4	25.00
99	6	37.50
104	7	43.75
107	9	56.25
109	10	62.50
110	11	68.75
111	12	75.00
113	13	81.25
115	14	87.50
134	15	93.75

of measurements *less than or equal to* that measurement. The third column gives the percentage cumulative frequency. This is calculated by using the formula:

$$\% \text{ cumulative frequency} = 100 \times \text{cumulative frequency} / (n + 1)$$

where n is the total number of measurements. (A divisor of $n + 1$ rather than n is used so that the % cumulative frequency of 50% falls at the middle of the data set, in this case at the eighth measurement. Note that two of the values, 99 and 107, occur twice.) If the data come from a normal population, a graph of percentage cumulative frequency against measurement yields an S-shaped curve, as shown in Fig. 3.3.

Normal probability paper has a non-linear scale for the percentage cumulative frequency axis, which will convert this S-shaped curve into a straight line. A graph plotted on such paper is shown in Fig. 3.4: the points lie approximately on a straight line, supporting the hypothesis that the data come from a normal distribution.

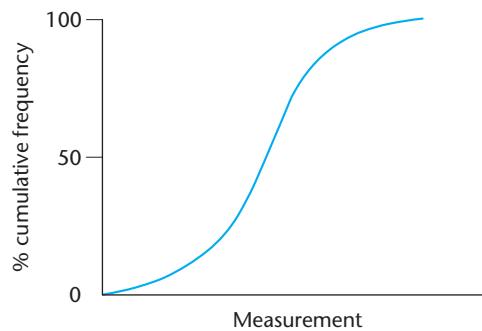


Figure 3.3 The cumulative frequency curve for a normal distribution.

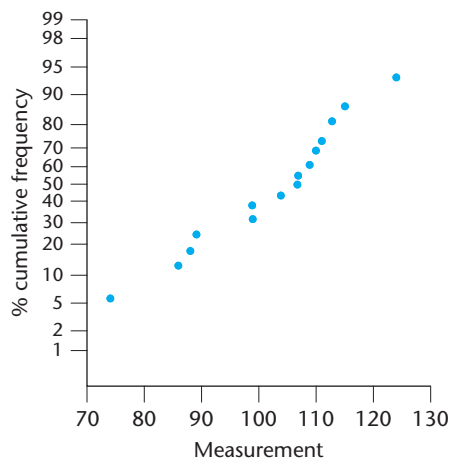


Figure 3.4 Normal probability plot for the example in Section 3.12.

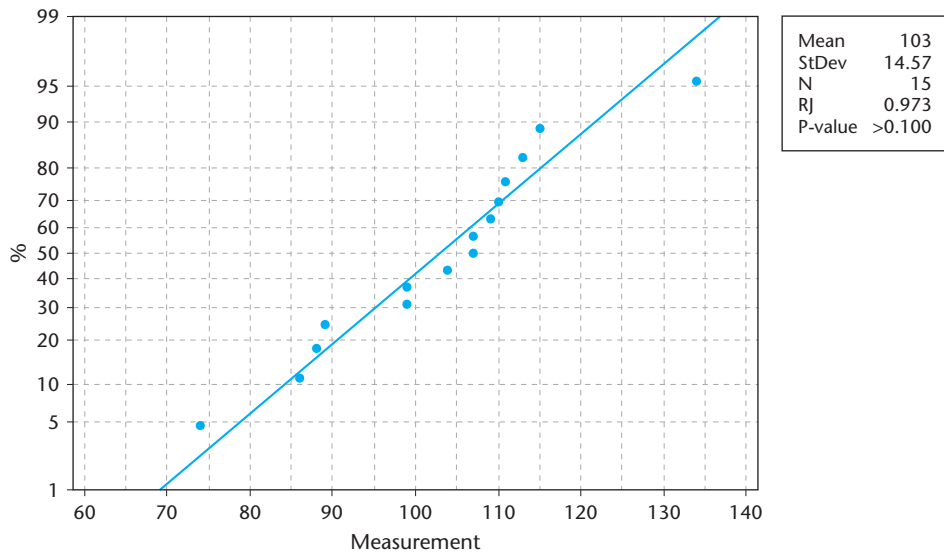


Figure 3.5 Normal probability plot obtained using Minitab®.

Minitab® will give a normal probability plot directly. The result is shown in Fig. 3.5. The program uses a slightly different method to calculate the percentage cumulative frequency, but the difference is not important.

One method of testing for normality is to measure how closely the points on a normal probability plot conform to a straight line. Minitab® gives a test for normality (the *Ryan–Joiner* (RJ) test) based on this idea. The value of this test statistic is given beside the graph in Fig. 3.5 ($RJ = 0.973$), together with a P value of >0.100 , indicating that the assumption of normality is justified.

The **Kolmogorov–Smirnov** method can be used to test for normality, among other applications. The principle of the method involves comparing the sample cumulative distribution function with the cumulative distribution function of the hypothesised distribution. The hypothetical and sample functions are drawn on the same graph. If the experimental data depart substantially from the expected distribution, the two functions will be widely separated over part of the diagram. If, however, the data are closely in accord with the expected distribution, the two functions will never be very far apart. The test statistic is given by the maximum vertical difference between the two functions and is compared in the usual way with a set of tabulated values.

When the Kolmogorov–Smirnov method is used to test whether a distribution is normal, we first transform the original data, which might have any values for their mean and standard deviation, into the **standard normal variable**, z (see Section 2.2). This is done by using the equation:

$$z = \frac{x - \mu}{\sigma} \quad (3.12.1)$$

where the terms have their usual meanings. The values of the mean and the standard deviation are estimated by the methods of Chapter 2. The data are next transformed by using Eq. (3.12.1) and then the Kolmogorov–Smirnov method is applied. This test is illustrated in the following example.

Example 3.12.2

Eight titrations were performed, with the results 25.13, 25.02, 25.11, 25.07, 25.03, 24.97, 25.14 and 25.09 ml. Could such results have come from a normal population?

First we estimate the mean and the standard deviation (with the aid of Eqs (2.1.1) and (2.1.2)) as 25.07 and 0.0593 ml respectively. Next we transform the x -values into z -values by using the relationship $z = (x - 25.07)/0.059$, obtained from Eq. (3.12.1). The eight results are transformed into 1.01, -0.84 , 0.67, 0, -0.67 , -1.69 , 1.18 and 0.34. These z -values are arranged in order of increasing size to give -1.69 , -0.84 , -0.67 , 0, 0.34, 0.67, 1.01, 1.18, and plotted as a stepped cumulative distribution function with a step height of $1/n$, where n is the number of measurements. Thus, in this case the step height is 0.125 (i.e. $1/8$). (Note that this is not quite the same approach as that used in Example 3.12.1.) Comparison with the hypothetical function for z (Table A.2) indicates (Fig. 3.6) that the maximum difference is 0.132 when $z = 0.34$. The critical values for this test are given in Table A.14. The table shows that, for $n = 8$ and $P = 0.05$, the critical value is 0.288. Since $0.132 < 0.288$ we can accept the null hypothesis that the data come from a normal population with mean 25.07 and standard deviation 0.059.

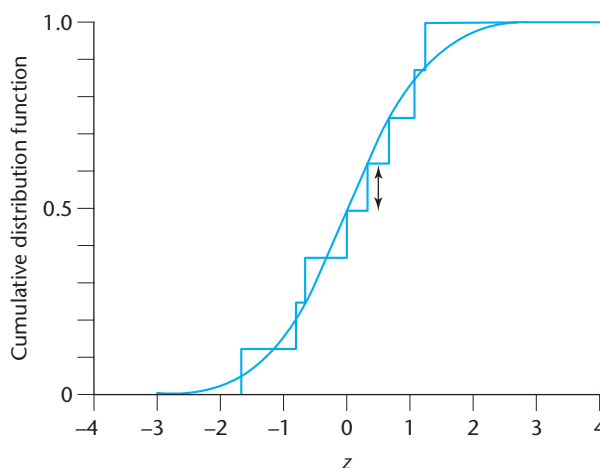


Figure 3.6 Kolmogorov's method used to test for the normal distribution. Maximum difference between the hypothetical and sample functions is shown by the arrow ($\updownarrow$).

The value of this Kolmogorov–Smirnov test statistic, together with its P value, can be obtained directly from Minitab® in conjunction with a normal probability plot. The P value is given as >0.150 , indicating again that, at $P = 0.05$, there is no significant difference between the continuous and stepped cumulative distribution plots.

3.13 Conclusions from significance tests

This section looks more closely at the conclusions which may be drawn from a significance test. As was explained in Section 3.2, a significance test at, for example, the $P = 0.05$ level involves a 5% risk that a null hypothesis will be rejected *even though it is true*. This type of error is known as a **Type I error**. The risk of such an error can be reduced by altering the significance level of the test to $P = 0.01$ or even $P = 0.001$. This, however, is not the only possible type of error: it is also possible to *retain* a null hypothesis *even when it is false*. This is called a **Type II error**. In order to calculate the probability of this type of error it is necessary to postulate an alternative to the null hypothesis, known as an **alternative hypothesis**, H_1 .

Consider the situation where a certain chemical product is meant to contain 3% of phosphorus by weight. It is suspected that this proportion has increased. To test this suspicion the composition is analysed by a standard method with a known standard deviation of 0.036%. Suppose four measurements are taken and a significance test is performed at the $P = 0.05$ level. A one-sided test is required, as we are interested only in an *increase* in the phosphorus level. The null hypothesis is:

$$H_0: \mu = 3.0\%$$

The solid line in Fig. 3.7 shows the sampling distribution of the mean if H_0 is true. This sampling distribution has mean 3.0 and standard deviation (i.e. standard error of the mean) $\sigma/\sqrt{n} = 0.036/\sqrt{4}\%$. If the sample mean lies above the indicated critical value, $\bar{x}_c$, the null hypothesis is rejected. Thus the darkest region, with area 0.05, represents the probability of a Type I error.

Suppose we take the alternative hypothesis:

$$H_1: \mu = 3.05\%$$

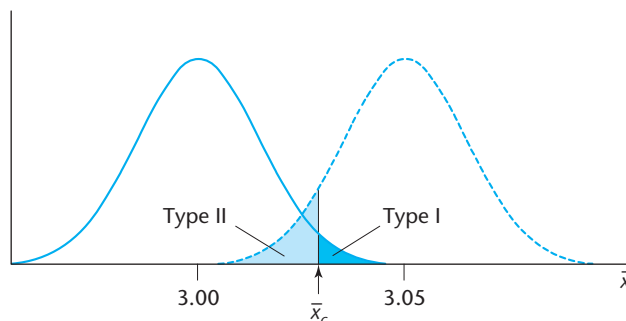


Figure 3.7 Type I and Type II errors.

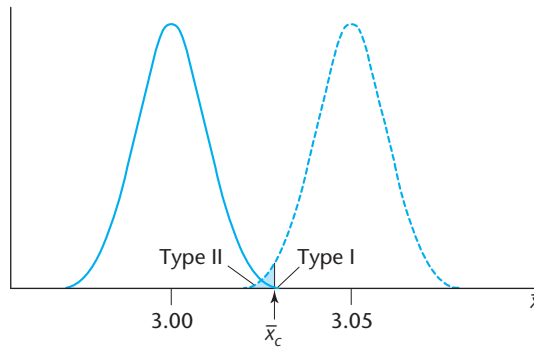


Figure 3.8 Type I and Type II errors for increased sample size.

The broken line in Fig. 3.7 shows the sampling distribution of the mean if the alternative hypothesis is true. *Even if this is the case*, the null hypothesis will be retained if the sample mean lies below $\bar{x}_c$. The probability of this Type II error is represented by the lightly shaded area. The diagram shows the interdependence of the two types of error. If, for example, the significance level is changed to $P = 0.01$ in order to reduce a risk of a Type I error, $\bar{x}_c$ will be increased and the risk of a Type II error is also increased. Conversely, a decrease in the risk of a Type II error can be achieved only at the expense of an increase in the probability of a Type I error. The only way in which both errors can be reduced (for a given alternative hypothesis) is by increasing the sample size. The effect of increasing n to 9, for example, is illustrated in Fig. 3.8: the resultant decrease in the standard error of the mean produces a decrease in both types of error, for a given value of $\bar{x}_c$.

The probability that a false null hypothesis is correctly rejected is known as the **power** of a test. That is, the power of a test is $(1 - \text{the probability of a Type II error})$. In the example above it is a function of the mean specified in the alternative hypothesis. It also depends on the sample size, the significance level of the test and whether the test is one- or two-sided. In some circumstances where two or more tests are available to test the same hypothesis, it may be useful to compare the powers of the tests in order to decide which is most appropriate.

Type I and Type II errors are also relevant when significance tests are applied sequentially. An example of this situation is the application of the t -test to the difference between two means, after first using the F -test to decide whether or not the sample variances can be pooled (see Sections 3.3 and 3.6). Both Type I and Type II errors can arise from the initial F -test, and the occurrence of either type will mean that the stated levels of significance for the subsequent t -test are incorrect, because the incorrect form of the t -test may have been applied.

3.14 Bayesian statistics

The above example emphasises the general conclusion that significance tests do not give clear-cut answers: instead they aid the interpretation of experimental data by giving the probabilities that certain conclusions are valid. Another problem is that

the conclusions drawn from significance tests may not be exactly those that we are seeking. Suppose, for example, that we need to know whether the sodium content of a bottle of mineral water exceeds the value of 8 ppm stated on its label. We could measure the sodium level several times and use the **t-test**, with the null hypothesis $H_0: \mu = 8$, to see whether the result is significantly higher than 8 ppm: note that this will be a *one-tailed test*, as we are only interested in values *above* 8 ppm. *The result of the test will provide the probability of obtaining the experimental results if there are 8 ppm of sodium in the water.* What it does not tell us is the inverse of this, i.e. *the probability that there are 8 ppm of sodium in the water, given the experimental data.*

It is possible to attach probabilities to hypotheses by using a different approach, called **Bayesian statistics**, named after an eighteenth-century clergyman, the Reverend Thomas Bayes. Bayesian analysis starts by stating that μ can take different possible values, each with an associated probability. It may seem strange that the population mean, which must have a definite fixed value, should be treated in this way, but the idea is to quantify our beliefs about μ . For the sodium in mineral water example in the previous paragraph we might say that we think that μ lies somewhere in the interval 7 to 9 ppm and that all possible values in that range are equally likely. Or we might postulate a more complicated distribution, for example a normal distribution, for the possible values. The distribution that we postulate initially is called the **prior distribution**. We can revise this distribution in the light of experimental data using **Bayes' theorem**, to find a **posterior distribution** for μ . Bayes' theorem states that:

$$\text{Posterior distribution} \propto \text{prior distribution} \times \text{probability of observed values} \quad (3.14.1)$$

where the $\propto$ symbol means 'proportional to'. The application of this equation to the sodium analysis is shown below.

Example 3.14.1

Measurements of the sodium concentration of a bottle of mineral water are normally distributed with a standard deviation of 0.2 ppm (due to measurement variation) and a mean μ which is unknown. A single measurement of the sodium content yields the value $x = 7.4$ ppm. Find the posterior distribution of μ , taking as a prior distribution that μ is uniformly distributed over the interval 7–9 ppm.

Figure 3.9(a) shows the prior distribution of μ . The prior distribution of μ is uniform so

$$\text{prior distribution} = C$$

where C is a constant.

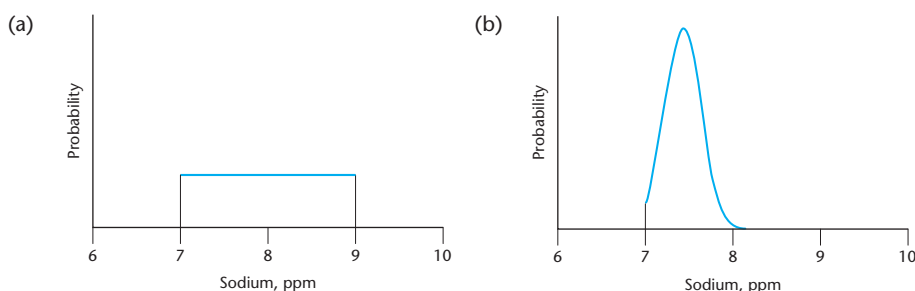


Figure 3.9 Bayesian calculation: (a) the prior distribution and (b) the posterior distribution.

The probability of an observed value of $x = 7.4$ ppm is given by the equation for the normal distribution (Eq. 2.2.1)

$$y = \frac{1}{\sigma\sqrt{2\pi}} \exp[-(x - \mu)^2/2\sigma^2]$$

so

$$\text{Probability of an observed value of } 7.4 \propto \exp[-(7.4 - \mu)^2/2 \times 0.2^2]$$

So from Eq. (3.14.1) we have

$$\text{Posterior distribution} \propto C \times \exp[-(7.4 - \mu)^2/2 \times 0.2^2]$$

where $7 \leq \mu \leq 9$.

This distribution is shown in Fig. 3.9(b). It is a normal distribution, truncated at 7 (and 9). It combines our prior beliefs and the experimental value of 7.4 ppm to give a distribution that is no longer uniform, but peaks at 7.4 ppm as the most likely value for μ . From this distribution we can calculate the probability that μ lies in a certain interval, for example that $\mu > 8$ ppm. This is given by the area under the curve to the right of $x = 8$ in Fig. 3.9(b) as a fraction of the total area under the curve.

The posterior distribution summarizes our current information about μ . It shows, for example, that μ is much more likely to lie between 7 and 8 ppm than between 8 and 9 ppm. To refine our knowledge about μ we could obtain additional experimental data and, *taking the posterior distribution as a new prior distribution*, update it to form a new posterior distribution of μ using Eq. (3.14.1). In our example a new posterior distribution would be centred on the mean of the experimental values up to that point, and its standard deviation would be $0.2/\sqrt{n}$ where n is the number of measurements. (Its calculation would involve the multiplication of two normal distributions.)

The ability to calculate exact probabilities associated with values of μ (or other population parameters), if necessary using an iterative approach as more data are obtained, is an attractive feature of Bayesian statistics. However, the drawback is that the choice of the prior distribution is subjective, so that two investigators using Bayesian methods with the same data but different prior distributions might obtain

significantly different posterior distributions, and hence draw different conclusions. This would make it difficult for scientists to compare and assess each other's results. Nonetheless Bayesian methods have been applied in such areas as the interpretation of clinical trial data, comparisons of spectra and chromatograms, uncertainty estimates, and in several applications of forensic methods, including the interpretation of DNA databases. The use of Bayesian statistics has become more widespread as the necessary computational processing power has become available to handle the often-difficult calculations involved.

Bibliography

- Barnett, V. and Lewis, T., 1994, *Outliers in Statistical Data*, 3rd edn, John Wiley, Chichester. A very comprehensive treatment of the philosophy of outlier rejection and the tests used.
- Box, G.E.P., Hunter, W.G. and Hunter, J.S., 2005, *Statistics for Experimenters: Design, Innovation and Discovery*, 2nd edn, John Wiley, New York. Gives further details of testing for significant differences between means as a follow-up to ANOVA.
- Davies, O.L. and Goldsmith, P.L., 1984, *Statistical Methods in Research and Production*, 4th edn, Longman, London. Gives more detail about Type 1 and Type 2 errors and other applications of the chi-squared test.
- Ellison, S.L.R., Barwick, V.J. and Farrant, T.J.D., 2009, *Practical Statistics for the Analytical Scientist*, 2nd edn, Royal Society of Chemistry, Cambridge. Summarises many statistical tests, including those for outliers, with comments and examples.
- Kanji, G.K., 2006, *100 Statistical Tests*, 3rd edn, Sage Publications, London. Covers all the common tests, with numerical examples and extensive statistical tables.
- Kleinbaum, D.G., Kupper, L.L. and Muller, K.E., 2007, *Applied Regression Analysis and Other Multivariable Methods*, 4th edn, Duxbury Press, Boston, MA. Gives further details of testing for significant differences between means as a follow-up to ANOVA.
- Mullins, E., 2003, *Statistics for the Quality Control Laboratory*, Royal Society of Chemistry, Cambridge. Extensive treatment of ANOVA methods, including tests for homogeneity of variance, differences between means, etc.

Exercises

- 1 Use a normal probability plot to test whether the following set of data could have been drawn from a normal population:
11.68, 11.12, 8.92, 8.82, 10.31, 11.88, 9.84, 11.69, 9.53, 10.30, 9.17, 10.04, 10.65, 10.91, 10.32, 8.71, 9.83, 9.90, 10.40
- 2 In order to evaluate a spectrophotometric method for the determination of titanium, the method was applied to alloy samples containing different certified amounts of titanium. The results (% Ti) are shown below.

Sample	Certified value	Mean	Standard deviation
1	0.496	0.482	0.0257
2	0.995	1.009	0.0248
3	1.493	1.505	0.0287
4	1.990	2.002	0.0212

For each alloy eight replicate determinations were made.

(Xing-chu, Q. and Ying-quen, Z., 1983, *Analyst*, **108**: 641)

For each alloy, test whether the mean value differs significantly from the certified value.

- 3 For the data in Example 3.3.3, concerning the concentration of thiol in blood lysate,
 - (a) verify that 2.07 is not an outlier for the 'normal' group;
 - (b) show that the variances of the two groups differ significantly.
- 4 The following data give the recovery of bromide from spiked samples of vegetable matter, measured using a gas-liquid chromatographic method. The same amount of bromide was added to each specimen.

Tomato:	777	790	759	790	770	758	764 $\mu\text{g g}^{-1}$
Cucumber:	782	773	778	765	789	797	782 $\mu\text{g g}^{-1}$

(Roughan, J.A., Roughan, P.A. and Wilkins, J.P.G., 1983, *Analyst*, **108**: 742)

- (a) Test whether the recoveries from the two vegetables have variances which differ significantly.
 - (b) Test whether the mean recovery rates differ significantly.
- 5 The following results show the percentage of the total available interstitial water recovered by centrifuging samples taken at different depths in sandstone.

Depth of sample (m)	Water recovered, %					
7	33.3	33.3	35.7	38.1	31.0	33.3
8	43.6	45.2	47.7	45.4	43.8	46.5
16	73.2	68.7	73.6	70.9	72.5	74.5
23	72.5	70.4	65.2	66.7	77.6	69.8

(Wheatstone, K.G. and Getsthorpe, D., 1982, *Analyst*, **107**: 731)

Show that the percentage of water recovered differs significantly at different depths. Use the least significant difference method described in Section 3.9 to find the causes of this significant result.

- 6 The following table gives the concentration of norepinephrine (μmol per gram creatinine) in the urine of healthy volunteers in their early twenties.

Male	0.48	0.36	0.20	0.55	0.45	0.46	0.47	0.23
Female	0.35	0.37	0.27	0.29				

(Yamaguchi, M., Ishida, J. and Yoshimura, M., 1998, *Analyst*, **123**: 307)

Is there any evidence that concentration of norepinephrine differs between the sexes?

- 7 In reading a burette to 0.01 ml the final figure has to be estimated. The following frequency table gives the final figures of 50 such readings. Carry out an appropriate significance test to determine whether some digits are preferred to others.

Digit	0	1	2	3	4	5	6	7	8	9
Frequency	1	6	4	5	3	11	2	8	3	7

- 8 The following table gives further results from the paper cited in Example 3.3.1 (Sahuquillo, A., Rubio, R. and Rauret, G., 1999, *Analyst*, 124: 1), in which the results of the determination of chromium in organic materials were compared for two different methods.

Pine needles:	Method 1	mean = 2.15	SD = 0.26
	Method 2	mean = 2.45	SD = 0.14
Beech leaves:	Method 1	mean = 5.12	SD = 0.80
	Method 2	mean = 7.27	SD = 0.44
Aquatic plant:	Method 1	mean = 23.08	SD = 2.63
	Method 2	mean = 32.01	SD = 4.66

In each case the mean is the average of five values.

For each material test whether the mean results obtained by the two methods differ significantly.

- 9 The data given in the example in Section 3.11 for the number of breakages by four different workers are reproduced below:

24, 17, 11, 9

Test whether

- the number of breakages by the first worker differs significantly from those of the other three workers;
 - the second, third and fourth workers differ significantly from each other in carefulness.
- 10 A new flow injection analysis enzymatic procedure for determining hydrogen peroxide in water was compared with a conventional method involving redox titration with potassium permanganate by applying both methods to samples of peroxide for pharmaceutical use. The table below gives the amount of hydrogen peroxide found in mg ml^{-1} . Each value is the mean of four replicate measurements.

Sample no.	Enzymatic method	Permanganate method
1	31.1	32.6
2	29.6	31.0
3	31.0	30.3

(da Cruz Vieira, I. and Fatibello-Filho, O., 1998, *Analyst*, 123: 1809)

Test whether the results obtained by the two different methods differ significantly.

- 11 Six analysts each made six determinations of the paracetamol content of the same batch of tablets. The results are shown below:

Analyst	Paracetamol content (% m/m)					
A	84.32	84.51	84.63	84.61	84.64	84.51
B	84.24	84.25	84.41	84.13	84.00	84.30
C	84.29	84.40	84.68	84.28	84.40	84.36
D	84.14	84.22	84.02	84.48	84.27	84.33
E	84.50	83.88	84.49	83.91	84.11	84.06
F	84.70	84.17	84.11	84.36	84.61	83.81

(Trafford, A.D., Jee, R.D., Moffat, A.C. and Graham, P., 1999, *Analyst*, **124**: 163)

Test whether there is any significant difference between the means obtained by the six analysts.

- 12 The following figures refer to the concentration of albumin, in g l^{-1} , in the blood sera of 16 healthy adults:

37, 39, 37, 42, 39, 45, 42, 39, 44, 40, 39, 45, 47, 47, 43, 41

(Foote, J.W. and Delves, H.T., 1983, *Analyst*, **108**: 492)

The first eight figures are for men and the second eight for women. Test whether the mean concentrations for men and women differ significantly.

- 13 A new flame atomic-absorption spectroscopic method of determining antimony in the atmosphere was compared with the recommended calorimetric method. For samples from an urban atmosphere the following results were obtained:

Sample no.	Antimony found (mg m^{-3})	
	New method	Standard method
1	22.2	25.0
2	19.2	19.5
3	15.7	16.6
4	20.4	21.3
5	19.6	20.7
6	15.7	16.8

(Castillo, J.R., Lanaja, J., Martinez, M.C. and Aznarez, J., 1982, *Analyst*, **107**: 1488)

Do the results obtained by the two methods differ significantly?

- 14 For the situation described in Section 3.13 ($H_0: \mu = 3.0\%$, $H_1: \mu = 3.05\%$, $\sigma = 0.03\%$) calculate the minimum size of sample required to make the probability of a Type I error and the probability of a Type II both equal to 0.01 at most.

- 15 The concentrations ($\text{g } 100 \text{ ml}^{-1}$) of immunoglobulin G in the blood sera of 10 donors were measured by radial immunodiffusion (RID) with the following results:

Donor	1	2	3	4	5	6	7	8	9	10
RID result	1.3	1.5	0.7	0.9	1.0	1.1	0.8	1.8	0.4	1.3

Use the Kolmogorov–Smirnov method to test the hypothesis that immunoglobulin G levels in blood serum are normally distributed.

- 16 This question refers to Example 3.14.1. Three further measurements of the sodium concentration in the mineral water are made, the results being 7.7, 7.3 and 7.7 ppm.
- (a) Calculate the posterior distribution of μ from all the information available.
 - (b) Show that the probability that μ lies between 7.3 and 7.7 ppm is approximately 0.95.

4

The quality of analytical measurements

Major topics covered in this chapter

- Sampling
- Quality control
- Control charts
- Proficiency testing schemes
- Method performance studies
- Uncertainty
- Acceptance sampling
- Method validation

4.1

Introduction

Chapter 1 showed that in analytical science quantitative studies predominate, so estimates of the inevitable errors are essential. The results of almost all analyses are supplied to a customer or user, and these users must be satisfied as far as possible with the **quality** – the *fitness for purpose* – of the measurements. This has important implications for analytical practice. First, any assessment of the measurement errors must take into account the whole analytical process – including the sampling steps, which often contribute to the overall error very significantly. Second, the performance of the analyses undertaken in each laboratory must be checked internally on a regular basis, usually by applying the methods to standard or reference materials. Third, in many cases the results from different laboratories must be compared with each other, so that the users can be satisfied that the performance of the laboratories meets statutory, regulatory and other requirements. Finally, the analytical results must be supplied with a realistic estimate of their uncertainty, i.e. the range within which the true value of the quantity being measured should lie. These are the major topics discussed in this chapter. The statistical methods used are often very simple, most of them being based on techniques described in Chapters 2 and 3. But their regular application has been a major advance in analytical sciences in recent years, with a

large improvement in the quality and acceptability of many analytical results. Some of the methods discussed here have broader applications. For example the principles used to monitor the performance of a single analysis in a single laboratory over a period of time can also be applied to the monitoring of an industrial process.

4.2 Sampling

In most analyses we rely on chemical samples to give us information about a whole object. Unless the sampling stages of an analysis are considered carefully, the statistical methods discussed in this book may be invalidated, as the samples studied may not be properly representative of the whole object under study. For example it is not possible to analyse all the water in a stream for a toxic pollutant, and it is not possible to analyse all the milk in a tanker lorry to see if it contains a prohibited steroid hormone. Sometimes a small sample has to be used because the analytical method is destructive, and we wish to preserve the remainder of the material. So in each case the sample studied must be taken in a way that ensures as far as possible that it is truly representative of the whole object.

To illustrate some aspects of sampling we can study the situation in which we have a large batch of tablets and wish to obtain an estimate for the mean weight of a tablet. Rather than weigh all the tablets, we take a few of them (say ten) and weigh each one. In this example the batch of tablets forms the *population* and the ten weighed tablets form a *sample* from this population (see Section 2.2). If the sample is to be used to deduce the properties of the population, it must be what is known statistically as a **random sample**, i.e. a sample taken in such a way that all the members of the population have an equal chance of inclusion. Only then will equations such as Eq. (2.7.1), which gives the confidence limits of the mean, be valid. Note that the term 'random' has, in the statistical sense, a different meaning from 'haphazard'. Although in practice an analyst might spread the tablets on a desk and attempt to pick a sample of ten in a haphazard fashion, such a method could conceal an unconscious bias. The best way to obtain a random sample is by the use of a random number table. Each member of the population is allocated a number in such a way that all the numbers have an equal number of digits, e.g. 001, 002, 003. Random numbers are then read off from a random number table (see Table A.8), starting at an arbitrary point to give, for example, 964, 173, etc., and the corresponding members of the population form the sample. An alternative (and much simpler) method which is sometimes used is to select the population members at regular intervals, for example to take every hundredth tablet off a production line. This is not wholly satisfactory, however, since there might be a coinciding periodicity in the weight of the tablets. The sampling method, which is not truly random, would then not reveal the true extent of the variations in weight. Similarly, if the last few tablets in the batch were taken and there had been a gradual decrease in weight during its production, then this sample would give a wholly misleading value for the mean weight of the batch.

In the tablet example the population is made up of obvious discrete members that are nominally the same. Sampling from materials for which this is not true, such as rocks, powders, gases and liquids, is called **bulk sampling**. If a bulk material were perfectly homogeneous, then only a small portion or **test increment** would be needed to determine the properties of the bulk. In practice bulk materials are

non-homogeneous for a variety of reasons. For example ores and sediments consist of macroscopic particles with different compositions, and these may not be uniformly distributed in the bulk. Fluids may be non-homogeneous on a molecular scale owing to concentration gradients. Such inhomogeneity can be detected only by taking a sample of test increments from different parts of the bulk. If possible this should be done randomly by considering the bulk as a collection of cells of equal size and selecting a sample of cells by using random numbers as described above.

From the random sample, the mean, $\bar{x}$, and variance, s^2 , can be calculated. There are two contributions to s^2 : the **sampling variance**, σ_1^2 , due to differences between the members of the sample, e.g. the tablets having different weights, and the **measurement variance**, σ_0^2 , e.g. the random errors in weighing each tablet. The next section describes how these two contributions can be separated and estimated by using ANOVA. For bulk materials the sampling variance is dependent on the size of the test increment relative to the scale of the inhomogeneities: as the test increment size increases, the inhomogeneities tend to be averaged out and so the sampling variance decreases.

4.3 Separation and estimation of variances using ANOVA

Sections 3.8–3.10 described the use of one-way ANOVA to test for differences between means when there was a possible variation due to a fixed-effect factor in addition to the measurement error. We now consider the situation where the additional factor is a *random-effect factor*, the sampling variation. In this case one-way ANOVA is used to separate and estimate the two sources of variation. Table 4.1 shows the results of the purity testing of a barrelful of sodium chloride. Five sample increments, A–E, were taken from different parts of the barrel chosen at random, and four replicate analyses were performed on each sample. As noted above, there are two possible sources of variation: that due to the random error in the measurement of purity, given by the measurement variance, σ_0^2 , and that due to real variations in the sodium chloride purity at different points in the barrel, given by the sampling variance, σ_1^2 . Since the within-sample mean square does not depend on the sample mean (Section 3.9) it can be used to give an estimate of σ_0^2 . The between-sample mean square *cannot* be used to estimate σ_1^2 directly, because the between-sample variation is caused both by the random error in measurement and by the possible variation in the purity. It can be shown that the between-sample mean square gives an estimate of $\sigma_0^2 + n\sigma_1^2$, where n is the number of replicate measurements of each sample, in this case four. However, before an estimate of σ_1^2 is made, a test should be carried out to see whether it differs significantly from 0. This is

Table 4.1 Purity testing of sodium chloride

Sample	Purity (%)	Mean
A	98.8, 98.7, 98.9, 98.8	98.8
B	99.3, 98.7, 98.8, 99.2	99.0
C	98.3, 98.5, 98.8, 98.8	98.6
D	98.0, 97.7, 97.4, 97.3	97.6
E	99.3, 99.4, 99.9, 99.4	99.5

done by comparing the within- and between-sample mean squares: if they do not differ significantly then $\sigma_1^2 = 0$ and both mean squares estimate σ_0^2 .

The one-way ANOVA output from Excel® for this example is shown below. It shows that the between sample mean square is greater than the within-sample mean square: the F -test shows that this difference is very significant, i.e. σ_1^2 does differ significantly from 0. The within-sample mean square, 0.0653, is the estimate of σ_0^2 , so we can estimate σ_1^2 using:

$$\begin{aligned}\sigma_1^2 &= (\text{between sample mean square} - \text{within sample mean square})/n \\ &= (1.96 - 0.0653)/4 \\ &= 0.47\end{aligned}$$

Sample A	Sample B	Sample C	Sample D	Sample E
98.8	99.3	98.3	98.0	99.3
98.7	98.7	98.5	97.7	99.4
98.9	98.8	98.8	97.4	99.9
98.8	99.2	98.8	97.3	99.4

One-Way Anova

SUMMARY

Groups	Count	Sum	Average	Variance
Sample A	4	395.2	98.8	0.006667
Sample B	4	396	99	0.086667
Sample C	4	394.4	98.6	0.06
Sample D	4	390.4	97.6	0.1
Sample E	4	398	99.5	0.073333

Source of Variation	SS	df	MS	F	P-value	F crit
Between-sample	7.84	4	1.96	30	5.34E-07	3.056
Within-sample	0.98	15	0.0653			
Total	8.82	19				

4.4

Sampling strategy

If one analysis is made on each of the h sample increments (example above, Section 4.3) then the confidence limits of the mean are given by Eq. (2.7.1):

$$\mu = \bar{x} \pm t_{n-1}s/\sqrt{h} \quad (4.4.1)$$

where $\bar{x}$ is the mean of the h measurements and s^2 is the variance of the measurements. The total variance, σ^2 , is estimated by s^2 and is the sum of the measurement and sampling variances, i.e. $\sigma_0^2 + \sigma_1^2$ (see Section 2.11): σ^2/h (estimated by s^2/h) is the variance of the mean, $\bar{x}$. If the value for each sample increment is the mean of n replicate measurements, then the variance of the mean is $(\sigma_0^2/n + \sigma_1^2)/h = \sigma_0^2/nh + \sigma_1^2/h$. Obviously, for maximum precision, we require the variance of the mean to be as small as possible. The term due to the measurement variance, σ_0^2/nh , can be reduced either by using a more precise method of analysis or by increasing n , the number of replicate measurements. However, there is no point in striving to make this measurement variance much less than (say) a tenth of the sampling variance, as any further reduction will not greatly improve the total variance, which is the sum of the two variances. Instead it is preferable to take a larger number of sample increments, since the confidence interval decreases with increasing h . If a preliminary sample has been used to estimate s , then the sample size required to achieve a given size of confidence interval can also be estimated (see Chapter 2, Exercise 4).

A possible sampling strategy with bulk material is to take h sample increments and *blend* them before making n replicate measurements. The variance of the mean of these replicate measurements is $\sigma_0^2/n + \sigma_1^2/h$. This total variance should be compared with that when each sample increment is analysed n times and the increment means are averaged, the variance then being $\sigma_0^2/nh + \sigma_1^2/h$ (see above). Obviously the latter variance is the smaller, resulting in greater precision of the mean, but more measurements (nh against h) are required. Knowledge of the values of σ_0^2 and σ_1^2 from previous experience, and the costs of sampling and analysis, can be used to calculate the relative costs of these sampling strategies. Improving the precision of the measurements to an unnecessary extent by increasing n and/or h will incur greater costs in terms of time, equipment use, etc., so in general the most economical scheme to give the required degree of precision will be used.

For bulk materials the sampling variance depends on the size of the sample increment relative to the scale of the inhomogeneities and decreases with increasing sample increment size. In some experiments it may be necessary to set an upper limit on the sampling variance so that changes in the mean can be detected. Preliminary measurements can be made to decide the minimum sample increment size required to give an acceptable level of sampling variance.

The heterogeneity of many materials encountered in analytical practice, and therefore the importance of using suitable sampling protocols, is closely related to the important topic of *sampling uncertainty*, which is considered in more detail in Section 4.13.

4.5

Introduction to quality control methods

If a laboratory is to produce analytical results of a quality that is acceptable to its clients, and allow it to perform well in proficiency tests or method performance studies (see below), it is obviously essential that its results should show excellent consistency from day to day. Checking for such consistency is complicated by the inevitable occurrence of random errors, so several statistical techniques have been developed to show whether or not time-dependent trends are occurring in

the results, alongside the random errors. These are referred to as **quality control** methods.

Suppose that a laboratory uses a chromatographic method for determining the level of a pesticide in fruits. The results may be used to determine whether a large batch of fruit is acceptable or not, and their quality is thus of great importance. The performance of the method will be checked at regular intervals by applying it, with a small number of replicate analyses, to a standard reference material (SRM), in which the pesticide level is certified by a regulatory authority. Alternatively an internal quality control (IQC) standard of known composition and high stability can be used. The SRM or IQC standard will probably be inserted at random into the sequence of materials analysed by the laboratory, so that the IQC materials are not separately identified to the laboratory staff, and are studied using exactly the same procedures as the routine samples. The known concentration of the pesticide in the SRM/IQC materials is the target value for the analysis, μ_0 . The laboratory needs to be able to stop and examine the analytical method quickly if it seems to be giving erroneous results. On the other hand, time and other resources will be wasted if the analyses are halted unnecessarily, so the quality control methods should allow their continued use as long as they are working satisfactorily. If the values for the IQC samples do not show significant time-dependent trends, and if the random errors in the measurements are not too large, the analytical process is said to be *under control* or *in control*.

Quality control methods are also very widely used to monitor industrial processes. Again it is important to stop a process if its output falls outside certain limits, but it is equally important not to stop the process if it is working well. For example, the weights of pharmaceutical tablets coming off a production line can be monitored by taking small samples of tablets from time to time. The tablet weights are bound to fluctuate around the target value μ_0 because of random errors, but if these random errors are not too large, and are not accompanied by time-dependent trends, the process is under control.

4.6

Shewhart charts for mean values

In Chapter 2 we showed how the mean, $\bar{x}$, of a sample of measurements could be used to provide an estimate of the population mean, μ , and how the sample standard deviation, s , provided an estimate of the population standard deviation, σ . For a small sample size, n , the confidence limits of the mean are normally given by Eq. (2.7.1), with the t -value chosen according to the number of degrees of freedom, $(n - 1)$, and the confidence level required. Similar principles can be applied to quality control work, but with one important difference. Over a long period, the *population* standard deviation, σ , of the pesticide level in the fruit (or, in the second example above, of the tablet weights), will become known from experience. In quality control work, σ is called the **process capability**. Equation (2.7.1) can be replaced by Eq. (2.6.3) with the estimate s replaced by the known σ . In practice $z = 1.96$ is often rounded to 2 for 95% confidence limits and $z = 2.97$ is rounded to 3 for 99.7% confidence limits.

$$\text{For 95\% confidence limits: } \mu = \bar{x} \pm \frac{2\sigma}{\sqrt{n}} \quad (4.6.1)$$

$$\text{For 99.7\% confidence limits: } \mu = \bar{x} \pm \frac{3\sigma}{\sqrt{n}} \quad (4.6.2)$$

These equations are used in the construction of the most common type of control chart, a **Shewhart** chart (Fig. 4.1). The vertical axis of a Shewhart chart displays the **process mean**, $\bar{x}$, of the measured values, e.g. of the pesticide concentration in the fruit, and the horizontal axis is a time axis, so that the variation of these $\bar{x}$ values with time can be plotted. The **target value**, μ_0 , is marked by a horizontal line. The chart also includes two further pairs of horizontal lines. The lines at $\mu_0 \pm 2\sigma/\sqrt{n}$ are called the **warning lines**, and those at $\mu_0 \pm 3\sigma/\sqrt{n}$ are called the **action lines**. The purpose of these lines is indicated by their names. Suppose a measured $\bar{x}$ value falls outside the action lines. The probability of such a result when the process is under control is known to be only 0.3%, i.e. 0.003, so in practice the process is usually stopped and examined if this occurs. There is a probability of ca. 5% (0.05) of a single point falling outside either warning line (but within the action lines) while the process remains in control. This alone would not cause the process to be stopped, but if *two successive* points fall outside the same warning line, the probability of such an occurrence ($p = 0.025^2 \times 2 = 0.00125$ in total for both warning lines) is again so low that the process is judged to be out of control. These two criteria – one point outside the action lines, or two successive points outside the same warning line – are the

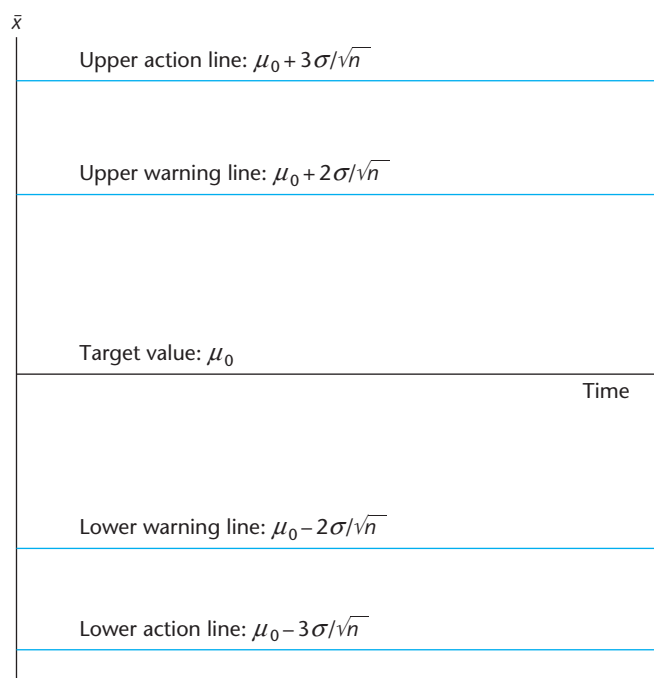


Figure 4.1 Shewhart chart for mean values.

ones most commonly applied in the interpretation of Shewhart charts. Others are often used in addition: for example the probability of eight successive points lying on one specific side of the target value line is clearly low, i.e. $0.5^8 = 0.0039$, and such an occurrence again suggests that the process is out of control. A process might also be stopped in cases where the plotted $\bar{x}$ values show a trend (e.g. six increasing or decreasing points in succession, even if the points are within the warning lines), or where they seem to oscillate (e.g. 14 successive points, alternating up and down). Users of control charts must establish clearly all the criteria to be used in declaring their own particular process out of control.

4.7

Shewhart charts for ranges

If a Shewhart chart for mean values suggests that a process is out of control, there are two possible explanations. The most obvious is that the process mean has changed: detecting such changes is the main reason for using control charts which plot $\bar{x}$ values. An alternative explanation is that the process mean has remained unchanged but that the variation in the process has increased. This means that the action and warning lines are too close together, giving rise to indications that changes in $\bar{x}$ have occurred when in fact they have not. Errors of the opposite kind are also possible. If the variability of the process has diminished (i.e. improved), then the action and warning lines will be too far apart, perhaps allowing real changes in $\bar{x}$ to go undetected. Clearly we must monitor the variability of the process as well as its mean value. This monitoring has its own intrinsic value: the variability of a process or an analysis is a measure of its quality, and in the laboratory situation is directly linked to the repeatability (within-laboratory precision) of the method (cf. Chapter 1).

The variability of a process can be displayed by plotting another Shewhart chart to display the **range**, R (= highest value – lowest value), of each of the samples taken. A typical control chart for the range is shown in Fig. 4.2. The general format of the chart is the same as for the mean values, with a line representing the target value and pairs of action and warning lines. The most striking difference between the two charts is that these pairs of lines are not symmetrical with respect to the target value for the range, $\bar{R}$. The value of $\bar{R}$ can be calculated using the value of σ , and the positions of the action and warning lines can be derived from $\bar{R}$, using multiplying factors obtained from statistical tables. These factors take values depending on the sample size, n . The relevant equations are:

$$\bar{R} = \sigma d_1 \quad (4.7.1)$$

$$\text{Lower warning line} = \bar{R}w_1 \quad (4.7.2)$$

$$\text{Upper warning line} = \bar{R}w_2 \quad (4.7.3)$$

$$\text{Lower action line} = \bar{R}a_1 \quad (4.7.4)$$

$$\text{Upper action line} = \bar{R}a_2 \quad (4.7.5)$$

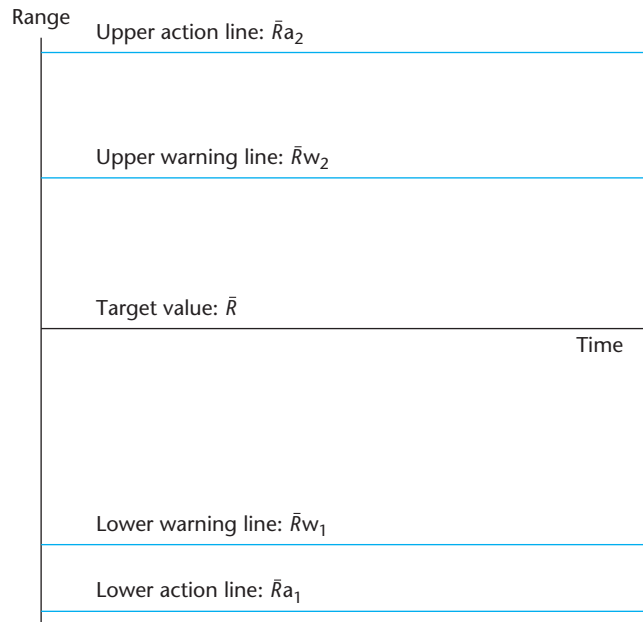


Figure 4.2 Shewhart chart for range.

Example 4.7.1

Determine the characteristics of the mean and range control charts for a process in which the target value is 57, the process capability is 5, and the sample size is 4.

For the control chart on which mean values will be plotted, the calculation is simple. From Eq. (4.6.1) the warning lines will be at $57 \pm 2 \times 5/\sqrt{4}$, i.e. at 57 ± 5 ; and from Eq. (4.6.2) the action lines will be at $57 \pm 3 \times 5/\sqrt{4}$, i.e. at 57 ± 7.5 . This chart is shown in Fig. 4.3(a).

For the control chart on which ranges are plotted, we must first calculate $\bar{R}$ using Eq. (4.7.1). This gives $\bar{R} = 5 \times 2.059 = 10.29$, where the d_1 value of 2.059 is taken from statistical tables for $n = 4$. (See for example the table in the collection by Neave, the details of which are given in the Bibliography for Chapter 1.) The value of $\bar{R}$ is then used to determine the lower and upper warning and action lines using Eqs (4.7.2)–(4.7.5). The values of w_1 , w_2 , a_1 and a_2 for $n = 4$ are 0.29, 1.94, 0.10, and 2.58 respectively, giving on multiplication by 10.29 positions for the four lines of 2.98, 19.96, 1.03 and 26.55 respectively. These lines are shown in Fig. 4.3(b).

It is not always the practice to plot the lower action and warning lines on a control chart for the range, as a reduction in the range is not normally a cause for concern. However, the variability of a process is one measure of its quality, and a reduction in $\bar{R}$ represents an improvement in the process capability; if this is not detected and

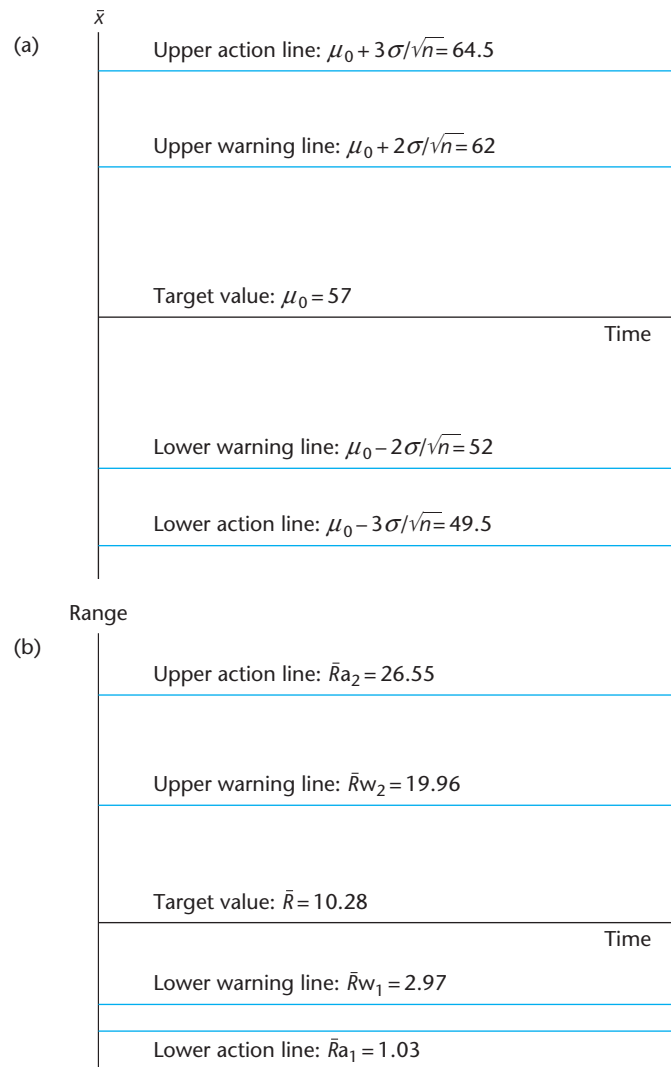


Figure 4.3 (a) Shewhart chart for mean values (example). (b) Shewhart chart for range (example).

acted upon, real changes in $\bar{x}$ might go undetected, as previously noted. So plotting both sets of warning and action lines is recommended.

4.8

Establishing the process capability

In the previous section we showed that, if the process capability, σ , is known, it is possible to construct control charts for both the sample mean and the sample range. Using these charts we can distinguish the situation where a process has gone out of control through a shift in the process mean from the one where the mean is

unchanged but an undesirable increase in the variability of the process has occurred. The establishment of a proper value for σ is therefore very important, and such a value should be based on a substantial number of measurements. But in making such measurements the same problem – distinguishing a change in the process mean from a change in the process variability – must be faced. If σ is calculated directly from a long sequence of measurements, its value may be overestimated by any changes in the mean that occur during that sequence, so proper control charts could not then be plotted.

The solution to this problem is to take a large number of small samples, measure the range, R , for each, and thus determine $\bar{R}$. This procedure ensures that only the inherent variability of the process is measured, any drift in the mean values being eliminated. The $\bar{R}$ -value can then be used with Eqs (4.7.2)–(4.7.5) to determine the action and warning lines for the range control chart. The warning and action lines for the control chart for the mean can be determined by calculating σ using Eq. (4.7.1), and then applying Eqs (4.6.1) and (4.6.2). In practice this two-stage calculation is unnecessary, as most statistical tables provide values of W and A , which give the positions of the warning and action lines for the mean directly from:

$$\text{Warning lines at } \bar{x} \pm W\bar{R} \quad (4.8.1)$$

$$\text{Action lines at } \bar{x} \pm A\bar{R} \quad (4.8.2)$$

These methods are illustrated by the following example.

Example 4.8.1

An internal quality control standard with an analyte concentration of 50 mg kg^{-1} is analysed in a laboratory for 25 consecutive days, the sample size being four on each day. The results are given in Table 4.2, which is in the form of an Excel® spreadsheet. Determine the value of $\bar{R}$ and hence plot control charts for the mean and range of the laboratory analyses.

When the results are studied there is some evidence that over the 25-day period the sample means are drifting up and down. All the sample means from days 3–15 inclusive are greater than the target value of 50, whereas four of the next six means are below the target value, and the last four are all above it. These are the circumstances in which it is important to estimate σ using the method described above. Using the R -values in the last column of data, $\bar{R}$ is found to be 4.31. Application of Eq. (4.7.1) estimates σ as $4.31/2.059 = 2.09$. Table 4.2 also shows that the standard deviation of the 100 measurements, treated as a single sample, is 2.43: because of the drifts in the mean this would be a significant overestimate of σ .

The control chart for the mean is then plotted with the aid of Eqs (4.8.1) and (4.8.2), with $W = 0.4760$, $A = 0.7505$, showing that the warning and

action lines are at 50 ± 2.05 and 50 ± 3.23 respectively. Figure 4.4 is the Excel® control chart, which shows that the process mean is not under control, since several of the points fall outside the upper action line. Similarly, Eqs (4.7.2)–(4.7.5) show that in the control chart for the range the warning lines are at 1.24 and 8.32 and the action lines are at 0.42 and 11.09. Excel® does not automatically produce control charts for ranges, though it does generate charts for standard deviations, which in some cases are used instead of range charts. However, with one exception, the range values in the last column of Table 4.2 all lie within the warning lines, indicating that the process variability is under control.

Minitab® can be used to produce Shewhart charts for the mean and the range. The program calculates the value of $\bar{R}$ directly from the data. Figure 4.5 shows the charts for the data in Table 4.2. Minitab® (like some texts) calculates the warning and action lines for the range by approximating the asymmetrical distribution of $\bar{R}$ by a normal distribution. This is why the positions of the lines differ from those calculated above using Eqs (4.8.1) and (4.8.2).

Table 4.2 Excel® Spreadsheet (example)

Sample Number	Sample Values				Chart Mean	Range
	1	2	3	4		
1	48.8	50.8	51.3	47.9	49.70	3.4
2	48.6	50.6	49.3	50.3	49.70	2.0
3	48.2	51.0	49.3	52.1	50.15	3.9
4	54.8	54.6	50.7	53.9	53.50	4.1
5	49.6	54.2	48.3	50.5	50.65	5.9
6	54.8	54.8	52.3	52.5	53.60	2.5
7	49.0	49.4	52.3	51.3	50.50	3.3
8	52.0	49.4	49.7	53.9	51.25	4.5
9	51.0	52.8	49.7	50.5	51.00	3.1
10	51.2	53.4	52.3	50.3	51.80	3.1
11	52.0	54.2	49.9	57.1	53.30	7.2
12	54.6	53.8	51.5	47.9	51.95	6.7
13	52.0	51.7	53.7	56.8	53.55	5.1
14	50.6	50.9	53.9	56.0	52.85	5.4
15	54.2	54.9	52.7	52.2	53.50	2.7
16	48.0	50.3	47.5	53.4	49.80	5.9
17	47.8	51.9	54.3	49.4	50.85	6.5
18	49.4	46.5	47.7	50.8	48.60	4.3
19	48.0	52.5	47.9	53.0	50.35	5.1
20	48.8	47.7	50.5	52.2	49.80	4.5
21	46.6	48.9	50.1	47.4	48.25	3.5
22	54.6	51.1	51.5	54.6	52.95	3.5
23	52.2	52.5	52.9	51.8	52.35	1.1
24	50.8	51.6	49.1	52.3	50.95	3.2
25	53.0	46.6	53.9	48.1	50.40	7.3
s.d. = 2.43					Mean = 4.31	

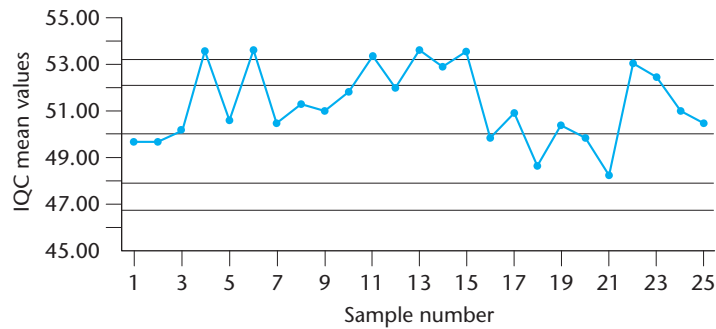


Figure 4.4 Shewhart chart for means (Table 4.2 example data).

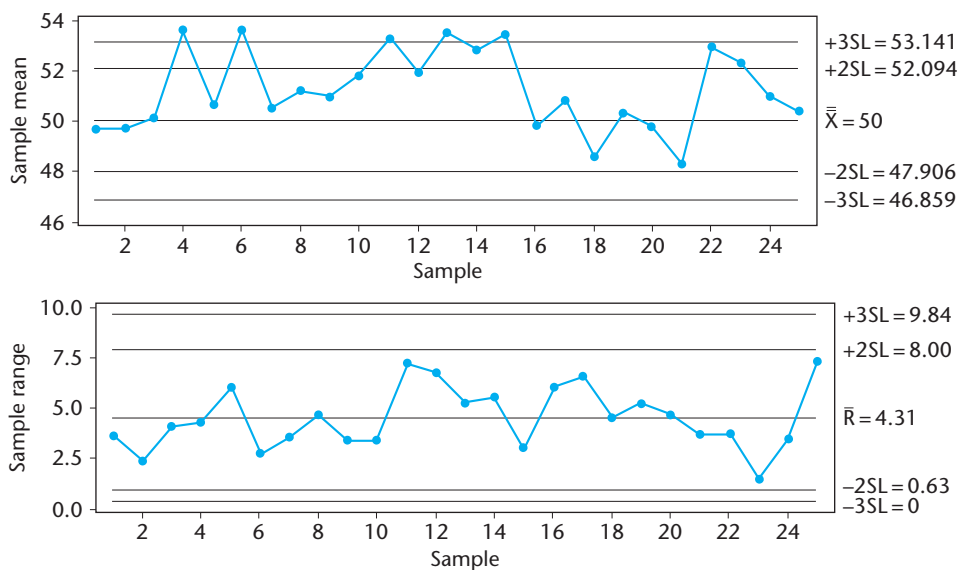


Figure 4.5 Shewhart charts for means and range produced using Minitab®. The abbreviation '3SL', for example, means three standard deviation control limits.

4.9 Average run length: CUSUM charts

An important property of a control chart is the speed with which it detects that a change in the process mean has occurred. The average number of measurements necessary to detect any particular change in the process mean is called the **average run length (ARL)**. Since the positions of the action and warning lines on a Shewhart chart for the process mean depend on the value of $\sigma/\sqrt{n}$, the ARL for that chart will depend on the size of the change in the mean compared with $\sigma/\sqrt{n}$. A larger change will be detected more rapidly than a smaller one, and the ARL will be reduced by using a larger sample size, n . It may be shown that if a change equal to $1\sigma/\sqrt{n}$ occurs, then the ARL is about 50 if only the action line criterion is used, i.e. about 50 samples will be measured before a value falls outside the action lines. If the

process is also stopped if two consecutive measurements fall outside the same warning line, then the ARL falls to ca. 25. These values are quite large: for example it would be serious if a laboratory continued a pesticide analysis for 25 days before noticing that the procedure had developed a systematic error. This represents a significant disadvantage of Shewhart charts. An example of the problem is shown in Table 4.3, a series of measurements for which the target value is 80, and $\sigma/\sqrt{n}$ is 2.5. When the sample means are plotted on a Shewhart chart (Fig. 4.6) it is clear that from about the seventh observation onwards a change in the process mean may well have occurred, but all the points remain on or inside the warning lines. (Only the lower warning and action lines are shown in the figure.)

The ARL can be reduced significantly by using a different type of control chart, a **CUSUM (cumulative sum)** chart. This approach is again illustrated by the data in Table 4.3. The calculation of the CUSUM is shown in the last two columns of the table, which show that the sum of the deviations of the sample means from the target value is carried forward cumulatively, careful attention being paid to the signs of the deviations. If a manufacturing or analytical process is under control, positive

Table 4.3 Example data for CUSUM calculation

Observation number	Sample mean	Sample mean – target value	CUSUM
1	82	2	2
2	79	–1	1
3	80	0	1
4	78	–2	–1
5	82	2	1
6	79	–1	0
7	80	0	0
8	79	–1	–1
9	78	–2	–3
10	80	0	–3
11	76	–4	–7
12	77	–3	–10
13	76	–4	–14
14	76	–4	–14
15	75	–5	–23

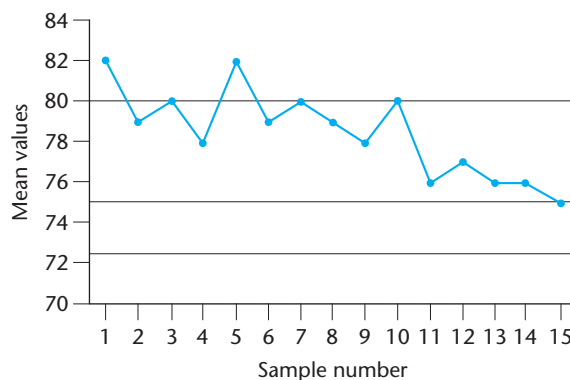


Figure 4.6 Shewhart chart for Table 4.3 data.

and negative deviations from the target value are equally likely and the CUSUM should oscillate about zero. If the process mean changes the CUSUM will move away from zero. In the example given, the process mean seems to fall after the seventh observation, so the CUSUM becomes more and more negative. The resulting control chart is shown in Fig. 4.7.

Proper interpretation of CUSUM charts, to show that a genuine change in the process mean has occurred, requires a **V-mask**. The mask is engraved on a transparent plastic sheet, and is placed over the control chart with its axis of symmetry horizontal and its apex a distance, d , to the right of the last observation (Fig. 4.8). If all the points on the chart lie within the arms of the V, then the process is in control

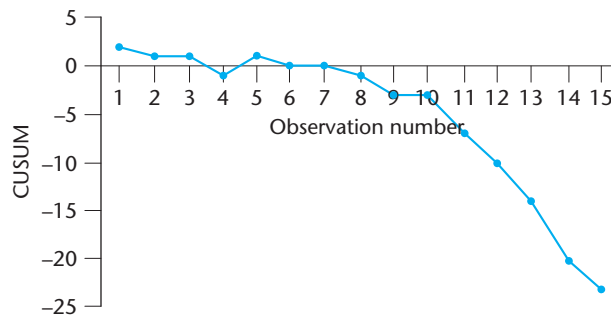


Figure 4.7 CUSUM chart for Table 4.3 data.

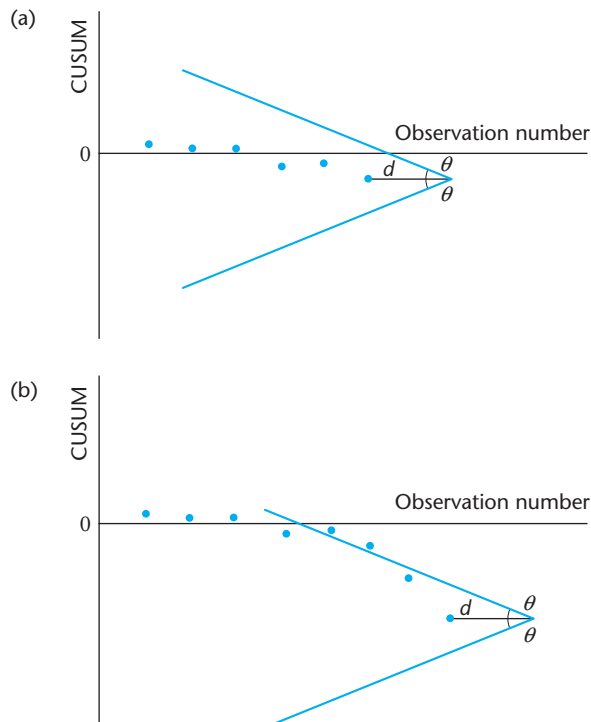


Figure 4.8 Use of a V-mask with the process (a) in control and (b) out of control.

(Fig. 4.8a). The mask is also characterised by $\tan \theta$, the tangent of the semi-angle, θ , between the arms of the V. Values of d and $\tan \theta$ are chosen so that significant changes in the process mean are detected quickly, but false alarms are few. The unit of d is the distance between successive observations. The value of $\tan \theta$ used clearly depends on the relative scales of the two axes on the chart: a commonly used convention is to make the distance between successive observations on the x -axis equal to $2\sigma/\sqrt{n}$ on the y -axis. Using this convention a V-mask with $d = 5$ units and $\tan \theta = 0.35$ gives an ARL of 10 if the process mean changes by $1\sigma/\sqrt{n}$ and only 4 if the change is $2\sigma/\sqrt{n}$. The ARL for a zero change in process mean, i.e. before a false alarm occurs, is ca. 350.

The corresponding figures for a Shewhart chart are ca. 25 (for a change in the mean of $1\sigma/\sqrt{n}$) and 320, so it is clear that the CUSUM chart is superior in both respects. The ARL provided by the CUSUM chart can be reduced to about 8 (for a change of $1\sigma/\sqrt{n}$) by using $\tan \theta = 0.30$, but inevitably the chance of a false alarm is then also increased, occurring once in ca. 120 observations.

In summary CUSUM charts have the advantage that they react more quickly than Shewhart charts to a change in the process mean (as Fig. 4.7 clearly shows), without increasing the chances of a false alarm. Moreover, the point of the slope change in a CUSUM chart indicates the point where the process mean has changed, and the value of the slope indicates the size of the change. Naturally, if a CUSUM chart suggests that a change in the process mean has occurred, we must also test for possible changes in σ . This can be done using a Shewhart chart, but CUSUM charts for ranges can also be plotted.

4.10 Zone control charts (J-charts)

The zone control chart (also known as the J-chart) is a control chart for the mean that combines features of the Shewhart chart and the CUSUM chart. It is simple to use, but effective. First it is necessary to establish a value for σ , as was done in Example 4.7.1. Then the chart is set up with horizontal lines at the target value, μ , and at $\mu \pm (\sigma/\sqrt{n})$, $\mu \pm 2(\sigma/\sqrt{n})$ and $\mu \pm 3(\sigma/\sqrt{n})$. These horizontal lines divide the chart into bands, or 'zones', of equal width, as shown in Fig. 4.9. This figure shows a zone control chart for the data in Table 4.2 (Example 4.8.1) obtained using Minitab®. At the right-hand side of the chart the horizontal lines are labelled with the values of μ , $\mu \pm (\sigma/\sqrt{n})$, $\mu \pm 2(\sigma/\sqrt{n})$ and $\mu \pm 3(\sigma/\sqrt{n})$, where $\mu = 50$, $\sigma = 2.09$ and $n = 4$. The sample means are plotted as circles in the appropriate zone and have been joined with straight lines. The chart thus looks similar to the Shewhart chart in Fig. 4.4. However, the sample means are also assigned scores, dependent on the zone in which they fall. These scores are indicated on the left-hand side of the chart. For example, a mean between $\mu - 2(\sigma/\sqrt{n})$ and $\mu - 3(\sigma/\sqrt{n})$ is assigned a score of 4.

The sample mean for the first sample scores 0, and this value has been written in its circle on the chart. As each sample mean is obtained it is assigned a score and the scores are added cumulatively. The total score is then written in the appropriate circle on the chart. The next two samples (2 and 3) each contribute 0 to the total. Sample number 4 contributes 8. Sample number 5 contributes 0, leaving the total at 8. Sample number 6 contributes 8, making a total of 16. This procedure is

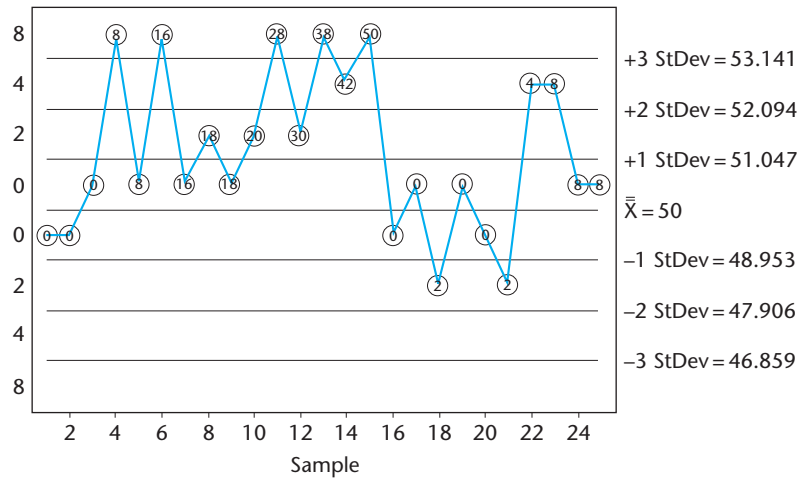


Figure 4.9 Zone control chart for the data in Table 4.2 produced using Minitab®.

continued until an observation falls on the opposite side of the centre line from the previous one, when the total is reset to 0. For example the total is reset for sample number 16.

The total performs the same function as the CUSUM. When the process is under control the total stays close to zero. However, a change in the process mean will result in a run of values on one side of the target value and a corresponding increase in the total. The system is deemed to be out of control if the total score equals or exceeds 8. The zone control chart in Fig. 4.9 confirms the suspicions aroused in Example 4.8.1 that the process mean is not under control. The zone chart suggests that the process mean has drifted upwards at sample 4. At sample 16 it has returned to the target value but then goes up again at sample 23.

It is also possible to set up a zone chart for single measurements, rather than for the means of replicates. Even if replicate measurements are not made, it is still possible to obtain an estimate of σ from the average range, $\bar{R}$. This is achieved by treating *each successive pair* of measurements as a sample, size 2. Taking successive pairs of measurements ensures that, as is required (see Section 4.8), the effect of any drift in the process mean is minimised. Considering only the values in column 1 of Table 4.2 (i.e. assuming that no replicate measurements are made), the first and second measurements (48.4, 48.6) have a range of 0.2, the second and third measurements (48.6, 48.2) have a range of 0.4, the third and fourth measurements (48.2, 54.8) have a range of 6.6, and so on. The sum of these ranges is 65 so $\bar{R} = 65/24 = 2.708$. (Note that although there are 25 values, there are only 24 differences.) Then from Eq. (4.7.1), $\sigma = \bar{R}/1.1284 = 2.40$, where the d_1 value is taken from statistical tables for $n = 2$. The zones for single measurements are then formed by drawing lines at μ , $\mu \pm \sigma$, $\mu \pm 2\sigma$ and $\mu \pm 3\sigma$.

A number of ISO publications deal with control charts. References in this area are given in the Bibliography at the end of this chapter.

4.11 Proficiency testing schemes

The quality of analytical measurements is enhanced by two types of testing scheme, in each of which a number of laboratories might participate simultaneously. The more common are **proficiency testing (PT) schemes**, in which aliquots from homogeneous materials are circulated to a number of laboratories for analysis at regular intervals (every few weeks or months), and the resulting data are reported to a central organiser. The circulated material is designed to resemble as closely as possible the samples normally submitted for analysis in the relevant field of application, and each laboratory analyses its portion *using its own usual method*. The results of all the analyses are circulated to all the participants, who thus gain information on how their measurements compare with those of others, how their own measurements improve or deteriorate with time, and how their own measurements compare with an external quality standard. In short, the aim of such schemes is the evaluation of the competence of analytical laboratories. PT schemes have now been developed for use in a wide range of application fields, including several areas of clinical chemistry, the analysis of water and various types of food and drink, forensic analysis, and so on. Experience shows that in such schemes widely divergent results will arise, even between experienced and well-equipped and well-staffed laboratories. In one of the commonest of clinical analyses, the determination of blood glucose at the mM level, most of the results obtained for a single blood sample approximated to a normal distribution with values between 9.5 and 12.5 mM, in itself a not inconsiderable range. But the complete range of results was from 6.0 to 14.5 mM, i.e. some laboratories obtained values almost 2.5 times those of others. The worrying implications of this discrepancy in clinical diagnosis are obvious. In more difficult areas of analysis the results can be so divergent that there is no real consensus between different laboratories. The importance of PT schemes in highlighting such alarming differences, and in helping to minimise them by encouraging laboratories to compare their performance, is very clear, and they have unquestionably helped to improve the quality of analytical results in many fields. Here we are concerned only with the statistical evaluation of the design and results of such schemes, and not with the administrative details of their organisation. Of particular importance are the methods of assessing participants' performance, and the need to ensure that the circulated aliquots are taken from a homogeneous bulk sample.

The recommended method for verifying homogeneity of the sample involves taking $n \geq 10$ portions of the test material at random, separately homogenising them if necessary, taking two test samples from each portion, and analysing the $2n$ portions by a method whose standard deviation under repeatability conditions is not more than 30% of the target standard deviation (i.e. the expected reproducibility, see below) of the proficiency test. If the homogeneity is satisfactory, one-way analysis of variance should then show that the between sample mean square is not significantly greater than the within-sample mean square (see Section 4.3).

The results obtained by the laboratories participating in a PT scheme are most commonly expressed as z-scores, where z is given by (see Section 2.2):

$$z = \frac{x - x_a}{\sigma} \quad (4.11.1)$$

In this equation the x -value is the result obtained by a single laboratory for a given analysis, x_a is the assigned value for the level of the analyte, and σ is the target value for the standard deviation of the test results. The assigned value x_a can be obtained by using a certified reference material (CRM), if one is available and suitable for distribution. If this is not feasible, the relevant ISO standard (which also provides many numerical examples – see Bibliography) recommends three other possible approaches. In order of decreasing rigour these are (i) a reference value obtained from one laboratory, by comparing random samples of the test material against a CRM; (ii) a consensus value obtained by analysis of random samples of the test material by expert laboratories; and (iii) a consensus value obtained from all the participants in a given round of the scheme. This last situation is of interest since, when many laboratories participate in a given PT scheme, there are bound to be a number of suspect results or outliers in an individual test. (Although many PT schemes provide samples and reporting facilities for more than one analyte, experience shows that a laboratory that scores well in one specific analysis does not necessarily score well in others.) This problem is overcome by the use of either the *median*, which is especially recommended for small data sets ($n < 10$), a *robust mean*, or the *mid-point of the inter-quartile range*. All these measures of location avoid or address the effects of dubious results (see Chapter 6). It is also recommended that the *uncertainty* of the assigned value is reported to participants in the PT scheme. This also may be obtained from the results from expert laboratories: estimating uncertainty is covered in more detail below (Section 4.13).

The target value for the standard deviation, σ , should be circulated in advance to the PT scheme participants along with a summary of the method by which it has been established. It will vary with analyte concentration, and one approach to estimating it is to use a functional relationship between concentration and standard deviation. The best known relationship is the **Horwitz trumpet**, dating from 1982, so-called because of its shape. Using many results from collaborative trials, Horwitz showed that the relative standard deviation of a method varied with the concentration, c as a mass ratio (e.g. $\text{mg g}^{-1} = 0.001$), according to the approximate and empirical equation:

$$\text{RSD} = \pm 2^{(1-0.5\log c)} \quad (4.11.2)$$

This equation leads to the trumpet-shaped curve shown in Fig. 4.10, which can be used to derive target values of σ for any analysis. Such σ values can also be estimated from prior knowledge of the standard deviations usually achieved in the analysis in question. Another approach uses fitness for purpose criteria: if the results of the analysis, used routinely, require a certain precision for the data to be interpreted properly and usefully, or to fulfil a legislative requirement, that precision provides the largest (worst) acceptable value of σ . It is poor practice to estimate σ from the results of previous rounds of the PT scheme itself, as this would conceal any improvement or deterioration in the quality of the results with time.

The results of a single round of a PT scheme are frequently summarised as shown in Fig. 4.11. If the results follow a normal distribution with mean x_a and standard deviation σ , the z -scores will be a sample from the standard normal distribution, i.e. a normal distribution with mean zero and variance 1. Thus a laboratory with a $|z|$ value of < 2 is generally regarded as having performed satisfactorily, a $|z|$ value between 2 and 3 is questionable, and $|z|$ values > 3 are unacceptable. Of course even the laboratories with satisfactory scores will strive to improve their values in the subsequent

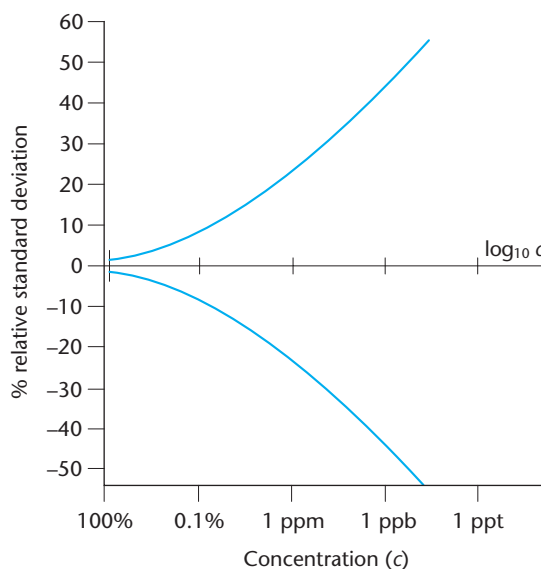


Figure 4.10 The Horwitz trumpet.

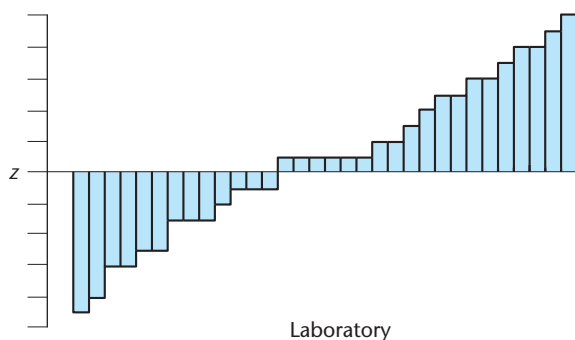


Figure 4.11 Summary of results of a single PT round.

rounds of the PT scheme. In practice it is not uncommon to find ‘heavy-tailed’ distributions, i.e. more results than expected with $|z| > 2$.

Some value has been attached to methods of combining z-scores. For example the results of one laboratory in a single PT scheme over a single year might be combined (though this would mask any improvement or deterioration in performance over the year). If the same analytical method is applied to different concentrations of the same analyte in each round of the same PT scheme, again a composite score might have limited value. Two functions used for this purpose are the **re-scaled sum of z-scores (RSZ)**, and the **sum of squared z-scores (SSZ)**, given by $RSZ = \sum_i z_i / \sqrt{n}$ and $SSZ = \sum_i z_i^2$ respectively. Each of these functions has disadvantages, and the use of combined z-scores is not to be recommended. In particular, combining scores from the results of *different* analyses is dangerous, as very high and very low z-scores might then cancel out to give a falsely optimistic result.

4.12 Method performance studies (collaborative trials)

Proficiency testing schemes allow the *competence of laboratories* to be monitored, compared and perhaps improved. By contrast a **method performance study** or **collaborative trial** aims to evaluate the precision of an *analytical method*, and sometimes its ability to provide results free from bias. It is normally a one-off experiment involving expert or competent laboratories, all of which by definition use the same technique.

A crucial preliminary experiment is the **ruggedness test**. As we saw in Chapter 1, even very simple analytical experiments involve several individual steps and perhaps the use of a number of reagents. Thus many experimental factors (e.g. temperature, solvent composition, pH, humidity, reagent purity and concentration) will affect the results, and it is essential that such factors are identified and studied before any collaborative trial is undertaken. In some cases a method is found to be so sensitive to small changes in one factor that is extremely difficult to control (e.g. very high reagent purity) that the method is rejected as impracticable before the performance study takes place. In other instances the study will continue, but the collaborators will be warned of the factors to be most carefully controlled. Although a more complete discussion of experimental design is deferred to Chapter 7, we can show here that much information on the most important factors can be obtained from a relatively small number of experiments. Suppose it is believed that seven experimental factors (A–G) might affect the results of an analysis. These factors must be tested at (at least) two values, called *levels*, to see whether they are really significant. Thus, if temperature is thought to affect the result, we must perform preliminary experiments at two temperatures (levels) and compare the outcomes. Similarly, if reagent purity may be important, experiments with high purity and lower purity reagent batches must be done. It might seem that 2^7 preliminary experiments, covering all the possible combinations of seven factors at two levels, will be necessary. In practice, however, just eight experiments can provide important information. The two levels of the factors are called + and –, and Table 4.4 shows how these levels are set in the eight experiments, the results of which are $y_1, y_2, \dots, y_8$. The effect of altering each factor from its high to its low level is easily calculated. Thus the effect of changing B from + to – is $(y_1 + y_2 + y_5 + y_6)/4 - (y_3 + y_4 + y_7 + y_8)/4$.

Table 4.4 Ruggedness test for seven factors

Experiment	Factors							Result
	A	B	C	D	E	F	G	
1	+	+	+	+	+	+	+	y_1
2	+	+	–	+	–	–	–	y_2
3	+	–	+	–	+	–	–	y_3
4	+	–	–	–	–	+	+	y_4
5	–	+	+	–	–	+	–	y_5
6	–	+	–	–	+	–	+	y_6
7	–	–	+	+	–	–	+	y_7
8	–	–	–	+	+	+	–	y_8

When the seven differences for factors A–G have all been calculated in this way, it is easy to identify any factors that have a worryingly large effect on the results. It may be shown that any difference that is more than twice the standard deviation of replicate measurements is significant and should be further studied. This simple set of experiments, technically known as an **incomplete factorial design**, has the disadvantage that *interactions* between the factors cannot be detected. This point is further discussed in Chapter 7.

In recent years international bodies have moved towards an agreement on how method performance studies should be conducted. At least eight laboratories ($k \geq 8$) should be involved. Since the precision of a method usually depends on the analyte concentration it should be applied to at least five different levels of analyte in the same sample matrix with duplicate measurements ($n = 2$) at each level. A crucial requirement of such a study is that it should distinguish between the repeatability standard deviation, s_r , and the reproducibility standard deviation, s_R . At each analyte level these are related by the equation:

$$s_R^2 = s_r^2 + s_L^2 \quad (4.12.1)$$

where s_L^2 is the variance due to inter-laboratory differences, which reflect different degrees of bias in different laboratories. Note that in this particular context, reproducibility refers to errors arising in different laboratories and equipment, but using the *same* analytical method: this is a more restricted definition of reproducibility than that used in other instances. As we saw in Section 4.3, one-way ANOVA can be applied (with separate calculations at each at each concentration level used in the study) to separate the sources of variance in Eq. (4.12.1). However, the proper use of the equation involves two assumptions: (1) that at each concentration level the means obtained in different laboratories are normally distributed; and (2) that at each concentration the repeatability variance among laboratories is equal. Both these assumptions are tested using standard methods before the ANOVA calculations begin. In practice the second assumption, that of homogeneity of variance, is tested first using **Cochran's test**. Strictly speaking this test is designed to detect outlying variances rather than testing for homogeneity of variance as a whole, but other more rigorous methods for the latter purpose are also more complex. Cochran's test calculates C by comparing the *largest range*, $w_{\max}$ (i.e. difference between the two results from a single laboratory) with the *sum of such ranges*, w_j , from all the laboratories (if $n > 2$ variances rather than ranges are compared, but here we assume that each participating laboratory makes just two measurements at each level):

$$C = \frac{w_{\max}^2}{\sum_j w_j^2} \quad (4.12.2)$$

where j takes values from 1 to k , the number of participating laboratories. The value of C obtained is compared with the critical values in Table A.15, and the null hypothesis, i.e. that the largest variance is not an outlier, is rejected if the critical value at the appropriate value of k is exceeded. When the null hypothesis is rejected, the results from the laboratory in question are discarded.

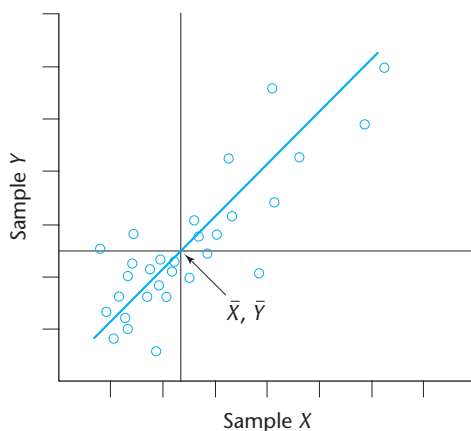


Figure 4.12 A Youden two-sample plot.

The first assumption is then tested using Grubbs' test (Section 3.7) which is applied first as a test for single outliers, and then (because each laboratory makes duplicate measurements) in a modified form as a test for paired outliers. In both cases all the results from laboratories producing outlying results are again dropped from the trial unless this would result in the loss of too much data. When these outlier tests are complete, the ANOVA calculation can proceed as in Section 4.3.

In many circumstances it is not possible to carry out a full method performance study as described above, for example when the test materials are not available with a suitable range of analyte concentrations. In such cases a simpler system can be used. This is the **Youden matched pairs** or **two-sample** method, in which each participating laboratory is sent *two materials of similar composition*, X and Y, and asked to make *one* determination on each. The results are plotted as shown in Fig. 4.12, each point on the plot representing a pair of results from one laboratory. The mean values for the two materials, $\bar{X}$ and $\bar{Y}$, are also determined, and vertical and horizontal lines are drawn through the point $(\bar{X}, \bar{Y})$, thus dividing the chart into four quadrants. This plot allows us to assess the occurrence of random errors and bias in the study. If only random errors occur the X and Y determinations may give results which are both too high, both too low, X high and Y low, or X low and Y high. These four outcomes would be equally likely, and the number of points in each of the quadrants would be roughly equal. But if a systematic error occurs in a laboratory, it is likely that its results for both X and Y will be high, or both will be low. So if systematic errors dominate, most of the points will be in the top-right and bottom-left quadrants. This is indeed the result that is obtained in most cases. In the impossible event that random errors were absent, all the results would lie on a line at 45° to the axes of the plot, so when in practice such errors do occur, the perpendicular distance of a point from that line is a measure of the random error of the laboratory represented by that point. Moreover the distance from the intersection of that perpendicular with the 45° line to the point $(\bar{X}, \bar{Y})$ measures the systematic error of the laboratory. This fairly simple approach to a method performance study is thus capable of yielding a good deal of information in a simple form. The Youden approach has the further advantages that participating laboratories are not tempted to censor one or more replicate determinations, and that more materials can be studied without large numbers of experiments.

Youden plots provide a good deal of information in an immediately accessible form, but we still need methods for calculating the variances s_R^2 and s_r^2 . The following example shows how this can also be done in a simple way.

Example 4.12.1

The lead levels (in ng g^{-1}) in two similar samples (X and Y) of solid milk formulations for infants were determined in nine laboratories (1–9) by graphite-furnace atomic-absorption spectrometry. The results were:

Sample	Laboratory								
	1	2	3	4	5	6	7	8	9
X	35.1	23.0	23.8	25.6	23.7	21.0	23.0	26.5	21.4
Y	33.0	23.2	22.3	24.1	23.6	23.1	21.0	25.6	25.0

Evaluate the overall inter-laboratory variation, and its random and systematic components.

In studies of this type there is a difference between the samples as well as the differences between laboratories. In the normal way, such a situation would be dealt with by two-way ANOVA (see Section 7.3), and in some cases this is done. However, in this instance there are only two samples, deliberately chosen to be similar in their analyte content, so there is little interest in evaluating the difference between them. The calculation can therefore be set out in a numerically and conceptually simpler way than a complete two-way ANOVA. We know that the result obtained by each laboratory for sample X may include a systematic error. The *same* systematic error will presumably be included in that laboratory's result for the similar sample Y . The difference $D (= X - Y)$ will thus have this error removed, so the spread of the D values will provide an estimate of the random or measurement errors. Similarly, X and Y can be added to give T , the spread of which gives an estimate of the overall variation in the results. The measurement variance is then estimated by:

$$s_r^2 = \frac{\sum_i (D_i - \bar{D})^2}{2(n - 1)} \quad (4.12.3)$$

and the overall variance, s_R^2 , due to all sources of error, is estimated by:

$$s_R^2 = \frac{\sum_i (T_i - \bar{T})^2}{2(n - 1)} \quad (4.12.4)$$

Notice that each of these equations includes a 2 in the denominator. This is because D and T each give estimates of errors in two sets of results, subtracted and

added in D and T respectively. The results of this trial can be expressed in a table as follows:

	1	2	3	4	5	6	7	8	9
X	35.1	23.0	23.8	25.6	23.7	21.0	23.0	26.5	21.4
Y	33.0	23.2	22.3	24.1	23.6	23.1	21.0	25.6	25.0
D	2.1	-0.2	1.5	1.5	0.1	-2.1	2.0	0.9	-3.6
T	68.1	46.2	46.1	49.7	47.3	44.1	44.0	52.1	46.4

From the third and fourth rows of the table, $\bar{D} = 0.244$ and $\bar{T} = 49.33$. Equations (4.12.3) and (4.12.4) then give the overall variance and the measurement variance as $(5.296)^2$ and $(1.383)^2$ respectively. These can be compared as usual using the F -test, giving $F = 14.67$. The critical value, $F_{8,8}$, is 3.44 ($P = 0.05$), so the inter-laboratory variation cannot simply be accounted for by random errors. The component due to bias, s_L^2 , is given here by

$$s_R^2 = 2s_L^2 + s_r^2 \quad (4.12.5)$$

Note again the appearance of the 2 in Eq. (4.12.5), because two sample materials are studied. Using this equation gives $s_L^2 = 3.615^2$. The mean of all the measurements is $49.33/2 = 24.665$, so the relative standard deviation is $(100 \times 5.296)/24.665 = 21.47\%$. This seems to be a high value, but the Horwitz trumpet relationship would predict an even higher value of ca. 28% at this concentration level. It should be noted that possible outliers are not considered in the Youden procedure, so the question of whether we should reject the high results from laboratory 1 does not arise.

4.13 Uncertainty

In Chapter 1 we learned that analytical procedures will be affected both by random errors and by bias. For some years now analytical chemists have recognised the importance of providing for each analysis a single number which describes their combined effect. The **uncertainty** of a result is a parameter that describes a range within which the value of the quantity being measured is expected to lie, taking into account all sources of error. Two symbols are used to express uncertainty. **Standard uncertainty** (u) expresses the concept as a standard deviation. **Expanded uncertainty** (U) defines a *range* that encompasses a large fraction of the values within which the quantity being measured will lie and is obtained by multiplying u by a **coverage factor**, k , chosen according to the degree of confidence required for the range, i.e. $U = u \times k$. Since u is analogous to a standard deviation, if k is 2 (this is generally taken as the default value if no other information is given), then U gives approximately one-half of the 95% confidence interval.

In principle, two basic approaches to estimating uncertainty are available. The **bottom-up** approach identifies each separate stage of an analysis, including sampling steps wherever possible (see below), assigns appropriate random and systematic errors

to each, and then combines these components using the rules summarised in Section 2.11 to give an overall u -value. However for a number of reasons this process may not be as simple as it seems. The first problem is that even simple analytical processes may involve many individual experimental steps and possible sources of error. It is easy to overlook some of these sources and thus arrive at an over-optimistic uncertainty value. If all the sources of error *are* fully identified, then the whole calculation process is liable to be long-winded. Examples of error sources that should be considered but are easily overlooked include operator bias; instrument bias, including sample carry-over; assumptions concerning reagent purity; use of volumetric apparatus at a temperature different from that at which it was calibrated; changes in the composition of the sample during the analysis, through contamination or because of inherent instability; use of calculators or computers with inadequate capabilities or with the wrong statistical model applied; and so on. All these factors may arise *in addition* to the random errors that inevitably occur in repeated measurements. Whereas the latter may be estimated directly by repeated measurements, some of the former may not be amenable to experiment, and may have to be estimated using experience, or equipment manufacturers' information such as calibration certificates or instrument specifications.

Another problem is that, as shown in Chapter 2, systematic errors do not immediately lend themselves to statistical treatment in the same way as random errors. How then can they be combined with random errors to give an overall u value? (It is naturally good practice to minimise systematic errors by the use of standards and reference materials, but the errors involved in that correction process should be included in the overall uncertainty estimate.) The usual method of tackling systematic errors is to treat them as coming from a *rectangular* distribution. Suppose for example that a manufacturer quotes the purity of a reagent as $99.9 \pm 0.1\%$. This does not mean that the purity of the reagent in its container varies randomly with a standard deviation of 0.1%: it means that the purity of the reagent in a single bottle is between 99.8% and 100.0%. That is, any single bottle provides a systematic error, and there is no reason to suppose that the actual purity is closer to 99.9% than to any other value in the range 99.8–100.0%. In other words the purity has a **uniform distribution** over this range. In such cases, the contribution to the standard uncertainty, u , is obtained by dividing the error by $\sqrt{3}$, giving a value of $0.1/\sqrt{3} = 0.0577$. Uncertainty contributions of this kind derived from uniform distributions (or from triangular distributions, where the corresponding division factor is $\sqrt{6}$) are referred to as type B uncertainties. Random errors that can be combined using the usual methods summarised in Chapter 2 are called type A contributions.

The following simplified example of a bottom-up uncertainty calculation shows some of these principles in operation. Further details, including a numerical calculation, are given in the Eurachem/CITAC guide (see Bibliography). Suppose we wish to determine the concentration of a solution of sodium hydroxide by titration with a standard acid such as potassium hydrogen phthalate (KHP). The molar concentration of NaOH given by this experiment will depend on the volume of NaOH solution used in the titration, and the mass, purity and molecular weight of the KHP. The uncertainty in the molecular weight of the acid can be computed from the atomic weights table of the International Union of Pure and Applied Chemistry. It would be treated as a type B uncertainty, but it is so small that it is negligible for most practical purposes. The mass of the KHP used would almost certainly be determined by difference, i.e. by weighing a container with the KHP in it, then weighing the container after

the KHP has been removed for dissolution. Each of these weighings would have an uncertainty derived as a type B estimate from the calibration certificate of the balance used. If the certificate indicated a balance error of ± 0.2 mg, the uncertainty in each weighing would be $0.2/\sqrt{3}$ mg = 0.1155 mg. The overall uncertainty in the weighing stage is then determined using Eq. (2.11.2), as $\sqrt{(0.1155)^2 + (0.1155)^2} = 0.1633$ mg. The contribution of the overall uncertainty of the uncertainty in the purity of the KHP is another type B estimate, again obtained by dividing the fractional impurity level by $\sqrt{3}$. The uncertainty contribution from the volume of NaOH used will have several sources, including temperature effects (i.e. using glassware at a temperature different from that at which it is calibrated), the calibration uncertainty of the burette (often assumed to derive from a triangular distribution) and possibly indicator end point errors. Finally, replicate titrations will show the extent of random errors during the analysis. Although in practice, most attention will be given to the major contributors to the uncertainty, it is clear that even in a simple analysis of this kind a full uncertainty estimate requires much care.

A further problem, the extent of which seems not to have been fully investigated, is that the rules for combining errors given in Chapter 2 assume that the sources of the errors are *independent*. In reality it seems quite possible that this is not always true. For example if a series of experiments is conducted over a period in which the laboratory temperature fluctuates, such fluctuations might have several effects, such as altering the capacity of volumetric apparatus, causing sample losses through volatility, affecting the sensitivity of optical or electrochemical detectors, and so on. Since all these errors would arise from a single source, they would be correlated, and strictly speaking could not be combined using the simple formulae. In such cases the actual uncertainty might be less than the u value calculated on the assumption of independent errors.

Overall the bottom-up approach to uncertainty estimates may be too time-consuming for many purposes. In some laboratories it may not be necessary to make such calculations very often, as an uncertainty estimate made in detail for one analysis may serve as a model for other closely similar analyses over a period of time. But in other instances, most obviously where legal or regulatory issues arise (see below), this will not be sufficient and an uncertainty estimate will have to be provided for each disputed sample. Despite this the bottom-up approach is the one currently recommended by many authorities.

A completely different approach is the **top-down** method, which seeks to use the results of PT schemes in a number of laboratories (see Section 4.11) to give estimates of the overall uncertainties of the measurements without necessarily trying to identify every individual source of error. The method is clearly only applicable in areas where data from properly run proficiency schemes are available, though such schemes are rapidly expanding in number and may thus provide a real alternative to bottom-up methods in many fields. It can be argued that uncertainty values calculated in this way are more realistic than bottom-up values, and there is a great saving of effort since PT scheme results provide uncertainty estimates directly. However, PT schemes use a variety of analytical methods, so it might reasonably be claimed that the uncertainty of results from a single laboratory that has long experience of a single well-established method might be better (smaller) than PT results would suggest. On the other hand, PT schemes strive to use single sample materials prepared with great care. Some sampling errors that would occur in a genuine analysis might thus be overlooked (see below).

These problems have led some bodies to propose simpler methods, explicitly designed to minimise the workload in laboratories that use a range of analytical procedures. One such approach uses the basic following principles:

- Systematic errors are not included in the uncertainty estimates, but are assessed using reference materials as usual and thus corrected or eliminated.
- At least ten replicate measurements are made on stable and well-characterised authentic samples or reference materials. (This again implies that sampling uncertainties are not included in the estimates.)
- Uncertainties are calculated from the standard deviations of measurements made in *internal reproducibility conditions*, i.e. with different analysts, using different concentrations (including any that are relevant to legal requirements), and in all relevant matrices.

These conditions are supposed to mimic those that would arise in a laboratory in everyday operation. Some provision is made for circumstances where the reproducibility conditions cannot be achieved (for example where samples are intrinsically unstable). This method seems to be very simple, but it may be adequate: indeed it may be the only practicable method in some instances.

Uncertainty estimates are important not only to anyone who has provided a sample for analysis and who requires a range of values in which the true analyte concentration should lie. They also have value in demonstrating that a laboratory has the capacity to perform analyses of legal or statutory significance. Once an uncertainty value for a particular analysis in a given laboratory is known, it is simple to interpret the results in relation to such statutory or other specification limits. Four possible situations are shown in Fig. 4.13, where it is assumed that a coverage factor of 2 has been used to determine U at the 95% level (the 95% interval is shown by the vertical double arrows), and where both upper and lower limits for the concentration of the analyte have been specified, as indicated by the horizontal lines. In case A the uncertainty interval lies completely between the upper and lower specified limits, so compliance with the specification has been achieved. In case B the 95% interval extends just beyond the upper limit, so although compliance is more likely than not, it cannot be fully verified at the 95% level. In case C compliance is very unlikely, though not impossible, and in case D there is a clear failure to comply.

Although none of the approaches to estimating uncertainties is ideal, and although the term itself still provokes controversy (some scientists think that the word ‘uncertainty’ is too negative or pessimistic in its implications for the lay public), uncertainty calculations are now an intrinsic part of most areas of modern analytical chemistry.

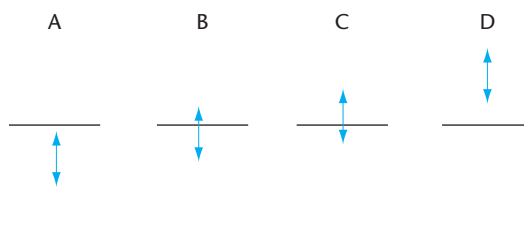


Figure 4.13 Use of uncertainty to test compliance with specification limits.

In recent years, a good deal of attention has been given to the component of the uncertainty arising from the sampling steps of an analysis. This concern arises from the realisation that in many cases the sampling uncertainty contribution might be the largest single component of the overall uncertainty of the analysis. The problem is likely to be most acute when the material under study may be grossly heterogeneous. This unsurprisingly occurs in the fields of geology or food science, where heterogeneous solids are commonly found. However liquid samples are often also heterogeneous, either with respect to time in the case of a flowing stream, or because their chemical composition may vary with depth below the surface. Recommendations on good sampling practice have been produced (see Bibliography) and as expected involve taking several samples from the target material. Each of these samples is divided into two, and duplicate measurements are made on each of the two sub-samples in repeatability conditions (see Section 1.3). ANOVA can then be used, as described in Section 4.3, to separate the contributions that the sample and measurement variations make to the uncertainty.

4.14 Acceptance sampling

Previous sections of this chapter have shown how the quality of the analytical results obtained in a laboratory can be monitored by internal quality control procedures and by participation in proficiency testing schemes. We have also shown how the concept of uncertainty is designed to help the interpretation of analytical results by the customers for analytical measurements, including regulatory authorities. In this section we consider a further important problem involving both analysts and their customers, called **acceptance sampling**. The simple statistical principles involved have been discussed in previous chapters. Suppose that the manufacturer of a chemical is required to ensure that it does not contain more than a certain level of a particular impurity. This is called the **acceptable quality level (AQL)** of the product and is given the symbol μ_0 . The manufacturer's intention to ensure that this impurity level is not exceeded is monitored by testing batches of the product. Each test involves n test portions, whose mean impurity level is found to be $\bar{x}$. The variation between portions, σ , is (as we have seen) normally known from previous experience. The practical problem that arises is that, even when a batch of manufactured material has an actual impurity level of μ_0 , and is thus satisfactory, values of $\bar{x}$ greater than μ_0 will be found in 50% of the analyses. Therefore the manufacturer establishes a critical value for $\bar{x}$, given the symbol $\bar{x}_0$. If a measured value of $\bar{x} > \bar{x}_0$ the batch is rejected. This critical value is higher than μ_0 , thus ensuring that the manufacturer runs only a small risk of rejecting a satisfactory batch.

At the same time the customer wishes to minimise the risk of accepting a batch with a mean impurity level greater than μ_0 . This can be achieved by setting an agreed **tolerance quality level (TQL)**, μ_1 , which has a small probability of acceptance. The aim of acceptance sampling is that the critical value $\bar{x}_0$ should minimise the risk to the customer as well as to the manufacturer. At the same time, to minimise the analytical effort involved, we wish to ensure that n is no larger than necessary. Both these aims can be achieved using the properties of the sampling distribution of the mean, given that σ is known. (The situation can be envisaged by analogy with Fig. 3.7, with the AQL and the TQL replacing the values 3% and 3.05% respectively in that figure.)

Suppose the manufacturer accepts a 5% risk of rejecting a batch of the chemical that is in fact satisfactory, i.e. a batch for which $\bar{x} > \bar{x}_0$, even though $\mu = \mu_0$. Then we can write

$$\frac{\bar{x}_0 - \mu_0}{\sigma/\sqrt{n}} = 1.64 \quad (4.14.1)$$

The value 1.64 can be found in Table A.1 as the z-value corresponding to $F(z) = 0.95$ (see also Section 2.2). Suppose also that the customer is prepared to accept a 10% risk of accepting a batch with the impurity at the TQL. Then we can similarly write:

$$\frac{\bar{x}_0 - \mu_1}{\sigma/\sqrt{n}} = -1.28 \quad (4.14.2)$$

This z-value of -1.28 (i.e. $-1.30 + 0.02$) from Table A.1 corresponds to $F(z) = 0.10$. Since in practice the values of μ_0 and μ_1 will have been agreed in advance, Eq. (4.14.1) and Eq. (4.14.2) provide simultaneous equations that can be solved for n and $\bar{x}_0$.

Example 4.14.1

Determine n and $\bar{x}_0$ for the case where the AQL and TQL are 1.00 g kg^{-1} and 1.05 g kg^{-1} impurity respectively, the manufacturer's and customer's risks are 5% and 10% respectively, and σ is 0.05 g kg^{-1} .

The solution to this problem involves the use of Eqs (4.14.1) and (4.14.2) with μ_0 and μ_1 taking values 1.00 and 1.05, the AQL and TQL respectively. Solving the equations simultaneously we can write:

$$\bar{x}_0 - 1 = 1.64\sigma/\sqrt{n}$$

and

$$\bar{x}_0 - 1.05 = -1.28\sigma/\sqrt{n}$$

which on subtraction to remove $\bar{x}_0$ and utilising $\sigma = 0.05$ yield:

$$0.05 = 2.92 \times 0.05/\sqrt{n}$$

from which $\sqrt{n} = 2.92$. Inserting this value into Eq. (4.14.1) gives:

$$\frac{\bar{x}_0 - 1}{0.05/2.92} = 1.64$$

hence $\bar{x}_0 = 1.028$. The value of n is $2.92^2 = 8.53$, which is rounded up to 9. Thus a critical value of 1.028% impurity and sample size of 9 will provide both manufacturer and customer with the necessary assurances.

4.15 Method validation

Good practice clearly demands that any analytical method in a field of application is suitable for its intended purpose. To demonstrate this fitness for purpose a method needs to be *validated* if it is:

- a new method, the viability of which needs to be tested;
- an existing method to be used in a particular laboratory for the first time; or
- an existing method applied to a new sample matrix, or used in a significantly different way, e.g. with different instrumentation or at different concentration ranges.

It is strictly more accurate to speak of validating an ‘analytical system’, the components of which are the method itself (with sampling steps), the matrix to which the method is applied, and the analyte concentration range over which it will be used. The validation of the methods used by a laboratory will normally be a prerequisite for the laboratory’s *accreditation*, i.e. recognition that it performs its work to an accepted national or international standard. *Method transfer*, the exporting of an analytical system from one laboratory to another, will in general be acceptable only for properly validated methods (though such transfers involve a variety of additional protocols, and have clear implications for issues such as staff training in the recipient laboratory). Validation is not a process that needs to be performed frequently (unless a part of the analytical system is changed): in this respect it clearly differs from the IQC procedures that all laboratories should use regularly.

In some cases a validation process is designed to support the work of a single laboratory and its immediate users and customers. But in other situations the extent and importance of the validation process are far broader. Examples of the latter include methods of food analysis that must meet legislative requirements, the methods supporting any submission to regulatory authorities of a possible new drug substance, the methods used to monitor the quality of approved drug products, and methods used in forensic work that are liable to be presented (and perhaps challenged) as evidence in court. The principal difference between a ‘single laboratory’ validation and a ‘full’ one is that the latter will normally involve the use of the method in a properly conducted method performance study (Section 4.12).

Much guidance on carrying out both full and single laboratory validations is provided by a range of statutory and international advisory bodies (some examples are given in the books listed in the Bibliography) but unfortunately such bodies are not in full agreement over the number of method performance characteristics to be specified, and their terminology and definitions differ. Here we follow the terminology and recommendations of the ‘Harmonized guidelines for single-laboratory validation of methods of analysis’. The characteristics which can be assessed using statistical methods include the following:

- **Trueness.** This is the measure of agreement between the analytical result when the method is applied to a reference material, and the accepted reference value of the analyte. A high level of trueness is equivalent to a lack of bias in the method (see Chapter 1). As we have seen, bias is tested with the aid of reference materials:

the method is used to measure the analyte content of such a material several times, and a significance test (Section 3.2) can be applied to the null hypothesis that the method is free from bias. If reference materials are not available, trueness is sometimes assessed by analysing a test portion of the sample before and after adding a known mass of analyte to it, a process known as 'spiking'. If the difference between the two measurements is not equivalent to the amount added, i.e. the *recovery* is not 100%, some bias in the method is indicated. Spiking methods and recovery calculations have some drawbacks. One is that the added analyte may not always behave in the same way as the analyte naturally present in the test material. For example, a hormone added to a blood plasma sample may not bind to the plasma proteins to the same extent as the hormone already present. For this reason it is wise to assume that, while a recovery of less than or greater than 100% indicates a degree of bias, a recovery close to 100% does not necessarily indicate the absence of bias. Trueness can also be tested by comparing the results from the method which is being validated with the results obtained when a standard or reference method is applied to the same test materials. If this approach is adopted a number of test materials containing different analyte levels should be used, and the results evaluated using the paired *t*-test (Section 3.4) or by regression methods (see Section 5.9).

- **Precision.** This reflects the level of agreement between replicate measurements and is normally expressed as a standard deviation or relative standard deviation. As shown in Section 1.3 it is important to distinguish between repeatability, i.e. within-run precision, and reproducibility, between-run precision. If the method to be validated is applied to the same test material on successive occasions, i.e. in between-run conditions, the observed variation will *include* the within-run component, and the two contributions to the random variation can be separated by one-way ANOVA (Sections 3.8–3.10). Good validation practice involves the use of typical 'real' test materials (preferably not reference materials, which may be atypically homogeneous), and precision should be measured at several concentrations covering the operating range of the method. A separate source of variation, which should be studied independently, is the extent to which different specimens of the intended sample matrix affect the results. Soils and blood plasma specimens are good examples of matrices showing wide variations in properties. The chemical composition, particle size and particle size distribution of soils may vary substantially, and in blood plasma studies variations in protein content and composition, viscosity, colour, etc. are inevitable. The effects such variations have on the method under study can be studied by collecting a number of specimens of the matrix, and for each one measuring the recoveries of spiked samples at different concentration levels covering the intended range of the method.
- **Calibration parameters.** Almost all analyses now involve the use of calibration graphs (see Chapter 5). These plot the responses of the analytical system (*y*) against the concentrations of a series of standards of known analyte composition (*x*). The graph is then used to obtain concentrations of test materials, studied under identical conditions to the standards, from their system responses. The statistical principles and implications of this approach are so important that the next chapter is devoted to them. Here we can simply note that the calibration graph can be used to assess the range of a method, and its limit of detection and limit of quantitation.

- **Ruggedness.** The ruggedness of a method is a measure of the extent to which its experimental conditions (e.g. pH, temperature, reagent purity, instrument, operator) can be changed without significantly affecting the results it gives. As we saw in Section 4.12 this property of a method can be studied economically by changing the conditions, i.e. studying each experimental *factor* (variable) at two levels and using a fractional factorial design to summarise the results. These show which factors need the closest control, and which are of less significance: it may be possible to estimate the ranges over which each factor can be allowed to vary in practice.
- **Performance in inter-laboratory comparisons.** A method is unlikely to be broadly acceptable to the analytical science community unless it has given satisfactory results over a significant period in an appropriate proficiency testing scheme, assuming that one is available (see Section 4.11). A Youden two-sample test (see Section 4.12) would give further information on its precision and freedom from bias.
- **Fitness for purpose and uncertainty.** A validation process must clearly show that the method being studied is suitable for its intended use and users. For example it must be able to provide satisfactory outcomes, with modest sample sizes, in terms of the acceptance sampling principles outlined in Section 4.14. In many cases the principal criterion that decides whether or not a method is fit for purpose will be its overall uncertainty, determined as described in Section 4.13.

Bibliography

- Ellison, S.L.R., Rossstein, M. and Williams, A., (eds.) 2000, *Quantifying Uncertainty in Analytical Measurement*, 2nd edn, Eurachem-CITAC.
- Harmonized guidelines for single-laboratory validation of methods of analysis, *Pure and Applied Chemistry*, 2002, **74**: 835–855.
- Lawn, R.E., Thompson, M. and Walker, R.F., 1997, *Proficiency Testing in Analytical Chemistry*, Royal Society of Chemistry, London. Provides a clear summary of the management of PT schemes and the handling of the resulting data.
- Massart, D.L., Vandeginste, B.G.M., Buydens, L.M.C., de Jong, S., Lewi, P.J. and Smeyers-Verbeke, J., 1997 *Handbook of Chemometrics and Qualimetrics*, Part A, Elsevier, Amsterdam. Comprehensive coverage of many quality-related topics.
- Montgomery, D.C., 2009, *Introduction to Statistical Quality Control*, 6th edn, John Wiley, New York. A major general text on quality control principles and statistics, with full coverage of acceptance quality sampling: business-oriented rather than laboratory-oriented.
- Prichard, E. and Barwick, V., 2007, *Quality Assurance in Analytical Chemistry*, John Wiley, Chichester. A fine introduction to all the crucial aspects of quality in the laboratory, with discussions of sampling methods and uncertainty calculations: it summarises fully the major protocols and recommendations of statutory bodies.
- Ramsey, M.H., and Ellison, S.L.R., (eds.) 2007, *Measurement Uncertainty Arising from Sampling*, Eurachem-CITAC.
- Wernimont, G.T. and Spendley, W., 1985, *Use of Statistics to Develop and Evaluate Analytical Methods*, AOAC, Arlington, VA. Authoritative sequel to Youden and Steiner's manual – see below.
- Youden, W.J. and Steiner, E.H., 1975, *Statistical Manual of the Association of Official Analytical Chemists*, AOAC, Arlington, VA. Classic text with much emphasis on collaborative studies.

Exercises

- 1 Two sampling schemes are proposed for a situation in which it is known, from past experience, that the sampling variance is 10 and the measurement variance 4 (arbitrary units).

Scheme 1: Take five sample increments, blend them and perform a duplicate analysis.

Scheme 2: Take three sample increments and perform a duplicate analysis on each. Show that the variance of the mean is the same for both schemes.

What ratio of the cost of sampling to the cost of analysis must be exceeded for the second scheme to be the more economical?

- 2 The data in the table below give the concentrations of albumin measured in the blood serum of one adult. On each of four consecutive days a blood sample was taken and three replicate determinations of the serum albumin concentration were made.

Day	Albumin concentrations (normalised, arbitrary units)		
1	63	61	62
2	57	56	56
3	50	46	46
4	57	54	59

Show that the mean concentrations for different days differ significantly. Estimate the variance of the day-to-day variation (i.e. 'sampling variation').

- 3 In order to estimate the measurement and sampling variances when the halofuginone concentration in chicken liver is determined, four sample increments were taken from different parts of the liver and three replicate measurements were made on each. The following results were obtained (mg kg^{-1}):

Sample	Replicate measurements		
A	0.25	0.22	0.23
B	0.22	0.20	0.19
C	0.19	0.21	0.20
D	0.24	0.22	0.22

Verify that the sampling variance is significantly greater than the measurement variance and estimate both variances.

Two possible sampling schemes are proposed:

Scheme 1: Take six sample increments, blend them and make four replicate measurements.

Scheme 2: Take three sample increments and make two replicate measurements on each.

Calculate the total variance of the mean for each scheme.

- 4 In order to estimate the capability of a process, measurements were made on six samples of size 4 as shown in the table below (in practice at least 25 such samples would be needed). Estimate the process capability, σ . If the target value is 50, calculate the positions of the action and warning lines for the Shewhart charts for the sample mean and the range.

Sample	Values			
1	48.8	50.8	51.3	47.9
2	48.6	50.6	49.3	49.7
3	48.2	51.0	49.3	50.3
4	54.8	54.6	50.7	53.9
5	49.6	54.2	48.3	50.5
6	54.8	54.8	52.3	52.5

- 5 In a collaborative trial, two closely similar samples of oil shale (A and B) were sent to 15 laboratories, each of which performed a single inductively coupled plasma determination of the cadmium level in each sample. The following results were obtained:

Laboratory	Cd levels (ppm)	
	A	B
1	8.8	10.0
2	3.8	4.7
3	10.1	12.1
4	8.0	11.0
5	5.0	4.7
6	5.2	6.4
7	6.7	8.7
8	9.3	9.6
9	6.9	7.5
10	3.2	2.8
11	9.7	10.4
12	7.2	8.3
13	6.5	6.8
14	9.7	7.2
15	5.0	6.0

Plot the two-sample chart for these data, and comment on the principal source of error in the collaborative trial. Estimate the overall variance, the measurement variance, and the systematic error component of the variance of the results.

- 6 The target value for a particular analysis is 120. If preliminary trials show that samples of size 5 give an $\bar{R}$ value of 7, set up Shewhart charts for the mean and range for samples of the same size.
- 7 An internal quality control sample of blood, used for checking the accuracy of blood alcohol determinations, contains 80.0 mg 100 ml⁻¹ of ethanol. Successive daily measurements of the alcohol level in the sample were made using four replicates. The precision (process capability) of the method was known to be 0.6 mg 100 ml⁻¹. The following results were obtained:

Day	Concentration (mg 100 ml ⁻¹)
1	79.8
2	80.2
3	79.4
4	80.3
5	80.4
6	80.1
7	80.4
8	80.2
9	80.0
10	79.9
11	79.7
12	79.6
13	79.5
14	79.3
15	79.2
16	79.3
17	79.0
18	79.1
19	79.3
20	79.1

Plot the Shewhart chart for the mean, and the CUSUM chart, for these results, and comment on the outcomes.

5

Calibration methods in instrumental analysis: regression and correlation

Major topics covered in this chapter

- Calibration in instrumental analysis
- Product-moment correlation coefficient
- Plotting the best straight line
- Errors and confidence limits in linear calibration
- Limits of detection
- Standard additions
- Regression lines used for method comparisons
- Weighted regression
- Intersecting straight lines
- ANOVA and regression calculations
- Curve fitting
- Outliers in regression

5.1

Introduction: instrumental analysis

Classical or 'wet chemistry' analysis techniques such as titrimetry and gravimetry remain in use in many laboratories. They are ideal for high-precision analyses, especially when small numbers of samples are involved, and are sometimes necessary for the analysis of standard materials. In practice, however, almost all analyses are now performed using instrumental methods. Techniques using absorption and emission spectrometry at various wavelengths, electrochemistry, mass spectrometry, chromatographic and electrophoretic separations, thermal and radiochemical methods probably account for at least 90% of all current analytical work. There are several reasons for this.

Instrumental methods can perform analyses that are difficult or impossible by classical methods. The latter can only rarely detect materials at sub-microgram levels, while instrumental methods can provide the ultimate in terms of sensitivity: in recent years fluorescence and electrochemical methods have been used to detect single organic molecules in very small volumes of solution. While ‘wet chemical’ methods can usually determine only one analyte at a time, many instrumental methods have multi-analyte capability, especially with the aid of modern chemometrics to interpret the data (see Chapter 8). Most classical methods only operate over a concentration range of two to three orders of magnitude (i.e. powers of 10), but many instrumental techniques work well over a range of six or more orders of magnitude: this characteristic has important implications for the statistical treatment of the results, as we shall see in the next section.

When there are many samples to be analysed instrumental methods are generally quicker and often cheaper than the labour-intensive manual methods. In clinical analysis, for example, there is frequently a requirement for the same analyses to be done on scores or even hundreds of whole blood or blood serum/plasma samples every day. Despite the high initial cost of the equipment, such work is generally performed using completely automatic systems. Automation has become such an important feature of analytical chemistry that the ease with which a particular technique can be automated often determines whether or not it is used. A typical automatic method may be able to process samples at the rate of 100 per hour or more. The equipment will take a measured volume of sample, dilute it appropriately, conduct one or more reactions with it, and determine and record the concentration of the analyte or of a derivative produced in the reactions. Other areas where the use of automated equipment is now crucial include environmental monitoring and the rapidly growing field of industrial process analysis. Special problems of error estimation will evidently arise in all these applications of automatic analysis: systematic errors, for example, must be identified and corrected as rapidly as possible.

An important trend in modern instrumental analysis is the development of miniaturised systems, which often combine modern micro-electronic components with micro-fluidic sample handling systems. Such tiny analytical systems have great potential in process analysis, *in vivo* diagnostics, security-related detection systems and many other areas.

Lastly, modern analytical instruments are invariably interfaced with personal computers to provide sophisticated system control, and the storage, treatment and reporting of data. Such systems can also evaluate the results statistically, and compare the analytical results with data libraries so as to match spectral and other information. All these facilities are now available from low-cost computers operating at high speeds. Also important is the use of ‘intelligent’ instruments, which incorporate automatic set-up and fault diagnosis and can perform optimisation processes (see Chapter 7).

Instrumental analysis methods – why?

- Extreme sensitivity, sometimes single molecule detection.
- Very wide concentration ranges.
- Multi-analyte capability in conjunction with suitable data handling.

- Automation: high sample throughput and low cost per sample.
- Miniaturised systems.
- Computer interfacing for control, rapid data handling, optimisation.

The statistical procedures used with instrumental analysis methods must provide, as always, information on the precision and accuracy of the measurements. They must also reflect the technical advantages of such methods, especially their ability to cover a great range of concentrations (including very low ones), and to handle many samples rapidly. (In this chapter we shall not cover methods that facilitate the simultaneous determination of more than one analyte: these topics are outlined in Chapter 8.) In practice instrumental analysis results are calculated and the errors evaluated using an approach that differs from that used when a single measurement is repeated several times.

5.2

Calibration graphs in instrumental analysis

The usual procedure is as follows. The analyst takes a series of samples (preferably at least six, and possibly several more) in which the concentration of the analyte is *known*. These calibration standards are measured in the analytical instrument under the same conditions as those subsequently used for the test (i.e. the ‘unknown’) samples. The results are used to plot a calibration graph, which is then used to determine the analyte concentrations in test samples by interpolation (Fig. 5.1). This general procedure raises several important statistical questions:

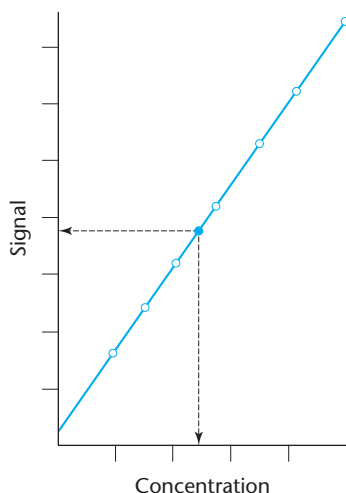


Figure 5.1 Calibration procedure in instrumental analysis: ○ calibration points; ● test sample.

- Is the calibration graph linear? If it is a curve, what is the form of the curve?
- Since each of the points on the calibration graph is subject to errors, what is the best straight line (or curve) through these points?
- Assuming that the calibration plot is actually linear, what are the errors and confidence limits for the slope and the intercept of the line?
- When the calibration plot is used for the analysis of a test sample, what are the errors and confidence limits for the determined concentration?
- What is the *limit of detection* of the method? That is, what is the least concentration of the analyte that can be detected with a predetermined level of confidence?

Before tackling these questions in detail, we must consider a number of aspects of plotting calibration graphs. First, it is usually essential that the calibration standards cover the whole range of concentrations required in the subsequent analyses. With the important exception of the 'method of standard additions', which is treated separately in a later section, concentrations of test samples are normally determined by interpolation and *not* by extrapolation. Second, it is crucially important to include the value for a 'blank' sample in the calibration curve. The blank contains no deliberately added analyte, but does contain the same solvent, reagents, etc., as the other test samples, and is subjected to exactly the same sequence of analytical procedures. The instrument signal given by the blank sample will sometimes not be zero. This signal is subject to errors like all the other points on the calibration plot, so it is wrong in principle to subtract the blank value from the other standard values before plotting the calibration graph. This is because, as shown in Chapter 2, when two quantities are subtracted, the error in the final result cannot also be obtained by simple subtraction. Subtracting the blank value from each of the other instrument signals before plotting the graph thus gives incorrect information on the errors in the calibration process.

Several assumptions are implicit in the usual methods for plotting calibration graphs. The calibration curve is always plotted with the instrument signals on the vertical (y) axis and the standard concentrations on the horizontal (x) axis. This is because many of the procedures to be described in the following sections *assume* that all the errors are in the y -values and that the standard concentrations (x -values) are error-free. In many routine instrumental analyses this assumption may well be justified. The standards can usually be made up with an error of ca. 0.1% or better (see Chapter 1), whereas the instrumental measurements themselves might have a coefficient of variation of 1–3% or worse. So the x -axis error is indeed negligible compared with the y -axis one. In recent years, however, the advent of high-precision automatic methods with coefficients of variation of 0.5% or better has put the assumption under question, and has led some users to make up their standard solutions by weight rather than by the less accurate combination of weight and volume. This approach is intended to ensure that the x -axis errors remain small compared with the y -axis ones. If for any reason this assumption is not valid, then separate, though more complex, approaches to plotting the calibration graph are available. These are discussed further in Section 5.9.

Other common assumptions are that if several measurements are made on a given standard material, the y -values obtained have a normal (Gaussian) error distribution, and that the magnitude of the random errors in the y -values is independent of the

analyte concentration. The first of these two assumptions is normally sound, but the second requires further discussion. If true, it implies that all the points on the graph should have equal *weight* in our calculations, i.e. that it is equally important for the line to pass close to points with high y -values and to those with low y -values. Such calibration graphs are said to be *unweighted*, and are treated in Sections 5.4–5.8 below. However, in practice the y -value errors often increase as the analyte concentration increases. This means that the calibration points should have unequal weights in our calculation, as it is more important for the line to pass close to the points where the errors are least. These *weighted* calculations are now becoming more common despite their additional complexity, and are treated in Section 5.10.

In subsequent sections we shall assume that straight line calibration graphs take the algebraic form:

$$y = a + bx \quad (5.2.1)$$

where b is the slope of the line and a its intercept on the y -axis. The individual points on the line will be referred to as (x_1, y_1) – normally the ‘blank’ reading), (x_2, y_2) , $(x_3, y_3) \dots (x_i, y_i) \dots (x_n, y_n)$, i.e. there are n points altogether. The mean of the x -values is, as usual, called $\bar{x}$, and the mean of the y -values is $\bar{y}$: the position $(\bar{x}, \bar{y})$ is then known as the ‘centroid’ of all the points.

5.3 The product–moment correlation coefficient

In this section we discuss the first problem listed in the previous section: is the calibration plot linear? A common method of estimating how well the experimental points fit a straight line is to calculate the **product–moment correlation coefficient**, r . This statistic is often referred to simply as the ‘correlation coefficient’ because in quantitative sciences it is much more commonly used than the other types of correlation coefficient that we shall meet in Chapter 6. The value of r is given by:

Product–moment correlation coefficient,

$$r = \frac{\sum_i [(x_i - \bar{x})(y_i - \bar{y})]}{\left\{ \left[\sum_i (x_i - \bar{x})^2 \right] \left[\sum_i (y_i - \bar{y})^2 \right] \right\}^{1/2}} \quad (5.3.1)$$

The numerator of Eq. (5.3.1) divided by n , that is $\sum_i [(x_i - \bar{x})(y_i - \bar{y})]/n$, is called the **covariance** of the two variables x and y : it measures their *joint* variation. If x and y are not related their covariance will be close to zero. The correlation coefficient r equals the covariance of x and y divided by the product of their standard deviations, so if x and y are not related r will also be close to zero. Covariances are also discussed in Chapter 8.

It can be shown that r can only take values in the range $-1 \leq r \leq +1$. As indicated in Fig. 5.2, an r value of -1 describes perfect negative correlation, i.e. all the experimental points lie on a straight line of negative slope. Similarly, when $r = +1$ we have

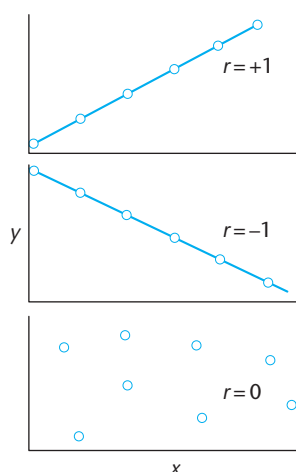


Figure 5.2 The product-moment correlation coefficient, r .

perfect positive correlation, all the points lying exactly on a straight line of positive slope. When there is no linear correlation between x and y the value of r is close to zero. In analytical practice, calibration graphs frequently give numerical r -values greater than 0.99, and r values less than about 0.90 are relatively uncommon. A typical example of a calculation of r illustrates a number of important points.

Example 5.3.1

Standard aqueous solutions of fluorescein are examined in a fluorescence spectrometer, and yield the following fluorescence intensities (in arbitrary units):

Fluorescence intensities: 2.1 5.0 9.0 12.6 17.3 21.0 24.7

Concentration, pg ml^{-1} 0 2 4 6 8 10 12

Determine the correlation coefficient, r .

In practice, such calculations will almost certainly be performed on a calculator or computer, alongside other calculations covered below, but it is important and instructive to examine a manually calculated result. The data are presented in a table, as follows:

	x_i	y_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$	$y_i - \bar{y}$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
	0	2.1	-6	36	-11.0	121.00	66.0
	2	5.0	-4	16	-8.1	65.61	32.4
	4	9.0	-2	4	-4.1	16.81	8.2
	6	12.6	0	0	-0.5	0.25	0
	8	17.3	2	4	4.2	17.64	8.4
	10	21.0	4	16	7.9	62.41	31.6
	12	24.7	6	36	11.6	134.56	69.6
Sums:	42	91.7	0	112	0	418.28	216.2

The figures below the line at the foot of the columns are in each case the sums of the figures in the table: note that $\sum(x_i - \bar{x})$ and $\sum(y_i - \bar{y})$ are both zero. Using these totals in conjunction with Eq. (5.3.1), we have:

$$r = \frac{216.2}{\sqrt{112 \times 418.28}} = \frac{216.2}{216.44} = 0.9989$$

Two observations follow from this example. Figure 5.3 shows that, although several of the points deviate noticeably from the 'best' straight line (which has been calculated as shown in the following section), the r -value is very close indeed to 1. Experience shows that even quite poor-looking calibration plots give very high r -values. In such cases the numerator and denominator in Eq. (5.3.1) are nearly equal. It is thus very important to perform the calculation with an adequate number of significant figures. In the example above, neglecting the figures after the decimal point would have given an obviously incorrect r -value of exactly 1, and the use of only one place of decimals would have given the incorrect r -value of 0.9991. So it is important when using a calculator or computer to determine r to ensure that it provides sufficient significant figures.

Correlation coefficients are simple to calculate, but are easily very misinterpreted. It must always be borne in mind that the use of Eq. (5.3.1) will generate an r -value even if the data do not warrant plotting a straight line graph. Figure 5.4 shows two examples in which a calculation of r would be misleading. In Fig. 5.4(a), the points of the calibration plot clearly lie on a curve; this curve is sufficiently gentle, however,

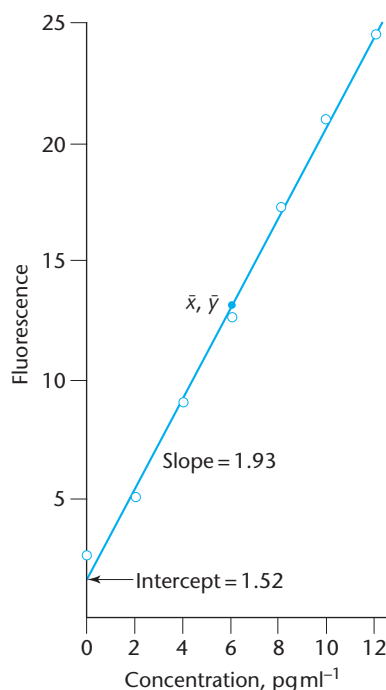


Figure 5.3 Calibration plot for the data in Example 5.3.1.

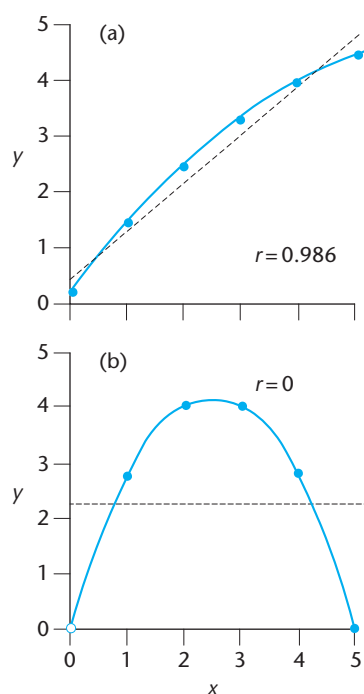


Figure 5.4 Misinterpretation of the correlation coefficient, r . Broken lines are “least Squares” straight lines, calculated as in Section 5.4.

to yield quite a high correlation coefficient when Eq. (5.3.1) is applied. The lesson of this example is that the calibration curve must *always* be plotted (i.e. on graph paper or a computer monitor) and inspected by eye: otherwise a straight-line relationship might wrongly be deduced from the calculation of r . Figure 5.4(b) is a reminder that a zero correlation coefficient does not mean that y and x are entirely unrelated; it only means that they are not *linearly* related.

As we have seen, r -values obtained in instrumental analysis are normally very high, so a calculated value, together with the calibration plot itself, is often sufficient to assure the analyst that a useful linear relationship has been obtained. In some circumstances, however, much lower r -values are obtained: one such situation is further discussed in Section 5.9. In these cases it will be necessary to use a proper statistical test to see whether the correlation coefficient is significant, bearing in mind the number of points used in the calculation. The simplest method of doing this is to calculate a t -value (see Chapter 3 for a fuller discussion of the t -test), using the following equation:

To test for a significant correlation, i.e. $H_0 =$ zero correlation, calculate

$$t = \frac{|r|\sqrt{n-2}}{\sqrt{1-r^2}} \quad (5.3.2)$$

The calculated value of t is compared with the tabulated value at the desired significance level, using a *two-sided* t -test and $(n-2)$ degrees of freedom. The null hypothesis in this case is that there is no correlation between x and y . If the calculated value of

t is greater than the tabulated value, the null hypothesis is rejected and we conclude that a significant correlation does exist. As expected, the closer $|r|$ is to 1, i.e. as the straight line relationship becomes stronger, the larger the values of t that are obtained.

5.4 The line of regression of y on x

In this section we assume that there is a linear relationship between the analytical signal (y) and the concentration (x), and show how to calculate the 'best' straight line through the calibration graph points, each of which is subject to experimental error. Since we are assuming for the present that all the errors are in y (cf. Section 5.2 above), we are seeking the line that minimises the deviations in the y -direction between the experimental points and the calculated line. Since some of these deviations (technically known as the y -residuals – see below) will be positive and some negative, it is sensible to try to minimise the **sum of the squares of the residuals**, since these squares will all be positive. This explains the frequent use of the term **method of least squares** for the procedure. The straight line required is calculated on this principle: as a result it is found that the line must pass through the centroid of the points, $(\bar{x}, \bar{y})$.

It can be shown that the least-squares straight line is given by:

$$\text{Slope of least-squares line, } b = \frac{\sum_i [(x_i - \bar{x})(y_i - \bar{y})]}{\sum_i (x_i - \bar{x})^2} \quad (5.4.1)$$

$$\text{Intercept of least-squares line, } a = \bar{y} - b\bar{x} \quad (5.4.2)$$

Notice that Eq. (5.4.1) contains some of the terms from Eq. (5.3.1), previously used to calculate r : this facilitates calculator or computer operations. The line determined from Eqs (5.4.1) and (5.4.2) is known as **the line of regression of y on x** , i.e. the line indicating how y varies when x is set to chosen values. It is very important to notice that the line of regression of x on y is *not the same line* (except in the highly improbable case where all the points lie exactly on a straight line, i.e. when $r = 1$ exactly). The line of regression of x on y (which also passes through the centroid of the points) assumes that all the errors occur in the x -direction. If we maintain rigidly the convention that the analytical signal is always plotted on the y -axis and the concentration on the x -axis, it is always the line of regression of y on x that we must use in calibration experiments.

Example 5.4.1

Calculate the slope and intercept of the regression line for the data given in the previous example (see Section 5.3).

In Section 5.3 we calculated that, for this calibration curve:

$$\sum_i (x_i - \bar{x})(y_i - \bar{y}) = 216.2; \quad \sum_i (x_i - \bar{x})^2 = 112; \quad \bar{x} = 6; \quad \bar{y} = 13.1$$

Using Eqs (5.4.1) and (5.4.2) we calculate that

$$b = 216.2/112 = 1.93$$

$$a = 13.1 - (1.93 \times 6) = 13.1 - 11.58 = 1.52$$

The equation for the regression line is thus $y = 1.93x + 1.52$.

The results of the slope and intercept calculations are depicted in Fig. 5.3. Again it is important to emphasise that Eqs (5.4.1) and (5.4.2) must not be misused – they will only give useful results when prior study (calculation of r and a visual inspection of the points) has indicated that a straight line relationship is realistic for the experiment in question.

Non-parametric methods (i.e. methods that make no assumptions about the nature of the error distribution) can also be used to calculate regression lines, and this topic is treated in Chapter 6.

5.5

Errors in the slope and intercept of the regression line

The line of regression calculated in the previous section will be used to estimate the concentrations of test samples by interpolation, and perhaps also to estimate the limit of detection of the analytical procedure. The random errors in the values for the slope and intercept are therefore important, and we need further equations to calculate them. We must first calculate the statistic $s_{y/x}$, which estimates the random errors in the y -direction.

$$s_{y/x} = \sqrt{\frac{\sum_i (y_i - \hat{y}_i)^2}{n - 2}} \quad (5.5.1)$$

It will be seen that this equation utilises the y -residuals, $y_i - \hat{y}_i$, where the $\hat{y}_i$ -values are the points on the calculated regression line corresponding to the individual x -values, i.e. the 'fitted' y -values (Fig. 5.5). The $\hat{y}_i$ -value for a given value of x is readily calculated from the regression equation. Equation (5.5.1) is clearly similar in form to the equation for the standard deviation of a set of repeated measurements (Eq. (2.1.2)). In Eq. (5.5.1) the deviations, $(y_i - \bar{y})$, are replaced by residuals, $y_i - \hat{y}_i$, and the denominator contains the term $(n - 2)$ rather than $(n - 1)$. In linear regression calculations the number of degrees of freedom (cf. Section 2.4) is $(n - 2)$. This reflects the obvious consideration that only one straight line can be drawn through two points.

Armed with a value for $s_{y/x}$ we can now calculate s_b and s_a , the standard deviations for the slope (b) and the intercept (a). These are given by:

$$\text{Standard deviation of slope: } s_b = \frac{s_{y/x}}{\sqrt{\sum_i (x_i - \bar{x})^2}} \quad (5.5.2)$$

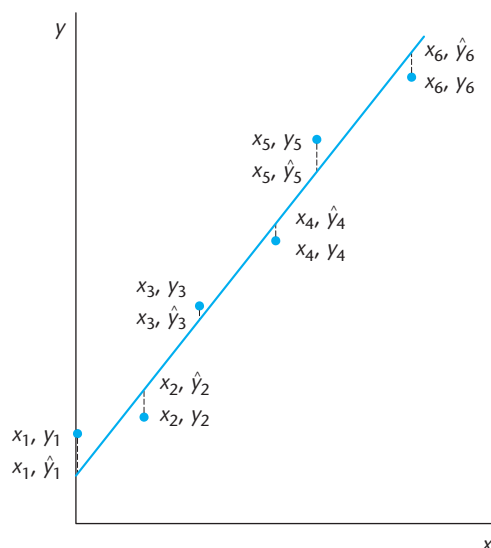


Figure 5.5 The y -residuals of a regression line.

$$\text{Standard deviation of intercept: } s_a = s_{y/x} \sqrt{\frac{\sum_i x_i^2}{n \sum_i (x_i - \bar{x})^2}} \quad (5.5.3)$$

Note that the term $\sum_i (x_i - \bar{x})^2$ appears again in both these equations. The values of s_b and s_a can be used in the usual way (cf. Chapter 2) to estimate confidence limits for the slope and intercept. Thus the confidence limits for the slope of the line are given by $b \pm t_{(n-2)} s_b$, where the t -value is taken at the desired confidence level and $(n - 2)$ degrees of freedom. Similarly the confidence limits for the intercept are given by $a \pm t_{(n-2)} s_a$.

Example 5.5.1

Calculate the standard deviations and confidence limits of the slope and intercept of the regression line calculated in Section 5.4.

This calculation may not be accessible on a simple calculator, but suitable computer software is available. Here we perform the calculation manually, using a tabular layout.

x_i	x_i^2	y_i	$\hat{y}_i$	$ y_i - \hat{y}_i $	$(y_i - \hat{y}_i)^2$
0	0	2.1	1.52	0.58	0.3364
2	4	5.0	5.38	0.38	0.1444
4	16	9.0	9.24	0.24	0.0576
6	36	12.6	13.10	0.50	0.2500
8	64	17.3	16.96	0.34	0.1156
10	100	21.0	20.82	0.18	0.0324
12	144	24.7	24.68	0.02	0.0004
$\sum_i x_i^2 = 364$ $\sum_i (y_i - \hat{y}_i)^2 = 0.9368$					

From the table and using Eq. (5.5.1) we obtain

$$s_{y/x} = \sqrt{0.9368/5} = \sqrt{0.18736} = 0.4329$$

From Section 5.3 we have $\sum_i (x_i - \bar{x})^2 = 112$, and Eq. (5.5.2) can be used to show that:

$$s_b = 0.4329/\sqrt{112} = 0.4329/10.58 = 0.0409$$

The t -value for $(n - 2) = 5$ and the 95% confidence level is 2.57 (Table A.1). The 95% confidence limits for b are thus

$$b = 1.93 \pm (2.57 \times 0.0409) = 1.93 \pm 0.11$$

Equation (5.5.3) requires knowledge of $\sum_i x_i^2$, calculated as 364 from the table. We can thus write:

$$s_a = 0.4329\sqrt{\frac{364}{7 \times 112}} = 0.2950$$

so the 95% confidence limits are

$$a = 1.52 \pm (2.57 \times 0.2950) = 1.52 \pm 0.76$$

In this example, the number of significant figures necessary was not large, but it is always a useful precaution to use the maximum available number of significant figures during such a calculation, rounding only at the end.

There is no necessity in practice for the manual calculation of all these results, which would clearly be too tedious for routine use. The application of a spreadsheet program to some regression data is demonstrated in Section 5.9. Every advantage should also be taken of the extra facilities provided by programs such as Minitab®, for example plots of residuals against x or $\hat{y}$ values, normal probability plots for the residuals, etc. (see also Section 5.15).

Error calculations are also minimised through the use of **single point calibration**, a simple method often used for speed and convenience. The analytical instrument in use is set to give a zero reading with a blank sample (see Section 5.2), and in exactly the same conditions is used to provide k measurements on a single reference material with analyte concentration x . The ISO recommends that k is at least 2, and that x is greater than any concentration to be determined using the calibration line. This line is obtained by joining the single point for the average of the k measurements, $(x, \bar{y})$, to the point $(0, 0)$, so its slope is $b = \bar{y}/x$. In this case the only measure of $s_{y/x}$ is the standard deviation of the k measurements, and the method clearly does not guarantee that the calibration plot is indeed linear over the concentration range 0 to x . It should be used only as a quick check on the stability of a properly established calibration line.

5.6

Calculation of a concentration and its random error

Once the slope and intercept of the regression line have been determined, it is very simple to calculate the concentration (x -value) corresponding to any measured instrument signal (y -value). But it will also be necessary to find the error associated

with this concentration estimate. Calculation of the x -value from the given y -value using Eq. (5.2.1) involves the use of both the slope (b) and the intercept (a) and, as we saw in the previous section, both these values are subject to error. Moreover, the instrument signal derived from any test sample is also subject to random errors. As a result, the determination of the overall error in the corresponding concentration is extremely complex, and most workers use the following approximate formula:

$$s_{x_0} = \frac{s_{y/x}}{b} \sqrt{1 + \frac{1}{n} + \frac{(y_0 - \bar{y})^2}{b^2 \sum_i (x_i - \bar{x})^2}} \quad (5.6.1)$$

In this equation, y_0 is the experimental value of y from which the concentration value x_0 is to be determined, s_{x_0} is the estimated standard deviation of x_0 , and the other symbols have their usual meanings. In some cases an analyst may make several readings to obtain the value of y_0 : if there are m such readings, then the equation for s_{x_0} becomes:

$$s_{x_0} = \frac{s_{y/x}}{b} \sqrt{\frac{1}{m} + \frac{1}{n} + \frac{(y_0 - \bar{y})^2}{b^2 \sum_i (x_i - \bar{x})^2}} \quad (5.6.2)$$

As expected, Eq. (5.6.2) becomes the same as Eq. (5.6.1) if $m = 1$. Confidence limits can be calculated as $x_0 \pm t_{(n-2)} s_{x_0}$, with $(n - 2)$ degrees of freedom. Again, a simple computer program will perform all these calculations, but most calculators will not be adequate.

Example 5.6.1

Using the data from Example 5.3.1, determine x_0 - and s_{x_0} -values and x_0 confidence limits for solutions with fluorescence intensities of 2.9, 13.5 and 23.0 units.

The x_0 -values are easily calculated by using the regression equation determined in Section 5.4, $y = 1.93x + 1.52$. Substituting the y_0 -values 2.9, 13.5 and 23.0, we obtain x_0 -values of 0.72, 6.21 and 11.13 pg ml^{-1} respectively.

To obtain the s_{x_0} -values corresponding to these x_0 -values we use Eq. (5.6.1), recalling from the preceding sections that $n = 7$, $b = 1.93$, $s_{y/x} = 0.4329$, $\bar{y} = 13.1$ and $\sum_i (x_i - \bar{x})^2 = 112$. The y_0 -values 2.9, 13.5 and 23.0 then yield s_{x_0} -values of 0.26, 0.24 and 0.26 respectively. The corresponding 95% confidence limits ($t_5 = 2.57$) are 0.72 ± 0.68 , 6.21 ± 0.62 and 11.13 ± 0.68 pg ml^{-1} respectively.

This example illustrates an important point. Although the confidence limits for the three concentrations are similar (we expect this, as we have used an unweighted regression calculation), the limits are rather smaller (i.e. better) for the

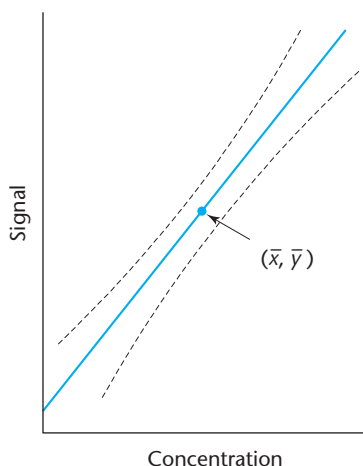


Figure 5.6 General form of the confidence limits for a concentration determined by using an unweighted regression line.

result $y_0 = 13.5$ than for the other two y_0 -values. Inspection of Eq. (5.6.1) confirms that as y_0 approaches $\bar{y}$, the third term inside the bracket approaches zero, and s_{x_0} thus approaches a minimum value. The general form of the confidence limits for a calculated concentration is shown in Fig. 5.6. Thus in a practice a calibration experiment of this type will give the most precise results when the measured instrument signal corresponds to a point close to the centroid of the regression line.

If we wish to improve (i.e. narrow) the confidence limits in this calibration experiment, Eqs (5.6.1) and (5.6.2) show that at least two approaches should be considered. We could increase n , the number of calibration points on the regression line, and/or we could make more than one measurement of y_0 , using the mean value of m such measurements in the calculation of x_0 . The results of these approaches can be assessed by considering the three terms inside the brackets in the two equations. In the example above, the dominant term in all three calculations is the first one – unity. It follows that in this case (and many others) an improvement in precision might be made by measuring y_0 several times and using Eq. (5.6.2) rather than Eq. (5.6.1). If, for example, the y_0 -value of 13.5 had been calculated as the mean of four determinations, then the s_{x_0} -value and the confidence limits would have been 0.14 and 6.21 ± 0.36 respectively, both results showing much improved precision. Of course, making too many replicate measurements (assuming that sufficient sample is available) generates much more work for only a small additional benefit: the reader should verify that eight measurements of y_0 would produce an s_{x_0} -value of 0.12 and confidence limits of 6.21 ± 0.30 .

The effect of n , the number of calibration points, on the confidence limits of the concentration determination is more complex. This is because we also have to take into account accompanying changes in the value of t . Use of a large number of calibration samples involves the task of preparing many accurate standards for only marginally increased precision (cf. the effects of increasing m described in the previous paragraph). On the other hand, small values of n are not permissible. In such cases $1/n$ will be larger and the number of degrees of freedom, $(n - 2)$, will become very small, necessitating the use of very large t -values in the calculation of the confidence

limits. As in the example above, six or so calibration points will be adequate in many experiments, the analyst gaining extra precision if necessary by repeated measurements of y_0 . If considerations of cost, time or availability of standards or samples limit the total number of experiments that can be performed, i.e. if $m + n$ is fixed, then it is worth recalling that the last term in Eq. (5.6.2) is often very small, so it is crucial to minimise $(1/m + 1/n)$. This is achieved by making $m = n$.

An entirely distinct approach to estimating s_{x_0} uses control chart principles (see Chapter 4). We have seen that these charts can be used to monitor the quality of laboratory methods used repeatedly over a period of time, and this chapter has shown that a single calibration line can in principle be used for many individual analyses. It thus seems natural to combine these two ideas, and to use control charts to monitor the performance of a calibration experiment, while at the same time obtaining estimates of s_{x_0} . The procedure recommended by the ISO involves the use of q ($=2$ or 3) standards or reference materials, which need not be (and perhaps ought not to be) from among those used to set up the calibration graph. These standards are measured at regular time intervals and the calibration graph is used to estimate their analyte content in the normal way. The differences, d , between these estimated concentrations and the known concentrations of the standards are plotted on a Shewhart type control chart, the upper and lower control limits of which are given by $0 \pm (ts_{y/x}/b)$. Here $s_{y/x}$ and b have their usual meanings as characteristics of the calibration line, while t has $(n - 2)$ degrees of freedom, or $(nk - 2)$ degrees of freedom if each of the original calibration standards was measured k times to set up the calibration graph. For a confidence level of α (commonly $\alpha = 0.05$), the two-tailed value of t at the $(1 - \alpha/2q)$ level is used. If any point derived from the monitoring standard materials falls outside the control limits, the analytical process is probably out of control, and may need further examination before it can be used again. Moreover, if the values of d for the lowest concentration monitoring standard, measured J times over a period, are called $d_{11}, d_{12}, \dots, d_{1J}$, and the corresponding values for the highest monitoring standards are called $d_{q1}, d_{q2}, \dots, d_{qJ}$, then s_{x_0} is given by:

$$s_{x_0} = \left[\frac{\sum_{j=1}^J (d_{1j}^2 + d_{qj}^2)}{2J} \right]^{1/2} \quad (5.6.3)$$

Strictly speaking this equation estimates s_{x_0} for the concentrations of the highest and lowest monitoring reference materials, so the estimate is a little pessimistic for concentrations between those extremes (see Fig. 5.6). As usual the s_{x_0} -value can be converted to a confidence interval by multiplying by t , which in this case has $2J$ degrees of freedom.

5.7 Limits of detection

As we have seen, one of the principal benefits of using instrumental methods of analysis is that they are capable of detecting and determining trace and ultra-trace quantities of analytes. These benefits have led to the appreciation of the importance

of very low concentrations of many materials, for example in biological and environmental samples, and thus to the development of many further techniques in which ever-lower limits of detection become available. Statistical methods for assessing and comparing limits of detection are therefore important. In general terms, the limit of detection of an analyte may be described as that concentration which gives an instrument signal (y) *significantly different* from the 'blank' or 'background' signal. This description gives the analyst a good deal of freedom to decide the exact definition of the limit of detection, based on a suitable interpretation of the phrase 'significantly different'. There is still no full agreement between researchers, publishers, and professional and statutory bodies on this point. But there is an increasing trend to define the limit of detection as the analyte concentration giving a signal equal to the blank signal, y_B , plus three standard deviations of the blank, s_B :

$$\text{Limit of detection} = y_B + 3s_B \quad (5.7.1)$$

The significance of this last definition is illustrated in more detail in Fig. 5.7. An analyst studying trace concentrations is confronted with two problems: it is important to avoid claiming the presence of the analyte when it is actually absent, but it is equally important to avoid reporting that the analyte is absent when it is in fact present. (The situation is analogous to the occurrence of Type I and Type II errors in significance tests – see Section 3.13.) The possibility of each of these errors must be minimised by a sensible definition of a limit of detection. In the figure, curve A represents the normal distribution of measured values of the blank signal. It would be possible to identify a point, $y = P$, towards the upper edge of this distribution, and claim that a signal greater than this was unlikely to be due to the blank (Fig. 5.7), whereas a signal less than P would be assumed to indicate a blank sample. However, for a sample giving an average signal P , 50% of the observed signals will be less than this, since the signal will have a normal distribution (of the same shape as that for the blank – see below) extending below P (curve B). The probability of concluding that this sample does not differ from the blank *when in fact it does* is therefore 50%. Point P , which has been called the limit of decision, is thus unsatisfactory as a limit of detection, since it solves the first of the problems mentioned above, but not the second. A more suitable point is at $y = Q$ (Fig. 5.7), such that Q is twice as far as P

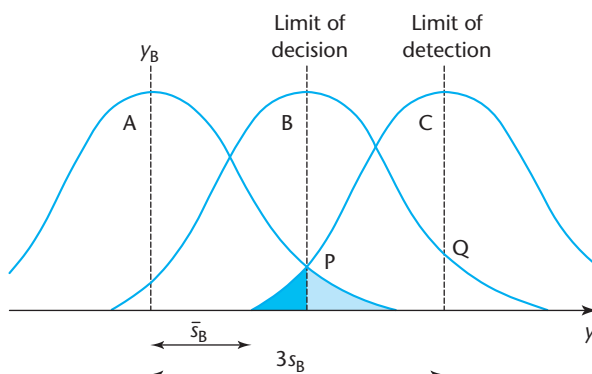


Figure 5.7 Definitions of the limit of decision and the limit of detection.

from y_B . It may be shown that if the distance from y_B to Q in the x -direction is 3.28 times the standard deviation of the blank, s_B , then the probability of each of the two kinds of error occurring (indicated by the shaded areas in Fig. 5.7) is only 5%. If, as suggested by Eq. (5.7.1), the distance from y_B to Q is only $3s_B$, the probability of each error is about 7%: many analysts would consider that this is a reasonable definition of a limit of detection.

It must be re-emphasised that this definition of a limit of detection is quite arbitrary, and it is entirely open to an analyst to provide an alternative definition for a particular purpose. For example, there may be occasions when an analyst is anxious to avoid at all costs the possibility of reporting the absence of the analyte when it is in fact present, but is relatively unworried about the opposite error. Evidently whenever a limit of detection is cited in a paper or report, the definition used to calculate it must also be provided. Some attempts have been made to define a further limit, the **limit of quantitation** (or **limit of determination**), which is regarded as the lower limit for precise quantitative measurements, as opposed to qualitative detection. A value of $y_B + 10s_B$ has been suggested for this limit.

We must now discuss how the terms y_B and s_B are found when a regression line is used for calibration as described in the preceding sections. A fundamental assumption of the unweighted least-squares method is that each point on the plot (including the point representing the blank or background) has a normally distributed variation (in the y -direction only) with a standard deviation estimated by $s_{y/x}$ (Eq. (5.5.1)). This is the justification for drawing the normal distribution curves with the same width in Fig. 5.7. It is therefore appropriate to use $s_{y/x}$ in place of s_B in the estimation of the limit of detection. It is, of course, possible to perform the blank experiment several times and obtain an independent value for s_B , and if our underlying assumptions are correct these two methods of estimating s_B should not differ significantly. But multiple determinations of the blank are time-consuming and the use of $s_{y/x}$ is quite suitable in practice. The value of a , the calculated intercept, can be used as an estimate of y_B , the blank signal itself; it should be a more accurate estimate of y_B than the single measured blank value, y_1 .

Example 5.7.1

Estimate the limit of detection for the fluorescein determination studied in the previous sections.

We use Eq. (5.7.1) with the values of $y_B (=a)$ and $s_B (=s_{y/x})$ previously calculated. The value of y at the limit of detection (L.o.d.) is found to be $1.52 + (3 \times 0.4329)$, i.e. 2.82. Use of the regression equation then yields a detection limit of 0.67 pg ml^{-1} . Figure 5.8 summarises all the calculations performed on the fluorescein determination data.

It is important to avoid confusing the limit of detection of a technique with its sensitivity. This very common source of confusion probably arises because there is no *single* and generally accepted English word synonymous with 'having a low limit of detection'. The word 'sensitive' is generally used for this purpose, giving rise to much ambiguity. The sensitivity of a technique is correctly defined as the *slope* of

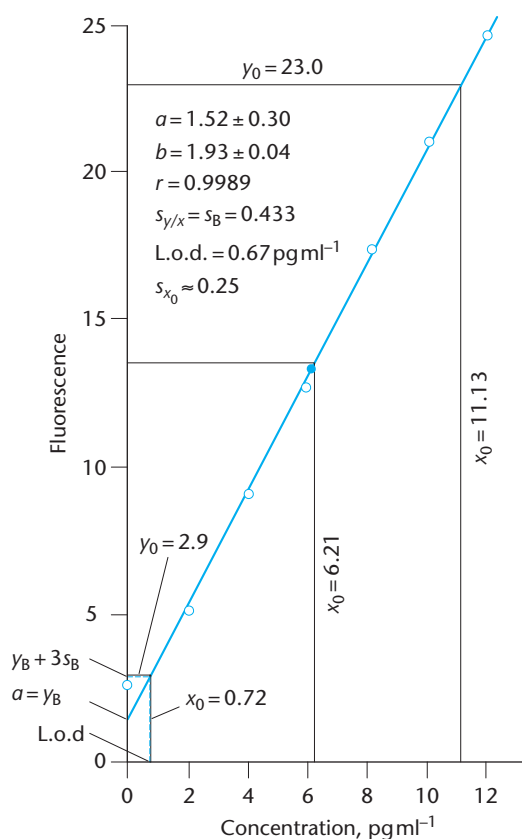


Figure 5.8 Summary of the calculations using the data in Example 5.3.1.

the calibration graph and, provided the plot is linear, can be measured at any point on it. In contrast, the limit of detection of a method is calculated with the aid of the section of the plot close to the origin, and utilises both the slope and the intercept.

5.8 The method of standard additions

Suppose that we wish to determine the concentration of silver in samples of photographic waste by atomic-absorption spectrometry. Using the methods of the previous sections, an analyst could calibrate the spectrometer with some aqueous solutions of a pure silver salt and use the resulting calibration graph in the determination of the silver in the test samples. This method is valid, however, only if a pure aqueous solution of silver and a photographic waste sample containing the same concentration of silver give the same absorbance values. In other words, in using pure solutions to establish the calibration graph it is assumed that there is no reduction (or enhancement) of the silver absorbance signal by other sample components. In many areas of analysis this assumption is often invalid. Matrix effects of this type

occur even with methods such as plasma spectrometry which have a reputation for being relatively free from interferences. Such effects are normally proportional to the analyte signal, and are hence often called *proportional effects*. Since they result in a change of the slope of the calibration graph, they are also called *rotational effects*.

The first possible solution to this problem might be to make up the calibration standards in a matrix that is similar to the test sample but free of the analyte. Thus in the example given above we might try to take a sample of photographic waste that is similar to the test sample, but free from silver, and add known amounts of a silver salt to it to make up the standard solutions. The calibration graph will then be set up using an apparently suitable matrix. In many cases, however, this *matrix matching* approach is impracticable. It will not eliminate matrix effects that differ in magnitude from one sample to another, and it may not be possible even to obtain a sample of the matrix that contains no analyte – for example a silver-free sample of photographic waste is unlikely to occur!

A better solution to the problem is that all the analytical measurements, including the establishment of the calibration graph, must in some way be performed *using the sample itself*. This is achieved in practice by using the **method of standard additions**. The method is widely practised in atomic-absorption and emission spectrometry and has also been applied in electrochemical analysis and many other areas. Equal volumes of the sample solution are taken, each is separately 'spiked' with known and different amounts of the analyte, and *all* are then diluted to the same volume. The instrument signals are then determined for all these solutions and the results plotted as shown in Fig. 5.9. As usual, the signal is plotted on the y -axis; in this case the x -axis is graduated in terms of the amounts of analyte *added* (either as an absolute weight or as a concentration). The (unweighted) regression line is calculated in the normal way, but space is provided for it to be extrapolated to the point on the x -axis at which $y = 0$. This negative intercept on the x -axis corresponds to the amount of the analyte in the test sample. Simple geometry shows that this value is given by a/b , the ratio of the intercept and the slope of the regression line. Since both a and b are subject to error (Section 5.5) the calculated concentration is clearly subject to error as well. However, this concentration is not predicted from a single measured value of y , so the formula for the standard

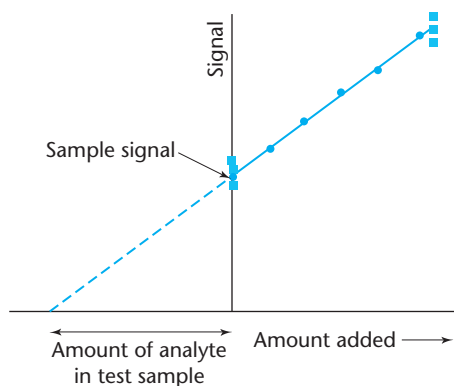


Figure 5.9 The method of standard additions (see text for details).

deviation, s_{x_E} , of the extrapolated x -value (x_E) is not the same as that in Eq. (5.6.1). Instead we use:

$$s_{x_E} = \frac{s_{y/x}}{b} \sqrt{\frac{1}{n} + \frac{\bar{y}^2}{b^2 \sum_i (x_i - \bar{x})^2}} \quad (5.8.1)$$

Confidence limits for x_E can as before be determined as $x_E \pm t_{(n-2)} s_{x_E}$. Increasing the value of n again improves the precision of the estimated concentration: in general at least six points should be used in a standard-additions experiment. Moreover, the precision is improved by maximising $\sum_i (x_i - \bar{x})^2$, so the calibration solutions should, if possible, cover a considerable range. To extend this principle, it can be argued that instead of using six separate calibration standards (the points marked • in Fig. 5.9), it is better to make (say) three measurements on the original sample, i.e. with no added analyte, and three replicate measurements on a spiked sample containing a substantial amount of added analyte (the points marked ■ in Fig. 5.9). The latter approach also significantly reduces the work involved in preparing the calibration graph for each sample (see below). It gives no information on, or confirmation of, the linear response of the system over the range of the graph, but in practice the prior validation of the method (see Section 4.15) will have established this linearity. (If the response of the system is non-linear, the extrapolation involved in the standard additions method is problematic, to say the least.) Provided that linearity has been demonstrated it has been suggested that the single spiked sample should have an analyte level at least five times that of the test sample. In all standard additions experiments the added analyte must obviously be in the same chemical form as the analyte in the test sample: thus, in analysis of Fe(III) ion levels in a material, the added iron must be in the form of Fe(III) ions, not Fe(II) ions or an Fe(III) complex.

Example 5.8.1

The silver concentration in a sample of photographic waste was determined by atomic-absorption spectrometry with the method of standard additions. The following results were obtained.

Added Ag: μg added per ml of <i>original</i> sample solution	0	5	10	15	20	25	30
Absorbance	0.32	0.41	0.52	0.60	0.70	0.77	0.89

Determine the concentration of silver in the sample, and obtain 95% confidence limits for this concentration.

Equations (5.4.1) and (5.4.2) yield $a = 0.3218$ and $b = 0.0186$. The ratio of these figures gives the silver concentration in the test sample as $17.3 \mu\text{g ml}^{-1}$. The confidence limits for this result can be determined with the aid of Eq. (5.8.1). Here $s_{y/x}$ is 0.01094, $\bar{y} = 0.6014$ and $\sum_i (x_i - \bar{x})^2 = 700$. The value of s_{x_E} is thus 0.749 and the confidence limits are $17.3 \pm 2.57 \times 0.749$, i.e. $17.3 \pm 1.9 \mu\text{g ml}^{-1}$.

Although it is an elegant approach to the common problem of matrix interference effects, the method of standard additions has a number of disadvantages. The principal one is that each test sample requires its own calibration graph, in contrast to conventional calibration experiments, where one graph can provide concentration values for many test samples. As well as requiring more effort for this reason, the standard additions method may also use larger quantities of sample than other methods. Moreover the interference in the measurements may be of a different type, i.e. when another component of the matrix affects the analyte signal by a *fixed* amount at all analyte concentrations. These so-called *translational* or *baseline* interferences (which may occur alongside other rotational interferences) do not affect the slope of a calibration graph, but simply shift the whole graph in the y -direction. In such cases the standard additions approach described above provides no correction (though more advanced methods do). Baseline interferences must usually be corrected quite separately, for example by comparing the results with those obtained using an entirely different method.

In statistical terms the standard additions approach is an extrapolation method, and although it is difficult to formulate a model that would allow exact comparisons it seems that it should in principle be less precise than interpolation techniques. In practice, the loss of precision is not very serious.

5.9

Use of regression lines for comparing analytical methods

If an analytical chemist develops a new method for the determination of a particular analyte, the method must be validated by (amongst other techniques) applying it to a series of materials already studied using another reputable or standard procedure. The main aim of such a comparison will be the identification of systematic errors – does the new method give results that are significantly higher or lower than the established procedure? In cases where an analysis is repeated several times over a very limited concentration range, such a comparison can be made using the statistical tests described in Sections 3.2 and 3.3. Such procedures will not be appropriate in instrumental analyses, which are often used over large concentration ranges.

When two methods are to be compared over a range of different analyte concentrations the procedure illustrated in Fig. 5.10 is normally adopted. One axis of a regression graph is used for the results obtained by the new method, and the other axis for the results obtained by applying the reference or comparison method to the same samples. (The question of which axis should be allocated to each method is further discussed below.) Each point on the graph thus represents a single sample analysed by the two separate methods. (Sometimes each method is applied just once to each test sample, while in other cases replicate measurements are used in the comparisons.) The methods of the preceding sections are then applied to calculate the slope (b), the intercept (a) and the product-moment correlation coefficient (r) of the regression line. Clearly if each sample yields an identical result with both analytical methods the regression line will have a zero intercept, a slope of 1 and a correlation coefficient of 1 (Fig. 5.10a). In practice this never occurs: even if systematic errors are entirely absent, random errors ensure that the two analytical procedures will not give results in exact agreement for all the samples.

Deviations from the 'ideal' situation ($a = 0$, $b = r = 1$) can occur in a number of different ways. First, it is possible that the regression line will have a slope of 1, but a

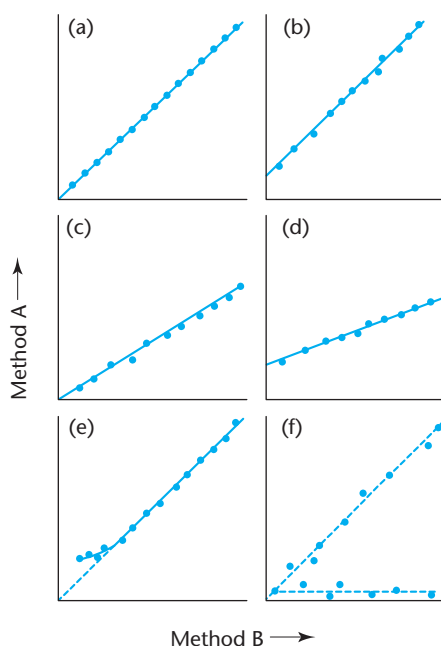


Figure 5.10 Use of a regression line to compare two analytical methods: (a) shows perfect agreement between the two methods for all the samples; (b)–(f) illustrate the results of various types of systematic error (see text).

non-zero intercept. That is, one method of analysis may yield a result higher or lower than the other by a fixed amount. Such an error might occur if the background signal for one of the methods was wrongly calculated (Fig. 5.10b). A second possibility is that the slope of the regression line is >1 or <1 , indicating that a systematic error may be occurring in the slope of one of the individual calibration plots (Fig. 5.10c). These two errors may occur simultaneously (Fig. 5.10d). Further possible types of systematic error are revealed if the plot is curved (Fig. 5.10e). Speciation problems may give surprising results (Fig. 5.10f). This last type of plot might arise if an analyte occurred in two chemically distinct forms, the proportions of which varied from sample to sample. One of the methods under study (here plotted on the y -axis) might detect only one form of the analyte, while the second method detected both forms.

In practice, the analyst most commonly wishes to test for an intercept differing significantly from zero, and a slope differing significantly from 1. Such tests are performed by determining the confidence limits for a and b , generally at the 95% significance level. The calculation is very similar to that described in Section 5.5, and is most simply performed by using a program such as Excel[®]. This spreadsheet is applied to the following example.

Example 5.9.1

The level of phytic acid in 20 urine samples was determined by a new catalytic fluorimetric (CF) method, and the results were compared with those

obtained using an established extraction photometric (EP) technique. The following data were obtained (all the results, in mg l^{-1} , are means of triplicate measurements).

(March, J.G., Simonet, B.M. and Grases, F., 1999, *Analyst*, **124**: 897)

Sample number	CF result	EP result
1	1.87	1.98
2	2.20	2.31
3	3.15	3.29
4	3.42	3.56
5	1.10	1.23
6	1.41	1.57
7	1.84	2.05
8	0.68	0.66
9	0.27	0.31
10	2.80	2.82
11	0.14	0.13
12	3.20	3.15
13	2.70	2.72
14	2.43	2.31
15	1.78	1.92
16	1.53	1.56
17	0.84	0.94
18	2.21	2.27
19	3.10	3.17
20	2.34	2.36

This set of data shows why it is inappropriate to use the paired t -test, which evaluates the differences between the pairs of results, in such cases (Section 3.3). The range of phytic acid concentrations (ca. $0.14\text{--}3.50 \text{ mg l}^{-1}$) in the urine samples is so large that a fixed discrepancy between the two methods will be of varying significance at different concentrations. Thus a difference between the two techniques of 0.05 mg l^{-1} would not be of great concern at a level of ca. 3.50 mg l^{-1} , but would be more disturbing at the lower end of the concentration range.

Table 5.1 shows the summary output of the Excel[®] spreadsheet used to calculate the regression line for the above data. The CF data have been plotted on the y -axis, and the EP results on the x -axis (see below). The output shows that the r -value (called 'Multiple R' by this program because of its potential application to multiple regression methods) is 0.9966. The intercept is -0.0497 , with upper and lower confidence limits of -0.1399 and $+0.0404$: this range includes the ideal value of zero. The slope of the graph, called 'X variable 1' because b is the coefficient of the x -term in Eq. (5.2.1), is 0.9924, with a 95% confidence interval of $0.9521\text{--}1.0328$: again this range includes the model value, in this case 1.0. (The remaining output data are not needed in this example, and are discussed further in Section 5.11.) Figure 5.11 shows the regression line with the characteristics summarised above.

Table 5.1 Excel output for Example 5.9.1

Regression statistics					
Multiple R	0.9966				
R square	0.9933				
Adjusted R square	0.9929				
Standard error	0.0829				
Observations	20				
ANOVA					
	df	SS	MS	F	Significance F
Regression	1	18.341	18.341	2670.439	5.02965E-21
Residual	18	0.124	0.007		
Total	19	18.465			
	Coefficients		Standard error	t stat	P-value
Intercept	-0.0497		0.0429	-1.158	0.262
X variable 1	0.9924		0.0192	51.676	5.03E-21
	Lower 95%		Upper 95%		
Intercept	-0.1399		0.0404		
X variable 1	0.9521		1.0328		

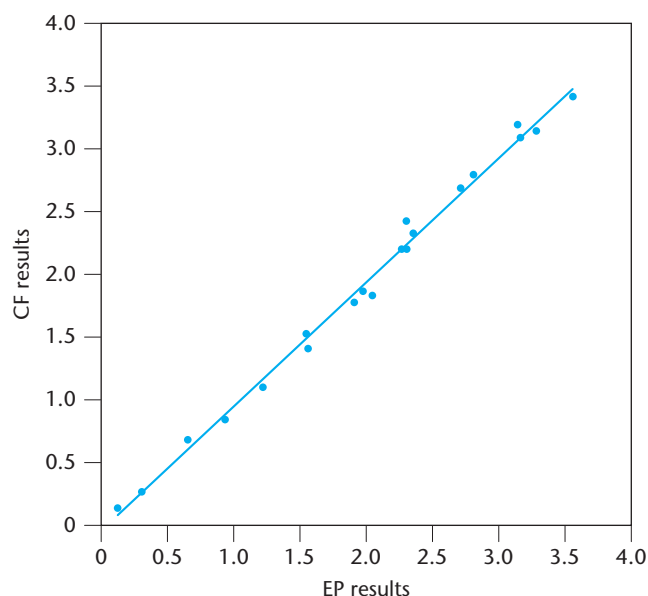


Figure 5.11 Comparison of two analytical methods: data from Example 5.9.1.

Two further points are important in connection with this example. First, the literature of analytical chemistry shows that authors frequently place great stress on the value of the correlation coefficient in such comparative studies. In the above example, however, this coefficient played no direct role in establishing whether or not

systematic errors had occurred. Even if the regression line had been slightly curved, the correlation coefficient might still have been close to 1 (cf. Section 5.3 above). This means that the calculation of r is less important in the present context than the establishment of confidence limits for the slope and the intercept. In some cases it may be found that the r -value is not very close to 1, even though the slope and the intercept are not significantly different from 1 and 0 respectively. Such a result would suggest very poor precision for either one or both of the methods under study. The precisions of the two methods can be determined and compared using the methods of Chapters 2 and 3. In practice it is desirable that this should be done *before* the regression line comparing the methods is plotted – the reason for this is explained below. The second point to note is that it is desirable to compare the methods over the full range of concentrations, as in the example given where the urine samples examined contained phytic acid concentrations that covered the range of interest fairly uniformly.

Although very widely adopted in comparative studies of instrumental methods, the approach described here is open to some theoretical objections. First, as has been emphasised throughout this chapter, the line of regression of y on x is calculated on the assumption that the errors in the x -values are negligible – all errors are assumed to occur in the y -direction. While generally valid in a calibration plot for a single analyte, this assumption is evidently not justified when the regression line is used for comparison purposes: it is certain that random errors will occur in both analytical methods, i.e. in both the x - and y -directions. Thus the equations used to calculate the regression line are not really valid for this application. However the regression method is still widely used, as the graphs obtained provide valuable information on the nature of any differences between the methods (Fig. 5.10). Simulations show, moreover, that the approach does give surprisingly acceptable results, provided that the more precise method is plotted on the x -axis (this is why we investigate the precisions of the two methods – see above), and that a reasonable number of points (ca. ten at least) uniformly covering the concentration range of interest is used. Since the confidence limit calculations are based on $(n - 2)$ degrees of freedom, it is particularly important to avoid small values of n .

A second objection to using the line of regression of y on x , as calculated in Sections 5.4 and 5.5, in the comparison of two analytical methods is that it also assumes that the error in the y -values is *constant*. Such data are said to be **homoscedastic**. As previously noted, this means that all the points have equal weight when the slope and intercept of the line are calculated. This assumption is obviously likely to be invalid in practice. In many analyses, the data are **heteroscedastic**, i.e. the standard deviation of the y -values increases with the concentration of the analyte, rather than having the same value at all concentrations (see below). This objection to the use of unweighted regression lines also applies to calibration plots for a single analytical procedure. In principle **weighted regression** lines should be used instead, as shown in the next section.

Both these problems are overcome by the use of a technique known as **functional relationship estimation by maximum likelihood (FREML)**. This calculation can be applied both to the comparison of analytical methods as discussed in this section, and to conventional calibration graphs in instrumental analyses if the assumption that the x -direction errors are negligible compared with the y -direction ones is not justifiable. The latter situation can arise in at least two ways. If solid reference materials are used for the calibration standards, they are quite likely to be heterogeneous,

so x -direction errors may be significant. In other cases, the opposite situation occurs, i.e. the y -direction errors are so small that they become comparable with the x -direction ones. Such data may occur in the use of highly automated flow analysis methods, such as flow injection analysis or high-performance liquid chromatography. The FREML method provides for both y - and x -direction errors. It assumes that the errors in the two directions are normally distributed, but that they may have unequal variances, and it minimises the sums of the squares of *both* the x - and y -residuals, divided by the corresponding variances. This result cannot be obtained by the simple calculation used when only y -direction errors are considered, but requires an iterative method, which can be implemented using, for example, a macro for Minitab® (see Bibliography). The method is reversible (i.e. in a method comparison it does not matter which method is plotted on the x -axis and which on the y -axis) and can also be used in weighted regression calculations (see Section 5.10).

5.10 Weighted regression lines

In this section the application of weighted regression methods is outlined. It is assumed that the weighted regression line is to be used for the determination of a single analyte rather than for the comparison of two separate methods. In any calibration analysis the overall random error of the result will arise from a combination of the error contributions from the several stages of the analysis (cf. Section 2.11). In some cases this overall error will be dominated by one or more steps in the analysis where the random error is not concentration-dependent. Such errors will also occur in the 'blank' calibration solution and can be regarded as baseline errors (as such they are related to limits of detection; see Section 5.7). In such cases we shall expect the y -direction errors in the calibration curve to be approximately equal for all the points (homoscedasticity), and an unweighted regression calculation is legitimate. In many calibration experiments there will also be errors that are approximately proportional to the analyte concentration: if these are predominant the *relative error* will be roughly constant. In the most common situation, both types of error will occur, with the result that the y -direction error will increase as x increases, but less rapidly than the concentration. Both these types of heteroscedastic data should be treated by weighted regression methods. Usually an analyst can only learn from experience whether weighted or unweighted methods are appropriate. Predictions are difficult: examples abound where two apparently similar methods show very different error behaviour. Weighted regression calculations are rather more complex than unweighted ones, and they require more information (or the use of more assumptions). Nonetheless they should be used whenever heteroscedasticity is suspected, and they are now more widely applied than formerly, partly as a result of pressure from regulatory authorities in the pharmaceutical industry and elsewhere.

Figure 5.12 shows – perhaps with some exaggeration! – the situation that arises when the y -direction error in a regression calculation gets larger as the concentration increases. The regression line must be calculated to give additional weight to those points where the error bars are smallest, i.e. it is more important for the calculated line to pass close to such points than to pass close to the points representing higher concentrations with the largest errors. This result is achieved by giving each point a *weighting* inversely

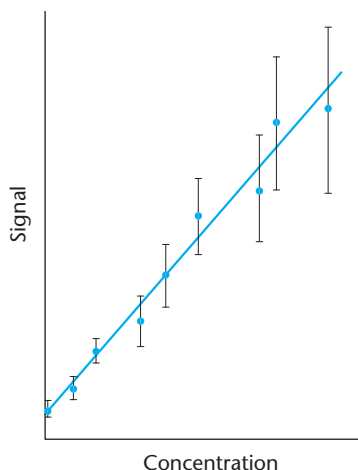


Figure 5.12 The weighting of errors in a regression calculation.

proportional to the corresponding y -direction variance, s_i^2 . If the individual points are denoted by (x_1, y_1) , (x_2, y_2) , etc. as usual, and the corresponding standard deviations are s_1 , s_2 , etc., then the individual weights, w_1 , w_2 , etc., are given by:

$$\text{Weights: } w_i = \frac{s_i^{-2}}{\sum_i s_i^{-2}/n} \quad (5.10.1)$$

By using the n divisor in the denominator of the equation the weights have been scaled so that their sum is equal to the number of points on the graph: this simplifies the subsequent calculations. The slope and the intercept of the regression line are then given by:

$$\text{Weighted slope: } b_w = \frac{\sum_i w_i x_i y_i - n \bar{x}_w \bar{y}_w}{\sum_i w_i x_i^2 - n \bar{x}_w^2} \quad (5.10.2)$$

and

$$\text{Weighted intercept: } a_w = \bar{y}_w - b \bar{x}_w \quad (5.10.3)$$

In these equations $\bar{y}_w$ and $\bar{x}_w$ represent the co-ordinates of the *weighted centroid*, through which the weighted regression line must pass. These co-ordinates are given as expected by $\bar{x}_w = \sum_i w_i x_i / n$ and $\bar{y}_w = \sum_i w_i y_i / n$.

Example 5.10.1

Calculate the unweighted and weighted regression lines for the following calibration data. For each line calculate also the concentrations of test samples with absorbances of 0.100 and 0.600.

Concentration, $\mu\text{g ml}^{-1}$	0	2	4	6	8	10
Absorbance	0.009	0.158	0.301	0.472	0.577	0.739
Standard deviation	0.001	0.004	0.010	0.013	0.017	0.022

Application of Eqs (5.4.1) and (5.4.2) shows that the slope and intercept of the *unweighted* regression line are respectively 0.0725 and 0.0133. The concentrations corresponding to absorbances of 0.100 and 0.600 are then found to be 1.20 and $8.09 \mu\text{g ml}^{-1}$ respectively.

The *weighted* regression line is a little harder to calculate: in the absence of a suitable computer program it is usual to set up a table as follows.

x_i	y_i	s_i	$1/s_i^2$	w_i	wix_i	wiy_i	$wixiy_i$	wix_i^2
0	0.009	0.001	10^6	5.535	0	0.0498	0	0
2	0.158	0.004	62 500	0.346	0.692	0.0547	0.1093	1.384
4	0.301	0.010	10 000	0.055	0.220	0.0166	0.0662	0.880
6	0.472	0.013	5 917	0.033	0.198	0.0156	0.0935	1.188
8	0.577	0.017	3 460	0.019	0.152	0.0110	0.0877	1.216
10	0.739	0.022	2 066	0.011	0.110	0.0081	0.0813	1.100
Sums			1 083 943	5.999	1.372	0.1558	0.4380	5.768

These figures give $\bar{y}_w = 0.1558/6 = 0.0260$, and $\bar{x}_w = 1.372/6 = 0.229$. By Eq. (5.10.2), b_w is calculated from

$$b_w = \frac{0.438 - (6 \times 0.229 \times 0.026)}{5.768 - [6 \times (0.229)^2]} = 0.0738$$

so a_w is given by $0.0260 - (0.0738 \times 0.229) = 0.0091$.

These values for a_w and b_w can be used to show that absorbance values of 0.100 and 0.600 correspond to concentrations of 1.23 and $8.01 \mu\text{g ml}^{-1}$ respectively.

Comparison of the results of the unweighted and weighted regression calculations is very instructive. The effects of the weighting process are clear. The weighted centroid ($\bar{x}_w, \bar{y}_w$) is much closer to the origin of the graph than the unweighted centroid ($\bar{x}, \bar{y}$) and the weighting given to the points nearer the origin (particularly to the first point (0, 0.009) which has the smallest error) ensures that the weighted regression line has an intercept very close to this point. The slope and intercept of the weighted line are remarkably similar to those of the unweighted line, however, with the result that the two methods give very similar values for the concentrations of samples having absorbances of 0.100 and 0.600. This is not simply because in this example the experimental points fit a straight line very well. In practice the weighted and unweighted regression lines derived from a set of calibration data have similar slopes and intercepts even if the scatter of the points about the line is substantial.

It thus seems on the face of it that weighted regression calculations have little value. They require more information (in the form of estimates of the standard deviation at various points on the graph), and are significantly more complex to execute, but they seem to provide results very similar to those obtained from the simpler unweighted regression method. This feeling may indeed account for some of the neglect of weighted regression calculations in practice. But we do not employ regression calculations simply to determine the slope and intercept of the calibration plot and the concentrations of test samples. There is also a need to obtain estimates of the errors or confidence limits of those concentrations, and it is here that the weighted regression method provides much more realistic results. In Section 5.6 we used Eq. (5.6.1) to estimate the standard deviation (s_{x_0}) and hence the confidence limits of a concentration calculated using a single y -value and an unweighted regression line. If we apply this equation to the data in the example above we find that the unweighted confidence limits for the solutions having absorbances of 0.100 and 0.600 are 1.20 ± 0.65 and $8.09 \pm 0.63 \mu\text{g ml}^{-1}$ respectively. As in Example 5.6.1, these confidence intervals are very similar. In the present example, however, such a result is entirely unrealistic. The experimental data show that the errors of the observed y -values increase as y itself increases, as expected for a method with a roughly constant relative standard deviation. We would expect that this increase in s_i with increasing y would also be reflected in the confidence limits of the determined concentrations: the confidence limits for the solution with an absorbance of 0.600 should be much greater (i.e. worse) than those for the solution with an absorbance of 0.100.

In weighted regression calculations, the standard deviation of a predicted concentration is given by

$$s_{x_{0w}} = \frac{s_{(y/x)w}}{b} \left\{ \frac{1}{w_0} + \frac{1}{n} + \frac{(y_0 - \bar{y}_w)^2}{b^2 \left(\sum_i w_i x_i^2 - n \bar{x}_w^2 \right)} \right\}^{1/2} \quad (5.10.4)$$

In this equation, $s_{(y/x)w}$ is given by:

$$s_{(y/x)w} = \left\{ \frac{\sum_i w_i (y_i - \hat{y}_i)^2}{n - 2} \right\}^{1/2} \quad (5.10.5)$$

and w_0 is a weighting appropriate to the value of y_0 . Equations (5.10.4) and (5.10.5) are clearly similar in form to Eqs (5.6.1) and (5.5.1). Equation (5.10.4) confirms that points close to the origin, where the weights are highest, and points near the centroid, where $(y_0 - \bar{y}_w)$ is small, will have the narrowest confidence limits (Fig. 5.13). The major difference between Eqs (5.6.1) and (5.10.4) is the term $1/w_0$ in the latter. Since w_0 falls significantly as y increases, this term ensures that the confidence limits increase with increasing y_0 , as we expect.

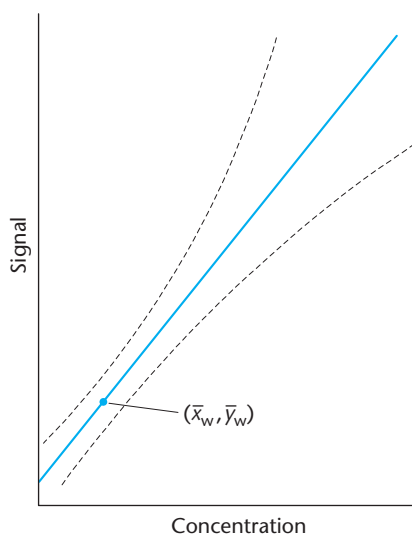


Figure 5.13 General form of the confidence limits for a concentration determined using a weighted regression line.

Applying Eq. (5.10.4) to the above example above we find that the test samples with absorbance of 0.100 and 0.600 have confidence limits for the calculated concentrations of 1.23 ± 0.12 and $8.01 \pm 0.72 \mu\text{g ml}^{-1}$ respectively: these confidence intervals are, as expected, proportional in size to the observed absorbances of the two solutions. The confidence interval for the less concentrated of the two samples is much smaller than in the unweighted regression calculation. By contrast the confidence limits for the higher of the two concentrations are quite similar in the unweighted and weighted calculations. This emphasises the particular importance of using weighted regression when the results of interest include those at low concentrations. Similarly detection limits may be more realistically assessed using the intercept and standard deviation obtained from a weighted regression graph. All these results accord much more closely with the reality of such a calibration experiment than do the results of the unweighted regression calculation.

Weighted regression calculations can also be applied in standard additions experiments. The equation for the standard deviation of a concentration obtained from a weighted standard additions calibration graph is:

$$s_{x_{\text{ew}}} = \frac{s_{(y/x)_w}}{b} \left[\frac{1}{\sum_i w_i} + \frac{\bar{y}_w^2}{b^2 \left(\sum_i w_i x_i^2 - \sum_i n \bar{x}_w^2 \right)} \right]^{1/2} \quad (5.10.6)$$

In addition, weighted regression methods may be essential when a straight line graph is obtained by algebraic transformations of an intrinsically curved plot (see below, Section 5.13). Computer programs for weighted regression calculations are available, mainly through the more advanced statistical software products, and this should encourage the more widespread use of this method.

5.11 Intersection of two straight lines

A number of problems in analytical science are solved by plotting two straight line graphs from the experimental data and determining the point of their intersection. Common examples include potentiometric and conductimetric titrations, the determination of the composition of metal–chelate complexes, and studies of ligand–protein and similar bio-specific binding interactions. If the equations of the two (unweighted) straight lines, $y_1 = a_1 + b_1x_1$ and $y_2 = a_2 + b_2x_2$ (with n_1 and n_2 points respectively) are known, then the x -value of their intersection, x_I , is easily shown to be given by:

$$\text{Intersection point: } x_I = \frac{\Delta a}{\Delta b} \quad (5.11.1)$$

where $\Delta a = a_1 - a_2$ and $\Delta b = b_1 - b_2$. Confidence limits for this x_I value are given by the two roots of the following quadratic equation:

$$x_I^2(\Delta b^2 - t^2 s_{\Delta b}^2) - 2x_I(\Delta a \Delta b - t^2 s_{\Delta a \Delta b}) + (\Delta a^2 - t^2 s_{\Delta a}^2) = 0 \quad (5.11.2)$$

The value of t used in this equation is chosen at the appropriate P -level and at $n_1 + n_2 - 4$ degrees of freedom. The standard deviations in Eq. (5.11.2) are calculated on the assumption that the $s_{y/x}$ values for the two lines, $s_{(y/x)1}$ and $s_{(y/x)2}$ are sufficiently similar to be pooled using an equation analogous to Eq. (3.3.1):

$$s_{(y/x)p}^2 = \frac{(n_1 - 2)s_{(y/x)1}^2 + (n_2 - 2)s_{(y/x)2}^2}{n_1 + n_2 - 4} \quad (5.11.3)$$

After this pooling process we can write:

$$s_{\Delta b}^2 = s_{(y/x)p}^2 \left[\frac{1}{\sum_i (x_{i1} - \bar{x}_1)^2} + \frac{1}{\sum_i (x_{i2} - \bar{x}_2)^2} \right] \quad (5.11.4)$$

$$s_{\Delta a}^2 = s_{(y/x)p}^2 \left[\frac{1}{n_1} + \frac{1}{n_2} + \frac{\bar{x}_1^2}{\sum_i (x_{i1} - \bar{x}_1)^2} + \frac{\bar{x}_2^2}{\sum_i (x_{i2} - \bar{x}_2)^2} \right] \quad (5.11.5)$$

$$s_{\Delta a \Delta b} = s_{(y/x)p}^2 \left[\frac{\bar{x}_1}{\sum_i (x_{i1} - \bar{x}_1)^2} + \frac{\bar{x}_2}{\sum_i (x_{i2} - \bar{x}_2)^2} \right] \quad (5.11.6)$$

These equations seem formidable, but if a spreadsheet such as Excel[®] is used to obtain the equations of the two lines, the point of intersection can be determined at once. The $s_{y/x}$ values can then be pooled, $s_{\Delta a}^2$, etc. calculated, and the confidence

limits found using the program's equation-solving capabilities. Some dedicated statistics software packages also provide facilities for studying intersecting straight lines and their associated errors and confidence limits.

5.12 ANOVA and regression calculations

When the least-squares criterion is used to determine the best straight line through a single set of data points there is one unique solution, so the calculations involved are relatively straightforward. However, when a curved calibration plot is calculated using the same criterion this is no longer the case: a least-squares curve might be described by polynomial functions such as ($y = a + bx + cx^2 + \dots$) containing different numbers of terms, a logarithmic or exponential function, or in other ways. So we need a method which helps us to choose the best way of plotting a curve from amongst the many that are available. Analysis of variance (ANOVA) provides such a method in all cases where we maintain the assumption that the errors occur only in the y -direction. In such situations there are two sources of y -direction variation in a calibration plot. The first is the *variation due to regression*, i.e. due to the relationship between the instrument signal, y , and the analyte concentration, x . The second is the random experimental error in the y -values, which is called the *variation about regression*. As we have seen in Chapter 3, ANOVA is a powerful method for separating two sources of variation in such situations. In regression problems, the average of the y -values of the calibration points, $\bar{y}$, is important in defining these sources of variation. Individual values of y_i differ from $\bar{y}$ for the two reasons given above. ANOVA is applied to separating the two sources of variation by using the relationship that the *total sum of squares (SS) about $\bar{y}$ is equal to the SS due to regression plus the SS about regression*:

$$\text{Additive sum of squares: } \sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i (y_i - \hat{y}_i)^2 \quad (5.12.1)$$

The total sum of squares, i.e. the left-hand side of Eq. (5.12.1), is clearly fixed once the experimental y_i values have been determined. A line fitting these experimental points closely will be obtained when the *variation due to regression* (the first term on the right-hand side of Eq. (5.12.1)) is as large as possible. The *variation about regression* (also called the *residual SS* as each component of the right-hand term in the equation is a single *residual*) should be as small as possible. The method is quite general and can be applied to straight line regression problems as well as to curvilinear regression. Table 5.1 (see p. 133) shows the Excel[®] output for a linear plot used to compare two analytical methods, including an ANOVA table set out in the usual way. The total number of degrees of freedom (19 in that example) is, as usual, one less than the number of measurements (20), as the y -residuals always add up to zero. For a straight line graph we have to determine only one coefficient (b) for a term that also contains x , so the number of degrees of freedom due to regression is 1. Thus there are $(n - 2) = 18$ degrees of freedom for the residual variation. The mean square (MS) values are determined as in previous ANOVA examples, and the F -test is applied to the two mean squares as usual. The F -value obtained is very large, as there is an obvious relationship between x and y , so the regression MS is much larger than the residual MS.

The Excel[®] output also includes ‘multiple R’, which as previously noted is in this case equal to the correlation coefficient, r , the standard error ($= s_{y/x}$), and the further terms ‘R square’ (R^2) and ‘adjusted R square,’ usually abbreviated R'^2 . The two latter statistics are given by Excel[®] as decimals, but are often given as percentages instead. They are defined as follows:

$$R^2 = \text{SS due to regression} / \text{total SS} = 1 - (\text{residual SS} / \text{total SS}) \quad (5.12.2)$$

$$R'^2 = 1 - (\text{residual MS} / \text{total MS}) \quad (5.12.3)$$

In the case of a straight line graph, R^2 is equal to r^2 , the square of the correlation coefficient, i.e. the square of ‘multiple R’. The applications of R^2 and R'^2 to problems of curve-fitting will be discussed below.

5.13

Introduction to curvilinear regression methods

In many instrumental analysis methods the instrument response is proportional to the analyte concentration over substantial concentration ranges. The simplified calculations that result encourage analysts to take significant experimental precautions to achieve such linearity. Examples of such precautions include the control of the emission line width of a hollow-cathode lamp in atomic absorption spectrometry, and the size and positioning of the sample cell to minimise inner filter artefacts in molecular fluorescence spectrometry. However, many analytical methods (e.g. immunoassays and similar competitive binding assays) produce calibration plots that are intrinsically curved. Particularly common is the situation where the calibration plot is linear (or approximately so) at low analyte concentrations, but becomes curved at higher analyte levels. When curved calibration plots are obtained we still need answers to the questions listed in Section 5.2, but those questions will pose rather more formidable statistical problems than occur in linear calibration experiments.

The first question to be examined is, how do we detect curvature in a calibration plot? That is, how do we distinguish between a plot that is best fitted by a straight line, and one that is best fitted by a gentle curve? Since the degree of curvature may be small, and/or occur over only part of the plot, this is not a straightforward question. Moreover, despite its widespread use for testing the goodness of fit of linear graphs, the product-moment correlation coefficient (r) is of little value in testing for curvature: we have seen (Section 5.3) that lines with obvious curvature may still give very high r values. An analyst would naturally hope that any test for curvature could be applied fairly easily in routine work without extensive calculations. Several such tests are available, based on the use of the y -residuals on the calibration plot.

We have seen (Section 5.5) that a y -residual, $y_i - \hat{y}_i$, represents the difference between an experimental value of y and the $\hat{y}$ -value calculated from the regression equation at the same value of x . If a linear calibration plot is appropriate, and if the random errors in the y -values are normally distributed, the residuals themselves should be normally distributed about the value of zero. If this turns out not to be true in practice, then we must suspect that the fitted regression line is not of the correct type. In the worked example given in Section 5.5 the y -residuals were shown to be +0.58, -0.38, -0.24, -0.50, +0.34, +0.18 and +0.02. These values sum to zero

(allowing for possible rounding errors, this must always be true), and are approximately symmetrically distributed about 0. Although it is impossible to be certain, especially with such small numbers of data points, that these residuals are normally distributed, there is certainly no contrary evidence in this case, i.e. no evidence to support a non-linear calibration plot. As previously noted, Minitab®, Excel® and other statistics packages provide extensive information, including graphical displays, on the sizes and distribution of residuals.

A second test uses the *signs* of the residuals given above. As we move along the calibration plot, i.e. as x increases, positive and negative residuals will be expected to occur in random order if the data are well fitted by a straight line. If, in contrast, we attempt to fit a straight line to a series of points that actually lie on a smooth curve, then the signs of the residuals will no longer have a random order, but will occur in *sequences* of positive and negative values. Examining again the residuals given above, we find that the order of signs is $+- -- ++$. To test whether these sequences of $+$ and $-$ residuals indicate the need for a non-linear regression line, we need to know the probability that such an order could occur by chance. Such calculations are described in the next chapter. Unfortunately the small number of data points makes it quite likely that these and other sequences could indeed occur by chance, so any conclusions drawn must be treated with caution. The choice between straight line and curvilinear regression methods is therefore probably best made by using the curve-fitting techniques outlined in the next section.

In the common situation where a calibration plot is linear at lower concentrations and curved at higher ones, it is important to be able to establish the range over which linearity can be assumed. Approaches to this problem are outlined in the following example.

Example 5.13.1

Investigate the linear calibration range of the following fluorescence experiment.

Fluorescence intensity	0.1	8.0	15.7	24.2	31.5	33.0
Concentration, $\mu\text{g ml}^{-1}$	0	2	4	6	8	10

Inspection of the data shows that the part of the graph near the origin corresponds rather closely to a straight line with a near-zero intercept and a slope of about 4. The fluorescence of the $10 \mu\text{g ml}^{-1}$ standard solution is clearly lower than would be expected on this basis, and there is some possibility that the departure from linearity has also affected the fluorescence of the $8 \mu\text{g ml}^{-1}$ standard. We first apply (unweighted) linear regression calculations to all the data. Application of the methods of Sections 5.3 and 5.4 gives the results $a = 1.357$, $b = 3.479$ and $r = 0.9878$. Again we recall that the high value for r may be deceptive, though it may be used in a comparative sense (see below). The y -residuals are found to be -1.257 , -0.314 , $+0.429$, $+1.971$, $+2.314$ and -3.143 , with the sum of squares of the residuals equal to 20.981. The trend in the values of the residuals suggests that the last value in the table is probably outside the linear range.

We confirm this suspicion by applying the linear regression equations to the first five points only. This gives $a = 0.100$, $b = 3.950$ and $r = 0.9998$. These slope

and intercept values are much closer to those expected for the part of the graph closest to the origin, and the r -value is higher than in the first calculation. The residuals of the first five points from this second regression equation are 0, 0, -0.2 , $+0.4$ and -0.2 , with a sum of squares of only 0.24. Use of the second regression equation shows that the fluorescence expected from a $10 \mu\text{g ml}^{-1}$ standard is 39.6, i.e. the residual is -6.6 . Use of a t -test (Chapter 3) would show that this last residual is significantly greater than the average of the other residuals: alternatively a test could be applied (Section 3.7) to demonstrate that it is an 'outlier' amongst the residuals (see also Section 5.15 below). In this example, such calculations are hardly necessary: the enormous residual for the last point, coupled with the very low residuals for the other five points and the greatly reduced sum of squares, confirms that the linear range of the method does not extend as far as $10 \mu\text{g ml}^{-1}$. Having established that the last data point can be excluded from the linear range, we can repeat the process to study the point (8, 31.5). We do this by calculating the regression line for only the first four points in the table, with the results $a = 0$, $b = 4.00$, $r = 0.9998$. The correlation coefficient value suggests that this line is about as good a fit of the points as the previous one, in which five points were used. The residuals for this third calculation are $+0.1$, 0, -0.3 and $+0.2$, with a sum of squares of 0.14. With this calibration line the y -residual for the $8 \mu\text{g ml}^{-1}$ solution is -0.5 : this value is larger than the other residuals but probably not by a significant amount. It can thus be concluded that it is reasonably safe to include the point (8, 31.5) within the linear range of the method. In making a marginal decision of this kind, the analytical chemist will take into account the accuracy required in their results, and the reduced value of a method for which the calibration range is very short. The calculations described above are summarised in Fig. 5.14. It will be seen that the lines calculated for the first four points and the first five points are in practice almost indistinguishable.

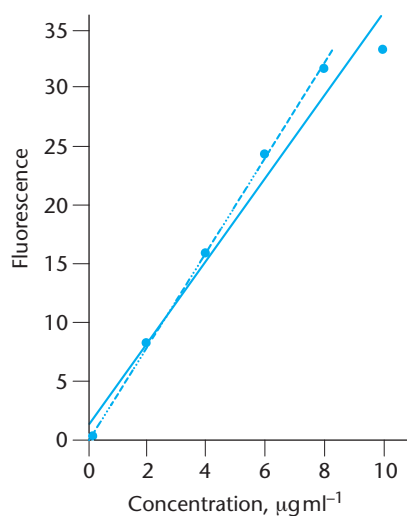


Figure 5.14 Curvilinear regression: identification of the linear range. The data in Example 5.13.1 are used; the unweighted linear regression lines through all the points (—), through the first five points (---) and through the first four points only (···) are shown.

Once a decision has been taken that a set of calibration points cannot be satisfactorily fitted by a straight line, the analyst can play one further card before using the more complex curvilinear regression calculations. It may be possible to *transform* the data so that a non-linear relationship is changed into a linear one. Such transformations are regularly applied to the results of certain analytical methods. For example, modern software packages for the interpretation of immunoassay data frequently offer a choice of transformations: commonly used methods involve plotting $\log y$ and/or $\log x$ instead of y and x , or the use of logit functions ($\text{logit } x = \ln [x/(1 - x)]$). Such transformations may also affect the nature of the errors at different points on the calibration plot. Suppose, for example, that in a set of data of the form $y = px^q$, the sizes of the random errors in y are independent of x . Any transformation of the data into linear form by taking logarithms will obviously produce data in which the errors in $\log y$ are *not* independent of $\log x$. In this case, and in any other instance where the expected form of the equation is known from theoretical considerations or from longstanding experience, it is possible to apply *weighted* regression equations (Section 5.10) to the transformed data. It may be shown that, if data of the general form $y = f(x)$ are transformed into the linear equation $Y = BX + A$, the weighting factor, w , used in Eqs (5.10.1)–(5.10.4) is obtained from the relationship:

$$w_i = \left(\frac{1}{dY_i/dx_i} \right)^2 \quad (5.13.1)$$

Unfortunately, there are not many cases in analytical chemistry where the exact mathematical form of a non-linear regression equation is known with certainty (see below), so this approach may not be very valuable.

In contrast to the situation described in the previous paragraph, experimental data can sometimes be transformed so that they can be treated by *unweighted* methods. Data of the form $y = bx$ with y -direction errors strongly dependent on x are sometimes subjected to a log-log transformation: the errors in $\log y$ then vary less seriously with $\log x$, so the transformed data can reasonably be studied by unweighted regression equations.

5.14 Curve fitting

In view of the difficulties that arise from transforming the data, and the increasing ease with which curves can be calculated to fit a set of calibration points, curvilinear regression methods are now relatively common in analytical chemistry. In practice curved calibration plots often arise from the combination of two or more physical or chemical phenomena. In molecular fluorescence spectrometry, for example, signal vs. concentration plots will often be approximately linear in very dilute solution, but will show increasing (negative) curvature at higher concentrations. This is because of (a) optical artefacts (inner filter effects) in the fluorescence detection method, (b) molecular interactions (e.g. quenching, excimer formation) and (c) the failure of the algebraic assumptions on which a linear plot is predicted. Effects (a)–(c) are independent of one another, so many curves of different shapes may appear in practice. This example shows why calibration curves of a known and predictable form are so rarely encountered in analytical work (see above). Thus the

analyst has little *a priori* guidance on which of the many types of equation that generate curved plots should be used to fit the calibration data in a particular case. In practice, much the most common strategy is to fit a curve which is a polynomial in x , i.e. $y = a + bx + cx^2 + dx^3 + \dots$. The mathematical problems to be solved are then (a) how many terms should be included in the polynomial and (b) what values must be assigned to the coefficients a , b , etc.? Computer software packages that tackle these problems are normally iterative: they fit first a straight line, then a quadratic curve, then a cubic curve, and so on, to the data, and present to the user the information needed to decide which of these equations is the most suitable. In practice quadratic or cubic equations are often entirely adequate to provide a good fit to the data: polynomials with many terms are almost certainly physically meaningless and do not significantly improve the analytical results. In any case, if the graph has n calibration points, the largest polynomial permissible is that of order $(n - 1)$.

To decide whether (for example) a quadratic or a cubic curve is the best fit to a calibration data set we can use the ANOVA methods introduced in Section 5.12. ANOVA programs generate values for R^2 , the **coefficient of determination**. Equation (5.12.2) shows that, as the least-squares fit of a curve (or straight line) to the data points improves, the value of R^2 will get closer to 1 (or 100%). It would thus seem that we have only to calculate R^2 values for the straight-line, quadratic, cubic, etc. equations, and end our search when R^2 no longer increases. Unfortunately it turns out that the addition of another term to the polynomial *always* increases R^2 , even if only by a small amount. ANOVA programs thus provide R'^2 ('adjusted R^2 ') values (Eq. (5.12.3)), which utilise mean squares (MS) rather than sums of squares. The use of R'^2 takes into account that the number of residual degrees of freedom in the polynomial regression (given by $(n - k - 1)$ where k is the number of terms in the regression equation containing a function of x) changes as the order of the polynomial changes. As the following example shows, R'^2 is always smaller than R^2 .

Example 5.14.1

In an instrumental analysis the following data were obtained (arbitrary units).

Concentration	0	1	2	3	4	5	6	7	8	9	10
Signal	0.2	3.6	7.5	11.5	15.0	17.0	20.4	22.7	25.9	27.6	30.2

Fit a suitable polynomial to these results, and use it to estimate the concentrations corresponding to signal of 5, 16 and 27 units.

Even a casual examination of the data suggests that the calibration plot should be a curve, but it is instructive nonetheless to calculate the least-squares straight line through the points using the method described in Section 5.4. This line turns out to have the equation $y = 2.991x + 1.555$. The ANOVA table in this case has the following form:

Source of variation	Sum of squares	d.f.	Mean square
Regression	984.009	1	984.009
Residual	9.500	9	1.056
Total	993.509	10	99.351

As already noted the number of degrees of freedom (d.f.) for the variation due to regression is equal to the number of terms (k) in the regression equation containing x , x^2 , etc. For a straight line, k is 1. There is only one constraint in the calculation (viz. that the sum of the residuals is zero, see above), so the total number of degrees of freedom is $n - 1$. Thus the number of degrees of freedom assigned to the residuals is $(n - k - 1) = (n - 2)$ in this case. From the ANOVA table R^2 is given by $984.009/993.509 = 0.99044$, i.e. 99.044%. An equation which explains over 99% of the relationship between x and y seems quite satisfactory but, as with the correlation coefficient, r , we must use great caution in interpreting absolute values of R^2 : we shall see that a quadratic curve provides a much better fit for the data. We can also calculate the R'^2 value from equation (5.12.3): it is given by $(1 - [1.056/99.351]) = 0.98937$, i.e. 98.937%.

As always an examination of the residuals provides valuable information on the success of a calibration equation. In this case the residuals are as follows:

x	y_i	$\hat{y}_i$	y -residual
0	0.2	1.0	-1.4
1	3.6	4.5	-0.9
2	7.5	7.5	0
3	11.5	10.5	1.0
4	15.0	13.5	1.5
5	17.0	16.5	0.5
6	20.4	19.5	0.9
7	22.7	22.5	0.2
8	25.9	25.5	0.4
9	27.6	28.5	-0.9
10	30.2	31.5	-1.3

In this table, the numbers in the two right-hand columns have been rounded to one decimal place for simplicity. The trend in the signs and magnitudes of the residuals, which are negative at low x -values, rise to a positive maximum, and then return to negative values, is a sure sign that a straight line is not a suitable fit for the data.

When the data are fitted by a curve of quadratic form the equation turns out to be $y = 0.086 + 3.970x - 0.098x^2$, and the ANOVA table takes the form:

Source of variation	Sum of squares	d.f.	Mean square
Regression	992.233	2	494.116
Residual	1.276	8	0.160
Total	993.509	10	99.351

The numbers of degrees of freedom for the regression and residual sources of variation have now changed in accordance with the rules described above, but the total variation is naturally the same as in the first ANOVA table. Here R^2 is $992.233/993.509 = 0.99872$, i.e. 99.872%. This figure is noticeably higher than the value of 99.044% obtained from the linear plot, and the R'^2 value is also higher at $[1 - (0.160/99.351)] = 0.99839$, i.e. 99.839%. When the y -residuals are

calculated, their signs (in increasing order of x -values) are $+ - - + + - - + - +$. There is no obvious trend here, so on all grounds we must prefer the quadratic over the linear fit.

Lastly we repeat the calculation for a cubic fit. Here, the best-fit equation is $y = -0.040 + 4.170x - 0.150x^2 + 0.0035x^3$. The cubic coefficient is very small, so it is questionable whether this equation is a significantly better fit than the quadratic one. The R^2 value is, inevitably, slightly higher than that for the quadratic curve (99.879% compared with 99.872%), but the value of R'^2 is slightly *lower* than the quadratic value at 99.827%. The order of the signs of the residuals is the same as in the quadratic fit. As there is no value in including unnecessary terms in the polynomial equation, we can be confident that a quadratic fit is satisfactory in this case.

When the above equations are used to estimate the concentrations corresponding to instrument signals of 5, 16 and 27 units, the results (x -values in arbitrary units) are:

	Linear	Quadratic	Cubic
$y = 5$	1.15	1.28	1.27
$y = 16$	4.83	4.51	4.50
$y = 27$	8.51	8.61	8.62

As expected, the differences between the concentrations calculated from the quadratic and cubic equations are insignificant, so the quadratic equation is used for simplicity.

Since non-linear calibration graphs often result from the simultaneous occurrence of a number of physicochemical and/or mathematical phenomena it is sensible to assume that no *single* mathematical function could describe the calibration curve satisfactorily. It thus seems logical to try to fit the points to a curve that consists of several linked sections whose mathematical forms may be different. This is the approach used in the application of **spline functions**. **Cubic splines** are most commonly used in practice, i.e. the final curve is made up of a series of linked sections of cubic form. These sections must clearly form a continuous curve at their junctions ('knots'), so the first two derivatives (dy/dx and d^2y/dx^2) of each curve at any knot must be identical. Several methods have been used for estimating both the number of knots and the equations of the curves joining them: these techniques are too advanced to be considered in detail here, but many commercially available statistics software packages now provide such facilities. Spline functions have been applied successfully to a variety of analytical methods, including gas-liquid chromatography, competitive binding immunoassays and similar receptor-based methods, and atomic-absorption spectrometry.

It is legitimate to ask whether we could use the spline idea in its simplest form, and just plot the curve (provided the curvature is not too great) as a series of straight lines joining the successive calibration points. The method is obviously non-rigorous, and would not provide any information on the precision of the interpolated x -values. However, its value as a simple initial data analysis (IDA) method (see Chapter 6) is indicated by applying it to the data in the above example. For y -values of 5, 16 and 27

this method of linear interpolation between successive points gives x -values of 1.36, 4.50 and 8.65 units respectively. Comparison with the above table shows that these results, especially the last two, would be quite acceptable for many purposes.

5.15 Outliers in regression

In this section we return to a problem already discussed in Chapter 3, the occurrence of suspect values – possible outliers – in our data. These anomalous results inevitably arise in calibration experiments, just as they occur in replicate measurements, but it is rather harder to deal with them in regression statistics. The least-squares method described in this chapter minimises the sum of the squares of the y -residuals, so a suspect point with a large y -residual can have a significant effect on the calculated slope and intercept of the regression line, and thus on the analytical information derived from the latter. In cases where an obvious error such as a transcription mistake or an instrument malfunction has occurred, it is both natural and permissible to reject the resulting measurement (and, if possible, to repeat it). If there are suspect measurements for which there are no obvious sources of error or explanation, three distinct approaches are available, just as in the case of replicate measurements. These are (a) the use of a significance test or similar method to decide whether a measurement should be accepted or rejected; (b) the use of median-based methods, in which suspect or outlying values are discounted; and (c) the use of robust methods, in which such values may be included in our calculations, but given less weight, i.e. importance, in plotting the regression line.

Unfortunately simple tests for outliers cannot be directly applied to the points forming regression lines. This is because, although the individual y_i -values in a calibration experiment are assumed to be independent of one another, the residuals ($y_i - \hat{y}_i$) are *not* independent of one another, as their sum is always zero. It is therefore not permissible to treat the residuals as if they were a conventional set of replicate measurements, and apply a familiar test such as the Grubbs' test to identify any outliers. (If the number of y_i -values is large, a condition not generally met in analytical work, this prohibition can be relaxed.) Most computer programs handling regression data provide residual diagnostics routines (see above). Some of these are simple, including plots of the individual residuals against y_i -values (Fig. 5.15). Such plots would normally be expected to show that, if the correct calibration model has been used, the residuals are approximately uniform in size across the range of y_i -values, and normally distributed about zero. The figure also illustrates cases where the y -direction errors increase with y_i (Section 5.10), and where the wrong regression equation has been used (Sections 5.11 and 5.12). Similarly, the y -residuals can be plotted against time if instrument drift or any other time-dependent effect is suspected. These plots show up suspect values very clearly, but do not provide criteria that can be immediately used to reject or accept them. Moreover, they are of limited value in many analytical chemistry experiments, where the number of calibration points is often small.

Some simple numerical criteria have been used in computer software to identify possible outliers. Some packages 'flag' calibration points where the y -residual is more than twice (or sometimes three times) the value of $s_{y/x}$. (A residual divided by $s_{y/x}$ is referred to as a *standardised residual*, so standardised residuals greater than 2 or 3 are

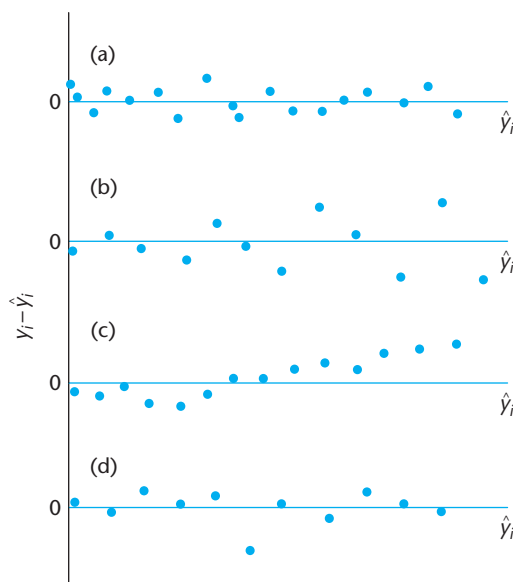


Figure 5.15 Residual plots in regression diagnosis: (a) satisfactory distribution of residuals; (b) the residuals tend to grow as y_i grows, suggesting that a weighted regression plot would be suitable; (c) the residuals show a trend, first becoming more negative, then passing through zero, and then becoming more positive as y_i increases, suggesting that a [different] curve should be plotted; and (d) a satisfactory plot, except that y_6 might be an outlier.

flagged.) Several more advanced methods have been developed, of which the best known is the estimation for each point of **Cook's squared distance**, CD^2 (sometimes abbreviated to 'Cook's distance'), first proposed in 1977. This is an example of an *influence function*, i.e. it measures the effect that rejecting the calibration point in question would have on the regression coefficients. For a straight line graph it can be calculated from:

$$CD^2 = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_j^{(i)})^2}{2s_{y/x}^2} \quad (5.15.1)$$

In this equation $\hat{y}_j$ is a predicted y -value obtained when all the data points are used, and $\hat{y}_j^{(i)}$ is the corresponding predicted y -value obtained when the i th point is omitted: $s_{y/x}^2$ is calculated using all the data points. Values of CD^2 greater than 1 justify the omission of the suspect point.

In practice the Cook's squared distance method turns out to be better at identifying some types of outlier than others: outliers in the middle of a data set are less readily detected than those at the extremes. However, the **alternative non-parametric and robust methods can be very effective in handling outliers in regression: robust regression methods have proved particularly popular in recent years**. These topics are covered in the next chapter.

Bibliography

Analytical Methods Committee, Royal Society of Chemistry, Cambridge. This body publishes a series of Technical Briefs on several aspects of regression and calibration methods, including weighted regression, errors and confidence limits, standard additions, etc. Along with associated software and datasets these short papers can be downloaded from www.rsc.org.

Draper, N.R. and Smith, H., 1998, *Applied Regression Analysis*, 3rd edn, John Wiley, New York. An established work with comprehensive coverage of many aspects of regression and correlation problems.

Kleinbaum, D.G., Kupper, L.L. and Muller, K.E. 2007. *Applied Regression Analysis and other Multivariable Methods*, 4th edn, Duxbury Press, Boston, MA. Extensive treatment of regression problems and the applications of ANOVA.

Mark, H. and Workman Jr, J., 2008, *Chemometrics in Spectroscopy*, Academic Press, London. A major work, with a substantial emphasis on calibration and regression methods. Also covers basic statistics, experimental designs and collaborative studies.

Snedecor, G.M. and Cochran, W.G., 1989, *Statistical Methods*, 8th edn, Iowa State University Press, Ames, IA. Gives an excellent general account of regression and correlation procedures.

Exercises

- 1 In a laboratory containing polarographic equipment, six samples of dust were taken at various distances from the polarograph and the mercury content of each sample was determined. The following results were obtained.

Distance from polarograph, m	1.4	3.8	7.5	10.2	11.7	15.0
Mercury concentration, ng g ⁻¹	2.4	2.5	1.3	1.3	0.7	1.2

Examine the possibility that the mercury contamination arose from the polarograph.

- 2 The response of a colorimetric test for glucose was checked with the aid of standard glucose solutions. Determine the correlation coefficient from the following data and comment on the result.

Glucose concentration, mM	0	2	4	6	8	10
Absorbance	0.002	0.150	0.294	0.434	0.570	0.704

- 3 The following results were obtained when each of a series of standard silver solutions was analysed by flame atomic-absorption spectrometry.

Concentration, ng ml ⁻¹	0	5	10	15	20	25	30
Absorbance	0.003	0.127	0.251	0.390	0.498	0.625	0.763

Determine the slope and intercept of the calibration plot, and their confidence limits.

- 4 Using the data of exercise 3, estimate the confidence limits for the silver concentrations in (a) a sample giving an absorbance of 0.456 in a single determination, and (b) a sample giving absorbance values of 0.308, 0.314, 0.347 and 0.312 in four separate analyses.
- 5 Estimate the limit of detection of the silver analysis from the data in exercise 3.
- 6 The gold content of a concentrated seawater sample was determined by using atomic-absorption spectrometry with the method of standard additions. The results obtained were as follows.

Gold added, ng per ml of concentrated sample	0	10	20	30	40	50	60	70
Absorbance	0.257	0.314	0.364	0.413	0.468	0.528	0.574	0.635

Estimate the concentration of the gold in the concentrated seawater, and determine confidence limits for this concentration.

- 7 The fluorescence of each of a series of acidic solutions of quinine was determined five times. The results are given below.

Concentration, ng ml ⁻¹	0	10	20	30	40	50
Fluorescence intensity	4	22	44	60	75	104
(arbitrary units)	3	20	46	63	81	109
	4	21	45	60	79	107
	5	22	44	63	78	101
	4	21	44	63	77	105

Determine the slopes and intercepts of the unweighted and weighted regression lines. Calculate, using both regression lines, the confidence limits for the concentrations of solutions with fluorescence intensities of 15 and 90 units.

- 8 An ion-selective electrode (ISE) determination of sulphide from sulphate reducing bacteria was compared with a gravimetric determination. The results obtained were expressed in milligrams of sulphide.

Sample:	1	2	3	4	5	6	7	8	9	10
Sulphide (ISE method):	108	12	152	3	106	11	128	12	160	128
Sulphide (gravimetry):	105	16	113	0	108	11	141	11	182	118

Comment on the suitability of the ISE method for this sulphide determination. (Al-Hitti, I.K., Moody, G.J. and Thomas, J.D.R, 1983, *Analyst*, **108**: 43)

- 9 In the determination of lead in aqueous solution by electrochemical atomic-absorption spectrometry with graphite-probe atomisation, the following results were obtained:

Lead concentration, ng ml ⁻¹	10	25	50	100	200	300
Absorbance	0.05	0.17	0.32	0.60	1.07	1.40

Investigate the linear calibration range of this experiment.

(Based on Giri, S.K., Shields, C.K., Littlejohn D. and Ottaway, J.M., 1983, *Analyst*, **108**: 244)

- 10 In a study of the complex formed between europium (III) ions and pyridine-2, 6-dicarboxylic acid (DPA), the absorbance values of solutions containing different DPA : Eu concentrations were determined, with the following results:

Absorbance	0.008	0.014	0.024	0.034	0.042	0.050	0.055	0.065
DPA : Eu	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6
Absorbance	0.068	0.076	0.077	0.073	0.066	0.063	0.058	
DPA : Eu	1.8	2.0	2.4	2.8	3.2	3.6	4.0	

Use these data to determine the slopes and intercepts of two separate straight lines. Estimate their intersection point and its standard deviation, thus determining the composition of the DPA–europium complex formed.

(Based on Arnaud, N., Vaquer, E. and Georges, J., 1998, *Analyst*, 123: 261)

- 11 In an experiment to determine hydrolysable tannins in plants by absorption spectroscopy the following results were obtained:

Absorbance		0.084	0.183	0.326	0.464	0.643
Concentration, mg ml ⁻¹		0.123	0.288	0.562	0.921	1.420

Use a suitable statistics or spreadsheet program to calculate a quadratic relationship between absorbance and concentration. Using R^2 and R'^2 values, comment on whether the data would be better described by a cubic equation.

(Based on Willis, R.B. and Allen, P.R., 1998, *Analyst*, 123: 435)

- 12 The following results were obtained in an experiment to determine spermine by high-performance thin layer chromatography of one of its fluorescent derivatives:

Fluorescence intensity	36	69	184	235	269	301	327
Spermine, ng	6	18	30	45	60	75	90

Determine the best polynomial calibration curve through these points.

(Based on Linares, R.M., Ayala, J.H., Afonso, A.M. and Gonzalez, V., 1998, *Analyst*, 123: 725)

6

Non-parametric and robust methods

Major topics covered in this chapter

- Occurrence of non-Gaussian error distributions
- Initial data analysis and median-based methods
- Sign test and Wald–Wolfowitz runs test
- Rank-based methods
- Mann–Whitney and Tukey tests
- Tests for more than two samples
- Rank correlation
- Non-parametric regression methods
- Robust statistics: trimming and winsorisation
- Robust measures of location and spread
- Robust ANOVA
- Robust regression methods
- Re-sampling methods; the bootstrap

6.1

Introduction

The statistical tests described in the previous chapters have all assumed that the data being analysed follow the normal (Gaussian) distribution. Some support for this assumption is provided by the **central limit theorem**, which shows that the sampling distribution of the mean may be approximately normal even if the parent population has quite a different distribution (see Section 2.5). However, the **theorem is not really valid for the very small data sets** (often only three or four readings) frequently used in analytical work.

Methods that do not require the assumption of normally distributed measurements are important for other reasons. Some sets of data occurring in the analytical sciences certainly have different distributions. For example the antibody concentrations in the

blood sera of a group of different people can be expressed approximately as a log-normal distribution (see Section 2.3): similar results are often obtained when a single analysis is made on each member of a group of human or animal subjects. More interestingly, there is growing evidence that, even when repeated measurements are made on a *single* test material, the distribution of the results is sometimes symmetrical but not normal: the data include more results than expected which are distant from the mean. Such **heavy-tailed** distributions can be regarded as normal distributions with the addition of outliers (see Chapter 3) arising from gross errors. Alternatively heavy-tailed data may arise from the superposition of two or more normal distributions with the same mean value, but with significantly different standard deviations. This result could occur if, for example, the measurements were made by more than one individual or by using more than one piece of equipment.

This chapter introduces two groups of statistical tests for handling data that may not be normally distributed. Methods which make no assumptions about the shape of the distribution from which the data are taken are called **non-parametric** or **distribution-free** methods. Many of them use greatly simplified calculations: with small data sets some non-parametric significance tests can be performed mentally. By contrast **robust** methods are based on the belief that the underlying population distribution may indeed be approximately normal, but with the addition of data such as outliers that may distort this distribution. Robust techniques in essence operate by *down-weighting* the importance of outliers, so are appropriate in the cases of heavy-tailed distributions, and their acceptance and use have increased dramatically in recent years. They differ from non-parametric methods in that they often involve iterative calculations that would be lengthy or complex without a computer, and their rise in popularity certainly owes much to the universal availability of desktop computers.

6.2 The median: initial data analysis

In previous chapters we have used the arithmetic mean or average as the ‘measure of central tendency’ or ‘measure of location’ of a set of results. This is logical enough when the (symmetrical) normal distribution is assumed, but in non-parametric statistics, the **median** is usually used instead. To calculate the median of n observations, we arrange them in ascending order: in the unlikely event that n is very large, this sorting process can be performed very quickly by programs available for most computers.

The median is the value of the $\frac{1}{2}(n + 1)$ th observation if n is odd, and the average of the $\frac{1}{2}n$ th and the $\frac{1}{2}(n + 1)$ th observations if n is even.

Determining the median of a set of experimental results usually requires little or no calculation. Moreover, in many cases it may be a more realistic measure of central tendency than the arithmetic mean.

Example 6.2.1

Determine the mean and the median for the following four titration values:

25.01, 25.04, 25.06, 25.21 ml

It is easy to calculate that the mean of these four observations is 25.08 ml, and that the median – in this case the average of the second and third values, the observations already being in numerical order – is 25.05 ml. The mean is greater than any of the three closely grouped values (25.01, 25.04 and 25.06 ml) and may thus be a less realistic measure of location than the median. Instead of calculating the median we could use the methods of Chapter 3 to test the value 25.21 as a possible outlier, and determine the mean according to the result obtained, but this approach involves extra calculation and assumes that the data come from a normal population.

This simple example illustrates one valuable property of the **median: it is unaffected by outlying values**. Confidence limits (see Chapter 2) for the **median can be estimated with the aid of the binomial distribution**. This calculation can be performed even when the number of measurements is small, but is not likely to be required in analytical chemistry, where the median is generally used only as a rapid measure of an average. The reader is referred to the Bibliography for further information.

In non-parametric statistics the usual measure of dispersion (replacing the standard deviation) is the interquartile range (IQR). As we have seen, the median divides the sample of measurements into two equal halves; if each of these halves is further divided into two the points of division are called the **upper and lower quartiles**. Several different conventions are used in making this calculation (the interested reader should again consult the bibliography): here we use the method adopted by the Minitab[®] program. The IQR is not widely used in analytical work, but various statistical tests can be performed on it.

The median and the IQR of a set of measurements are just two of the statistics which feature strongly in **initial data analysis (IDA), often also called exploratory data analysis (EDA)**. This is an aspect of statistics that has grown rapidly in popularity in recent years. One reason for this is, yet again, the ability of modern computers and dedicated software to present data almost instantly in a wide range of graphical formats: as we shall see, such pictorial representations form an important element of IDA. A second reason for the rising importance of IDA is the increasing acceptance of statistics as a practical and pragmatic subject not necessarily restricted to the use of techniques whose theoretical soundness is unquestioned: some IDA methods seem almost crude in their principles, but have nonetheless proved most valuable.

The main advantage of IDA methods is their ability to indicate which (if any) further statistical methods are most appropriate to a given data set.

Several simple presentation techniques are obviously useful. We have already used **dot-plots** to summarise small data sets (see Chapters 1 and 3). These plots help in

the visual identification of outliers and other unusual features of the data. Here is a further example illustrating their value.

Example 6.2.2

In an experiment to determine whether Pb^{2+} ions interfered with the enzymatic determination of glucose in various foodstuffs, nine food materials were treated with a 0.1 mM solution of $\text{Pb}(\text{II})$, while four other materials (the control group) were left untreated. The rates (arbitrary units) of the enzyme-catalysed reaction were then measured for each food and corrected for the different amounts of glucose known to be present. The results were:

Treated foods	21	1	4	26	2	27	11	24	21
Controls	22	22	32	23					

Comment on these data.

Written out in two rows as above, the data do not convey much immediate meaning, and an unthinking analyst might proceed straight away to perform a *t*-test (Chapter 3), or perhaps one of the non-parametric tests described below, to see if the two sets of results are significantly different. But when the data are presented as two dot-plots, or as a single plot with the two sets of results given separate symbols, it is clear that the results, while interesting, are so inconclusive that little can be deduced from them without further measurements (Fig. 6.1).

The medians of the two sets of data are similar: 21 for the treated foods and 22.5 for the controls. But the range of reaction rates for the $\text{Pb}(\text{II})$ -treated materials is enormous, with the results apparently failing into at least two groups: five of the foods seem not to be affected by the lead (perhaps because in these cases $\text{Pb}(\text{II})$ is complexed by components other than the enzyme in question), while three others show a large inhibition effect (i.e. the reaction rate is much reduced), and another lies somewhere in between these two extremes. There is the further problem that one of the control group results is distinctly different from the rest, and might be considered as an outlier (see Chapter 3). In these circumstances it seems most unlikely that a conventional significance test will reveal chemically useful information: the use of the simplest IDA method has guided us away from thoughtless and valueless significance testing and (as so often happens) towards more experimental measurements.

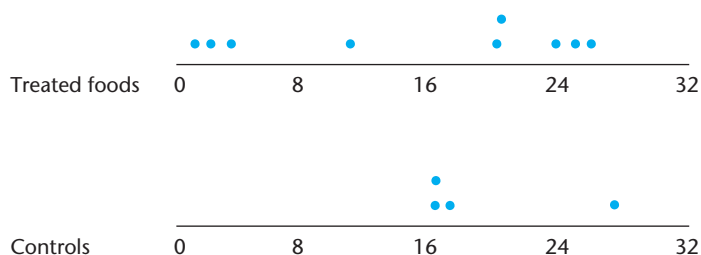


Figure 6.1 Dot-plots for Example 6.2.2.

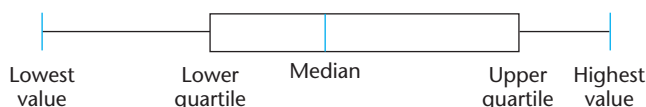


Figure 6.2 Box-and-whisker plot.

Another simple data representation technique, of greater value when rather larger samples are studied, is the **box-and-whisker plot**. In its normal form such a diagram consists of a rectangle (the box) with two lines (the whiskers) extending from opposite edges of the box, and a further line in the box, crossing it parallel to the same edges. The ends of the whiskers indicate the range of the data, the edges of the box from which the whiskers protrude represent the upper and lower quartiles, and the line crossing the box represents the median of the data (Fig. 6.2).

The box-and-whisker plot, with a numerical scale, is a graphical representation of the **five-number summary**: the data set is described by its extremes, its lower and upper quartiles, and its median. The plot shows at a glance the spread and the symmetry of the data.

Some computer programs enhance box-and-whisker plots by identifying possible outliers separately, the outliers often being defined as data points which are lower than the lower quartile, or higher than the upper quartile, by more than 1.5 times the inter-quartile range. The whiskers then only extend to these upper and lower limits or **fences** and outlying data are shown as separate points. (These refinements are not shown in Fig. 6.2.)

Example 6.2.3

The levels of a blood plasma protein in 20 men and 20 women ($\text{mg } 100 \text{ ml}^{-1}$) were found to be:

Men	3	2	1	4	3	2	9	13	11	3
	18	2	4	6	2	1	8	5	1	14
Women	6	5	2	1	7	2	2	11	2	1
	1	3	11	3	2	3	2	1	4	8

What information can be gained about any differences between the levels of this protein in men and women?

As in the previous example, the data as presented convey very little, but the use of two box-and-whisker plots or five-number summaries is very revealing. The five-number summaries are:

	Min.	Lower quartile	Median	Upper quartile	Max.
Men	1	2	3.5	8.75	18
Women	1	2	2.5	5.5	11

It is left as a simple sketching exercise for the reader to show that (a) the distributions are very skewed in both men and women, so statistical methods that assume a normal distribution are not appropriate (as we have seen this is often true when a single measurement is made on a number of *different* subjects, particularly when the latter are living organisms); (b) the median concentrations for men and women are similar; and (c) the range of values is considerably greater for men than for women. The conclusions suggest that we might apply the Siegel–Tukey test (see below, Section 6.6) to see whether the greater variation in protein levels amongst men is significant.

While it is usual for analysts to handle relatively small sets of data there are occasions when a larger set of measurements is to be examined. Examples occur in the areas of clinical and environmental analysis, where in many instances there are large natural variations in analyte levels. Table 6.1 shows, in numerical order, the levels of a pesticide in 30 samples of butter beans. The individual values range from 0.03 to 0.96 mg kg^{-1} . They might be expressed as a histogram. This would show that, for example, there are four values in the range $0\text{--}0.095 \text{ mg kg}^{-1}$, four in the range $0.10\text{--}0.195 \text{ mg kg}^{-1}$, and so on. But a better IDA method uses a **stem-and-leaf diagram**, as shown in Fig. 6.3.

The left-hand column of figures – the stem – shows the first significant digit for each measurement, while the remaining figures in each row – the leaves – provide the second significant digit. The length of each row thus corresponds to the length of the bars on the corresponding histogram, but the advantage of the stem-and-leaf diagram is that it retains the value of each measurement. The leaves use only whole numbers, so some indication of the scale used must always be given. In this case a key is used to provide this information. Minitab® provides facilities for stem-and-leaf diagrams.

Table 6.1 Levels of pp-DDT in 30 butter bean specimens (mg kg^{-1})

0.03	0.05	0.08	0.08	0.10	0.11	0.18	0.19	0.20	0.20
0.22	0.22	0.23	0.29	0.30	0.32	0.34	0.40	0.47	0.48
0.55	0.56	0.58	0.64	0.66	0.78	0.78	0.86	0.89	0.96

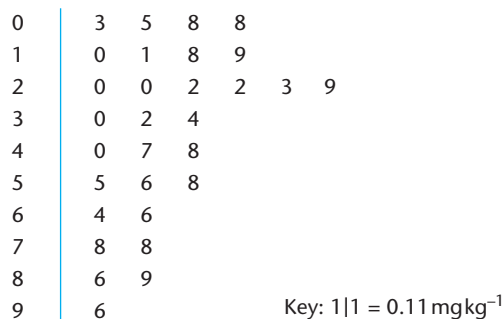


Figure 6.3 Stem-and-leaf diagram for data from Table 6.1.

In summary, IDA methods are simple, readily handled by computers, and most valuable in indicating features of the data not apparent on initial inspection. They are helpful in deciding the most suitable significance tests or other statistical procedures to be adopted in further work, and sometimes even in suggesting that statistics has no further role to play until more data are obtained.

They can of course be extended to the area of calibration and other regression techniques: the very crude method of plotting a curved calibration graph suggested at the end of the previous chapter can be regarded as an IDA approach. Many IDA methods are described in the books by Chatfield and by Velleman and Hoaglin listed in the Bibliography at the end of this chapter.

6.3 The sign test

The sign test is amongst the simplest of all non-parametric statistical methods, and was first discussed in the early eighteenth century. It can be used in a number of ways, the simplest of which is demonstrated by the following example.

Example 6.3.1

A pharmaceutical preparation is claimed to contain 8% of a particular component. Successive batches were found in practice to contain 7.3, 7.1, 7.9, 9.1, 8.0, 7.1, 6.8 and 7.3% of the constituent. Are these results consistent with the manufacturer's claim?

In Section 3.2 it was shown that such problems could be tackled by using the *t*-test after calculation of the mean and standard deviation of the experimental data. The *t*-test assumes, however, that the data are normally distributed. The sign test avoids this assumption and is much easier to perform. The same underlying principles are used as in other significance tests: a null hypothesis is established, the probability of obtaining the experimental results is determined, and the null hypothesis is rejected if this probability is less than a certain critical level. In this case the null hypothesis is that the data come from a population with a median value of 8.0% of the constituent. This postulated median is subtracted from each experimental value in turn, and the *sign* of each result is considered. Values equal to the postulated median are ignored *entirely*. In this case, therefore, we effectively have seven experimental values, six of them lower than the median and hence giving minus signs, and one higher than the median and hence giving a plus sign. To test whether this preponderance of minus signs is significant we use the binomial theorem. This theorem shows that the probability of *r* out of *n* signs being minus is given by

$$P(r) = {}^nC_r p^r q^{(n-r)} \quad (6.3.1)$$

where nC_r is the number of combinations of r items from a total of n items (calculated using ${}^nC_r = n!/r!(n-r)!$), p is the probability of getting a minus sign in a single result, and q is the probability of not getting a minus sign in a single result, i.e. $q = 1 - p$. Since the median is defined so that half the experimental results lie above it and half below it, it is clear that if the median is 8.0 in this case then both p and q should be $\frac{1}{2}$. Using Eq. (6.3.1) we find that $P(6) = {}^7C_6 \times (\frac{1}{2})^6 \times \frac{1}{2} = 7 \times (\frac{1}{2})^6 \times \frac{1}{2} = 7/128$. Similarly we can calculate that the chance of getting seven minus signs, $P(7)$, is $1 \times (\frac{1}{2})^7 \times 1 = 1/128$. Overall, therefore, the probability of getting *six or more* negative signs in our experiment is $8/128$. We are only asking, however, whether the data *differ* significantly from the postulated median, so we perform a *two-sided* test (see Chapter 3). We must find the probability of obtaining six or more identical signs (i.e. ≥ 6 plus *or* ≥ 6 minus signs) when seven results are taken at random. This is clearly $16/128 = 0.125$. Since this value is >0.05 , the critical probability level usually used, the null hypothesis that the data come from a population with median 8.0 cannot be rejected. As pointed out in Chapter 3, we have not proved that the data *do* come from such a population; we have only concluded that such a hypothesis cannot be rejected.

This example shows that the sign test will involve using the binomial distribution with $p = q = \frac{1}{2}$. This approach to non-parametric statistics is so common that most sets of statistical tables include the necessary data, allowing such calculations to be made instantly (see Table A.9). Moreover, in many practical situations, an analyst will always take the same number of readings or samples, and will be able to memorise easily the probabilities corresponding to the various numbers of plus or minus signs.

The sign test can also be used as a non-parametric alternative to the paired t -test (Section 3.4) to compare two sets of results for the same samples. Thus if ten samples are examined by each of two methods, A and B, we can test whether the two methods give significantly different readings by calculating for each sample (result obtained by method A – result obtained by method B). The null hypothesis will be that the two methods do not give significantly different results – in practice this will again mean that the probability of obtaining a plus sign (or a minus sign) for each difference is $\frac{1}{2}$. The number of plus or minus signs actually obtained can be compared with the probability derived from Eq. (6.3.1). An example of this application of the sign test is given in the exercises at the end of the chapter.

Yet another use of the sign test is to indicate a trend. This application is illustrated by the following example.

Example 6.3.2

The level of a hormone in a patient's blood plasma is measured at the same time each day for 10 days. The resulting data are:

Day	1	2	3	4	5	6	7	8	9	10
Level, ngml ⁻¹	5.8	7.3	4.9	6.1	5.5	5.5	6.0	4.9	6.0	5.0

Is there any evidence for a trend in the hormone concentration?

Using parametric methods, it would be possible to make a linear regression plot of such data and test whether its slope differed significantly from zero (Chapter 5). Such an approach would assume that the errors were normally distributed, and that any trend that did occur was linear. The non-parametric approach is again simpler. The data are divided into two equal sets, the sequence being retained:

5.8	7.3	4.9	6.1	5.5
5.5	6.0	4.9	6.0	5.0

(If there is an odd number of measurements, the middle one in the time sequence is ignored.) The result for the sixth day is then subtracted from that for the first day, that for the seventh day from that for the second day, etc. The signs of the differences between the pairs of values in the five columns are determined in this way to be +, +, 0, +, +. As usual the zero is ignored completely, leaving four results, all positive. The probability of obtaining four identical signs in four trials is $2 \times (1/16) = 0.125$. (Note that a two-sided test is again used, as the trend in the hormone level might be upwards or downwards.) The null hypothesis, that there is no trend in the results, can therefore not be rejected at the $P = 0.05$ probability level.

The price paid for the extreme simplicity of the sign test is some loss of statistical power. The test does not utilise all the information offered by the data, so it is not surprising to find that it also provides less discriminating information. In later sections, non-parametric methods that do use the magnitudes of the individual results as well as their signs will be discussed.

6.4

The Wald–Wolfowitz runs test

In some instances we are interested not merely in whether observations generate positive or negative signs, but also in whether these signs occur in a random *sequence*. In Section 5.13, for example, we showed that if a straight line is a good fit to a set of calibration points, positive and negative residuals will occur more or less at random. By contrast, attempting to fit a straight line to a set of points that actually lie on a curve will yield non-random sequences of positive or negative signs: there might, for example, be a sequence of + signs, followed by a sequence of – signs, and then another sequence of + signs. Such sequences are technically known as **runs** – the word being used here in much the same way as when someone refers to ‘a run of bad luck’ or ‘a run of high scores’. In the curve-fitting case, it is clear that a non-random sequence of + and – signs will lead to a smaller number of runs than a random sequence.

The Wald–Wolfowitz method tests whether the number of runs is small enough for the null hypothesis of a random distribution of signs to be rejected.

The number of runs in the experimental data is compared with the numbers in Table A.10, which refers to the $P = 0.05$ probability level. The table is entered by using the appropriate values for N , the number of + signs, and M , the number of – signs. If the experimental number of runs is *smaller* than the tabulated value, then the null hypothesis can be rejected.

Example 6.4.1

Linear regression equations are used to fit a straight line to a set of 12 calibration points. The signs of the resulting residuals in order of increasing x value are: + + + + – – – – – + +. Comment on whether it would be better to attempt to fit a curve to the points.

Here $M = N = 6$, and the number of runs is 3. Table A.10 shows that, at the $P = 0.05$ level, the number of runs must be < 4 if the null hypothesis is to be rejected. So in this instance we can reject the null hypothesis, and conclude that the sequence of + and – signs is not a random one. The attempt to fit a straight line to the experimental points is therefore unsatisfactory, and a curvilinear regression plot is indicated instead.

The Wald–Wolfowitz test can be used with any results that can be divided or converted into just two categories. Suppose, for example, that it is found that 12 successively used spectrometer light sources last for 450, 420, 500, 405, 390, 370, 380, 395, 370, 370, 420 and 430 hours. The median lifetime, in this case the average of the sixth and seventh numbers when the data are arranged in ascending order, is 400 hours. If the lamps with lifetimes less than the median are given a minus sign, and those with longer lifetimes are given a plus sign, then the sequence becomes: + + + + – – – – – + +. This is the same sequence as in the regression example above, where it was shown to be significantly non-random. In this case, the significant variations in lifetime might be explained if the lamps came from different batches or different manufacturers.

We may be concerned with unusually large numbers of short runs, as well as unusually small numbers of long runs. If six plus and six minus signs occurred in the order + – + – + – + – + – + – we would strongly suspect a non-random sequence. Table A.10 shows that, with $N = M = 6$, a total of 11 or 12 runs indicates that the null hypothesis of random order should be rejected, and some periodicity in the data suspected.

6.5 The Wilcoxon signed rank test

Section 6.3 described the use of the sign test. Its value lies in the minimal assumptions it makes about the experimental data. The population from which the sample is taken is not assumed to be normal, or even to be symmetrical. On the other hand a disadvantage of the sign test is that it uses so little of the information provided.

The only material point is whether an individual measurement is greater than or less than the median – the size of this deviation is not used at all.

For many data sets the measurements may be approximately *symmetrically* distributed, but not necessarily normally distributed (see Section 6.1). In such cases the mean and the median of the population will be equal, allowing more powerful significance tests to be developed. Important advances were made by Wilcoxon, and his signed rank test has several applications. Its mechanism is best illustrated by an example.

Example 6.5.1

The blood lead levels (in pg ml^{-1}) of seven children were found to be 104, 79, 98, 150, 87, 136 and 101. Could such data come from a population, assumed to be symmetrical, with a median/mean of 95 pg ml^{-1} ?

On subtraction of the reference concentration (95) the data give values of

9, -16, 3, 55, -8, 41, 6

These values are first arranged in order of magnitude without their signs:

3, 6, 8, 9, 16, 41, 55

Their signs are then restored to them (in practice these last two steps can be combined):

3, 6, -8, 9, -16, 41, 55

The numbers are then **ranked**: in this process they keep their signs but are assigned numbers indicating their order (or rank):

1, 2, -3, 4, -5, 6, 7

The positive ranks add up to 20, and the negative ones to 8. The *lower* of these two figures (8) is taken as the test statistic. If the data came from a population with median 95 the sums of the negative and positive ranks would be expected to be approximately equal numerically; if the population median was very different from 95 the sums of the negative and positive ranks would be unequal. The probability of a particular sum occurring in practice is given by a set of tables (see Table A.11). In this test the null hypothesis is rejected if the experimental value is *less than or equal to* the tabulated value, i.e. the opposite of the situation encountered in most significance tests. For this example Table A.11 shows that, for $n = 7$, the test statistic must be less than or equal to 2 before the null hypothesis – that the data do come from a population of median (mean) 95 – can be rejected at a significance level of $P = 0.05$. Since the test statistic is 8 the null hypothesis must be retained. As usual, a two-sided test is used, though there may be occasional cases where a one-sided test is more appropriate.

An important advantage of the signed rank test is that it can also be used on paired data, as they can be transformed into the type of data given in the previous example. The signed rank method can thus be used as a non-parametric alternative to the paired t -test (Section 3.4).

Example 6.5.2

The following table gives the percentage concentration of zinc, determined by two different methods, for each of eight samples of health food.

Sample	EDTA titration	Atomic spectrometry
1	7.2	7.6
2	6.1	6.8
3	5.2	4.6
4	5.9	5.7
5	9.0	9.7
6	8.5	8.7
7	6.6	7.0
8	4.4	4.7

Is there any evidence for a systematic difference between the results of the two methods?

The approach to this type of problem is simple. If there is no systematic difference between the two methods, then we would expect that the differences between the results for each sample, i.e. titration result – spectrometry result, should be symmetrically distributed about zero. The signed differences are:

$$-0.4, -0.7, 0.6, 0.2, -0.7, -0.2, -0.4, -0.3$$

Arranging these values in numerical order while retaining their signs, we have:

$$-0.2, 0.2, -0.3, -0.4, -0.4, 0.6, -0.7, -0.7$$

The ranking of these results presents an obvious difficulty, that of *tied ranks*. There are two results with the numerical value 0.2, two with a numerical value of 0.4, and two with a numerical value of 0.7. How are the ranks to be calculated? This problem is resolved by giving the tied values *average* ranks, with appropriate signs. Thus the ranking for the present data is:

$$-1.5, 1.5, -3, -4.5, -4.5, 6, -7.5, -7.5$$

In such cases, it is worth verifying that the ranking has been done correctly by calculating the sum of all the ranks without regard to sign. This sum for the numbers above is 36, which is the same as the sum of the first eight integers (for the first n integers the sum is $n(n + 1)/2$), and therefore correct. The sum of the positive ranks is 7.5, and the sum of the negative ranks is 28.5. The test statistic is thus 7.5. Inspection of Table A.11 shows that, for $n = 8$, the test statistic has to be ≤ 3 before the null hypothesis can be rejected at the level $P = 0.05$. In the present case, the null hypothesis must be retained – there is no evidence that the median (mean) of the difference is not zero, and hence no evidence for a systematic difference between the two analytical methods.

The signed rank test is seen from these examples to be a simple and valuable method. Its principal limitation is that it cannot be applied to very small sets of data: for a two-tailed test at the significance level $P = 0.05$, n must be at least 6.

6.6 Simple tests for two independent samples

The signed rank test described in the previous section is valuable for the study of single sets of measurements, and for paired sets that can readily be reduced to single sets. In many instances, however, it is necessary to compare two independent samples that cannot be reduced to a single set of data, and may contain different numbers of measurements. Several non-parametric tests to tackle such problems have been devised. The simplest to understand and perform is the **Mann–Whitney *U*-test**, the operation of which is most easily demonstrated by an example.

Example 6.6.1

A sample of photographic waste was analysed for silver by atomic absorption spectrometry, five successive measurements giving values of 9.8, 10.2, 10.7, 9.5 and 10.5 $\mu\text{g ml}^{-1}$. After chemical treatment, the waste was analysed again by the same procedure, five successive measurements giving values of 7.7, 9.7, 8.0, 9.9 and 9.0 $\mu\text{g ml}^{-1}$. Is there any evidence that the treatment produced a significant reduction in the levels of silver?

The Mann–Whitney procedure involves finding the number of results in one sample that exceeds each of the values in the other sample. In the present case, we believe that the silver concentration of the treated solution should, if anything, be *lower* than that of the untreated solution so a one-sided test is appropriate. We thus expect to find that the number of cases in which a treated sample has a higher value than an untreated one should be small. Each of the values for the untreated sample is listed, and the number of instances where the values for the treated sample are greater is counted in each case.

Untreated sample	Higher values in treated sample	Number of higher values
9.8	9.9	1
10.2	–	0
10.7	–	0
9.5	9.7, 9.9	2
10.5	–	0

The total of the numbers in the third column, in this case 3, is the test statistic. Table A.12 is used for the Mann–Whitney *U*-test: again the critical values leading to the rejection of the null hypothesis are those which are *less than or equal to* the tabulated numbers. The table shows that for a one-sided test at $P = 0.05$, with five measurements of each sample, the test statistic must be ≤ 4 if the null hypothesis is to be rejected. In our example we can thus reject H_0 : the treatment of the silver-containing material probably does reduce the level of the metal.

When, as in this example, the numbers of measurements are small the Mann–Whitney calculation can be done mentally, a great advantage. If ties (identical values) occur in the U -test, each tie is assigned a value of 0.5 in the count of U .

A further convenient method with some interesting features is **Tukey's quick test**. Its use can be shown using the same example.

Tukey's quick test involves counting the total number of measurements in the two independent samples that are not included in the overlap region of the two data sets.

Example 6.6.2

Apply Tukey's quick test to the data of the previous example.

The test can be regarded as having two stages, though when only a few results are available, these two steps can be combined in one rapid mental calculation. In the first step, the number of results in the second set of data that are *lower than all the values in the first set* are counted. If there are no such values, the test ends at once, and the null hypothesis of equal medians is accepted. In the present example, there are three such values, the readings 7.7, 8.0 and 9.0 being lower than the lowest value from the first set (9.5). The test thus continues to the second step, in which we count all the values in the first data set that *are higher than all the values in the second set*. Again, if there are no such values, the test ends and the null hypothesis is accepted. Here, there are again three such values, the readings 10.2, 10.5 and 10.7 exceeding the highest value in the second set (9.7). (This approach contrasts with that of the Mann–Whitney U -test, which identifies *high* values in the sample that might be expected to have the *lower* median.) Overall there are thus six values that are not within the range over which the two samples overlap. This total (often called T) is the test statistic. The most interesting and valuable aspect of Tukey's quick test is that statistical tables are not normally needed to interpret this result. Provided that the number of readings in each sample does not exceed about 20, and that the two sample sizes are not greatly different (conditions that would probably be valid in most analytical experiments), the critical values of T for a particular level of significance are *independent of sample size*. For a one-sided test the null hypothesis may be rejected if $T \geq 6$ (for $P = 0.05$), ≥ 7 ($P = 0.025$), ≥ 10 ($P = 0.005$) and ≥ 14 ($P = 0.0005$). (For a two-tailed test the critical T values at $P = 0.05$, 0.025, 0.005 and 0.0005 are 7, 8, 11 and 15 respectively.) In the present example, therefore, the experimental T value is big enough to be significant at $P = 0.05$ in a one-sided test. We can thus reject the null hypothesis and report that the treatment does reduce the silver content of the photographic waste significantly, a result in accord with that of the Mann–Whitney U -test.

If ties occur in Tukey's quick test (i.e. if one of the values in the hypothetically higher sample is equal to the highest value in the other sample, or if one of the

values in the 'lower' sample is equal to the lowest value in the 'higher' sample) then each tie counts 0.5 towards the value of T .

A test which is distantly related to the Mann–Whitney method has been developed by Siegel and Tukey to compare the *spread* of two sets of results: it thus offers a genuinely non-parametric alternative to the F -test (see Section 3.6). The data from the two sets of measurements are first pooled and arranged in numerical order, but with one set of results distinguished by underlining. Then they are ranked in an ingenious way: the lowest measurement is ranked one, the highest measurement ranked two, the highest but one measurement ranked three, the lowest but one ranked four, the lowest but two ranked five, and so on. (If the total number of measurements is odd, the central measurement is ignored.) This *paired alternate ranking* produces a situation in which the low and high results receive low ranks, and the central results receive high ranks. If one data set has a significantly wider spread than the other, its sum of ranks should thus be much lower, while if the dispersion of the two sets of results is similar, their rank sums will be similar. Application of this method to the data from Example 6.6.1 gives the following rankings:

Data	<u>7.7</u>	<u>8.0</u>	<u>9.0</u>	9.5	<u>9.7</u>	9.8	<u>9.9</u>	10.2	10.5	10.7
Ranks	<u>1</u>	<u>4</u>	<u>5</u>	8	<u>9</u>	10	<u>7</u>	6	3	2

Two *rank sums* are then calculated. The sum of the underlined ranks (treated silver-containing samples) is 26, and the rank sum for the untreated samples is 29. In this example the sample sizes for the two sets of measurements are equal, but this will not always be the case. Allowance is made for this by subtracting from the rank sums the number $n_i(n_i + 1)/2$, where the n_i values are the sample sizes. In our example $n_i = 5$ in each case, so 15 must be subtracted from each rank sum. The lower of the two results is the one used in the test, and the critical values are the same as those used in the Mann–Whitney test (Table A.12). The test statistic obtained in this example is $(26 - 15) = 11$, much higher than the critical value of 2 (for a two-tailed test at $P = 0.05$). The null hypothesis, in this case that the spread of the results is similar for the two sets of data, is thus retained.

The Siegel–Tukey test pools the two data samples with identification, ranks them, applies paired alternate ranking to generate rank sums, and allows for the sample sizes, to provide a test statistic that can be evaluated using the same tables as for the Mann–Whitney U -test.

A little thought will show that the validity of this useful test will be reduced if the average values for the two sets of data are substantially different. In the extreme case where all the measurements in one sample are lower than all the measurements in the other sample, the rank sums will always be as similar as possible, whatever the spread of the two samples. If it is feared that this effect is appreciable, it is permissible to estimate the means of the two samples, and add the difference between the means to each of the measurements of the lower set. This will remove any effect due to the different means, while preserving the dispersion of the sample. An exercise of the application of this test is provided at the end of the chapter.

6.7

Non-parametric tests for more than two samples



The previous section described tests in which two statistical samples were compared with each other. Non-parametric methods are not, however, limited to two sets of data: several methods which compare three or more samples are available. Before two of these tests are outlined, it is important to mention one pitfall that should be avoided in all multi-sample comparisons. When (for example) three sets of measurements are examined to see whether or not their medians are similar, there is a great temptation to compare only the two samples with the highest and lowest medians. This simplistic approach can give misleading results. When several samples are taken from the same parent population, there are cases where the highest and lowest medians, considered in isolation, appear to be significantly different. This is because, as the number of samples increases, the difference between the highest and the lowest medians will tend to increase. The correct approach is to perform first a test that considers all the samples together: if it shows that they might not all come from the same population, then separate tests may be performed to try to identify where the significant differences occur. Here we describe in outline the principles of two non-parametric tests for three or more sets of data; further details are given in the books listed in the Bibliography.

The Kruskal–Wallis test is applied to the comparison of the medians of three or more unmatched samples. (An extension of the silver analysis described in the previous section, with three samples of photographic waste, one untreated and the other two treated by different methods, would provide an instance where the test would be useful.) The results from the three (or more) samples are pooled and arranged in rank order. The rank totals for the data from the different samples are determined: tied ranks are averaged, as shown above, though a special correction procedure is advisable if there are numerous ties. If each sample has the same number of measurements (this is not a requirement of the test), and if the samples have similar medians, then the rank totals for each sample should be similar, and the sum of their squares should be a minimum. For example, if we have three samples, each with five measurements, the rankings will range from 1 to 15 and the sum of all the ranks will be 120. Suppose that the three medians are very similar, and that the rank totals for each sample are thus equal, each being 40. The sum of the squares of these totals will thus be $40^2 + 40^2 + 40^2 = 4800$. If the medians are significantly different, then the rank totals will also be different from one another – say 20, 40 and 60. The sum of the squares of such totals will always be larger than 4800 ($20^2 + 40^2 + 60^2 = 5600$).

The probability of obtaining any particular sum of squares can be determined by using the **chi-squared** statistic (see Chapter 3). If the samples are referred to as A, B, C, etc. (with k samples in all), with numbers of measurements n_A, n_B, n_C , etc. and rank totals R_A, R_B, R_C , etc., then the value of χ^2 is given by:

$$\chi^2 = \frac{12}{N^2 + N} \left(\frac{R_A^2}{n_A} + \frac{R_B^2}{n_B} + \frac{R_C^2}{n_C} + \dots \right) - 3(N + 1) \quad (6.7.1)$$

where $N = n_A + n_B + n_C$, etc. This χ^2 value is compared as usual with tabulated values. The latter are identical to the usual values when the total number of measurements is greater than ca. 15, but special tables are used for smaller numbers

of measurements. The number of degrees of freedom is $k - 1$. Experimental values of χ^2 that exceed the tabulated values allow the null hypothesis (that the medians of the samples are not significantly different) to be rejected. As already noted, in the latter situation further tests can be performed on individual pairs of samples; texts listed in the Bibliography provide more details. Minitab® provides facilities for performing the Kruskal–Wallis test.

We have already seen (Sections 3.4 and 6.3) that when *paired* results are compared, special statistical tests can be used. These tests use the principle that, when two experimental methods that do not differ significantly are applied to the *same* chemical samples, the differences between the matched pairs of results should be close to zero. This principle can be extended to three or more matched sets of results by using a non-parametric test devised in 1937 by Friedman. In analytical chemistry, the main application of **Friedman's test** is in the comparison of three (or more) experimental methods applied to the same chemical samples. The test again uses the χ^2 statistic, in this case to assess the differences that occur between the total rank values for the different methods. The following example illustrates the simplicity of the approach:

Example 6.7.1

The levels of a pesticide in four plant extracts were determined by (A) high-performance liquid chromatography, (B) gas-liquid chromatography and (C) radioimmunoassay. The following results (all in ng ml^{-1}) were obtained:

Sample	Method		
	A	B	C
1	4.7	5.8	5.7
2	7.7	7.7	8.5
3	9.0	9.9	9.5
4	2.3	2.0	2.9

Do the three methods give values for the pesticide levels that differ significantly?

This problem is solved by replacing the values in the table by ranks. In each row the method with the lowest result is ranked 1, and that with the highest result is ranked 3:

Sample	Method		
	A	B	C
1	1	3	2
2	1.5	1.5	3
3	1	3	2
4	2	1	3

The use of an average value is necessary in the case of tied ranks in sample 2 (see Section 6.5). The sums of the ranks for the three methods A, B and C are 5.5, 8.5 and 10 respectively. These sums should total $nk(k + 1)/2$ ($= 24$ here), where k is

the number of methods (3 here) and n the number of samples (4 here). The rank sums are squared, yielding 30.25, 72.25 and 100 respectively, and these squares are added to give the statistic R , which here is 202.5. The experimental value of χ^2 is then calculated from:

$$\chi^2 = \frac{12R}{nk(k+1)} - 3n(k+1) \quad (6.7.2)$$

which gives a result of 2.625. At the level $P = 0.05$, and with $k = 3$, the critical values of χ^2 are 6.0, 6.5, 6.4, 7.0, 7.1 and 6.2 for $n = 3, 4, 5, 6, 7$ and 8 respectively. (More extensive data are given in many sets of statistical tables, and when $k > 7$ the usual χ^2 tables can be used at $k - 1$ degrees of freedom.) In this instance, the experimental value of χ^2 is much less than the critical value, and we must retain the null hypothesis: the three methods give results that do not differ significantly.

The Friedman test could alternatively be used in the reverse form: assuming that the three analytical methods give indistinguishable results, the same procedure could be used to test differences between the four plant extracts. In this case k and n are 4 and 3 respectively, and the reader may care to verify that R is 270 and that the resulting χ^2 value is 9.0. This is higher than the critical value for $P = 0.05$, $n = 3$, $k = 4$, which is 7.4. So in this second application of the test we can reject the null hypothesis, and state that the four samples do differ in their pesticide levels. Further tests, which would allow selected comparisons between pairs of samples, are then available.

Friedman's test, which is also available in Minitab®, is clearly much simpler to perform in practice than the ANOVA method (Sections 3.8–3.10), though it does not have the latter's ability to study interaction effects (see Chapter 7).

6.8

Rank correlation

Ranking methods can also be applied to correlation problems. The Spearman rank correlation coefficient method described in this section is the oldest application of ranking methods in statistics, dating from 1904. Like other ranking methods, it is particularly useful when one or both of the sets of observations studied can be expressed only in terms of a rank order rather than in quantitative units. In the following example, the possible correlation between the sulphur dioxide concentrations in a series of table wines and their taste quality is investigated. The taste quality of a wine is not easily expressed quantitatively, but it is relatively simple for a panel of wine-tasters to rank the wines in order of preference. Examples of other attributes that are easily ranked, but not easily quantified, include the condition of experimental animals, the quality of laboratory accommodation, and the efficiency of laboratory staff. If either or both the sets of data under study should happen to be quantitative, then (in contrast to the methods described in Chapter 5) there is no need for them to be normally distributed. Like other non-parametric statistics, the Spearman rank correlation coefficient, r_s , is easy to determine and interpret. This is shown in the following example.

Example 6.8.1

Seven different table wines are ranked in order of preference by a panel of experts. The best wine is ranked 1, the next best 2, and so on. The sulphur dioxide content (in parts per million) of each wine is then determined by flow injection analysis with colorimetric detection. Use the following results to determine whether there is any relationship between perceived wine quality and sulphur dioxide content.

Wine	A	B	C	D	E	F	G
Taste ranking	1	2	3	4	5	6	7
SO ₂ content	0.9	1.8	1.7	2.9	3.5	3.3	4.7

The first step is to convert the sulphur dioxide concentrations from absolute values into ranks (any tied ranks can be averaged as described in previous sections):

Wine	A	B	C	D	E	F	G
Taste ranking	1	2	3	4	5	6	7
SO ₂ content	1	3	2	4	6	5	7

The differences, d_i , between the two ranks are then calculated for each wine. They are 0, -1, 1, 0, -1, 1, 0. The correlation coefficient, r_s , is then given by:

$$r_s = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)} \quad (6.8.1)$$

where n is the number of pairs.

In this example, r_s is $1 - (24/336)$, i.e. 0.929. Theory shows that, like the product-moment correlation coefficient, r_s can vary between -1 and +1. When $n = 7$, r_s must exceed 0.786 if the null hypothesis of no correlation is to be rejected at the significance level $P = 0.05$ (Table A.13). Here, we can conclude that there is a correlation between the sulphur dioxide content of the wines and their perceived quality: given the way the taste rankings were assigned, there is strong evidence that higher sulphur dioxide levels produce less palatable wines!

Another rank correlation method, due to Kendall, was introduced in 1938. It claims to have some theoretical advantages over the Spearman method, but is harder to calculate (especially when tied ranks occur) and is not so frequently used.

6.9 Non-parametric regression methods

In the discussion of linear regression methods in Chapter 5, the difficulties in dealing with suspect or outlier calibration points were described and the assumption of normally distributed y -direction errors was noted. To tackle these issues, several rapid and non-parametric methods for fitting a straight line to a set of points are available, of which the simplest is perhaps **Theil's 'incomplete' method** (the reason for this rather puzzling title is explained below).

Theil's method determines the slope of a regression line as the median of the slopes calculated from selected and evenly spaced pairs of points: the intercept of the line is the median of the intercept values calculated from the slope and the coordinates of the individual points.

The method assumes that a series of points $(x_1, y_1), (x_2, y_2)$, etc. is fitted by a line of the form $y = a + bx$. The first step in the calculation involves ranking the points in order of increasing x . If the number of points, x , is odd, the middle point, i.e. the median value of x , is deleted: the calculation always requires an even number of points. For any pair of points $(x_i, y_i), (x_j, y_j)$ where $x_j > x_i$, the slope, b_{ij} , of the line joining the points can be calculated from:

$$b_{ij} = \frac{(y_j - y_i)}{(x_j - x_i)} \quad (6.9.1)$$

Slopes b_{ij} are calculated for the pair of points (x_1, y_1) and the point immediately after the median x -value, for (x_2, y_2) and the second point after the median x -value, and so on until the slope is calculated for the line joining the point immediately before the median x -value with the last point. Thus, if the original data contained 11 points, five slopes would be estimated (the median point having been omitted). For eight original points there would be four slope estimates, and so on. These slope estimates are arranged in ascending order and their median is the estimated slope of the straight line. With this value of b , intercept values a_i are estimated for each point using the equation $y = a + bx$. Again the estimates of a are arranged in ascending order and the median value is chosen as the best estimate of the intercept of the line. The method is illustrated by the following example.

Example 6.9.1

The following results were obtained in a calibration experiment for the absorptiometric determination of a metal chelate complex:

Concentration, $\mu\text{g ml}^{-1}$	0	10	20	30	40	50	60	70
Absorbance	0.04	0.23	0.39	0.59	0.84	0.86	1.24	1.42

Use Theil's method to estimate the slope and the intercept of the best straight line through these points.

This calculation is simplified by the occurrence of an even number of observations, and by the fact that the x -values (i.e. the concentrations) occur at regular intervals and are already in ranking order. We thus calculate slope estimates from four pairs of points:

$$b_{15} = (0.84 - 0.04)/40 = 0.0200$$

$$b_{26} = (0.86 - 0.23)/40 = 0.0158$$

$$b_{37} = (1.24 - 0.39)/40 = 0.0212$$

$$b_{48} = (1.42 - 0.59)/40 = 0.0208$$

We now arrange these slope estimates in order, obtaining 0.0158, 0.0200, 0.0208, 0.0212. The median estimate of the slope is thus the average of 0.0200 and 0.0208, i.e. 0.0204. Using this value of b to estimate the intercept, a , the eight individual a_i values are:

$$a_1 = 0.04 - (0.0204 \times 0) = +0.040$$

$$a_2 = 0.23 - (0.0204 \times 10) = +0.026$$

$$a_3 = 0.39 - (0.0204 \times 20) = -0.018$$

$$a_4 = 0.59 - (0.0204 \times 30) = -0.022$$

$$a_5 = 0.84 - (0.0204 \times 40) = +0.024$$

$$a_6 = 0.86 - (0.0204 \times 50) = -0.160$$

$$a_7 = 1.24 - (0.0204 \times 60) = +0.016$$

$$a_8 = 1.42 - (0.0204 \times 70) = -0.008$$

Arranging these intercept estimates in order, we have $-0.160, -0.022, -0.018, -0.008, +0.016, +0.024, +0.026, +0.040$. The median estimate is $+0.004$. So the best straight line is given by $y = 0.0204x + 0.004$. The 'least-squares' line, calculated by the methods of Chapter 5, is $y = 0.0195x + 0.019$. Figure 6.4 shows that the two lines are quite similar when plotted. However, Theil's method has three distinct advantages over the least-squares approach: it does not assume that all the errors are in the y -direction; it does not assume that either the x - or y -direction errors are normally distributed; and it is not affected by the presence of outlying results. This last advantage is clearly illustrated by the point (50, 0.86) in the present example. It seems likely to be an outlier, but its value does not directly influence the Theil calculation, since neither b_{26} nor a_6 affects the median estimates of the slope and intercept respectively. In the least-squares calculation, however, this outlying point carries as much weight as the other points. This is reflected in the calculated results; the least-squares line has a smaller slope and passes closer to the outlier than does the non-parametric line.

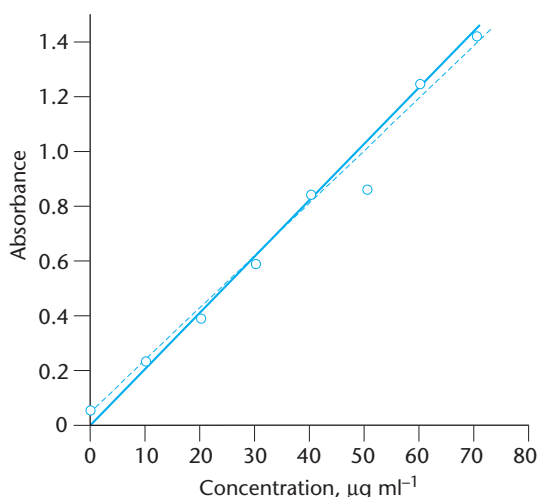


Figure 6.4 Straight-line calibration graph calculated by Theil's method (—) and by the least-squares method of Chapter 5 (---).

Unlike most non-parametric methods, Theil's method involves some tedious manual calculations, so a simple computer program or a spreadsheet macro is necessary in practice. Other, still more complex median-based methods have been described, including **Theil's 'complete' method, in which the median slope is obtained from the slopes of all the possible pairs of points.** An experiment such as the one in the example above with eight points generates 28 such pairs, though programs such as Minitab® readily perform the calculation of the median slope (in this case it is 0.0199, very similar to the value obtained from the incomplete method). Such methods are open to the objection that it may not be appropriate to assign equal importance to the slope of the line joining two adjacent points and the slope of the line joining two well-separated points. However, because each point on the graph is involved in the calculation several times, these methods may have the advantage of having a superior **breakdown point**, i.e. the proportion of outliers amongst the data they can handle. Non-parametric methods for fitting curves are also available, but these are beyond the scope of the present book.

6.10 Introduction to robust methods



We have seen (Section 6.1) that there is substantial evidence for the occurrence in the experimental sciences of *heavy-tailed* error distributions. These can be regarded as normal (Gaussian) distributions with the addition of outliers arising from gross errors, or as the result of the superposition of several normal distributions with similar means but different variances. In either case, and in other instances where the departure from a normal distribution is not great, it seems to be a waste of information to use non-parametric methods, which make no assumptions at all about the underlying error distribution. **A better approach would be to develop methods which do not entirely exclude suspicious results, but which reduce the weight given to such measurements.** This is the philosophy underlying the robust methods to be summarised in the rest of this chapter, which can be applied to repeated measurements, and to calibration/regression data. Many robust methods have been developed, so only a fairly brief survey of this important field is possible here; the reader is referred to the Bibliography for sources of further material.

Robust statistical methods can be applied to samples from symmetrical but heavy-tailed distributions, or when outliers may occur. They should not be applied in situations where the underlying distribution is bi-modal, multi-modal or very asymmetrical, for example log-normal distributions.

An obvious problem occurs in virtually all these methods. If we are to downgrade the significance of some of our measurements, one or more criteria are needed on which to base such decisions, but we cannot use such criteria unless we initially consider all the data, i.e. before we know whether some measurements are suspect or not. This problem is solved by using *iterative* methods: we estimate or guess a starting value for some property of the data, use this initial estimate with our weighting

criteria to arrive at a second estimate, then we re-apply our criteria, etc. Such methods are practicable only using a computer, though it must be stressed that many otherwise excellent suites of statistics software do not yet include programs for robust methods.

6.11 Simple robust methods: trimming and winsorisation

There are some very simple robust methods that do not require iterations because they arbitrarily eliminate, rather than down-weighting, a proportion of the data. For example, the **trimmed mean** for any set of data is found very easily by omitting r observations at the top and at the bottom of the range of measurements and calculating the mean of the remainder. This principle can be applied to the set of data in Example 3.7.2, which represented seven replicate measurements of nitrite ion in river water (mg l^{-1}):

0.380, 0.400, 0.401, 0.403, 0.408, 0.410, 0.413

The data have been arranged in numerical order for convenience: this emphasises that the obvious question is whether or not the measurement 0.380 is an outlier. If the value 0.380 is retained, the mean of the seven measurements is 0.4021, and their standard deviation is 0.0109. If, as Grubbs' and Dixon's tests (see Section 3.7) suggest, it is permissible to reject the result 0.380 (at $P = 0.05$), the mean and standard deviation then become 0.4058 and 0.0053 respectively. This comparison confirms, as we noted in Section 3.3, that the mean and (especially) the standard deviation are vulnerable to the occurrence of outliers. Now suppose that we omit the smallest (0.380) and the largest (0.413) of the measurements, and recalculate the mean. This produces a number technically known as the 14.28% trimmed mean, the percentage being calculated as $100r/n$, where r top and bottom measurements have been omitted from n results. The trimmed mean and its standard deviation are 0.4044 and 0.0044 respectively, clearly closer to values obtained after rejection of the suspect value and possible outlier 0.380. The robustness of this trimmed mean is obvious: it would have been the same, *whatever* the values taken by the smallest and largest results. But this example also illustrates the crudity of the trimmed mean method. Why should we omit the value 0.413, except for reasons of symmetry? Is it acceptable to calculate a statistic that *completely* ignores the suspect result(s) as well as one or more that seem to be valid? What percentage of the data should be removed by trimming? (Opinions on this last question have been divided, but in practice 10–25% trimming is common.)

A rather less arbitrary robust approach is provided by **winsorisation**. In its simplest form, this process *reduces* the importance of the measurements giving the largest positive and negative deviations from the mean or median by moving the measurements so that these deviations become equal to the next largest or smallest ones (or perhaps the third largest ones). The advantage of this approach is that if it is applied to a data set lacking suspect values or actual outliers, the effects on the calculated measures of location and spread are small, so no harm is done. For the nitrite data listed above this simple symmetrical winsorisation would move the initially

suspect value 0.380 to the next smallest value, 0.400, and the largest value, 0.413, would be moved to 0.410. The mean and standard deviation would then be 0.4046 and 0.0046 respectively, values similar to those obtained after rejection of the outlier 0.380, and to those obtained after trimming. The principles of winsorisation are applied in a more sophisticated way in Huber's approach to robust estimation (see Section 6.12).



A simple robust estimate of the standard deviation is provided by the IQR (see Section 6.2 above). For a normal error distribution, the IQR is ca. 1.35σ , i.e. 50% of the measurements occur in an overall range of 1.35σ . This relationship supplies a standard deviation estimate that is not affected by any value taken by the largest or smallest measurements. Unfortunately, the IQR is not a very meaningful concept for very small data sets. Moreover, and somewhat surprisingly, there are several different conventions for calculating it. For large samples the method chosen makes little difference, but for small samples the differences in the calculated IQR values are large, so the IQR has found little application in analytical chemistry.

6.12 Further robust estimates of location and spread

A more logical approach to robust estimation can be based on the concept of a **distance function**. Suppose we have a series of n replicate measurements, $x_1 \dots x_i \dots x_n$, and we wish to estimate μ , the mean of the 'reliable' results. The conventional estimate of μ , to which in this discussion we give the symbol $\hat{\mu}$, is the mean, which is found by minimising the sum of squares $\sum_i (x_i - \mu)^2$. (This sum of *squared* terms is the source of the sensitivity of the mean to large errors.) The expression $(x_i - \mu)^2$ is called a **distance function**, since it measures the distance of a point from μ . An obvious alternative distance function is $|x_i - \mu|$. One widely used method to test measurements for down-weighting (winsorisation) is to compare $|x_i - \mu|$ with $c\sigma$, where c is usually taken to be 1.5 and σ is a robust estimate of the standard deviation. We consider first the estimation of σ , and then discuss the down-weighting procedure used to estimate the robust mean, to which we give the symbol $\hat{\mu}$.

The robust variance estimate can be derived from a statistic related to the unfortunately abbreviated **median absolute deviation** (MAD!), which is calculated from:

$$\text{MAD} = \text{median} [|x_i - \text{median}(x_i)|] \quad (6.12.1)$$

where $\text{median}(x_i)$ is the median of all the x_i values, i.e. all the measurements.



The MAD is an extremely useful statistic: one rough method for evaluating outliers (x_0) is to reject them if $||x_0 - \text{median}(x_i)||/\text{MAD} > 5$. It can be shown that $\text{MAD}/0.6745$ is a useful robust estimate of σ , often called the **standard deviation based on MAD**, the standardised MAD, or SMAD. The SMAD is sometimes used unchanged during the iterative estimates of the robust mean, but in the following example we show the effect of iterating the robust estimate of σ , which is given the symbol $\hat{\sigma}$.

Example 6.12.1

Apply these techniques to the measurements discussed above (0.380, 0.400, 0.401, 0.403, 0.408, 0.410, 0.413).

First it is necessary to calculate the MAD. The median of these numbers is 0.403 (i.e. the fourth of the seven ordered values), so the individual deviations (without regard to their signs) are 0.023, 0.003, 0.002, 0, 0.005, 0.007 and 0.010. Rewriting these in numerical order we have: 0, 0.002, 0.003, 0.005, 0.007, 0.010 and 0.023. The MAD is the median of these seven numbers, i.e. 0.005, so $\hat{\sigma} = \text{MAD}/0.6745 = 0.005/0.6745 = 0.0074$, and $1.5 \hat{\sigma}$ is 0.0111.

We are now in a position to begin iterative estimates of $\hat{\mu}$. This process is begun by taking any reasonable estimate for $\hat{\mu}$ and calculating $|x_i - \hat{\mu}|$ values for each measurement. In this example, suppose the initial $\hat{\mu}$ -value is the median, 0.403. As we have seen, the individual deviations from this value are (in numerical order, but neglecting their signs) 0, 0.002, 0.003, 0.005, 0.007, 0.010 and 0.023. In the first iteration for $\hat{\mu}$ the original measurements are retained if these deviations from the median are ≤ 0.0111 . This applies to all the deviations listed except the last. In the event that the deviation is > 0.0111 , the original value in question is *changed* to become $\hat{\mu} - c\hat{\sigma}$ or $\hat{\mu} + c\hat{\sigma}$, depending on whether it was originally below or above the median respectively. In the present example the value 0.380, which gives rise to the large deviation of 0.023, has to be changed to $\hat{\mu} - c\hat{\sigma}$, i.e. $0.403 - 0.0111 = 0.3919$.

There is thus now a new data set, with the measurement 0.380 in the original set having been replaced by 0.3919. This new set of numbers is called a set of *pseudo-values* ($\tilde{x}_i$), and the calculation is repeated using this new set. The first step is to calculate the mean of the new values (note that although the initial value of $\hat{\mu}$ can be based on the mean or the median or any other sensible estimate, subsequent steps in the iteration always use the mean): this gives the result 0.4038. The individual deviations from this new estimate of $\hat{\mu}$ are, in ascending numerical order and without signs, 0.0008, 0.0028, 0.0038, 0.0042, 0.0062, 0.0092 and 0.0119. Since the range of the data set has been reduced by the process of winsorisation, we can also calculate a new value for $\hat{\sigma}$. This is given by $1.134s$, where s is the standard deviation of the pseudo-values, in this instance 0.0071. The number 1.134 is derived from the properties of the normal distribution and is used if c is 1.5. It provides a reminder that our robust method assumes that we are working with a sample taken from a normal population, but with added outliers or heavy tails. Thus $\hat{\sigma} = 0.0071 \times 1.134 = 0.0081$, and $1.5\hat{\sigma} = 0.0121$. In the next iteration of the process, only deviations from $\hat{\mu}$ larger than this need further adjustment. The largest deviation in the pseudo-data set is 0.0119, so no further steps are necessary in this case. (Had the largest deviation been > 0.0121 one or more further winsorisation steps would have been necessary, with the calculation of a new mean and standard deviation at each stage.)

The Huber method demonstrated in this example has produced a robust estimate of $\hat{\mu}$, 0.4038, with a robust standard deviation, $\hat{\sigma}$, of 0.0071. This robust mean is similar to the values obtained after rejection of the suspect measurement as an outlier, and after trimming or symmetrical winsorisation

(see above). The robust standard deviation is, as expected, smaller than the value calculated when the suspect value is included, but larger than that found when the suspect value is rejected. These results have been obtained after a single application of the Huber approach, but in practice, when the data set is larger and/or there is more than one suspect measurement, several iterations may be required. As noted above, it is possible to simplify the process using the initial $\hat{\sigma}$ estimate (0.0074 in this case) for all the iterations if a reliable robust estimate of spread is not important. A Minitab[®] algorithm is available for the full Huber procedure (see Bibliography).

This calculation deserves several comments. The first is that, like so many iterative procedures, it is much more tedious to describe and explain than it is to perform! The second point to note is that in this example we have applied the conventional value $c = 1.5$ to define the distance function. Other values for c have been explored: if it is too large potential outliers are not winsorised, and if it is too small only the measurements near the centre of the data carry any weight. Other weighting methods have also been suggested, including some that reject extreme outliers (for example, those $>4\sigma$ from the mean) entirely, and apply winsorisation to moderate outliers. Lastly it is worth re-emphasising that these robust methods do not have the worries and ambiguities of outlier tests. In the example just examined, Dixon's test (Section 3.7) suggested that the value 0.380 could be rejected as an outlier ($P = 0.05$), but the simple MAD-based test (see above) suggests that it should not, as $[|x_0 - \text{median}(x_i)|]/\text{MAD} = [|0.380 - 0.403|]/0.005 = 4.6$, below the (rough) critical value of 5. Such concerns and contradictions disappear in robust statistics, where the outliers are neither wholly rejected nor accepted unchanged, but accepted in a changed or down-weighted form.

6.13 Robust ANOVA

We have seen in Chapters 3, 4 and 5 that analysis of variance (ANOVA) is a very powerful method for separating and analysing the several sources of variation in a range of different types of experiment. However, in common with the example discussed in the previous section, conventional ANOVA methods rely on the use of squared terms, so the occurrence of suspect or outlying results can seriously distort the conclusions drawn from the calculations. One area where such problems frequently arise is in the performance of inter-laboratory trials (see Chapter 4). If a single material is sent for analysis to several laboratories, and each laboratory analyses it a number of times, ANOVA will be able to separate inter- and intra-laboratory sources of variation, but suspect values may occur in two ways: one or more laboratories may obtain results out of line with the rest, or one or more results obtained by a single laboratory may be suspect compared with the remaining results from that laboratory. It is in such situations that the use of robust ANOVA is extremely helpful. It is advisable to compare the results of classical and robust ANOVA in interpreting the data. The approach used in the calculations, which again are iterative, is

analogous to the one used in the example in the previous section, i.e. results that are more than 1.5 times the SMAD are down-weighted. In essence the mean values from the laboratories are taken, and robust estimates of their mean (i.e. the mean of the means) and the sample mean variance are calculated. The results from each laboratory are then taken and treated similarly: this gives a robust estimate of the variance due to the experimental error, σ_0^2 , and of the mean for each laboratory. The method used in Section 4.3 is then applied to get an estimate of the between-laboratory variance: this utilises the relationship that the between-sample mean square gives an estimate of $\sigma_0^2 + n\sigma_1^2$, where n is the number of measurements taken by each laboratory and σ_1^2 is the between-laboratory variance. This is an example of (robust) one-way ANOVA, i.e. there is only one source of variation apart from the inevitable experimental error. Robust ANOVA has also been applied where there is more than one such source (see Chapter 7). Facilities for these calculations are available as a short program (see Bibliography) but, like other robust procedures, many established statistics packages do not incorporate them.



6.14 Robust regression methods

The problems caused by possible outliers in regression calculations have been outlined in Sections 5.13 and 6.9, where rejection using a specified criterion and non-parametric approaches respectively were described. Robust approaches will evidently be useful in regression statistics as well as in the statistics of repeated measurements, and there has indeed been a rapid growth of interest in robust regression methods amongst analytical scientists. Most of the approaches used in the study of sets of replicate measurements have been adapted to regression problems. For example the conventional least-squares method, which seeks to minimise the sum of the squares of the y -residuals, has been modified so that the points giving the largest residuals are trimmed or winsorised.

In Section 6.9 we saw that a single suspect measurement has a considerable effect on the a and b values calculated for a straight line by the least-squares method. This is because, just as in the nitrite determination example studied in Section 6.12, the use of squared terms causes such suspect points to have a big influence on the sum of squares. A clear and obvious alternative is to seek to minimise the *median* of the squared residuals, which will be much less affected by large residuals. This **least median of squares (LMS)** method is very robust: its *breakdown point*, i.e. the proportion of outliers amongst the data that it can tolerate, is 50%, the theoretical maximum value. (If the proportion of 'suspect' results exceeds 50% it clearly becomes impossible to distinguish them from the 'reliable' results.) Simulations using data sets with deliberately included outliers show that this is a much better performance than that obtained with Theil's incomplete method. The LMS method also works well in the situation discussed in Section 5.11, where we wish to characterise the straight-line portion of a set of data which are linear near to the origin but non-linear at higher x - and y -values: this is because it effectively treats the points in the non-linear portion as outliers. LMS also handles both x - and y -direction outliers, a useful characteristic when the regression approach is used to compare analytical methods (see Section 5.9). The LMS slope and intercept can be calculated in a number of

ways, the most general of which uses a re-sampling technique similar to the *bootstrap* method, to be described in the next section. The disadvantage of this method is that it involves an iterative calculation which converges rather slowly, i.e. many iterations may be required before the estimated *a*- and *b*-values become more or less constant. A simpler method is to consider all the lines joining all the possible pairs of points on the graph (for *n* points there will be $n(n - 1)/2$ pairs). For each line the median of the squared residuals is determined. The smallest of these medians then defines the LMS line: of course this means that the latter will have the same slope and intercept as one of the individual lines, i.e. it will pass exactly through two of the points on the graph. The LMS method also provides a robust estimate of R^2 (see Section 5.12), which is given by:

$$R^2 = 1 - \left[\frac{\text{median}|r_i|}{\text{MAD}(y_i)} \right] \quad (6.14.1)$$

where r_i and y_i are respectively the y -residuals and the individual y -values of the points of the LMS plot.

Other robust regression methods are being increasingly used. The **iteratively re-weighted least squares** method begins with a straightforward least-squares estimate of the parameters of a line. The resulting residuals are then given different weights, usually via a **biweight** approach. The biweight method (which is also used in the treatment of sets of replicate data) rejects completely the points with very large residuals (e.g. those at least six times greater than the median residual value), while the remaining points are assigned weights which increase as their residuals get smaller. A *weighted* least-squares calculation (Section 5.10) is then applied to the new data set, and these steps are repeated until the values for *a* and *b* converge to stable levels. In this method convergence is usually fairly rapid.

Trials with a wide range of chemical data sets confirm that, while the conventional least-squares method described in Chapter 5 is the best approach if the measurements fulfil all its requirements and there are no outliers, the robust methods give better results if suspect values occur, and may thus be more appropriate in many real-world situations.



6.15 Re-sampling statistics

The development of high-speed computers provides access to a further group of very useful statistical methods, generally referred to as re-sampling techniques. These approaches again involve iterative calculations, but are mostly quite distinct from the robust methods described in the preceding sections. The best-known re-sampling method is known as the **bootstrap**. (The title refers to people or organisations succeeding from small beginnings by 'pulling themselves up by their bootstraps'.) The method operates as follows. Suppose we have a series of measurements $x_1, x_2, x_3, \dots, x_n$, and we wish to find a statistical parameter such as the 95% confidence limits of the mean. As shown in Chapter 2, we can do this by calculating the mean and the standard deviation of the data, and then applying the properties of the normal distribution. The bootstrap does not require the assumption of the normal (or any other) distribution, and involves taking a large number of samples of the same size

with replacement from the original data. So if the original data were (units irrelevant) 1, 2, 3, 4, 5, then one bootstrap sample of the same size might be 2, 4, 3, 5, 2. Note that the same measurement might well appear more than once in a given bootstrap sample, since the sampling is done with replacement. That is, when the number 2 has been taken as the first member of the sample, it is replaced in the original set of five measurements and is thus available for random selection again. As a result some of the measurements might not appear at all in a single bootstrap sample (e.g. 1 in this simple example). The number of possible bootstrap samples from n measurements is n^n , so even with only five measurements 3125 such samples are possible. In practice it is commoner to take only a few hundred samples. Once the samples have been taken, their means are determined and can be plotted as a histogram (or sorted into numerical order) by the computer. The 95% confidence limits can then be determined by inspection. For example, with 200 bootstrap samples the 95% confidence limits of the mean would be given by the 5th and 195th of the mean values sorted into numerical order. Confidence limits obtained in this way do not depend on any assumptions about the underlying error distribution, and would be expected to reflect (for example) any skewness in the measurements.

Example 6.15.1

The levels of a blood plasma protein in ten men were found to be (in numerical order) 1, 1, 2, 2, 3, 6, 8, 13, 14 and 18 mg 100 ml⁻¹. Find the 95% confidence limits for the mean of these values.

Inspection of the data shows that the values are skewed towards the lower end of the measurement range. Using the methods of Chapter 2 we find that the mean, $\bar{x}$, of the ten measurements is 6.80, and the standard deviation, s , is 6.20. Using Eq. (2.7.1) we find that the 95% confidence interval is $6.80 \pm (2.26 \times 6.20)/\sqrt{10} = 6.80 \pm 4.43$ so the 95% confidence limits are 2.40 and 11.23.

Using the bootstrap approach we take 500 samples with replacement, and obtain the results summarised in the histogram in Fig. 6.5. The mean value of these 500 samples is 6.84, and the 95% confidence limits, defined by the average of the 12th and 13th and the average of the 487th and 488th ordered values, are 3.5 and 10.9. The confidence interval defined by these values is narrower than the conventionally calculated one, and is asymmetrical relative to the mean, so both the interval and the histogram correctly reflect the negative skewness of the data. No assumptions have been made in the bootstrap method about the error distribution in the original data, i.e. the method is non-parametric.

The main principles of bootstrapping are easily understood, and the iterative calculations are simple, for example using a macro written for Minitab® (see Bibliography) or add-ins available for Excel®. The most important applications in analytical practice are likely to be in more complex situations than the one in the above example. We have noted that bootstrap principles can be used to generate the LMS line

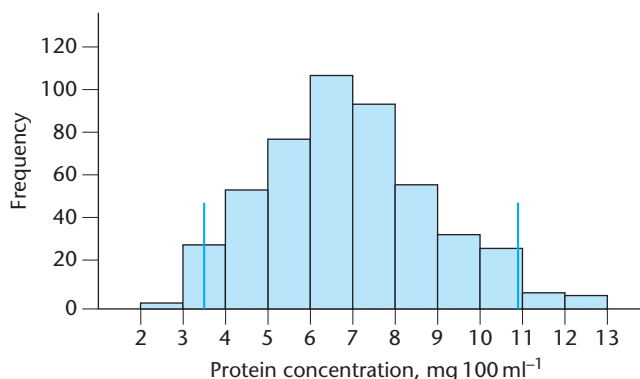


Figure 6.5 Histogram of 500 bootstrap samples from data in Example 6.15.1. The additional vertical lines show the 95% confidence limits from the bootstrap method.

when regression methods are applied, and they have been used in curve-fitting as applied to a range of spectroscopic and other analytical methods. Additional suggested uses have included the estimation of between-laboratory precision in method performance studies (see Chapter 4) and the determination of the best model to use in multivariate calibration (see Chapter 8).

6.16 Conclusions

The robust and non-parametric tests described in this chapter are only a small fraction of the total available number of such methods. The examples given exemplify their strengths and weaknesses. In many cases the speed and convenience of **non-parametric tests** give them a distinct advantage over conventional approaches, and they do not involve the assumption of a normal distribution. They are thus ideally suited to the preliminary examination of small numbers of measurements, and to quick calculations made – often without the need for statistical tables – while the analyst is at the bench or on the shop floor. They can also be used when three or more samples are studied, and in experiments using regression methods, though in each case the calculations are inevitably more complex. **The power** (i.e. the probability that a false null hypothesis is correctly rejected: see Section 3.13) **of a non-parametric method may be less than that of the corresponding parametric test, but the difference is only rarely serious.** For example, many comparisons have been made of the comparative powers of the Mann–Whitney *U*-test and the *t*-test, using various population distributions and sample sizes. The *U*-test performs very well in almost all circumstances and is only marginally less powerful than the *t*-test even when the data come from a normally distributed population. Many computer programs now include several non-parametric tests, so it is possible for a particular set of data to be evaluated rapidly by two or more methods.

Robust methods are not normally so easy to use, in view of the need in most cases for iterative calculations, but they represent the best way of tackling one of the most common and difficult problems for practising analysts, **the occurrence of suspicious or outlying results** superimposed on an error distribution which is

approximately normal. Their popularity is growing, at least in selected areas such as inter-laboratory comparisons, but they are still not implemented in many popular statistics programs.

Overall a great variety of significance tests – parametric non-parametric and robust – are available, and often the most difficult task in practice is to decide which method is best suited to a particular problem. The diagram in Appendix 1 is designed to make such choices easier, though inevitably it cannot cover all possible practical situations.

Bibliography and resources

Books

- Chatfield, C. 1995, *Problem Solving: A Statistician's Guide*, 2nd edn, Chapman & Hall, London. A thoroughly practical and useful book containing much valuable advice: lengthy appendices summarise the background theory.
- Conover, W.J., 1999, *Practical Non-parametric Statistics*, 3rd edn, John Wiley, New York. Probably the best-known general text on non-parametric methods.
- Efron, B. and Tibshirani, R.J., 1998, *An Introduction to the Bootstrap*, Chapman & Hall/CRC Press, New York. The classic in this area, written by two leading researchers and advocates of resampling methods.
- Rousseeuw, P.J. and Leroy, A.M., 2003, *Robust Regression and Outlier Detection*, WileyBlackwell, New York. An important and remarkably readable book with many illustrative examples.
- Sprent, P. and Smeeton, N.C., 2007, *Applied Nonparametric Statistical Methods*, 4th edn, Chapman & Hall/CRC Press, London. Covers a wide range of significance tests in a practical way, with a good discussion of robust techniques and bootstrapping and other resampling methods.
- Velleman, P.F. and Hoaglin, D.C., 1991, *Applications, Basics and Computing of Exploratory Data Analysis*, Duxbury Press, Boston, MA. An excellent introduction to IDA/EDA: computer programs in BASIC and FORTRAN are provided.

Software and Internet resources

- Analytical Methods Committee, 2001, *Robust Statistics: A Method of Coping with Outliers*, Royal Society of Chemistry, Cambridge. One of a series of technical briefs that can be downloaded from www.rsc.org. The site also provides a Minitab[®] implementation of the Huber algorithm and a stand-alone robust ANOVA program that can be freely downloaded.
- Analytical Methods Committee, 2001, *The Bootstrap: A Simple Approach to Estimating Standard Errors and Confidence Intervals when Theory Fails*, Royal Society of Chemistry, Cambridge. Another technical brief that can be downloaded from www.rsc.org. It shows how to write a Minitab[®] macro for the bootstrap calculation.
- www.resample.com. A website that provides much information and software on re-sampling statistics. There is an extensive and free online tutorial, and an online course. Software downloads include time-limited versions for Excel[®], with full versions available for on-line purchase.

Exercises

- 1 A titration was performed four times, with the results: 9.84, 9.91, 9.89, 10.20 ml. Calculate and comment on the median and the mean of these results.
- 2 The level of sulphur in batches of an aircraft fuel is claimed by the manufacturer to be symmetrically distributed with a median value of 0.10%. Successive batches are found to have sulphur concentrations of 0.09, 0.12, 0.10, 0.11, 0.08, 0.17, 0.12, 0.14 and 0.11%. Use the sign test and the signed rank test to check the manufacturer's claim.
- 3 The concentrations ($\text{g } 1000 \text{ ml}^{-1}$) of immunoglobulin G in the blood sera of ten donors are measured by radial immunodiffusion (RID) and by electroimmuno-diffusion (EID), with the following results:

Donor:	1	2	3	4	5	6	7	8	9	10
RID result:	1.3	1.5	0.7	0.9	1.0	1.1	0.8	1.8	0.4	1.3
EID result:	1.1	1.6	0.5	0.8	0.8	1.0	0.7	1.4	0.4	0.9

Are the results of the two methods significantly different?

- 4 Ten carbon rods used successively in an electrothermal atomic-absorption spectrometer were found to last for 24, 26, 30, 21, 19, 17, 23, 22, 25 and 25 samples. Test the randomness of these rod lifetimes.
- 5 After each drinking three pints of beer, five volunteers were found to have blood alcohol levels of 104, 79, 88, 120 and 90 $\text{mg } 100 \text{ ml}^{-1}$. A further set of six volunteers drank three pints of lager each, and were found to have blood alcohol levels of 68, 86, 71, 79, 91 and 66 $\text{mg } 100 \text{ ml}^{-1}$. Use Tukey's quick test or the Mann-Whitney *U*-test to investigate the suggestion that drinking lager produces a lower blood alcohol level than drinking the same amount of beer. Use the Siegel-Tukey method to check whether the spreads of these two sets of results differ significantly.
- 6 A university chemical laboratory contains seven atomic-absorption spectrometers (A-G). Surveys of the opinions of the research students and of the academic staff show that the students' order of preference for the instruments is B, G, A, D, C, F, E, and that the staff members' order of preference is G, D, B, E, A, C, F. Are the opinions of the students and the staff correlated?
- 7 Use Theil's incomplete method to calculate the regression line for the data of exercise 1 in Chapter 5.
- 8 The nickel levels in three samples of crude oil were determined (six replicates in each case) by atomic-absorption spectrometry, with the following results:

Sample	Measurements (Ni, ppm)					
1	14.2	16.8	15.9	19.1	15.5	16.0
2	14.5	20.0	17.7	18.0	15.4	16.1
3	18.3	20.1	16.9	17.7	17.9	19.3

Use the Kruskal-Wallis method to decide whether the nickel levels in the three oils differ significantly.

7

Experimental design and optimisation

Major topics covered in this chapter

- Experimental design: introduction and nomenclature
- Randomisation and blocking in experimental design
- Two-way analysis of variance
- Latin squares
- Interactions in experimental design and their estimation
- Identifying the important factors
- Complete and fractional (incomplete) factorial designs
- Optimisation: aims and principles
- Optimising a single factor
- Alternating variable search and steepest ascent optimisation methods
- Optimisation using simplex and simulated annealing approaches

7.1

Introduction

A recurring theme in this book has been that statistical methods are invaluable not only in the analysis of experimental data, but also in designing and optimising experiments. So many experiments fail in their purpose because they are not properly thought out and designed in the first place, and in such cases even the best data analysis procedures cannot compensate for the fatal lack of foresight and planning. Failure to design an experiment properly may mean that insufficient results – or at least insufficient results of the right kind – are obtained, or that an unnecessarily large number of measurements are taken. It is even possible to fall into both traps, i.e. to take too many measurements of one kind, and not enough of another. In all such cases the quality of the conclusions drawn will be reduced, and invaluable resources of time, samples, reagents, etc. wasted. This chapter introduces the basic concepts of experimental design and optimisation, and summarises the methods that should be carefully considered and used before any new experimental procedure is started.

In Chapter 3 we introduced the idea of a **factor**, i.e. any aspect of the experimental conditions which affects the result obtained from an experiment. Section 3.9 described the dependence of a fluorescence signal on one factor, the conditions in which a solution was stored. This factor was called a **controlled factor** because it could be altered at will by the experimenter. In another example in Section 4.3, in which salt from different parts of a barrel was tested for purity, the factor of interest, i.e. the part of the barrel from which the salt was taken, was chosen at random, so that factor was called an **uncontrolled factor**. In both these examples the factors were **qualitative** since their possible 'values' could not be arranged in numerical order. A factor for which the possible values can be arranged in numerical order, e.g. temperature or pH, is a **quantitative** one. The different values that a factor takes are called **levels**, whether the factor is quantitative or qualitative.

These two examples from previous chapters provided an introduction to the calculations involved in the analysis of variance (ANOVA). In each case only a single factor (apart from the inevitable measurement errors) was considered. But in reality, even that simple fluorescence experiment might be affected by many additional factors such as the ambient temperature, pH, ionic strength and chemical composition of the buffer in which the fluorophore is dissolved, the use of the same or a different fluorescence spectrometer for each measurement, and the dates, times and staff used in making the measurements. Any of these factors might have influenced the results to contribute to the observed behaviour, thus invalidating the conclusions concerning the effect of the storage conditions. Clearly, if the correct conclusions are to be drawn from an experiment, the various factors affecting the result must be identified in advance and, if possible, controlled.

The term **experimental design** is usually used to describe the stages of:

- 1 identifying the factors which may affect the result of an experiment;
- 2 designing the experiment so that the effects of uncontrolled factors are minimised;
- 3 using statistical analysis to separate and evaluate the effects of the various factors involved.

Before beginning any experimental design it is crucial to establish the *exact* purpose of the proposed experiment. This seems entirely obvious, but in many instances a failure to define sufficiently the object of an experiment has led to the failure of a subsequent design. If we wish to analyse a single drug in a urine extract by high-performance liquid chromatography, the resolution of the chromatogram we obtain will be affected by many factors, including those related to the mobile phase and stationary phase properties, the detector, and so on. But the best set of factor levels will not be the same if, for example, we wish to analyse the drug *and* its major metabolite, or the drug *and* as many as possible of its metabolites; so we need to be quite certain what the aim of the experiment is. In many cases an experiment will not be a completely new one, so there will be a great deal of literature on the factors studied by other experimenters aiming for the same result, and the factor levels used by them. It is of course legitimate to use such information to obtain guidance on the factors likely to be important, and their levels. At the same time experience shows that it is

very difficult to reproduce exactly in one laboratory the results obtained in another (see Chapter 4). In some cases this is because authors omit or obscure (deliberately or accidentally!) some vital aspect of their described experimental conditions; and in some cases the results in different laboratories may be affected significantly by uncontrolled factors, such as reagent or solvent purity, humidity, etc. Taking over the factor levels previously used by others is thus something to be attempted with caution.

Since many factors will affect experimental results quite complex experimental designs may be necessary. (This perhaps helps to explain why experimental design is not used as much as it should be.) The choice of the best practical levels of these factors, i.e. the optimisation of the experimental conditions, will also require detailed study. These methods, along with other multivariate methods covered in the next chapter, are amongst those given the general term **chemometrics**.

7.2 Randomisation and blocking

One of the assumptions of the one-way ANOVA calculations such as those in Chapters 3 and 4 is that the uncontrolled variation is truly random. This is also true of other ANOVA calculations. However, in measurements made over a period of time, variation in an uncontrolled factor such as pressure, temperature, photodecomposition of the sample and deterioration of apparatus may produce a trend in the results. As a result the errors due to uncontrolled variation are no longer random since the errors in successive measurements are *correlated*. This can lead to a systematic error in the results. This problem can be overcome by using the technique of **randomisation**. Suppose we wish to compare the effect of a single factor, the concentration of perchloric acid in aqueous solution, at three different levels or treatments (0.1 M, 0.5 M and 1.0 M) on the fluorescence intensity of quinine (which is widely used as a primary standard in fluorescence spectrometry). Let us suppose that four replicate intensity measurements are made for each treatment, i.e. in each perchloric acid solution. Instead of making the four measurements in 0.1 M acid, followed by the four in 0.5 M acid, then the four in 1 M acid, we make the 12 measurements in a random order, decided by using a table of random numbers. Each treatment is assigned a number for each replication as follows:

0.1 M				0.5 M				1 M			
01	02	03	04	05	06	07	08	09	10	11	12

(Note that each number has the same number of digits.) We then enter a random number table (see Table A.8) at an arbitrary point and read off pairs of digits, discarding the pairs 00, 13–99, and also discarding repeats. Suppose this gives the sequence 02, 10, 04, 03, 11, 01, 12, 06, 08, 07, 09, 05. Then, using the numbers assigned above, the measurements would be made at the different acid levels in the following order: 0.1 M, 1 M, 0.1 M, 0.1 M, 1 M, 0.1 M, 1 M, 0.5 M, 0.5 M, 0.5 M, 1 M, 0.5 M. This random order of measurement ensures that the errors at each acid level due to uncontrolled factors are random.

One disadvantage of complete randomisation is that it fails to take advantage of any natural subdivisions in the experimental material. Suppose, for example, that the 12 measurements in this example could not be made on the same day but were divided among four consecutive days. Using the same order as before would give:

Day 1	0.1 M,	1 M,	0.1 M
Day 2	0.1 M,	1 M,	0.1 M
Day 3	1 M,	0.5 M,	0.5 M
Day 4	0.5 M,	1 M,	0.5 M

With this design all the measurements using 0.1 M perchloric acid as the quinine solvent occur (by chance) on the first two days, whereas those using 0.5 M perchloric acid happen to be made on the last two days. If it seemed that there was a difference between the effects of these two acid levels it would not be possible to tell whether this difference was genuine or was caused by the effect of using the two treatments on different pairs of days. A better design is one in which each treatment is used once on each day, with the order of the treatments randomised on each day. For example:

Day 1	0.1 M,	1 M,	0.5 M
Day 2	0.1 M,	0.5 M,	1 M
Day 3	1 M,	0.5 M,	0.1 M
Day 4	1 M,	0.1 M,	0.5 M

A group of results containing one measurement for each treatment (here, the data obtained on each day) is known as a **block**, so this design is called a **randomised block** design. Further designs that do not use randomisation are considered in Section 7.4 below.

7.3 Two-way ANOVA

When two factors may affect the results of an experiment, two-way ANOVA must be used to study their effects. Table 7.1 shows the general form of a layout for this method. Each of the N measurements, x_{ij} , is classified under the terms **treatment levels** and **blocks**; the latter term was introduced in the previous section. (These terms are derived from the original use of ANOVA by R.A. Fisher in agricultural experiments, but are still generally adopted.) Using the conventional symbols there are c treatment levels and r blocks, so $N = cr$. The row totals (T_1, T_2 , etc.) and the column totals ($T_{.1}, T_{.2}$, etc.), and the grand total, T , are also given as they are used in the calculations. (The dots in the column and row totals remind us that in each case only one of the two factors is being studied.) The formulae for calculating the variation from the three different sources, viz. between-treatment, between-block and experimental error, are given in Table 7.2. Their derivation will not be given in detail here: the principles are similar to those for one-way ANOVA (Section 3.9) and the texts listed in the Bibliography provide further details.

Table 7.1 General form of table of two-way ANOVA

	Treatment						Row total
	1	2	...	j	...	c	
Block 1	x_{11}	x_{12}	...	x_{1j}	...	x_{1c}	$T_{1.}$
Block 2	x_{21}	x_{22}	...	x_{2j}	...	x_{2c}	$T_{2.}$
•	•	•	...	•	...	•	•
Block i	x_{i1}	x_{i2}	...	x_{ij}	...	x_{ic}	$T_{i.}$
•	•	•	...	•	...	•	•
Block r	x_{r1}	x_{r2}	...	x_{rj}	...	x_{rc}	$T_{r.}$
Column total	$T_{.1}$	$T_{.2}$		$T_{.j}$		$T_{.c}$	$T = \text{grand total}$

Table 7.2 Formulae for two-way ANOVA

Source of variation	Sum of squares	Degrees of freedom
Between-treatment	$\sum T_{.j}^2/r - T^2/N$	$c - 1$
Between-block	$\sum T_{i.}^2/c - T^2/N$	$r - 1$
Residual	by subtraction	by subtraction
Total	$\sum \sum x_{ij}^2 - T^2/N$	$N - 1$

As in one-way ANOVA, the calculations are simplified by the repeated appearance of the term T^2/N , and by the fact that the residual (random experimental) error is obtained by subtraction. Note that an estimate of this experimental error can be obtained, even if only one measurement is made at each combination of treatment level and block, as in the example below.

Example 7.3.1

In an experiment to compare the percentage efficiency of different chelating agents in extracting a metal ion from aqueous solution the following results were obtained:

Chelating agent				
Day	A	B	C	D
1	84	80	83	79
2	79	77	80	79
3	83	78	80	78

On each day a fresh solution of the metal ion (with a specified concentration) was prepared and the extraction performed with each of the chelating agents taken in a random order.

In this experiment the use of different chelating agents is a controlled factor since the chelating agents are chosen by the experimenter. The day on which the experiment is performed introduces uncontrolled variation, caused both by changes in laboratory temperature, pressure, etc., and by slight differences in the concentration of the metal ion solution, i.e. the day is a random factor. In previous chapters it was shown that ANOVA can be used either to test for a significant effect due to a controlled factor, or to estimate the variance of an uncontrolled factor. In this case, where both types of factor occur, two-way ANOVA can be used in both ways: (i) to test whether the different chelating agents have significantly different efficiencies, and (ii) to test whether the day-to-day variation is significantly greater than the variation due to the random error of measurement and, if it is, to estimate the variance of this day-to-day variation. As in one-way ANOVA, the calculations can be simplified by subtracting an arbitrary number from each measurement. The table below shows the measurements with 80 subtracted from each.

Blocks	Treatments				Row totals, T_i	T_i^2
	A	B	C	D		
Day 1	4	0	3	-1	6	36
Day 2	-1	-3	0	-1	-5	25
Day 3	3	-2	0	-2	-1	1
Column totals, T_j	6	-5	3	-4	0 = Grand total, T	$\sum T_i^2 = 62$
$\sum T_j^2$	36	25	9	16	$\sum T_j^2 = 86$	

We also have $r = 3$, $c = 4$, $N = 12$, and $\sum \sum x_{ij}^2 = 54$

The calculation of the ANOVA table gives the following results:

Source of variation	Sum of squares	Degrees of freedom	Mean square
Between-treatment	$86/3 - 0^2/12 = 28.6667$	3	$28.6667/3 = 9.5556$
Between-block	$62/4 - 0^2/12 = 15.5$	2	$15.5/2 = 7.75$
Residual	by subtraction = 9.8333	6	$9.8333/6 = 1.6389$
Total	$54 - 0^2/12 = 54.0$	11	

Since the residual mean square is obtained by subtraction, it is important to use many significant figures initially in the table to avoid significant errors if this difference is small.

We can verify that this calculation does indeed separate the between-treatment and between-block effects. For example, if all the values in one block are increased by a fixed amount and the sums of squares re-calculated, the between-block and total sums of squares are changed, but the between-treatment and residual sums of squares are not.

If there is no difference between the efficiencies of the chelators, and no day-to-day variation, then all three mean squares should give an estimate of σ_0^2 , the variance of the random variation due to experimental error (cf. Section 3.9). As in one-way ANOVA, the F -test is used to see whether the variance estimates differ significantly. Comparing the between-treatment mean square with the residual mean square gives:

$$F = \frac{9.5556}{1.6389} = 5.83$$

From Table A.3 the critical value of $F_{3,6}$ (1-tailed, $P = 0.05$) is 4.76, so we find that there is a difference between the two variances, i.e. between the efficiencies of the chelating agents, at the 5% level. Comparing the between-block (i.e. between-day) and residual mean squares gives:

$$F = \frac{7.75}{1.6389} = 4.73$$

In this case the critical value of $F_{2,6}$ (one-tailed, $P = 0.05$) is 5.14, so there is no significant difference between the days. Nevertheless the between-block mean square is considerably larger than the residual mean square, and had the experiment been 'unblocked', so that these two effects were combined in the estimate of experimental error, the experiment would probably have been unable to detect whether different *treatments* gave significantly different results. If the difference between days *had* been significant it would indicate that other factors such as temperature, pressure and the preparation of the solution were having an effect. It can be shown that the between-block mean square gives an estimate of $\sigma_0^2 + c\sigma_b^2$, where σ_b^2 is the variance of the random day-to-day variation. Since the residual mean square gives an estimate of σ_0^2 , an estimate of σ_b^2 can be obtained.

This example illustrates clearly the benefits of considering carefully the design of an experiment before it is performed. Given a blocked and an unblocked experiment with the same number of measurements in each, the former is more sensitive and yields more information. The sensitivity of the experiment depends on the size of the random variation: the smaller this is, the smaller the difference between the treatments that can be detected. In an unblocked experiment the random variation would be larger since it would include a contribution from the day-to-day variation, so the sensitivity would be reduced.

The two-way ANOVA calculation performed above is based on the assumption that the effects of the chelators and the days, if any, are *additive*, not *interactive*. This point is discussed further in Section 7.5.

7.4 Latin squares and other designs

In some experimental designs it is possible to take into account an extra factor without a large increase in the number of experiments performed. A simple example is provided by the study of the chelating agents in the previous section, where an uncontrolled factor not taken into account was the time of day at which the

measurements were made. Systematic variation during the day due to deterioration of the solutions or an increase in laboratory temperature could have produced a trend in the results.

In such cases, when there is an *equal number* of blocks and treatments (this was not the case in the previous example) it is possible to use an experimental design which allows the separation of such an additional factor. Suppose that the treatments are simply labelled A, B and C, then a possible design would be:

Day 1	A	B	C
Day 2	C	A	B
Day 3	B	C	A

This block design, in which each treatment appears once in each row and once in each column, is known as a **Latin square**. It allows the separation of the variation into the between-treatment, between-block, between-time-of-day and random experimental error components. More complex designs are possible which remove the constraint of equal numbers of blocks and treatments. If there are more than three blocks and treatments a number of Latin square designs are obviously possible (one can be chosen at random). Experimental designs of the types discussed so far are said to be **cross-classified** designs, as they provide for measurements for every possible combination of the factors. But in other cases (for example when samples are sent to different laboratories, and are analysed by two or more different experimenters in each laboratory) the designs are said to be **nested** or **hierarchical**, because the experimenters do not make measurements in laboratories other than their own. Mixtures between nested and cross-classified designs are also possible.

7.5

Interactions

In the example in Section 7.3 the two-way ANOVA calculations used assumed that the effects of the two factors (chelating agents and days) were additive. This means that if, for example, we had had only two chelating agents, A and B, and studied them both on each of two days, the results might have been something like:

	Chelating agents	
	A	B
Day 1	80	85
Day 2	73	78

That is, using chelating agent B instead of A produces an increase of 5% in extraction efficiency on both days; and the extraction efficiency on day 2 is lower than that on day 1 by 7%, whichever chelating agent is used. In a simple table of the kind shown, this means that when three of the measurements are known, the fourth can easily be deduced. Suppose, however, that the extraction efficiency on day 2 for chelating agent B had been 75% instead of 78%. Then we would conclude that the difference between the two agents depended on the day of the measurements, or that the difference between the results on the two days depended on which agent was in use.

That is, there would be an **interaction** between the two factors affecting the results. Such interactions are in practice extremely important: a recent estimate suggests that at least two-thirds of the processes in the chemical industry are affected by interacting, as opposed to additive, factors.

Unfortunately the detection of interactions is not quite as simple as the above example implies, as the situation is complicated by the presence of random errors. If a two-way ANOVA calculation is applied to the very simple table above, the residual sum of squares is found to be zero. But if the extraction efficiency for agent B on day 2 had been, say, 77% instead of 78% because of experimental error, or indeed if any of the four values in the table had changed, this is no longer so. With this design of experiment we cannot tell whether a non-zero residual sum of squares is due to random measurement errors, to an interaction between the factors, or to both effects. **To resolve this problem the measurement in each cell must be made at least twice.** It is important that the replicate measurements must be performed in such a way that all the sources of random error are present in every case. For example, if different equipment items have been used in experiments on the different chelating agents, then the duplicate or replicate measurements applied to each chelating agent on each day must also use different apparatus. If the *same* equipment is used for these replicates, the random error in the measurements may be underestimated. If the replicates are performed properly the method by which the interaction sum of squares and the random error can be separated is illustrated by the following new example.

Example 7.5.1

In an experiment to investigate the validity of a material as a liquid absorbance standard, the value of the molar absorptivity, ϵ , of solutions of three different concentrations was calculated at four different wavelengths. Two replicate absorbance measurements were made for each combination of concentration and wavelength. The order in which the measurements were made was randomised.

The results are shown in Table 7.3; for simplicity of calculation the calculated ϵ values have been divided by 100.

Table 7.4 shows the result of the Minitab[®] calculation for these results. (NB In using this program for two-way ANOVA calculations with interaction, it is essential to avoid the option for an additive model: the latter excludes the desired interaction effect. Excel[®] also provides facilities for including interaction effects in two-way ANOVA.) Here we explain in more detail how this ANOVA table is obtained. The first stage of the calculation is to find the cell totals. This

Table 7.3 Molar absorptivity values for a possible absorbance standard

Concentration, g l ⁻¹	Wavelength, nm			
	240	270	300	350
0.02	94, 96	106, 108	48, 51	78, 81
0.04	93, 93	106, 105	47, 48	78, 78
0.06	93, 94	106, 107	49, 50	78, 79

Table 7.4 Minitab® output for Example 7.5.1

Two-way analysis of variance			
Analysis of Variance for Response			
Source	DF	SS	MS
Conc.	2	12.33	6.17
Wavelength	3	11059.50	3686.50
Interaction	6	2.00	0.33
Error	12	16.00	1.33
Total	23	11089.83	

is done in Table 7.5, which also includes other quantities needed in the calculation. As before, $T_{i.}$ denotes the total of the i th row, $T_{.j}$ the total of the j th column and T the grand total.

As before, the between-row, between-column and total sums of squares are calculated. Each calculation requires the term T^2/nrc (where n is the number of replicate measurements in each cell, in this case 2, r is the number of rows and c is the number of columns). This term is sometimes called the **correction term**, C . Here we have:

$$C = \frac{T^2}{nrc} = \frac{1966^2}{2 \times 3 \times 4} = 161\,048.17$$

The sums of squares are now calculated:

$$\begin{aligned} \text{Between-row sum of squares} &= \sum_i T_{i.}^2 / nc - C \\ &= \frac{1\,288\,484}{2 \times 4} - 161\,048.17 \\ &= 12.33 \end{aligned}$$

with $r - 1 = 2$ degrees of freedom

$$\begin{aligned} \text{Between-column sum of squares} &= \sum_j T_{.j}^2 / nr - C \\ &= \frac{1\,032\,646}{2 \times 3} - 161\,048.17 \\ &= 11\,059.50 \end{aligned}$$

Table 7.5 Cell totals for two-way ANOVA calculation

	240 nm	270 nm	300 nm	350 nm	$T_{i.}$	$T_{i.}^2$
0.02 g l ⁻¹	190	214	99	159	662	438 244
0.06 g l ⁻¹	186	211	95	156	648	419 904
0.10 g l ⁻¹	187	213	99	157	656	430 336
$T_{.j}$	563	638	293	472	$T = 1966$	
$T_{.j}^2$	316 969	407 044	85 849	222 784		
$\sum_j T_{.j}^2 = 1\,032\,646$ $\sum_i T_{i.}^2 = 1\,288\,484$						

with $c - 1 = 3$ degrees of freedom.

$$\text{Total sum of squares} = \sum x_{ijk}^2 - C$$

where x_{ijk} is the k th replicate in the i th row and j th column, i.e. $\sum x_{ijk}^2$ is the sum of the squares of the individual measurements in Table 7.3.

$$\begin{aligned}\text{Total sum of squares} &= 172\,138 - 161\,048.17 \\ &= 11\,089.83\end{aligned}$$

with $nrc - 1 = 23$ degrees of freedom.

The variation due to random error (usually called the **residual variation**) is estimated from the within-cell variation, i.e., the variation between replicates.

The residual sum of squares = $\sum x_{ijk}^2 - \sum T_{ij}^2/n$, where T_{ij} is the total for the cell in the i th row and j th column, i.e. the sum of the replicate measurements in the i th row and j th column.

$$\begin{aligned}\text{Residual sum of squares} &= \sum x_{ijk}^2 - \sum T_{ij}^2/n \\ &= 172\,138 - (344\,244/2) \\ &= 16\end{aligned}$$

with $(n - 1)rc = 12$ degrees of freedom.

The interaction sum of squares and number of degrees of freedom can now be found by subtraction. Each source of variation is compared with the residual mean square to test whether it is significant.

- 1 **Interaction.** This is obviously not significant since the interaction mean square is less than the residual mean square.
- 2 **Between-column** (i.e. between-wavelength). This is highly significant since we have:

$$F = \frac{3686.502}{1.3333} = 2765$$

The critical value of $F_{3,12}$ is 3.49 ($P = 0.05$). In this case a significant result would be expected since absorbance is wavelength-dependent.

- 3 **Between-row** (i.e. between-concentration). We have:

$$F = \frac{6.17}{1.3333} = 4.63$$

The critical value of $F_{2,12}$ is 3.885 ($P = 0.05$), indicating that the between-row variation is too great to be accounted for by random variation. So the solution is not suitable as an absorbance standard. Figure 7.1 shows the molar absorptivity plotted against concentration, with the values for the same wavelength joined by straight lines. This illustrates the results of the analysis above in the following ways:

- the lines are parallel, indicating no interaction;
- the lines are not quite horizontal, indicating that the molar absorptivity varies with concentration;
- the lines are at different heights on the graph, indicating that the molar absorptivity is wavelength-dependent.

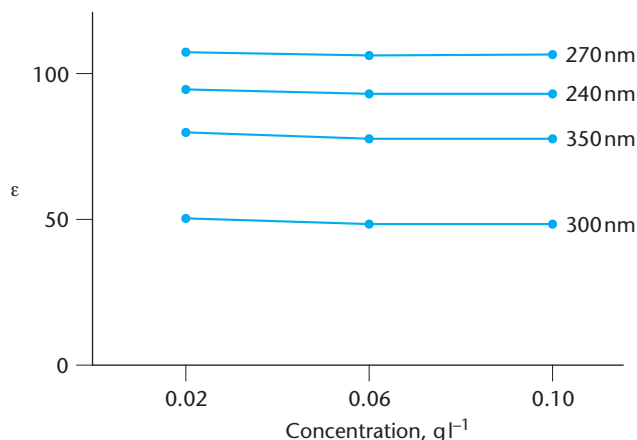


Figure 7.1 Relationships in the two-way ANOVA example (Example 7.5.1).

The formulae used in the calculation above are summarised in Table 7.6.

In this experiment both factors, i.e. the wavelength and the concentration of the solution, are controlled factors. In analytical chemistry an important application of ANOVA is the investigation of two or more controlled factors and their interactions in optimisation experiments. This is discussed in Section 7.7.

As discussed in Section 4.11, another important application of ANOVA is in collaborative investigations of precision and accuracy between laboratories. In full-scale method performance studies (collaborative trials) several different types of sample are sent to a number of laboratories, and each laboratory performs a number of replicate analyses on each sample. Statistical analysis of the results would yield the following sums of squares: between-laboratory, between-samples, laboratory-sample interaction and residual. The purpose of such an experiment would be to test first whether there is any interaction between laboratory and sample, i.e. whether some laboratories showed unexpectedly high or low results for some samples. This is done by comparing the interaction and residual sums of squares. If there is no interaction, then we could test whether the laboratories obtained significantly different results, i.e. if there is any systematic difference between laboratories. If there is, then the inter-laboratory variance can be estimated. However, if there is a significant interaction, the testing for a significant difference between laboratories has little relevance.

Table 7.6 Formulae for two-way ANOVA with interaction

Source of variation	Sum of squares	Degrees of freedom
Between-row	$\sum_i T_{i.}^2 / nc - C$	$r - 1$
Between-column	$\sum_j T_{.j}^2 / nr - C$	$c - 1$
Interaction	by subtraction	by subtraction
Residual	$\sum x_{ijk}^2 - \sum T_{ij.}^2 / n$	$rc(n - 1)$
Total	$\sum x_{ijk}^2 - C$	$rcn - 1$

For two-way ANOVA to be valid the following conditions must be fulfilled (see also Section 3.10):

- The random error is the same for all combinations of the levels of the factors.
- The random errors are approximately normally distributed.

7.6

Identifying the important factors: factorial designs

In Section 7.1 we used the examples of fluorescence measurements in solution, and the analysis of a drug using high-performance liquid chromatography, to demonstrate that in many analytical techniques the response of the measurement system depends on several or many factors under the control of the operator. For a particular application it is important to set the levels of these factors such that the system is giving the best possible results. The process of finding the most suitable factor levels is known as **optimisation** (see Sections 7.8–7.12). Before an optimisation process can begin we must determine which factors, and which interactions between them, are important in affecting the response: it is also obviously valuable to know which factors have little or no effect, so that time and resources are not wasted on unnecessary experiments.

An experiment such as the example in the previous section, where the response variable (i.e. the molar absorptivity) is measured for all possible combinations of the chosen factor levels, is known as a **complete factorial design**. This type of design is quite different from an approach which is perhaps more obvious, a **one-at-a-time design**, in which the effect of changing the level of a single factor on the response, with all the other factors held at constant levels, is investigated for each factor in turn. There are two reasons for preferring a factorial design to a one-at-a-time approach. The fundamental reason is that a suitable factorial design can detect and estimate the interactions between the factors, while a one-at-a-time methodology cannot. Second, even if interactions are absent, a factorial design needs fewer measurements than the one-at-a-time approach to give the same precision. This is again exemplified by the molar absorptivity experiment, in which 24 measurements were used to estimate the effect of varying the wavelength, and the same 24 used to estimate the effect of varying the concentration. In a one-at-a-time experiment, first the concentration would have been fixed and, to obtain the same precision for the effect of varying the wavelength, six measurements would have been needed *at each wavelength*, i.e. 24 in all. Then the wavelength would have been fixed and another eight measurements made at each of the three different concentrations, making a total of 48 altogether. In general, for k factors, a one-at-a-time approach involves k times as many measurements as a factorial one with the same precision. However, as we shall see, complete factorial designs still involve a large number of experiments if the number of factors is substantial (Section 7.7).

In many factorial designs each factor is studied at just two levels, usually called 'low' and 'high', to minimise the need for numerous experiments. For this reason, two-level designs are sometimes called **screening designs**. For a quantitative variable the terms 'low' and 'high' have their usual meaning. The exact choice of levels is determined mainly by the experience and knowledge of the experimenter and the physical constraints of the system, e.g. in aqueous solutions only temperatures in the range 0–100 °C are practicable. Some problems affecting the choice of

Table 7.7 Complete factorial design for three factors

Combination	A	B	C	Response
1	–	–	–	y_1
a	+	–	–	y_2
b	–	+	–	y_3
c	–	–	+	y_4
bc	–	+	+	y_5
ac	+	–	+	y_6
ab	+	+	–	y_7
abc	+	+	+	y_8

levels are discussed below. For a qualitative variable ‘high’ and ‘low’ refer to a pair of different conditions, such as the presence or absence of a catalyst, the use of mechanical or magnetic stirring, or taking the sample in powdered or granular form. Since we have already considered two-factor experiments in some detail we will turn to one with three factors: A, B and C. This means that there are $2^3 = 8$ possible combinations of factor levels, as shown in Table 7.7. A plus sign denotes that the factor is at the high level and a minus sign that it is at the low level. (In three-level designs, the symbols +1, 0 and –1 are often used to denote the levels.) The first column gives a notation often used to describe the combinations, where the presence of the appropriate lower case letter indicates that the factor is at the high level and its absence that the factor is at the low level. The number 1 is used to indicate that all factors are at the low level. Sometimes the experiments in tables summarising experimental designs are simply given in numerical order (see Section 7.7).

The method by which the effects of the factors and their interactions are estimated is illustrated by the following example.

Example 7.6.1

In a high-performance liquid chromatography experiment, the dependence of the retention parameter, k' , on three factors was investigated. The factors were pH (factor P), the concentration of a counter-ion (factor T) and the concentration of the organic solvent in the mobile phase (factor C). Two levels were used for each factor and two replicate measurements made for each combination. The measurements were randomised. The table below gives the average value for each pair of replicates.

Combination of factor levels	k'
1	4.7
p	9.9
t	7.0
c	2.7
pt	15.0
pc	5.3
tc	3.2
ptc	6.0

Effect of individual factors

The effect of changing the level of P can be found from the average difference in response when P changes from high to low level with the levels of C and T fixed. There are four pairs of responses that give an estimate of the effect of the level of P as shown in the table below.

Level of C	Level of T	Level of P		Difference
		+	−	
−	−	9.9	4.7	5.2
+	−	5.3	2.7	2.6
−	+	15.0	7.0	8.0
+	+	6.0	3.2	2.8
Total = 18.6				

$$\text{Average effect of altering the level of P} = 18.6/4 = 4.65$$

The average effects of altering the levels of T and C can be found similarly to be:

$$\text{Average effect of altering the level of C} = -4.85$$

$$\text{Average effect of altering the level of T} = 2.15$$

Interaction between two factors

Consider now the two factors P and T. If there is no interaction between them, then the change in response between the two levels of P should be independent of the level of T. The first two figures in the last column of the table above give the change in response when P changes from high to low level with T at low level. Their average is $(5.2 + 2.6)/2 = 3.9$. The last two figures in the same column give the effect of changing P when T is at high level. Their average is $(8.0 + 2.8)/2 = 5.4$. If there is no interaction and no random error (see Section 7.5) these estimates of the effect of changing the level of P should be equal. The convention is to take half their difference as a measure of the interaction:

$$\text{Effect of PT interaction} = (5.4 - 3.9)/2 = 0.75$$

It is important to realise that this quantity estimates the degree to which the effects of P and T are not additive. It could equally well have been calculated by considering how far the change in response for the two levels of T is independent of the level of P.

The other interactions are calculated in a similar fashion:

$$\text{Effect of CP interaction} = -1.95$$

$$\text{Effect of CT interaction} = -1.55$$

Interaction between three factors

The PT interaction calculated above can be split into two parts according to the level of C. With C at low level the estimate of interaction would be

$(8.0 - 5.2)/2 = 1.4$ and with C at high level it would be $(2.8 - 2.6)/2 = 0.1$. If there is no interaction between all three factors and no random error, these estimates of the PT interaction should be equal. The three-factor interaction is estimated by half their difference $[= (0.1 - 1.4)/2 = -0.65]$. The three-factor interaction measures the extent to which the effect of the PT interaction and the effect of C are not additive: it could equally well be calculated by considering the difference between the PC estimates of interaction for low and high levels of T or the difference between the TC estimates of interaction for low and high levels of P.

These results are summarised in the table below:

	Effect
Single factor (main effect)	
P	4.65
T	2.15
C	-4.85
Two-factor interactions	
TP	0.75
CT	-1.55
CP	-1.95
Three-factor interactions	
PTC	-0.65

These calculations have been presented in some detail in order to make the principles clear. An algorithm due to Yates, which is available as an Excel® macro, greatly simplifies the calculation in practice.

In order to test which effects, if any, are significant, ANOVA may be used (provided that there is homogeneity of variance). It can be shown that in a two-level experiment, such as this one, the required sums of squares can be calculated from the estimated effects by using

$$\text{Sum of squares} = N \times \frac{(\text{estimated effect})^2}{4}$$

where N is the total number of measurements, including replicates. In this case N is 16 since two replicate measurements were made for each combination of factor levels. The calculated sums of squares are given below.

Factor(s)	Sum of squares
P	86.49
T	18.49
C	94.09
PT	2.25
TC	9.61
PC	15.21
PCT	1.69

It can be shown that each sum of squares has one degree of freedom. Since the mean square is given as usual by:

$$\text{Mean square} = \frac{\text{sum of squares}}{\text{number of degrees of freedom}}$$

each mean square is simply the corresponding sum of squares. To test for the significance of an effect, the mean square is compared with the error (residual) mean square. This is calculated from the individual measurements by the method described in the molar absorptivity example in Section 7.5. In the present experiment the calculated residual mean square was 0.012 with eight degrees of freedom. Testing for significance, starting with the highest-order interaction, we have for the PTC interaction:

$$F = \frac{1.69}{0.012} = 141$$

which is obviously significant. If there is interaction between all three factors there is no point in testing whether the factors taken in pairs or singly are significant, since all factors will have to be considered in any optimisation process. A single factor should be tested for significance only if it does not interact with other factors.

One serious problem with a complete factorial experiment such as this is that the number of experiments required rises rapidly with the number of factors. For k factors at two levels, with two replicates for each combination of levels to allow the estimation of experimental errors, 2^{k+1} experiments are necessary. So only five factors generate as many as 64 experiments. Such a workload is normally unacceptable: some more practicable alternative approaches are outlined in the next section.

Another obvious problem in using a factorial design is that for factors which are continuous variables, the observed effect depends on the high and low levels used. If the chosen high and low levels of a factor are too close to each other the effect of the factor may be found to be not significant, despite the fact that over the whole possible range of factor levels its effect may be substantial. On the other hand, if the factor levels are chosen to be too far apart the responses may fall on either side of a maximum value and thus give a difference that is not significant. An experimenter's experience and prior literature may help to avoid these difficulties. The problem can also be tackled in principle by studying each factor at three rather than two levels. Such designs are sometimes called **response surface designs**, as they can be used to model curved response surfaces. The main problem with three-level designs is, as expected, the large number of experiments involved. A complete factorial design with only two factors studied at three levels requires $3^2 = 9$ experiments, and with more than two factors the size of such a design would often be quite impracticable.

When a factorial design involves more than three factors some economy in the number of experiments is possible by assuming that three-way and higher-order interactions are negligible. The sums of squares corresponding to these interactions can then be combined to give an estimate of the residual sum of squares, and replicate measurements are no longer necessary. The rationale for this approach is that

higher-order effects are usually much smaller than main effects and two-factor interaction effects. If higher-order interactions can be assumed negligible, a suitable fraction of all possible combinations of factor levels may be sufficient to provide an estimate of the main and two-factor interaction effects. As mentioned in Section 4.12, such an experimental design is called a **fractional (or incomplete) factorial design**.

7.7 Fractional factorial designs



For a given number of factors, fractional factorial designs use one-half, one-quarter, one-eighth, etc. of the number of experiments that would be used in the complete factorial design. The individual experiments in the fractional design must be carefully chosen to ensure that they give the maximum information. A three-factor design provides a simple example that can be illustrated graphically. In this case the complete factorial design involves, as we have seen, eight experiments, which can be represented by the vertices of a cube (Fig. 7.2a): in this view the lower left-hand vertex, for example, corresponds to the experiment in which the three factors A, B and C are all at their low level. A half-factorial design for three factors involves $2^{k-1} = 4$ experiments, and these can be represented by taking four corners of the cube that form a tetrahedron (Fig. 7.2b), so that each factor is represented twice at its high level and twice at its low level. (Of course, it would be equally legitimate to perform the four experiments represented by the other four vertices of the cube.) This half-factorial design provides information only on the main effects of the factors A, B and C and gives no information on any interactions between them.

As the number of factors studied increases, additional complications arise. If we study four factors, A–D, again at two levels, then a full factorial design without duplication will involve 16 experiments and a half-factorial design will require 8 experiments. Four factor designs are not easily shown graphically but the half-factorial design can be summarised as in Table 7.8.

Again the factor levels are chosen so that each factor is studied at its high and low levels in half of the experiments (which should be performed in a random order to minimise the effects of uncontrolled factors). The main effect for each factor is

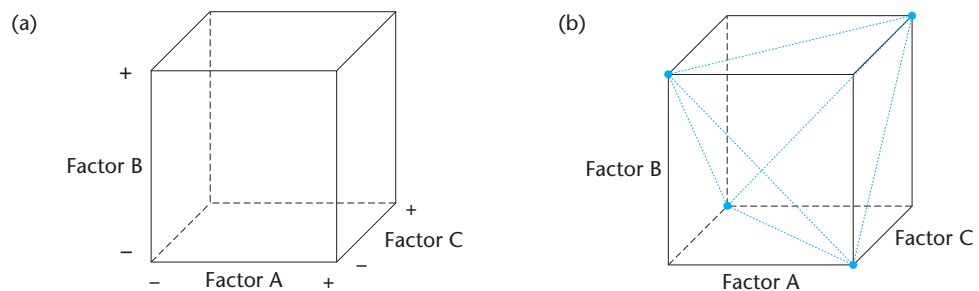


Figure 7.2 Representation of (a) a complete factorial and (b) a half-factorial design for three factors, each at two levels.

Table 7.8 Half-factorial design for four factors at two levels

Experiment	A	B	C	D	Response
1	+	+	+	+	y_1
2	+	+	−	−	y_2
3	+	−	+	−	y_3
4	+	−	−	+	y_4
5	−	+	+	−	y_5
6	−	+	−	+	y_6
7	−	−	+	+	y_7
8	−	−	−	−	y_8

easily found. For example the effect of factor C is given by $(y_1 + y_3 + y_5 + y_7 - y_2 - y_4 - y_6 - y_8)/4$. Unlike the half-factorial design for three factors, these eight experiments also give some information on some of the interactions between the factors A to D. But this is where the complications occur. It can be shown, for example, that the expression for the three-fold interaction between factors A, B and D (usually just called ABD for short) is the *same* as the formula above for the main effect of C, so when the value of the expression (i.e. $y_1 + y_3 + \dots$ as above) is calculated it is actually the *sum* of C and ABD. Such pairs of effects are called **aliases** of each other and the problem is called **confounding**. In the design above the main effect for each factor is confounded with the three-fold interaction of the other three factors, and confounding is a general feature of 2^{k-1} designs. In some cases the confounding may not be important (in our example, the three-fold interactions might be neglected entirely) but in other cases it will require careful study. However, it is not surprising that complications of this kind arise. With four factors, there are four main effects, six two-factor interactions, four three-factor interactions, and one four-factor interaction: we can hardly expect to resolve all these effects perfectly in just eight experiments.

The extent of the confounding problem in any fractional factorial design is given by the **resolution**, **R**, of the design, which by convention is usually expressed in Roman numerals. The meaning of R is that there is no confounding between a p -factor effect and an effect containing $<(R - p)$ factors. In the above four-factor example the resolution is IV (i.e. four), so there is no confounding between the main effects ($p = 1$) and *less than* three-fold ($4 - 1$), i.e. two-fold interactions. If $p = 2$ (i.e. the two-fold interactions), $R - p = 2$, so there is no confounding between these two-fold interactions and the main effects ($<(R - p) = 1$).

Amongst the simplest and most popular incomplete factorial designs are the **Plackett–Burman designs**, which provide information on the main effects of the factors, but not on their interactions. A feature of these methods is that they all depend on performing $4n$ experiments, when $n = 1, 2, 3$, etc., giving 4, 8, 12, etc. experiments. They thus avoid the limitations of factorial and fractional factorial designs where the number of runs is 2^n . A Plackett–Burman design with $4n$ experiments is suitable for the study of up to $4n - 1$ factors. But these designs are widely used when the number of factors of interest is less than the maximum for a given design. Suppose, for example, that we wish to study four factors. Since four experiments can handle only three factors we would use the Plackett–Burman design with eight experiments, which would accommodate up to seven factors. The three remaining factors are called **dummy factors** and have no chemical meaning at all. They are

valuable nonetheless, because their apparent effects, determined as shown below, can be used to estimate the measurement error. This attractive feature of the Plackett–Burman designs, which is analogous to ignoring higher order interactions in some other designs (see above), allows us to determine which of the real factors are significant at any given probability level. This method of estimating the measurement error can be extended by using larger designs. Suppose for example that six factors are being studied. A Plackett–Burman design with eight experiments could be used, as it would accommodate up to seven factors, but this would leave only one dummy factor to use to estimate the measurement error. If a 12-experiment design was used instead, five dummy factors would be available, so the estimate of the measurement error, and hence of the significance of the effect of each ‘real’ factor, would be improved, though at the cost of doing four more experiments.

The way in which a Plackett–Burman design is laid out is simple to understand. Suppose that we use a 12-experiment design, allowing us to study up to 11 factors. As usual each factor’s high level is indicated by a plus sign and the low level by a minus sign. In the first experiment the 11 factors A–K (including dummy ones) are set using a **generating vector** as follows:

A	B	C	D	E	F	G	H	I	J	K
+	+	–	+	+	+	–	–	–	+	–

(Details of generating vectors for different n can be found, for example, in Minitab®.) In the next experiment the factor levels are set by moving the last sign to the beginning of the line, giving:

A	B	C	D	E	F	G	H	I	J	K
–	+	+	–	+	+	+	–	–	–	+

This cyclical process is repeated for the first 11 experiments, in all of which there are therefore six factors at their high level and five factors at their low level. The twelfth and last experiment balances this out by setting all the factors to their low level, so that overall the table contains 66 plus signs and 66 minus signs. The effect of each factor is then given by:

$$1/6[\sum(y+) - \sum(y-)]$$

where $y+$ represents the system responses when the factor in question is set to its high level and $y-$ the responses when the factor is set to its low level. As described earlier in this chapter (see Example 7.6.1), the required sums of squares are given by $N \times (\text{estimated effect})^2/4$, where N is the total number of experiments, in this case 12.

Suppose that this experiment was being used to determine the effect of eight factors. In this case there would be three dummy factors: the mean of their sums of squares gives an estimate of the measurement variance, s^2 , with three degrees of freedom. The significance of each factor can then be tested by comparing its mean square with s^2 using an F -test, as in other ANOVA examples. Alternatively we can calculate a critical difference which is given by $2st/\sqrt{N}$, where N is the total number of experiments and t has the value appropriate to the required probability level and with the number of degrees of freedom equal to the number of dummy variables. Any effect for a real factor that exceeds this critical value is then taken to be a significant effect.

Minitab® provides substantial facilities in the area of factorial designs.

7.8 Optimisation: basic principles and univariate methods



When the various factors and interactions affecting the results of an experiment have been identified, separate methods are needed to determine the combination of factor levels which will provide the optimum response. In Section 7.1 we noted that an experimental design is likely to fail unless the exact aim of the experiment is carefully defined first. Similarly, it is essential to define carefully what is meant by the 'optimum response' in a given analytical procedure. In some cases the aim will be to ensure that the measurement system gives a *maximum* response signal, i.e. the largest possible absorbance, current, emission intensity, etc. However, in many other cases the optimum outcome of an experiment may be the *maximum signal to noise* or *signal to background ratios*, the best *resolution* in separation methods, or even a *minimum* response, for example when the removal of an interfering signal is being studied. (In mathematical terms, finding maxima and minima are virtually identical processes, so the last example causes no additional problems.) If the exact aim of an optimisation experiment is not carefully defined in advance, the optimisation process may fail simply because the target was not sufficiently clearly laid down.

A good optimisation method has two qualities. It produces a set of experimental conditions that provides the optimum response, or at least a response that is close to the optimum; and it does so with the smallest possible number of trial experimental steps. In practice the speed and convenience of the optimisation procedure are extremely important, and it may be sufficient in some cases to use a method that gets reasonably close to the true optimum in a small number of steps.

Even the optimisation of a single factor may present some interesting problems. Suppose we wish to find the optimum pH of an enzyme-catalysed reaction within the pH range 2–12, the best pH being that at which the reaction rate is a maximum. Each rate measurement will be a separate experiment with a different buffer solution and taking considerable time and effort, so it is particularly important to get the maximum information from the smallest possible number of experiments. Two approaches suggest themselves. One is to make a fixed number of rate measurements, for example by dividing the pH interval of interest up into a number of equal regions. The second and more logical method is to make the measurements sequentially, so that the pH for each experiment depends on the results of the previous experiments.

Figure 7.3 shows the result of making four rate measurements at pH values of 4, 6, 8 and 10. In considering the four outcomes we shall assume, as in most of our other optimisation examples, that there is only one maximum within the range of the factor level(s) under study. (Inevitably, this is not always true, and we return to the point later.) The four points on the graph show that the highest reaction rate is obtained at pH 10, and the next highest at pH 8. But even with the assumption of a single maximum it is possible to draw two types of curve through the points: the maximum may occur between pH 8 and 10, or between pH 10 and 12. So the result of the four experiments is that, starting with the pH range between 2 and 12, we

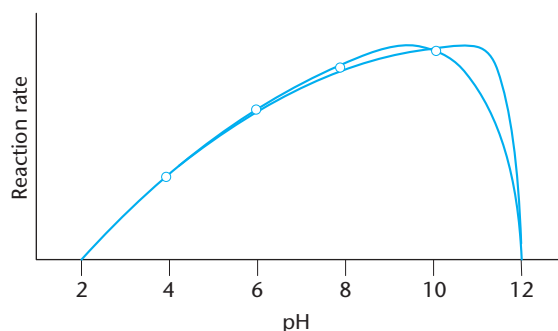


Figure 7.3 Optimisation experiment with equally spaced factor levels.

conclude that the optimum pH is actually between 8 and 12, i.e. we have narrowed the possible range for the optimum by a factor of $4/10$. This is an example of the general result that, if n experiments are done using equal intervals of the factor level, the range for the optimum is narrowed by a factor of $2/(n + 1)$ or $2/5$ here. This is not a very impressive result! The weakness of the method is emphasised by the fact that, if we wished to define the optimum pH within a range of 0.2 units, i.e. a 50-fold reduction of the original range of 10 units, 99 experiments would be needed, an obvious impossibility.

The principle of the **superior step-wise approach** is shown in Fig. 7.4, which shows a possible relationship between reaction rate and pH. (This curve would of course not be known in advance to the experimenter.) In brief, the procedure is as follows. The **first two experiments are carried out at pH A and B, equidistant from the extremes of the pH range, 2 and 12.** (The choice of pH values for these first experiments is discussed below.) The experiment at B will give the higher reaction rate so, since there is only one maximum in the curve, the portion of the curve between pH 2 and A can be rejected. The remainder of the pH range, between A and pH 12, certainly includes the maximum, and it already has one reading, B, within it. A new measurement, C, is then made at a pH such that the pH difference between C and A is the same as that between B and pH 12. The pH at C gives a higher reaction rate than B, so the interval between B and pH 12 can now be rejected, and a new measurement, D, made so that the A–D and C–B distances are equal. Further measurements use the same principle, so it only remains to establish how many steps are necessary, and where the starting points A and B should be.

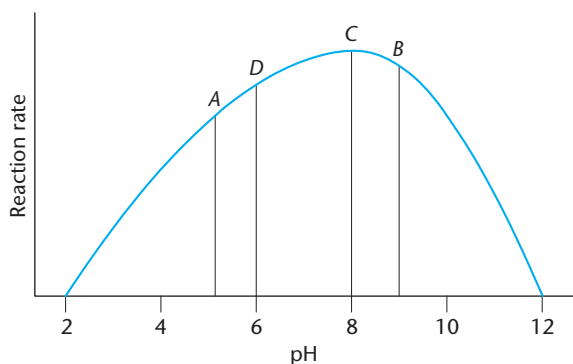


Figure 7.4 Step-wise approach to univariate searching.

In one approach the distances between the pairs of measurements and the extremes of the corresponding ranges are related to the **Fibonacci series**. This series of numbers, known since the thirteenth century, starts with 1 and 1 (these terms are called F_0 and F_1), with each subsequent term being the sum of the two previous ones. Thus F_2, F_3 , etc. are 2, 3, 5, 8, 13, 21, 34, 55, 89. . . . To use this series to optimise a single factor over a defined range we begin by deciding *either* on the degree of optimisation required, which automatically determines the number of experiments necessary, *or* on the number of experiments we can perform, which automatically determines the degree of optimisation obtained. Suppose that, as before, we require the optimum pH to be known within 0.2 units, a 50-fold reduction of the original pH interval of 10 units. We must then take the first Fibonacci number above 50: this is 55, F_9 . The subscript tells us that nine experiments will be needed to achieve the desired result. The spacing of the first two points, *A* and *B*, within the range, is also given by the series. We use F_9 and the member of the series *two below it*, F_7 , to form the fraction F_7/F_9 , i.e. 21/55. Point *A* is then at pH $[2 + (10 \times 21/55)]$, and point *B* is at pH $[12 - (10 \times 21/55)]$, i.e. 5.8 and 8.2 respectively. (The number 10 appears in these expressions because the pH range of interest is 10 units in width.) Once these first points are established, the positions of *C*, *D*, etc. follow automatically by symmetry.

It is striking that the Fibonacci search method achieves in just nine experiments a degree of optimisation that requires 99 experiments using the 'equal intervals' method. This method is the most efficient univariate search procedure for a given range when the degree of optimisation is known or decided in advance. In some other optimisation methods it is not necessary to decide in advance either the number of experiments or the degree of optimisation needed. Further details of such methods are given in the texts listed in the Bibliography.

The success of the Fibonacci and other optimisation procedures depends on the assumption that the random measurement errors (of the reaction rates in the example above) are significantly smaller than the rate of change of the response with the factor level (pH). This assumption is most likely to fail near to the optimum value of the response, where the slope of the response curve is close to zero. This confirms that in many practical cases an optimisation method which gets fairly close to the optimum in a few experimental steps will be most valuable. Trying to refine the optimum by extra experiments might fail if the experimental errors give misleading results.

7.9

Optimisation using the alternating variable search method

When the response of an analytical system depends on two factors which are continuous variables, the relationship between the response and the levels of the two factors can be represented by a surface in three dimensions as shown in Fig. 7.5. This surface is known as the **response surface**, with the target optimum being the top of the 'mountain'. A more convenient representation is a **contour diagram** (Fig. 7.6). Here the response on each contour is constant, and the target optimum is close to the centre of the contours. The form of the contour lines is, of course, unknown to the experimenter who wishes to determine the optimum levels, x_0 and y_0 for the factors *X* and *Y* respectively. A search method using a one-at-a-time approach would set the initial level of *X* to a fixed value at x_1 , say, and vary the level of *Y* to give a

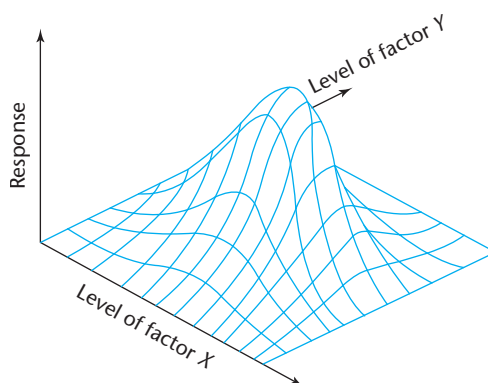


Figure 7.5 A response surface for two factors.

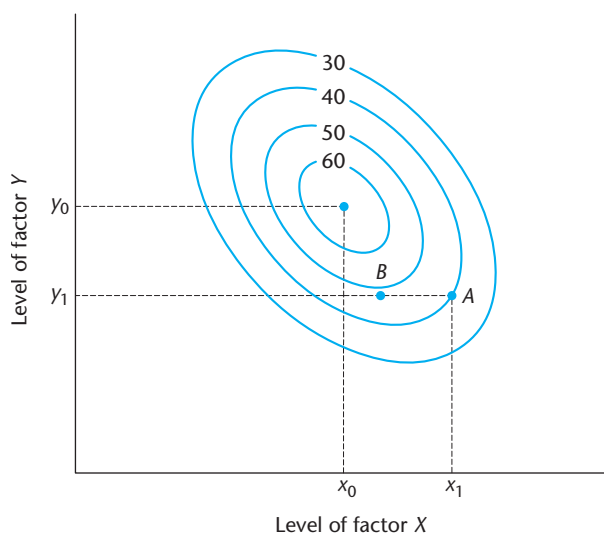


Figure 7.6 The contour diagram for a two-factor response surface.

maximum response at the point A, where the level of Y is y_1 . Next, holding the level of Y at y_1 and varying the level of X would give a maximum at B. Obviously this is not the true maximum, as the position obtained depends on the initial value chosen for x_1 . A better response can be obtained by repeating the process, varying the levels of X and Y alternately. This method is known as the **alternating variable search (AVS)** or the **iterative univariate method**. When there is no interaction between the two factors this method is extremely efficient. In such a case the response surface has the form of Fig. 7.7(a) or (b) and varying X and then Y just once will lead to the maximum response. If, however, there is interaction between the two variables then the response surface has the form of Fig. 7.7(c) and X and Y must then be varied in turn a number of times. In some cases, even this will not lead to the true maximum: this is illustrated in Fig. 7.8 where, although C is not the true maximum, the response falls on either side of it in both the X and the Y directions. The AVS method is used infrequently in analytical chemistry. It is practicable only if the response can be monitored continuously as the factor level is altered easily, for example in spectrometry when the monochromator wavelength or slit width is readily changed.

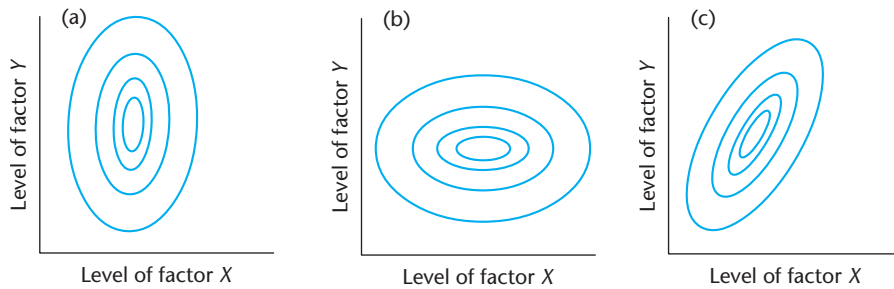


Figure 7.7 Simplified contour diagrams: (a) and (b) show no X-Y interaction; (c) shows significant X-Y interaction.

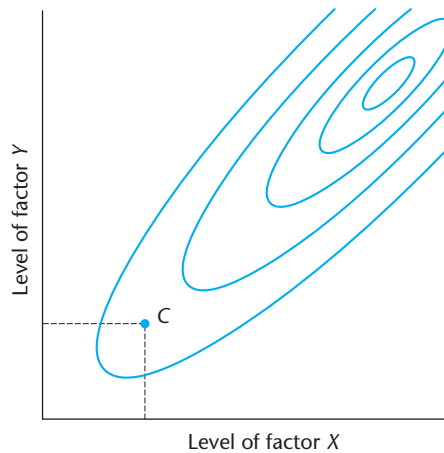


Figure 7.8 Contour diagram: a situation in which the one-at-a-time method fails to locate the maximum.

Otherwise a choice of step size has to be made for the change in each of the factors. A more sophisticated method would allow changes in these step sizes depending on the observed change in response, but in practice other optimisation methods involving fewer separate experiments are superior. A number of other methods are used to overcome the problems of one-at-a-time optimisation. All of them can be applied to any number of factors, but the response surface cannot easily be visualised for three or more factors: our remaining discussion of optimisation methods will thus largely be confined to experiments involving two factors.

7.10 The method of steepest ascent

The process of optimisation can be visualised in terms of a person standing on a mountain (Fig. 7.5) in thick fog, with the task of finding the summit! In these circumstances an obvious approach is to walk in the direction in which the gradient is steepest. This is the basis of the **method of steepest ascent**. Figure 7.9 shows two possible contour maps. The direction of steepest ascent at any point is at right angles

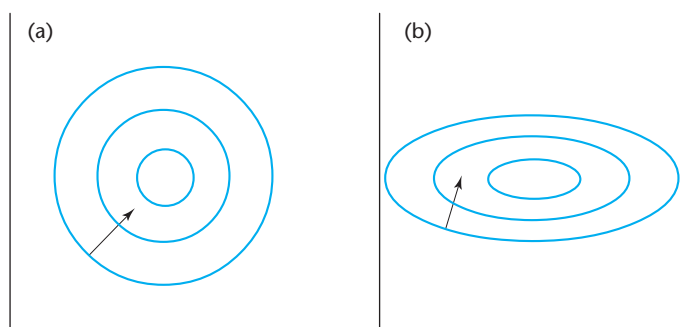


Figure 7.9 Contour diagrams: the arrow in each diagram indicates the path of steepest ascent. In (a) it goes close to the maximum but in (b) it does not.

to the contour lines at that point, as indicated by the arrows. When the contour lines are circular this will be towards the summit but when the contour lines are elliptical it may not. The shape of the contour lines depends on the scales chosen for the axes: the best results are obtained from the method if the axes are scaled so that a unit change in either direction gives a roughly equal change in response. The first step is to perform a factorial experiment with each factor at two levels. The levels are chosen so that the design forms a square as shown in Fig. 7.10. Suppose, for example, that the experiment is an enzyme-catalysed reaction in which the reaction rate, which in this case is the response, is to be maximised with respect to the pH (X) and the temperature (Y). The table below gives the results (reaction rate measured in arbitrary units) of the initial factorial experiment.

		<i>pH</i> (X)	
		6.8	7.0
Temperature, °C (Y)	20	30	35
	25	34	39

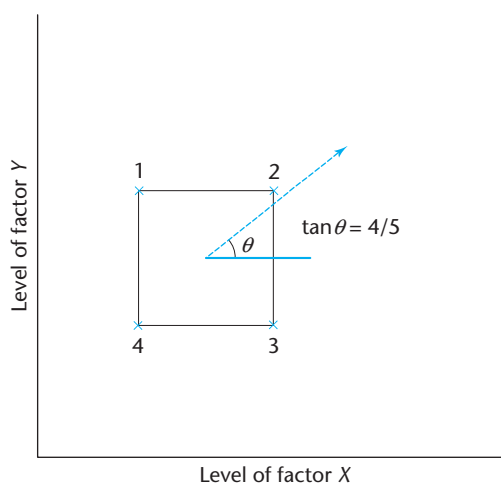


Figure 7.10 A 2×2 factorial design to determine the direction of steepest ascent, indicated by the broken line.

The effects of the two factors can be separated as described in Section 7.6. Rewriting the table above, in the notation of that section, gives:

Combination of levels	Rate of reaction
1	30
x	35
y	34
xy	39
Average effect of change in level of $X = [(35 - 30) + (39 - 34)]/2 = 5$	
Average effect of change in level of $Y = [(34 - 30) + (39 - 35)]/2 = 4$	

The effects of X and Y indicate that in Fig. 7.10 we should seek for the maximum response to the right and above the original region. Since the change in the X direction is greater than that in the Y direction the distance moved in the X direction should be in the ratio 5:4 to the distance moved in the Y direction, as indicated by the broken line in Fig. 7.10.

The next step in the optimisation is to carry out further experiments in the direction indicated by the dotted line in Fig. 7.11, at (for example) the points numbered 5, 6 and 7. These experiments would indicate point 6 as a rough position for the maximum in this direction. Another factorial experiment is carried out in this region to determine the new direction of steepest ascent.

This method gives satisfactory progress towards the maximum provided that, over the region of the factorial design, the contours are approximately straight. This is equivalent to the response surface being a plane which can be described mathematically by a linear combination of terms in x and y . Nearer the summit terms in xy , x^2 and y^2 are also needed to describe the surface. The xy term represents the interaction between X and Y and can be estimated by using replication as described

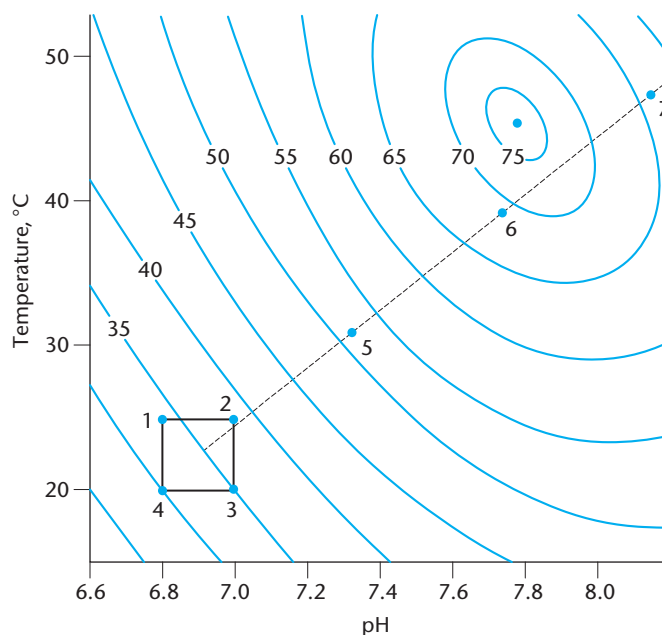


Figure 7.11 Contour diagram: the initial direction of steepest ascent is shown by the broken line. Further experiments are done at points 5, 6 and 7.

in Section 7.6. The squared terms, which represent the curvature of the surface, can be estimated by comparing the response at the centre of the factorial design with the average of the responses at the corners. When interaction and curvature effects become appreciable compared with the experimental error (estimated by replication) a more elaborate factorial design is used which allows the form of the curved surface, and thus the approximate position of the maximum to be determined.

We have seen that factorial designs can be very complicated when several factors are involved, so the same is true of the method of steepest ascent. The next section describes a method of optimisation which is conceptually much simpler.

7.11 Simplex optimisation



Simplex optimisation may be applied when all the factors are continuous variables. A **simplex** is a geometrical figure which has $k + 1$ vertices when a response is being optimised with respect to k factors. For example, in the optimisation of two factors the simplex will be a triangle. The method of optimisation is illustrated by Fig. 7.12. The initial simplex is defined by the points labelled 1, 2 and 3. In the first experiments the response is measured at each of the three combinations of factor levels given by the vertices of this triangle. The worst response in this case would be found at point 3 and it would be logical to suggest that a better response might be found at a point which is the reflection of 3 with respect to the line joining 1 and 2, i.e. at 4. The points 1, 2 and 4 form a new simplex and the response is measured for the combination of factor levels given by 4. (We immediately notice a major advantage of the simplex method: at each stage of the optimisation, only a *single* additional experiment is required.) Comparing the responses for the points 1, 2 and 4 will show that 1 now gives the worst response. The reflection process is repeated to give the simplex defined by 2, 4 and 5. The continuation of this process is shown in the figure. It can be seen that no further progress is possible beyond the stage shown, since points 6 and 8 both give a worse response than 5 and 7, so the simplex oscillates across the

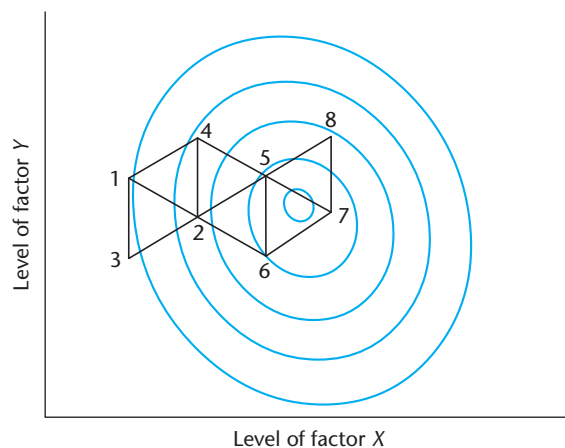


Figure 7.12 Simplex optimisation.

Table 7.9 Simplex optimisation example

	Factors					Response
	A	B	C	D	E	
Vertex 1	1.0	3.0	2.0	6.0	5.0	7
Vertex 2	6.0	4.3	9.5	6.9	6.0	8
Vertex 3	2.5	11.5	9.5	6.9	6.0	10
Vertex 4 (rejected)	2.5	4.3	3.5	6.9	6.0	6
Vertex 5	2.5	4.3	9.5	9.7	6.0	11
Vertex 6	2.5	4.3	9.5	6.9	9.6	9
(i) Sum (excluding vertex 4)	14.50	27.40	40.00	36.40	32.60	
(ii) Sum/k (excluding vertex 4)	2.90	5.48	8.00	7.28	6.52	
(iii) Rejected vertex (i.e. 4)	2.50	4.30	3.50	6.90	6.00	
(iv) Displacement = (ii) – (iii)	0.40	1.18	4.50	0.38	0.52	
(v) Vertex 7 = (ii) + (iv)	3.30	6.66	12.50	7.66	7.04	

region of the true optimum. Depending on the shape of the response surface, oscillations of this kind may occur even when the simplex is not close to the optimum. Further improvements can then sometimes be made by reflecting the *next-worst* point rather than the worst one to continue the simplex in a new direction.

The position of the new vertex of a simplex is in practice found by calculation rather than drawing: this is essential when there are more than two factors. The calculation (using constant step sizes) is most easily set out as shown in Table 7.9, the calculation lines being labelled (i)–(v). In this example there are five factors and hence the simplex has six vertices. (Note that it is not essential for each simplex to have a different level for each of the vertices: for example factor A takes the value 2.5 for each of the vertices 3–6.) In the initial simplex the response for vertex 4 is the lowest and so this vertex is to be replaced. The co-ordinates of the centroid of the vertices which are to be *retained* are found by summing the co-ordinates for the retained vertices and dividing by the number of factors, k . The displacement of the new point from the centroid is given by (iv) = (ii) – (iii), and the co-ordinates of the new vertex, vertex 7, by (v) = (ii) + (iv).

An obvious question in using the simplex method is the choice of the initial simplex. If this is taken as a *regular* figure in k dimensions, then the positions taken by the vertices in order to produce such a figure will depend on the scales used for the axes. As with the method of steepest ascent these scales should be chosen so that unit change in each factor gives about the same change in response. If there is insufficient information available to achieve this, the difference between the highest and lowest feasible value of each factor can be represented by the same distance. One obvious problem with the method is that, if the initial simplex is too small, too many experiments may be needed to approach the optimum. If the initial simplex is too big, the accuracy with which the optimum is determined will be poor (see Fig. 7.12). The size of the initial simplex is not so critical if it can be expanded or contracted as the method proceeds (see below). Algorithms that can be used to calculate the initial positions of the vertices have been developed: one vertex is normally positioned at the currently accepted levels of the factors. This is a reminder that the analyst is

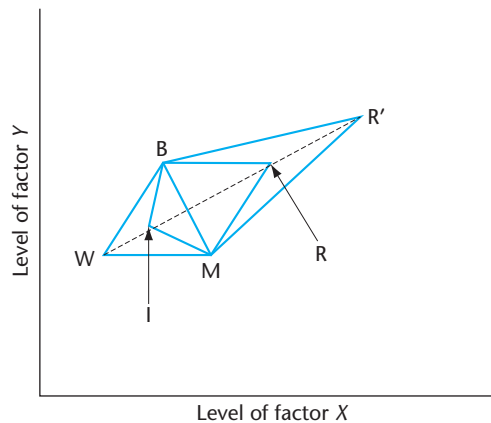


Figure 7.13 Optimisation using variable size simplexes. See text for details.

rarely completely in the dark at the start of an optimisation process: previous experience will provide some guidance on feasible values for the vertices of the starting simplex.

In order to improve the performance of the simplex method many modifications have been proposed. In particular it is possible to move towards the optimum by varying the step size according to how the response for the new vertex in a simplex compares with the other vertices. The principle is illustrated in Fig. 7.13, in which the three vertices are called W (giving the worst response), B (best response) and M (middle response). When W is reflected in the line joining B and M the response is called R. If the response at R is better than that at B, i.e. R gives a new best response, this indicates that the simplex is moving in the right direction, so the reflection is extended, normally by a factor of 2, to give a new vertex R'. If the response at R' is also better than that at B, R' becomes one vertex of a new simplex, BMR'. If the R' response is not better than that at B, then the expansion of the simplex has evidently failed, and the conventional simplex BMR is used as the basis of the next step. In some cases the point R might produce a response that is poorer than that at B, but still better than that at M, so again the simplex BMR is used for the next reflection. If the response at R is poorer than that at M, the simplex has apparently moved too far. In that case a new vertex, C, is used, the reflection being restricted to (usually) half its normal extent: the new simplex is then BMC. Lastly, if the response at R is worse even than that at W, then the new vertex, I, should be *inside* the original simplex, giving a new simplex BMI. All these changes can be calculated by insertion of the appropriate positive or negative numerical factors in row (iv) of Table 7.9.

The effect of these variable step sizes is that (when two factors are being studied) the triangles making up each simplex are not necessarily equilateral ones. The benefit of the variable step sizes is that initially the simplex is large, and gives rapid progress towards the optimum. Near the optimum it contracts to allow the latter to be found more accurately. When several factors are studied, it may be helpful to alter some of them by a constant step size, but change others with a variable step size.

It can be seen that in contrast to a factorial design the number of experiments required in the simplex method does not increase rapidly with the number of factors. For this reason all the factors which might reasonably be thought to have a bearing on the response should be included in the optimisation.

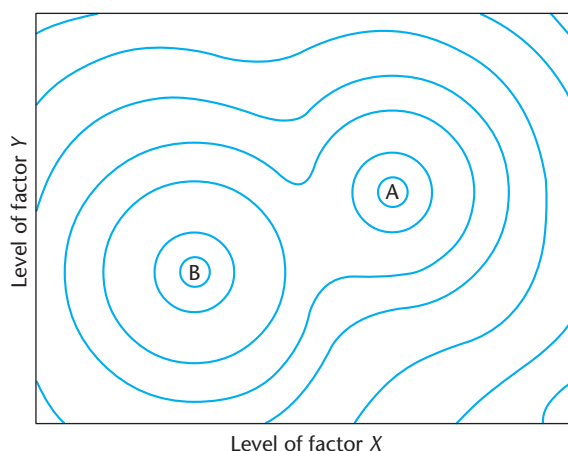


Figure 7.14 Contour diagram showing localised optimum (A) and true optimum (B).

Once an optimum has been found, the effect on the response when one factor is varied while the others are held at their optimum levels can be investigated for each factor in turn. This procedure can be used to check the optimisation. It also indicates how important deviations from the optimum level are for each factor: the sharper the response peak in the region of the optimum, the more critical any variation in factor level.

Simplex optimisation has been used with success in many areas of analytical science, e.g. atomic-absorption spectrometry, gas chromatography, colorimetric methods of analysis, plasma spectrometry, and the use of centrifugal analysers in clinical chemistry. When an instrument is interfaced with a computer, the results of simplex optimisation can be used to initiate automatic improvements in the instrument variables.

Simplex optimisation has some disadvantages. As always, difficulties may arise if the random measurement errors are larger than the slope of the response surface near the optimum (see above). Moreover, the small number of experiments performed, while usually advantageous in practice, means that little information is gained on the overall shape of the response surface. Occasionally response surfaces with more than one maximum occur, such as that shown in Fig. 7.14. Both the alternating variable search and simplex optimisation methods may then locate a local optimum such as A rather than the true optimum B. Starting the optimisation process in a second region of the factor space and verifying that the same optimum conditions are obtained is the preferred method for checking this point. Again the simplex method is valuable here, as it minimises the extra work required.

7.12 Simulated annealing

In recent years there has been much interest in the application of calculation methods that mimic natural processes: these are collectively known as **natural computation** methods. Neural networks (see Chapter 8) are now being applied more frequently in

analytical chemistry, and in the area of optimisation **simulated annealing** (SA) has found some recent applications. Annealing is the process by which a molten metal or other material cools slowly in controlled conditions to a low-energy state. In principle the whole system should be in equilibrium during the whole of the annealing process, but in practice random processes occur which result in short-lived and/or local *increases* in energy. When an analogous process is applied to an optimisation problem the algorithm used allows access to positions in factor space that give a *poorer* response than the previous position. The result is that, unlike the AVS and simplex methods, which almost inevitably lead to the identification of an optimum which is closest to the starting point, SA methods can handle any local optima which occur, and successfully identify the true overall optimum.

In simple terms the method operates as follows. The first step is to identify, either at random or from experience, starting values for the levels of the k factors. These values give an initial response, R_1 . In the second step a random vector, obtained using k random numbers, is added to the starting values, and a new set of experimental conditions generated: these yield a new response, R_2 . As in other optimisation methods, if R_2 is a better response than R_1 , that is a good outcome, and the random addition step is repeated. The crucial characteristic of the method, however, is that *even if R_2 is a poorer response than R_1* , it is accepted as long as it is not much worse. (Clearly, numerical rules have to be applied to make this decision.) Eventually a situation arises in which a response is rejected, and (for example) five alternatives generated at random are also rejected as giving unacceptably poorer responses. In that situation it is assumed that the previous response was the optimum one.

A major application area of SA in analytical science has been in the selection of suitable wavelengths for multi-component analysis using UV-visible and near-IR spectroscopy. It has also been used for the refinement of molecular structures determined by nuclear magnetic resonance (NMR) spectroscopy and X-ray crystallography, and in the study of chromatographic systems. Applications in other areas of chemistry include extensive studies of quantitative structure-activity relationships (QSAR).

Bibliography and resources

Books

- Brereton, R.G., 2003, *Chemometrics: Data Analysis for the Laboratory and Chemical Plant*, John Wiley, Chichester. Extensive treatment of experimental design, with many examples, and a shorter treatment of optimisation.
- Massart, D.L., Vandeginste, B.G.M., Buydens, L.M.C., De Jong, S., Lewi, P.J. and Smeyers-Verbeke, J., 1997, *Handbook of Chemometrics and Qualimetrics*, Part A, Elsevier, Amsterdam. Comprehensive coverage of optimisation and experimental design, with many examples.
- Otto, M. 2007. *Chemometrics: Statistics and Computer Applications in Analytical Chemistry*, 2nd edn, Wiley-VCH, Weinheim. An excellent treatment of many topics covered in Chapters 7 and 8 of this book.

Software and internet resources

Analytical Methods Committee, Royal Society of Chemistry, Cambridge. A series of short technical briefs on experimental design and optimisation can be downloaded from www.rsc.org.

www.statease.com. Stat-Ease, Inc., Minneapolis, USA, offers two experimental design programs for Windows, Design-Expert, which is the more advanced one, and Design-Ease, which is an entry-level program on the use of screening factorial designs. These programs are easy to use, and there is tutorial support. Free 45-day trial versions can be downloaded.

Exercises

- Four standard solutions were prepared, each containing 16.00% (by weight) of chloride. Three titration methods, each with a different technique of end point determination, were used to analyse each standard solution. The order of the experiments was randomised. The results for the chloride found (% w/w) are shown below:

Solution	Method		
	A	B	C
1	16.03	16.13	16.09
2	16.05	16.13	16.15
3	16.02	15.94	16.12
4	16.12	15.97	16.10

Test whether there are significant differences between (a) the concentration of chloride in the different solutions, and (b) the results obtained by the different methods.

- A new microwave-assisted extraction method for the recovery of 2-chlorophenol from soil samples was evaluated by applying it to five different soils on each of three days. The percentage recoveries obtained were:

Soil	Day		
	1	2	3
1	67	69	82
2	78	66	76
3	78	73	75
4	70	69	87
5	69	71	80

Determine whether there were any significant differences in percentage recovery (a) between soils, and/or (b) between days.

(Data adapted from Egizabal, A., Zuloaga, O., Extebarria, N., Fernández, L.A. and Madariaga, J.M., 1998, *Analyst*, 123: 1679)

- 3 In studies of a fluorimetric method for the determination of the anionic surfactant sodium dodecyl sulphate (SDS) the interfering effects of four organic compounds at three different SDS:compound molar ratios were studied. The percentage recoveries of SDS were found to be:

Organic compound	Molar ratios		
	1:1	1:2	1:3
2,3-Naphthalene dicarboxylic acid	91	84	83
Tannic acid	103	104	104
Phenol	95	90	94
Diphenylamine	119	162	222

Determine whether the SDS recovery depends on the presence of the organic compounds and/or on the molar ratios at which they are present. How should the experiment be modified to test whether any interaction effects are present? (Recalde Ruiz, D.L., Carvalho Torres, A.L., Andrés García, E. and Díaz García, M.E., 1998, *Analyst*, **123**: 2257)

- 4 Mercury is lost from solutions stored in polypropylene flasks by combination with traces of tin in the polymer. The absorbance of a standard aqueous solution of mercury stored in such flasks was measured for two levels of the following factors:

Factor	Low	High
A – Agitation of flask	Absent	Present
C – Cleaning of flask	Once	Twice
T – Time of standing	1 hour	18 hours

The following results were obtained. Calculate the main and interaction effects.

Combination of factor levels	Absorbance
1	0.099
a	0.084
c	0.097
t	0.076
ac	0.082
ta	0.049
tc	0.080
atc	0.051

(Adapted from Kuldvere, A., 1982, *Analyst*, **107**: 179)

- 5 In an inter-laboratory collaborative experiment on the determination of arsenic in coal, samples of coal from three different regions were sent to each of three

laboratories. Each laboratory performed a duplicate analysis on each sample, with the results shown below (measurements in $\mu\text{g g}^{-1}$).

Sample	Laboratory		
	1	2	3
A	5.1, 5.1	5.3, 5.4	5.3, 5.1
B	5.8, 5.4	5.4, 5.9	5.2, 5.5
C	6.5, 6.1	6.6, 6.7	6.5, 6.4

Verify that there is no significant sample–laboratory interaction, and test for significant differences between the laboratories.

- 6 The optimum pH for an enzyme-catalysed reaction is known to lie between 5 and 9. Determine the pH values at which the first two experiments of an optimisation process should be performed in the following circumstances:
 - (a) The optimum pH needs to be known with a maximum range of 0.1 pH units.
 - (b) Only six experiments can be performed.
 In (b) what is the degree of optimisation obtained?
- 7 If the response at vertex 7 in the example on simplex optimisation (p. 214) is found to be 12, which vertex should be rejected in forming the new simplex and what are the co-ordinates of the new vertex?

8

Multivariate analysis

Major topics covered in this chapter

- Representing multivariate data: initial analysis
- Principal component analysis (PCA)
- Cluster analysis (CA) methods
- Linear discriminant analysis and canonical variate analysis
- The *K*-nearest neighbour (KNN) method
- Disjoint class modelling
- Multiple linear regression (MLR)
- Principal component regression (PCR)
- Partial least-squares (PLS) regression
- Natural computation methods: artificial neural networks (ANN)

8.1

Introduction

Modern automatic analysis methods provide opportunities to collect large amounts of data very easily. For example, in clinical chemistry it is routine to determine many analytes for each specimen of blood, urine, etc. A number of chromatographic and spectroscopic methods can provide analytical data on many components of a single specimen. The widespread use of multi-channel array detectors in spectroscopy and the development of miniature sensor arrays based on solid state or bio-specific detection methods have further encouraged the use of multi-analyte measurements, and extended their applications to areas such as process analysis. Situations such as these, where several variables are measured for each specimen, yield **multivariate data**. One use of such data in analytical chemistry is in discrimination, for example determining whether an oil-spill comes from a particular source by analysing the fluorescence spectrum. Another use is classification, for example dividing the stationary phases used in gas-liquid chromatography into groups with similar properties by studying the retention properties of a variety of solutes

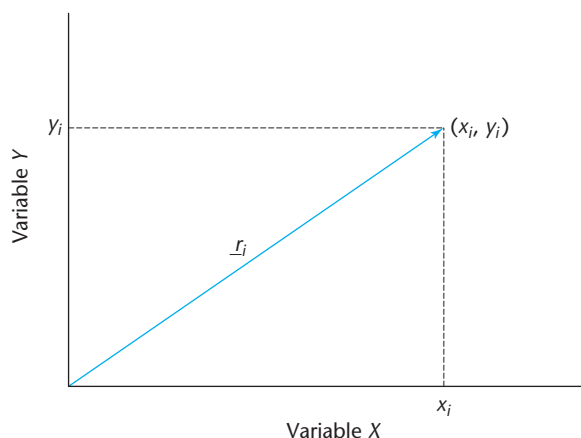


Figure 8.1 A diagram to illustrate a data vector. r_i , x_i and y_i are the values taken by the variables X and Y respectively.

with different chemical properties. In each case it would be possible to compare specimens by considering each variable in turn but modern computers allow more sophisticated processing methods where all the variables are considered simultaneously. The ANOVA approach described in previous chapters can be extended to multivariate data (the method is called multivariate ANOVA or MANOVA), but in practice this method seems to have been applied to chemical problems less than those summarised in the sections below.

Each specimen, or, to generalise, each object, in the methods we shall study is characterised by a set of measurements. When only two variables are measured this information can be represented graphically, as shown in Fig. 8.1, where the co-ordinates of the point give the values taken by the two variables. The point can also be defined by a vector, called a **data vector**, drawn to it from the origin. Objects which have similar properties will have similar data vectors; they will lie close to each other in the space defined by the variables. Such a group is called a **cluster**.

A graphical representation is less easy for three variables and no longer possible for four or more: it is here that computer analysis is particularly valuable in finding patterns and relationships. Matrix algebra is needed in order to describe the methods of multivariate analysis fully. No attempt will be made to do this here. The aim is to give an appreciation of the purpose and power of multivariate methods. Simple data sets will be used to illustrate the methods and some practical applications will be described.

8.2

Initial analysis

Table 8.1 shows an example of some multivariate data. This gives the relative intensities of fluorescence emission at four different wavelengths (300, 350, 400, 450 nm) for 12 compounds, A–L. In each case the emission intensity at the wavelength of maximum fluorescence would be 100. As a first step it may be useful to calculate the mean and standard deviation for each variable. These are also shown in the table.

In addition, since we have more than one variable, it is possible to calculate a product-moment (Pearson) correlation coefficient for each pair of variables. These are summarised in the **correlation matrix** in Table 8.2, obtained using Minitab®.

Table 8.1 The intensity of the fluorescence spectrum at four different wavelengths for a number of compounds

Compound	Wavelength, nm			
	300	350	400	450
A	16	62	67	27
B	15	60	69	31
C	14	59	68	31
D	15	61	71	31
E	14	60	70	30
F	14	59	69	30
G	17	63	68	29
H	16	62	69	28
I	15	60	72	30
J	17	63	69	27
K	18	62	68	28
L	18	64	67	29
Mean	15.75	61.25	68.92	29.25
Standard deviation	1.485	1.658	1.505	1.485

Table 8.2 The correlation matrix for the data in Table 8.1

Correlations (Pearson)				
	300	350	400	450
350	0.914			
400	-0.498	-0.464		
450	-0.670	-0.692	0.458	

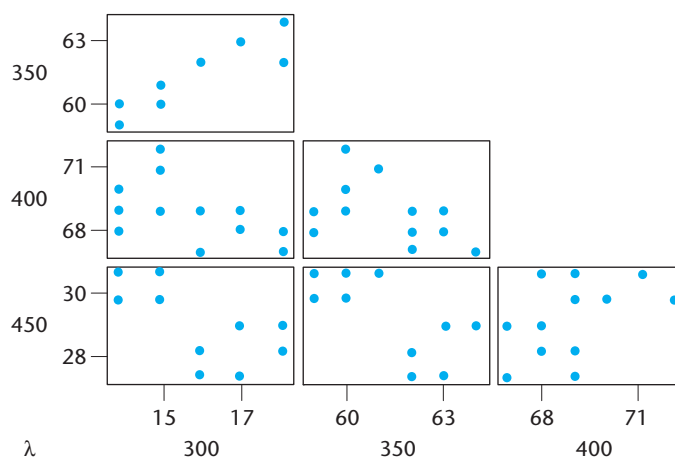


Figure 8.2 Draftsman plot for the data in Table 8.1.

This shows that, for example, the correlation coefficient for the intensities at 300 and 350 nm is 0.914. The relationships between pairs of variables can be illustrated by a **draftsman plot** as shown in Fig. 8.2. This gives scatter diagrams for each pair of



variables. Both the correlation matrix and the scatter diagrams indicate that there is some correlation between some of the pairs of variables.

8.3

Principal component analysis



One problem with multivariate data is that the sheer volume may make it difficult to see patterns and relationships. For example, a spectrum would normally be characterised by several hundred intensity measurements rather than just four as in Table 8.1 and in this case the correlation matrix would contain hundreds of values. Thus the aim of many methods of multivariate analysis is data reduction. Quite frequently there is some correlation between the variables, as there is for the data in Table 8.1, so some of the information is redundant. Principal component analysis (PCA) is a technique for reducing the amount of data when there is correlation present. It is worth stressing that it is not a useful technique if the variables are uncorrelated.

The idea behind PCA is to find **principal components** $Z_1, Z_2, \dots, Z_n$ which are linear combinations of the original variables describing each specimen, $X_1, X_2, \dots, X_n$, i.e.

$$\begin{aligned} Z_1 &= a_{11}X_1 + a_{12}X_2 + a_{13}X_3 + \dots + a_{1n}X_n \\ Z_2 &= a_{21}X_1 + a_{22}X_2 + a_{23}X_3 + \dots + a_{2n}X_n \\ &\text{etc.} \end{aligned}$$

For example, for the data in Table 8.1 there would be four principal components Z_1, Z_2, Z_3 and Z_4 , each of which would be a linear combination of X_1, X_2, X_3 and X_4 , the fluorescence intensities at the given wavelengths. The coefficients, a_{11}, a_{12} , etc. are chosen so that the new variables, unlike the original variables, are *not* correlated with each other. Creating a new set of variables in this way may seem a pointless exercise since we obtain n new variables in place of the n original ones, and hence no reduction in the amount of data. However, the principal components are also chosen so that the first principal component (PC1), Z_1 , accounts for most of the variation in the data set, the second (PC2), Z_2 , accounts for the next largest variation and so on. Hence, when significant correlation occurs the number of *useful* PCs is much less than the number of original variables.

Figure 8.3 illustrates the method when there are only two variables and hence only two principal components. In Fig. 8.3(a) the principal components are shown by the dotted lines. The principal components are at right angles to each other, a property known as **orthogonality**. Figure 8.3(b) shows the points referred to these two new axes and also the projection of the points on to PC1 and PC2. We can see that in this example Z_1 accounts for most of the variation and so it would be possible to reduce the amount of data to be handled by working in one dimension with Z_1 rather than in two dimensions with X_1 and X_2 . (In practice we would not need to use PCA when there are only two variables because such data are relatively easy to handle.)

Figure 8.3 shows that PCA is equivalent to a rotation of the original axes in such a way that PC1 is in the direction of maximum variation, but with the angle between the axes unchanged. With more than two variables it is not possible to illustrate the method diagrammatically but again we can think of PCA as a rotation of the axes in such a way that PC1 is in the direction of maximum variation, PC2 is in the direction

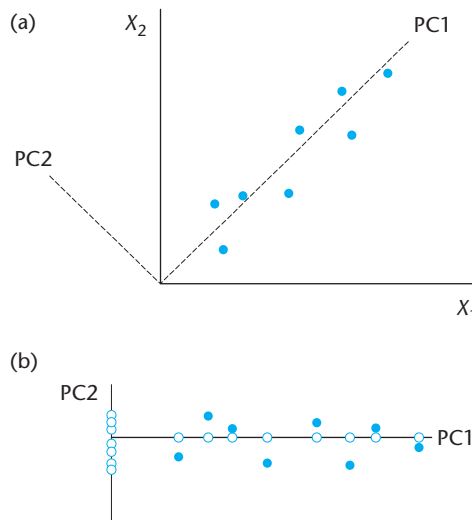


Figure 8.3 (a) Diagram illustrating the two principal components, PC1 and PC2, for the two variables X_1 and X_2 . (b) Points referred to the principal component axes. ● indicates data points, ○ their projection on to the axes.

of next greatest variation, and so on. It is often found that PC1 and PC2 then account between them for most of the variation in the data set. As a result the data can be represented in only two dimensions instead of the original n .

The principal components are obtained from the covariance matrix. The term ‘covariance’ (see Section 5.3) is a measure of the joint variance of two variables. The covariance matrix for the data in Table 8.1 is:

	300	350	400	450
300	2.20455			
350	2.25000	2.75000		
400	-1.11364	-1.15909	2.26515	
450	-1.47727	-1.70455	1.02273	2.20455



This shows that, for example, the covariance for the fluorescence intensities at 350 and 400 nm is -1.15909 . The table also gives the variances of the fluorescence intensities at each wavelength along the leading diagonal of the matrix. For example, for the fluorescence intensities at 350 nm the variance is 2.75. In mathematical terms the principal components are the *eigenvectors* of the covariance matrix and the technique for finding these eigenvectors is called *eigenanalysis*. Corresponding to each principal component (i.e. eigenvector) is an *eigenvalue*, which gives the amount of variance in the data set which is explained by that principal component.

Example 8.3.1

Carry out a principal component analysis of the data in Table 8.1.

This can be done using a variety of computer packages (for example, Minitab®, SAS®, The Unscrambler®). The printout below was obtained from Minitab®.

Principal Component Analysis

Eigenanalysis of the Covariance Matrix

Eigenvalue	6.8519	1.4863	0.8795	0.2066
Proportion	0.727	0.158	0.093	0.022
Cumulative	0.727	0.885	0.978	1.000
Variable	PC1	PC2	PC3	PC4
300	0.529	-0.218	-0.343	0.745
350	0.594	-0.319	-0.324	-0.664
400	-0.383	-0.917	0.100	0.050
450	-0.470	0.099	-0.876	-0.041

The sum of the variances of the original variables can be calculated from the covariance matrix. It is equal to $2.20455 + 2.75000 + 2.26515 + 2.20455 = 9.42425$. It can be seen that the variances of the variables are similar, so each one accounts for about 25% of the total. The first line of the top table shows how the total variance is shared among the four principal components and the second line shows the proportion as a proportion of the total. Thus PC1 has a variance of 6.8519, which is 72.7% of the total. This is a much greater proportion than any of the original variables. PC2 accounts for 15.8% of the total variance. The last line of the first block gives the cumulative proportion. It shows, for example, that between them PC1 and PC2 account for 88.5% of the variation.

The bottom table gives the coefficients of the principal components. For example, the first principal component is $Z_1 = 0.529X_1 + 0.594X_2 - 0.383X_3 - 0.470X_4$, where X_1 , X_2 , X_3 and X_4 are the intensities at 300, 350, 400 and 450 nm respectively. The values that the principal components take for each of the compounds can be calculated by substituting the relevant values of X_1 , X_2 , X_3 and X_4 into this formula. For example, the value of PC1 for compound A is equal to $(0.529 \times 16) + (0.594 \times 62) - (0.383 \times 67) - (0.470 \times 27) = 6.941$. This value is sometimes referred to as a 'score' for PC1. Figure 8.4 plots the scores of the first two principal components, calculated in this way, for the compounds A–L. This diagram reveals that the compounds fall into two distinct groups, a fact which is not readily apparent from the original data.

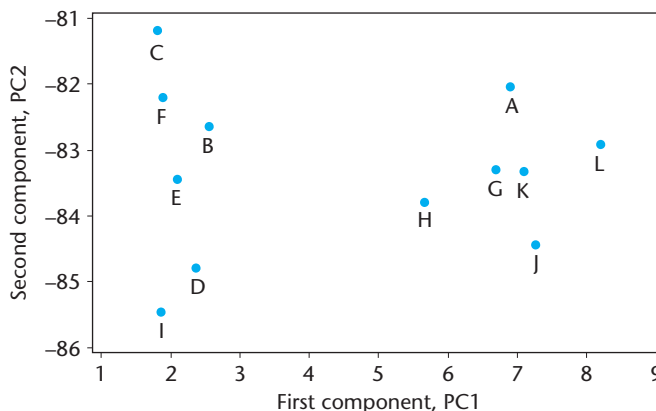


Figure 8.4 The scores of the first two principal components for the data in Table 8.1.

Table 8.3 The data in Table 8.1 rearranged so that compounds with similar spectra are grouped together

Compound	Wavelength, nm			
	300	350	400	450
A	16	62	67	27
G	17	63	68	29
H	16	62	69	28
J	17	63	69	27
K	18	62	68	28
L	18	64	67	29
B	15	60	69	31
C	14	59	68	31
D	15	61	71	31
E	14	60	70	30
F	14	59	69	30
I	15	60	72	30

Table 8.3 shows the original data rearranged so that compounds with similar spectra are grouped together. The differences between the two groups are now apparent. There is a difference at all four wavelengths, and the sizes of these differences are similar. This corresponds to the fact that the coefficients for the first principal component are similar in size. The top group in Table 8.3 has higher intensities than the bottom group at 300 and 350 nm and the opposite is true at 400 and 450 nm. This corresponds to the fact the first two coefficients of Z_1 have the opposite sign from the second two. Once two or more groups have been identified by using PCA, it may be possible to explain the differences between them in terms of chemical structure. Sometimes it may be possible to give a physical interpretation to the principal components. For this reason, principal components are sometimes referred to as **latent** (i.e. hidden) **variables**.

In this example the values of the coefficients show that each variable contributes to PC1 and at least three of them contribute to PC2. In other cases it is found that some variables do not contribute significantly even to PC1. An important benefit of PCA is that such variables can then be rejected.

Sometimes the PCA is carried out by analysing the correlation matrix rather than the covariance matrix, as was done in Example 8.3.1. The effect of using the correlation matrix is to standardise each variable to zero mean and unit variance. For standardised data, each variable has a variance of 1 and thus the sum of the eigenvalues is equal to the number of variables. Standardisation is desirable when the variables are measured on different scales. Another reason for standardising would be that one variable has a much larger variance than the others and as a result dominates the first principal component: standardising avoids this by making all variables carry equal weight. Neither of these considerations applies to the data in Example 8.3.1. It should be noted that standardisation can have a considerable effect on the results of a PCA when the original variables have very different variances.

PCA is primarily a mathematical method for data reduction and it does not assume that the data have any particular distribution. We have seen how PCA can be used to reduce the dimensionality of a data set and how it may thus reveal clusters.



It has been used, for example, on the results of Fourier transform spectroscopy in order to reveal differences between hair from different racial groups and for classifying different types of cotton fibre. In another example the concentrations of a number of chlorobiphenyls were measured in specimens from a variety of sea mammals. A PCA of the results revealed differences between species, differences between males and females, and differences between young and adult individuals. PCA has been applied in recent years to a very large number of analytical methods and problems, and it also finds application in multiple regression (see Section 8.10).

8.4 Cluster analysis

Although PCA may reveal groups of like objects, it is not always successful in doing so: Fig. 8.5 shows a situation in which the first principal component does not give a good separation between two groups. We now turn to methods whose explicit purpose is to search for groups.

Cluster analysis (CA) is a method for dividing a group of objects into classes so that similar objects are in the same class. As in PCA, the groups are not known prior to the mathematical analysis and no assumptions are made about the distribution of the variables. Cluster analysis searches for objects which are close together in the variable space. The distance, d , between two points in n -dimensional space with co-ordinates $(x_1, x_2, \dots, x_n)$ and $(y_1, y_2, \dots, y_n)$ is usually taken as the **Euclidian distance defined by**

$$d = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2} \quad (8.4.1)$$

For example the distance between the compounds E and F in Table 8.3 (if the unstandardised variables are used) is given by:

$$d = \sqrt{(14 - 14)^2 + (60 - 59)^2 + (70 - 69)^2 + (30 - 30)^2} = \sqrt{2}$$

As in PCA, a decision has to be made as to whether or not the data are standardised. Standardising the data will mean that all the variables are measured on a common scale so that one variable does not dominate the others.

There are a number of methods for searching for clusters. One method starts by considering each object as forming a 'cluster' of size one, and compares the distances between these clusters. The pair of points which are closest together are joined to

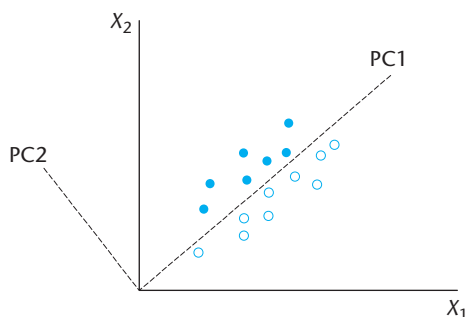


Figure 8.5 A situation in which the first principal component does not give a good separation between two groups.

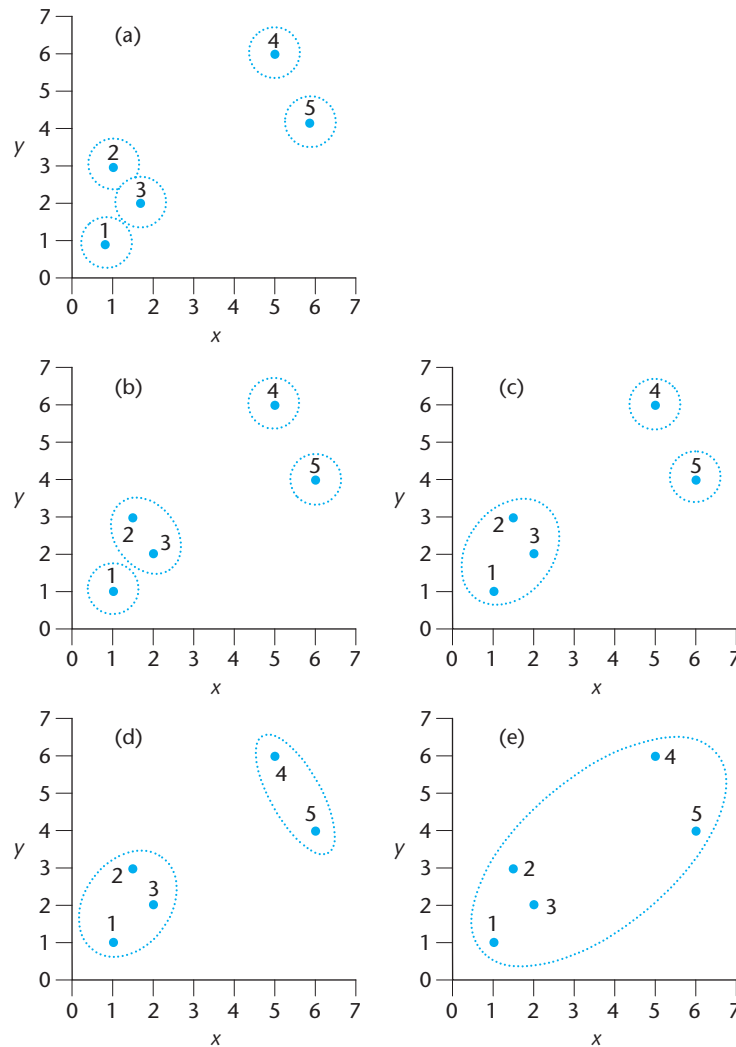


Figure 8.6 Stages in clustering: the dotted lines enclose clusters.

form a new cluster. The distances between the clusters are again compared and the two nearest clusters combined. This procedure is repeated and, if continued indefinitely, will group all the points together. There are a variety of ways of computing the distance between two clusters which contain more than one member. Conceptually the simplest approach is to take the distance between two clusters as the distance between nearest neighbours. This is called the **single linkage method**. It is illustrated in Fig 8.6. The successive stages of grouping can be shown on a **dendrogram** as in Fig. 8.7. The vertical axis can show either the distance, d_{ij} , between two points i and j when they are joined or alternatively the **similarity**, s_{ij} , defined by $s_{ij} = 100(1 - d_{ij}/d_{\max})$ where $d_{\max}$ is the maximum separation between any two points. The resulting diagrams look the same but their vertical scales differ. The stage at which the grouping is stopped, which determines the number of clusters in the final classification, is a matter of judgement for the person carrying out the analysis.

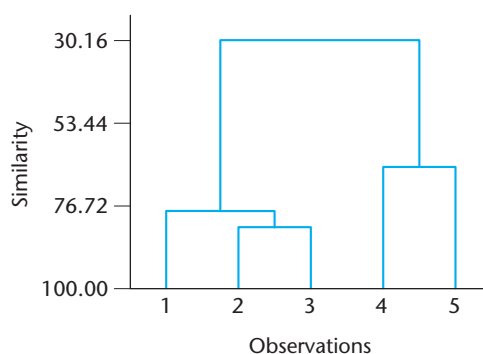


Figure 8.7 A dendrogram illustrating the stages of clustering for Fig. 8.6.

Example 8.4.1

Apply the single linkage method to the (unstandardised) data in Table 8.1.

The printout below was obtained using Minitab®. With this software the linkages continue until there is only one cluster, unless the user specifies otherwise.

Hierarchical Cluster Analysis of Observations

Euclidean Distance, Single Linkage

Amalgamation Steps

Step	Number of clusters	Similarity level	Distance level	Clusters joined		New cluster	Number of Obs in new cluster
1	11	80.20	1.414	5	6	5	2
2	10	80.20	1.414	3	5	3	3
3	9	75.75	1.732	7	12	7	2
4	8	75.75	1.732	7	11	7	3
5	7	75.75	1.732	8	10	8	2
6	6	75.75	1.732	4	9	4	2
7	5	75.75	1.732	2	3	2	4
8	4	71.99	2.000	7	8	7	5
9	3	71.99	2.000	2	4	2	6
10	2	68.69	2.236	1	7	1	6
11	1	49.51	3.606	1	2	1	12

The dendrogram in Fig. 8.8 illustrates the stages of the linkage. The vertical scale gives the distance between the two groups at the point when they were combined. The table above shows that the first two points to be joined were 5 (compound E) and 6 (compound F) with a separation of 1.414 ($= \sqrt{2}$ as calculated earlier). The reader can verify that the distance of C from F is also $\sqrt{2}$ so the next stage is to join point 3 to the cluster consisting of points 5 and 6. The process continues until all the points are in one cluster. However, if we 'cut the tree', i.e. stop the grouping, at the point indicated by the dotted line in Fig. 8.8, this analysis suggests that the compounds A–L fall into two distinct groups. Not surprisingly, the groups contain the same members as they did when PCA was used.

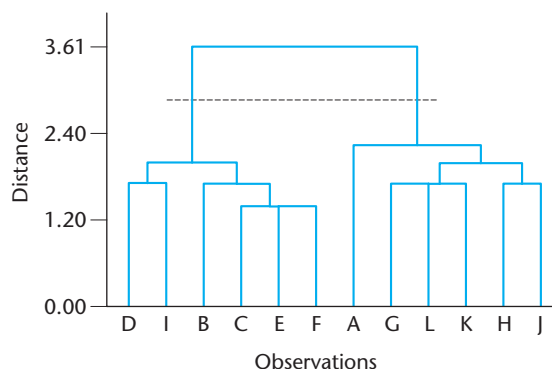


Figure 8.8 A dendrogram for the data in Table 8.1.

The CA method described here is **hierarchical**, meaning that once an object has been assigned to a group the process cannot be reversed. For non-hierarchical methods the opposite is the case. One such method is the **k-means method** which is available, for example, in Minitab®. This starts by either dividing the points into k clusters or alternatively choosing k 'seed points'. Then each individual is assigned to the cluster (or seed point) whose centroid is nearest. When a cluster loses or gains a point the position of the centroid is recalculated. The process is continued until all the points are in the cluster whose centroid is nearest.

This method has the disadvantage that the final grouping reflects the initial choice of clusters or seed points. Another disadvantage is that the value of k has to be chosen in advance. Many methods have been suggested for deciding on the best value of k but none of them is really satisfactory.

Analytical applications of CA are now numerous. It has been used to classify the many phases used in gas-liquid chromatography. A small preferred set of phases can then be selected by taking one phase from each cluster: this provides a range of stationary phases, each with distinctive separation characteristics. Another application is the classification of antibiotics in terms of their activity against various types of bacteria in order to elucidate the relationship between biological activity and molecular structure. Further recent applications of CA include the classification of wines and wine vinegars on the basis of a variety of organic and inorganic constituents, organic vapours detected by an array of semiconductor sensors, ligand-protein interactions, the properties of foodstuffs such as wheat, rice, tea, coffee and olive oil, protein patterns in human diseases, and metallic elements in oil and petroleum products.

8.5 Discriminant analysis

The methods described so far in this chapter have helped us to see whether objects fall into groups when we have no prior knowledge of the groups to be expected. Such methods are sometimes called **unsupervised pattern recognition**. We will now turn to so-called **supervised pattern recognition**. Here we start with a number of objects whose group membership is *known*, for example apple juices extracted from different

varieties of fruit. These objects are sometimes called the **learning or training objects**. The aim of supervised pattern recognition methods is to use these objects to find a rule for allocating a new object of unknown group to the correct group.

The starting point of **linear discriminant analysis (LDA)** is to find a **linear discriminant function (LDF)**, Y , which is a linear combination of the original measured variables:

$$Y = a_1X_1 + a_2X_2 + \cdots + a_nX_n$$

The original n measurements for each object are combined into a single value of Y , so the data have been reduced from n dimensions to one dimension. The coefficients of the terms are chosen in such a way that Y reflects the *difference* between groups as much as possible: objects in the same group will have similar values of Y and objects in different groups will have very different values of Y . Thus the **LDF provides a means of discriminating between the two groups**.

The simplest situation is that in which there are two classes and two variables, X_1 and X_2 , as illustrated in Fig. 8.9(a). This diagram also shows the distribution of the individual variables for each group in the form of dot-plots. For both the variables, there is a considerable overlap in the distributions for the two groups. It can be shown that the LDF for these data is $Y = 0.91X_1 + 0.42X_2$. This LDF is shown by the line labelled Y in Fig. 8.9(b) and the value which the function takes for a given point is given by the projection of the point on to this line. Figure 8.9(b) shows the dot-plots of the LDF, Y , for each group. It can be seen that there is no overlap between the distributions of Y for the two groups, so Y is clearly better at discriminating between the groups than the original variables.

An unknown object will be classified according to its Y value. An initial common-sense approach would be to compare Y with $\bar{Y}_1$ and $\bar{Y}_2$, the Y values for the means of the two groups. If Y is closer to $\bar{Y}_1$ than to $\bar{Y}_2$ then the object belongs to group 1, otherwise it belongs to group 2. For these data, $\bar{Y}_1 = 3.15$ and $\bar{Y}_2 = 10.85$. So if $Y - 3.15 < 10.85 - Y$, that is $Y < 7.0$, we classify the object in group 1, otherwise we classify it in group 2. This method is satisfactory only if the two groups have similarly shaped distributions. Also, if experience shows that a single object is more

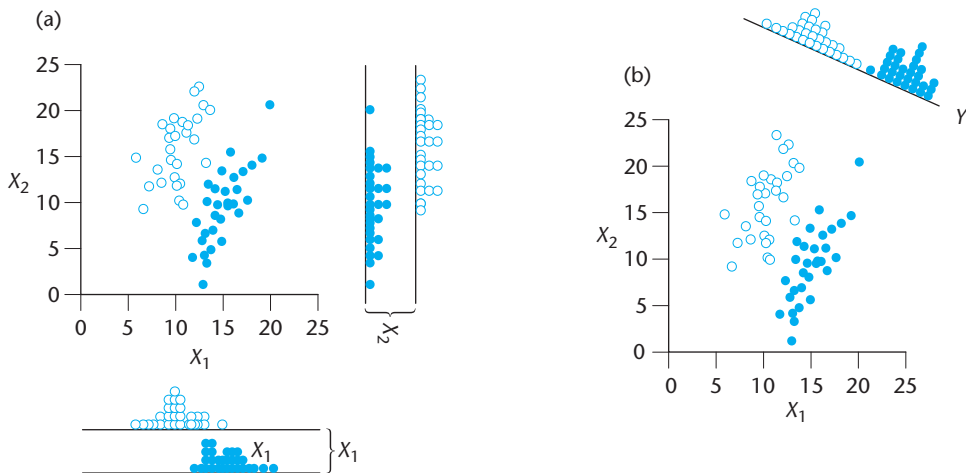


Figure 8.9 (a) Two groups and the distributions of each variable for the group. (b) The distribution of the linear discriminant function for each group.

likely to belong to one of the groups rather than the other, then the decision rule will need to be modified. Software such as Minitab® permits such a modification.

The success of LDA at allocating an object correctly can be tested in several ways. The simplest is to use the classification rule to classify each object in the group and to record whether the resulting classification is correct. The table summarising the results of this procedure is sometimes called the **confusion matrix** (always displayed in Minitab®). This method tends to be overoptimistic since the object being classified was part of the set which was used to form the rule. A better method divides the original data into two groups chosen at random. The first group, known as the **training set**, is used to find the LDF. Then the objects in the second group (the **test set**) are allocated using this function and a success rate found. A third method, which uses the data more economically, is **cross-validation**, sometimes called the 'leave-one-out method'. As the latter name suggests, this finds the LDF with one object omitted and checks whether this LDF then allocates the omitted object correctly. The procedure is then repeated for each object in turn and again a success rate can be found. This method is an option in Minitab®.

If the distributions do not have similar shapes, then a modification of LDA, known as **quadratic discriminant analysis (QDA)**, may be used. This method assumes that the two groups have multivariate normal distributions but with different variances.

LDA and QDA can both be extended to the situation where there are more than two groups of objects. To avoid complex decision rules of the type given above (if $y - 3.15 < 10.85 - y$, etc.) many programs assume a multivariate normal distribution and find a new function, which includes a constant term, for each group. From these functions a score is calculated for each new object and the object is assigned to the group for which the score is highest. This is illustrated in the following example.

Example 8.5.1

The table below gives the concentration in g l^{-1} of sucrose, glucose, fructose and sorbitol in apple juice from three different sources, A, B and C. Carry out an LDA and evaluate the method using cross-validation.

Variety	Sucrose	Glucose	Fructose	Sorbitol
A	20	6	40	4.3
A	27	11	49	2.9
A	26	10	47	2.5
A	34	5	47	2.9
A	29	16	40	7.2
B	6	26	49	3.8
B	10	22	47	3.5
B	14	21	51	6.3
B	10	20	49	3.2
B	8	19	49	3.5
C	8	17	55	5.3
C	7	21	59	3.3
C	15	20	68	4.9
C	14	19	74	5.6
C	9	15	57	5.4

Classify an apple juice with 11, 23, 50 and 3.9 g l⁻¹ of sucrose, glucose, fructose and sorbitol respectively.

The analysis below was obtained using Minitab®.

Discriminant Analysis

```
Linear Method for Response:      Variety
Predictors:  Sucrose  Glucose  Fructose  Sorbitol

Group      A      B      C
Count      5      5      5
```

Summary of Classification

```
Put into      ....True Group....
Group          A      B      C
A              5      0      0
B              0      5      0
C              0      0      5
Total N        5      5      5
N Correct      5      5      5
Proportion     1.000  1.000  1.000

N =      15      N Correct = 15      Proportion Correct = 1.000
```

Summary of Classification with Cross-validation

```
Put into      ....True Group....
Group          A      B      C
A              5      0      0
B              0      5      0
C              0      0      5
Total N        5      5      5
N Correct      5      5      5
Proportion     1.000  1.000  1.000

N =      15      N Correct = 15      Proportion Correct = 1.000
```

Linear Discriminant Function for Group

```
          A      B      C
Constant -44.19 -74.24 -114.01
Sucrose   0.39  -1.66  -2.50
Glucose   0.42   1.21   0.54
Fructose  1.46   2.53   3.48
Sorbitol  2.19   3.59   5.48
```

The 'summary of classification' gives the confusion matrix and shows a 100% success rate. The 'summary of classification with cross-validation' also shows a 100% success rate.

For the new apple juice the linear discriminant scores for each group have values:

Group A: $-44.19 + 0.39 \times 11 + 0.42 \times 23 + 1.46 \times 50 + 2.19 \times 3.9 = 51.301$

Group B: $-74.24 - 1.66 \times 11 + 1.21 \times 23 + 2.53 \times 50 + 3.59 \times 3.9 = 75.831$

Group C: $-114.01 - 2.5 \times 11 + 0.54 \times 23 + 3.48 \times 50 + 5.48 \times 3.9 = 66.282$

The score for group B is highest, so the unknown apple juice is presumed to have come from source B.

Unlike the other procedures described in this chapter, standardising the variables has no effect on the outcome of LDA: it merely re-scales the axes. It may, however, be useful to work with standardised variables in order to decide which variables are important in providing discrimination between the groups: in general it will be those variables which have the larger coefficients in the linear discriminant functions. Once these important variables have been identified, the performance of the method with fewer variables can be studied to see whether a satisfactory discrimination between the groups can still be achieved (see exercise 1 at the end of this chapter).

Some recent applications of LDA include the classification of vegetable oils using the data obtained from an array of gas sensors, and the use of proton magnetic resonance spectra to discriminate between normal and cancerous ovarian tissue. Brain tumours of different types studied by infrared spectrometry have been classified with a high level of accuracy, and studies of many types of food and wine have been successfully performed, in many cases with near-infrared spectroscopy as the analytical method in view of its ease of use in the study of solids and suspensions as well as conventional liquid samples.

Although the above method appears to analyse all the groups simultaneously, the method is actually equivalent to analysing the groups pairwise. An alternative method for more than two groups which genuinely analyses them simultaneously is **canonical variate analysis**. This is an extension of LDA which finds a number of **canonical variates** Y_1 , Y_2 , etc. (which are again linear combinations of the original variables). As with LDA, Y_1 is chosen in such a way that it reflects the difference between the groups as much as possible. Then Y_2 is chosen so that it reflects as much of the remaining difference between the groups as possible, subject to the constraint that there is no correlation between Y_1 and Y_2 , and so on. CVA could be thought of as PCA for groups but, unlike PCA, the results are not dependent on scale, so no pre-treatment of the data is necessary.

The following section describes an alternative method which can be used when there are two or more groups.

8.6

K-nearest neighbour method

The K-nearest neighbour (KNN) method is a conceptually simple way to classify an unknown object when there are two or more groups of objects of *known* class, often called a training set. It makes no assumptions about the distributions in the classes, and can be used when the groups cannot be separated by a plane, as illustrated in Fig. 8.10. In its simplest form the method involves assigning the members of the training set to their known classes. The data should not contain outliers or samples with an ambiguous classification. In addition the classes should be approximately equal in size, to avoid bias when an unknown sample is assigned to a class. The distances of an unknown object from all the members of the training set are then found. The Euclidian distance, Eq. (8.4.1), in multi-dimensional space is usually used, though other distance measures are available. The K smallest distances between the unknown object and the training set samples are then identified, K normally being a small odd number, and the unknown is allocated to the class with the

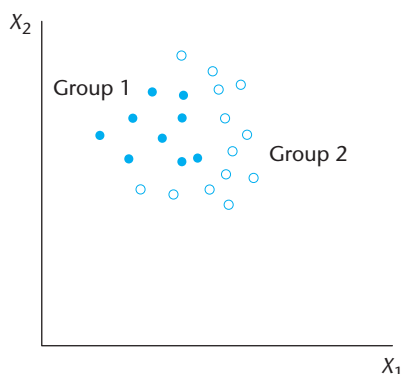


Figure 8.10 Two groups which cannot be separated by a plane.

majority of these K distances. (Clearly, the simplest version of the method uses $K = 1$.) It may be useful to use more than one value of K : if different K -values result in changes in an object's classification, the latter is evidently not very secure. In more sophisticated versions of the method, voting schemes other than the simple majority can be used: this may be appropriate if, for example, the different classes in the training set have notably different variances. In some applications each object is characterised by many variables, some of which may be strongly correlated, while others may have little value in the classification process. (An example would be the absorption or emission intensities recorded at scores or hundreds of wavelengths in spectroscopy.) In such cases a smaller number of variables may be selected before the KNN method is applied.

KNN methods have been applied to many problems in analytical chemistry and related areas, including, as expected, classifications based on chromatographic or spectroscopic data of foodstuffs, soil, water and other environmental samples. In many cases a range of voting schemes and feature selection methods have been compared, to enhance the practical value of this attractive and easily understood approach to classification.

8.7

Disjoint class modelling

The emphasis in the methods described in Sections 8.5 and 8.6 has been on trying to find a boundary between two or more classes, so that an unknown object may be allocated to the correct class. However, the situation may arise when the unknown object does not belong to any of the classes being considered. For example, in Example 8.4.1, it was assumed that the unknown apple juice came from one of the sources A, B or C. However, it might have come from none of these sources but we would have still (incorrectly) allocated it to one of them. A different approach is needed if this sort of error is to be avoided. Instead of having a rule which discriminates between classes, we need a rule which allows us to discriminate between membership and non-membership of a given class. This is done by making a separate model for each class



and using the model in order to test whether the unknown object could be a member of the class. This is called **disjoint class modelling**. For example, if the number of variables is small, each class might be modelled by a multivariate normal distribution. With more variables, some data reduction needs to be carried out first. One such method, called **SIMCA (soft independent modelling of class analogy)**, makes a model of each class in terms of the first few principal components for that class.

8.8 Regression methods

We turn now to the situation in which the variables for each sample can be divided into two groups: response variables and predictor variables. Such a situation arises in multivariate calibration: an example is the determination of the concentration of constituents in a mixture of analytes by spectral analysis. **Here the concentrations of the analytes are the predictor variables and the absorbances at the different wavelengths are the response variables.** Multivariate analysis is appropriate when the spectra of the constituents overlap so that their concentrations cannot be determined without previous chemical separation. In order to calibrate the system a number of specimens containing different mixtures of the analytes are taken and the spectrum is measured for each specimen. Table 8.4 gives an illustrative data set. It shows the UV absorbance ($\times 100$) at six different wavelengths for ten calibration specimens containing different concentrations of three constituents of interest. (In practice, of course, the absorbance would be recorded at hundreds of wavelengths.) What we require is a relationship between these two groups of variables that allows the concentrations of the analytes in a new specimen to be predicted from the spectrum of the new specimen.

Table 8.4 The UV absorbance ($\times 100$) recorded at six different wavelengths, A_1 , A_2 , etc., of ten specimens (1–10) and the measured concentrations (mM), c_1 , c_2 and c_3 of three constituents of interest

Specimen	c_1	c_2	c_3	A_1	A_2	A_3	A_4	A_5	A_6
A	0.89	0.02	0.01	18.7	26.8	42.1	56.6	70.0	83.2
B	0.46	0.09	0.24	31.3	33.4	45.7	49.3	53.8	55.3
C	0.45	0.16	0.23	30.0	35.1	48.3	53.5	59.2	57.7
D	0.56	0.09	0.09	20.0	25.7	39.3	46.6	56.5	57.8
E	0.41	0.02	0.28	31.5	34.8	46.5	46.7	48.5	51.1
F	0.44	0.17	0.14	22.0	28.0	38.5	46.7	54.1	53.6
G	0.34	0.23	0.20	25.7	31.4	41.1	50.6	53.5	49.3
H	0.74	0.11	0.01	18.7	26.8	37.8	50.6	65.0	72.3
I	0.75	0.01	0.15	27.3	34.6	47.8	55.9	67.9	75.2
J	0.48	0.15	0.06	18.3	22.8	32.8	43.4	49.6	51.1

In the classical approach to this problem, the intensities would be treated as the dependent variables and the concentrations as the independent variables. The techniques of linear regression, which were described in Chapter 5, can be used to find a set of regression equations relating the absorbance, A_i , at each wavelength to the concentrations of the analytes. Assuming that absorbance at each wavelength is the sum of the absorbances of the individual constituents, the regression equations take the form $A_i = b_{0i} + b_{1i}c_1 + b_{2i}c_2 + b_{3i}c_3$, where the coefficients for each constituent are dependent on wavelength.

In practice this simple additive model may not describe the situation completely. There are two reasons for this. The first is that the substances of interest may interfere with each other chemically in a way that affects their spectra. The second is that the specimens from 'real-life' sources may well contain substances other than those of interest, which make a contribution to the absorbance. In these cases it is better to use **inverse calibration** and calibrate with 'real-life' specimens. The term 'inverse calibration' means that the analyte concentration is modelled as a function of the spectrum (i.e. the reverse of the classical method). For the data in Table 8.4 the regression equations take the form $c_i = b_{0i} + b_{1i}A_1 + b_{2i}A_2 + \dots + b_{6i}A_6$. Inverse calibration is appropriate because the concentrations can no longer be considered as controlled variables.

The following sections describe a number of methods for predicting one set of variables from another set of variables. In each case the inverse calibration method is illustrated using the data in Table 8.4.

8.9

Multiple linear regression

Multiple linear regression (MLR) involves finding regression equations in the form $c_i = b_{0i} + b_{1i}A_1 + b_{2i}A_2 + \dots + b_{6i}A_6$. **In order to carry out MLR the number of calibration specimens must be greater than the number of predictors.** This is true for the data in Table 8.4 where there are ten specimens and six predictors.

Example 8.9.1

Find the regression equations for predicting c_1 , c_2 and c_3 from A_1 , A_2 , etc. for the data in Table 8.4.

The printout below was obtained using Minitab®.

Regression Analysis: c1 versus A1, A2, A3, A4, A5, A6

The regression equation is

$$c1 = 0.0501 + 0.00252A1 - 0.00939A2 + 0.00375A3 - 0.00920A4 \\ - 0.00106A5 + 0.0179A6$$

Predictor	Coef	SE Coef	T	P
Constant	0.05010	0.08945	0.56	0.615
A1	0.002525	0.008376	0.30	0.783
A2	-0.009387	0.008811	-1.07	0.365
A3	0.003754	0.005852	0.64	0.567
A4	-0.009197	0.005140	-1.79	0.172
A5	-0.001056	0.005373	-0.20	0.857
A6	0.017881	0.002249	7.95	0.004

S = 0.0188690 R-Sq = 99.6% R-Sq(adj) = 98.9%

PRESS = 0.0274584 R-Sq(pred) = 90.55%

This printout gives the regression equation for predicting c_1 from A_1, A_2 , etc. as

$$c_1 = 0.0501 + 0.00252A_1 - 0.00939A_2 + 0.00375A_3 - 0.00920A_4 \\ - 0.00106A_5 + 0.0179A_6$$

A similar analysis can be carried out in order to find the equations for predicting c_2 and c_3 . These are

$$c_2 = 0.027 + 0.0067A_1 - 0.0007A_2 - 0.0184A_3 + 0.0141A_4 + 0.0160A_5 \\ - 0.0152A_6$$

$$c_3 = -0.0776 + 0.00168A_1 + 0.00754A_2 + 0.00668A_3 + 0.00221A_4 \\ - 0.00510A_5 - 0.00237A_6$$

As with univariate regression, an analysis of the residuals is important in evaluating the model. The residuals should be randomly and normally distributed. Figure 8.11 shows a plot of the residuals against the fitted value for c_1 : the residuals do not show any particular pattern. Figure 8.12 plots the predicted values against the measured values. The points are reasonably close to a straight line with no obvious outliers.

The prediction performance can be validated by using a cross-validation ('leave-one-out') method. The values for the first specimen (specimen A) are omitted from the data set and the values for the remaining specimens (B-J) are used to find the regression equation of, for example, c_1 on A_2, A_3 , etc. Then this new equation is used to obtain a predicted value of c_1 for the first specimen. This procedure is repeated, leaving each specimen out in turn. Then for each specimen the difference between the actual and predicted value is calculated. The sum of the squares of these differences is called the **predicted residual error sum of squares** or **PRESS** for short: the closer the value of the PRESS statistic to zero, the better the predictive power of the model. It is particularly useful for comparing the predictive powers of different models. For the model fitted here Minitab® gives the value of PRESS as 0.0274584.

Minitab® also gives values of t (called ' T ' in the Minitab printout) and associated P -values for each of the coefficients in the regression equation. This tests the null hypothesis that each coefficient is zero, *given that all the other variables are present*

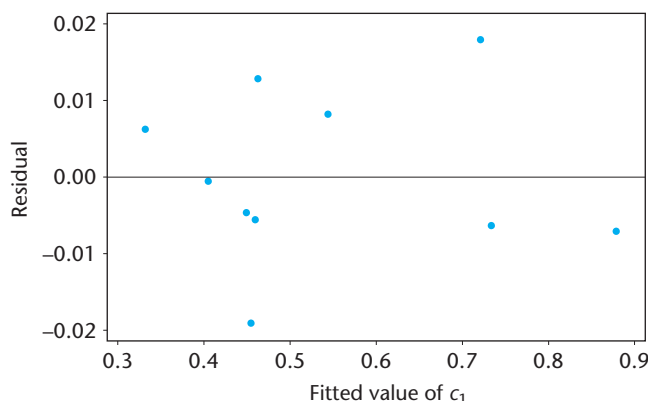


Figure 8.11 A plot of the residuals against the fitted values for c_1 for Example 8.9.1.

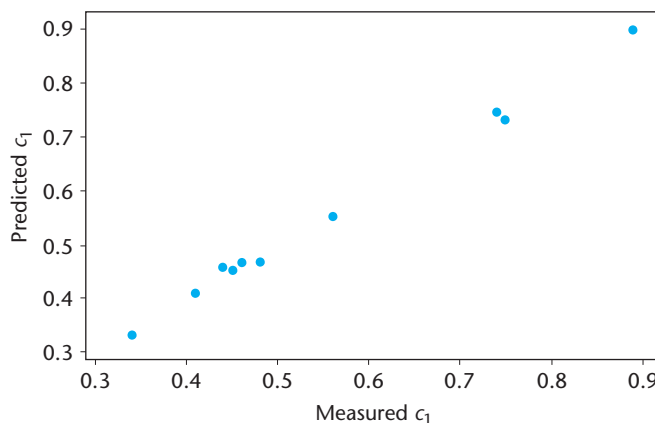


Figure 8.12 A plot of the predicted values of c_1 against the measured values for Example 8.9.1.

in the model. Inspection of these suggests that any one of A_1 to A_5 could be left out of the model without reducing its effectiveness. We could, if we wished, try all possible combinations of predictor variables and find the model that predicts most successfully with the minimum number of predictor variables, using the PRESS statistic to compare the models. This might seem to be the course that we would have to follow when we are dealing with a spectrum containing measurements at hundreds of wavelengths because in this case the number of predictor variables far exceeds the number of specimens. In order to form a regression equation we would have to select the absorbances at only a small proportion of the wavelengths. However, it is not the best way to proceed because it means that a large amount of information is discarded. The next section describes a method that makes better use of the data.



8.10 Principal component regression

The basis of principal component regression (PCR) is to reduce the number of predictor variables by using their first few principal components rather than the original variables. The method works well when there is a considerable degree of correlation between the predictor variables. This is usually the case in inverse calibration: it is true for the data in Table 8.4, as can be seen from the correlation matrix in Table 8.5. In this case only a few of the principal components are needed to describe most of the variation in the data. These principal components are uncorrelated (see Section 8.3).

PCR is also a useful technique when the predictor variables are *very* highly correlated: this can cause mathematical complications problems with MLR, resulting in unreliable predictions.

The following example shows the steps involved in carrying out PCR using the data in Table 8.4. Obviously there is no need for PCR when there are so few predictor variables: the purpose of the example is to illustrate the method.

Table 8.5 Correlation matrix for the data in Table 8.4

	c_1	c_2	c_3	A_1	A_2	A_3	A_4	A_5
c_2	-0.637							
c_3	-0.717	0.088						
A_1	-0.482	-0.116	0.947					
A_2	-0.260	-0.194	0.832	0.941				
A_3	-0.001	-0.413	0.677	0.841	0.936			
A_4	0.625	-0.355	-0.096	0.148	0.422	0.598		
A_5	0.899	-0.434	-0.541	-0.293	-0.002	0.227	0.857	
A_6	0.977	-0.608	-0.603	-0.346	-0.089	0.161	0.771	0.960

Example 8.10.1

Carry out a PCR of the data in Table 8.4 in order to obtain an equation for predicting c_1 from the spectrum.

(The reader needs to be familiar with the material in Section 8.3 before reading the solution to this example.)

This can be done using a variety of computer packages (for example, The Unscrambler®). In Minitab® it is necessary to first carry out a principal components analysis (PCA) and then perform the regression.

The printout below was obtained from Minitab® and shows the results of a PCA of the absorbances in Table 8.4.

Principal Component Analysis: A1, A2, A3, A4, A5, A6

Eigenanalysis of the Covariance Matrix

Eigenvalue	210.01	73.86	4.62	0.93	0.79	0.28
Proportion	0.723	0.254	0.016	0.003	0.003	0.001
Cumulative	0.723	0.977	0.993	0.996	0.999	1.000
Variable	PC1	PC2	PC3	PC4	PC5	PC6
A1	-0.124	-0.592	-0.253	-0.048	0.340	0.672
A2	-0.017	-0.513	0.048	0.196	0.493	-0.673
A3	0.066	-0.571	-0.102	0.128	-0.793	-0.118
A4	0.244	-0.239	0.575	-0.743	-0.002	-0.002
A5	0.510	-0.042	0.545	0.602	0.059	0.276
A6	0.813	0.043	-0.544	-0.168	0.091	-0.075

This shows that between them the first three principal components account for over 99% of the variation in the absorbances and so the regression will be carried out on these three components.

The scores (see Example 8.3.1) for these three principal components can be calculated using Minitab® and are given below.

Specimen	Z_1	Z_2	Z_3
A	117.1	-61.7	17.7
B	83.0	-73.4	16.6
C	89.0	-76.1	20.8
D	86.8	-58.4	18.3
E	76.2	-74.0	14.5
F	81.9	-60.5	19.0
G	78.7	-67.0	22.3
H	104.0	-58.1	17.9
I	108.6	-74.1	18.1
J	76.9	-51.5	17.3

Then the regression equation of c_1 on Z_1 , Z_2 and Z_3 is found (using Minitab®):

Regression Analysis: c1 versus z1, z2, z3

The regression equation is

$$c_1 = 0.0685 + 0.0119Z_1 + 0.00419Z_2 - 0.0171Z_3$$



Predictor	Coef	SE Coef	T	P
Constant	0.06849	0.06571	1.04	0.337
Z1	0.0118502	0.0003480	34.05	0.000
Z2	0.0041884	0.0005868	7.14	0.000
Z3	-0.017058	0.002345	-7.27	0.000

S = 0.0151299 R-Sq = 99.5% R-Sq(adj) = 99.3%

PRESS = 0.00301908 R-Sq(pred) = 98.96%

As for MLR, an analysis of the residuals should be carried out. The PRESS statistic is 0.00301908, lower than it was for the MLR. Here the T -values indicate that all the coefficients other than the constant term are significantly different from zero. (At this stage the possibility of fitting a model with zero intercept could be explored.)

The regression equation is

$$c_1 = 0.0685 - 0.0119Z_1 + 0.00419Z_2 - 0.171Z_3$$

If required an expression for c_1 in terms of the absorbances can be obtained by substituting expressions for Z_1 , Z_2 and Z_3 in terms of A_1 , A_2 , etc. For example, referring back to the PCA gives:

$$Z_1 = -0.124A_1 - 0.017A_2 + 0.066A_3 + 0.244A_4 + 0.510A_5 + 0.813A_6$$

and similarly for Z_2 and Z_3 . This leads to the equation:

$$c_1 = 0.06849 + 0.00037A_1 - 0.00317A_2 + 0.00014A_3 - 0.00792A_4 \\ - 0.00343A_5 + 0.01909A_6$$

A similar analysis can be carried out to predict c_2 and c_3 .

As already noted, this example illustrates the method very simply. However, even with a spectrum containing absorbances at several hundred wavelengths we would expect to find that only a few principal components are needed to describe most of the variation, provided that the absorbances at the different wavelengths are correlated.

PCR only utilises the correlations between the predictor variables. If we look at Table 8.5 we see that there is also considerable correlation between the predictor and the response variables. The following section describes a regression method that makes use of both types of correlation.



8.11 Partial least-squares regression

Like PCR, partial least-squares (PLS) regression uses linear combinations of the predictor variables rather than the original variables. However, the way in which these linear combinations is chosen is different. In PCR the principal components are chosen so that they describe as much of the variation in the *predictors* as possible, irrespective of the strength of the relationships between the predictor and the response variables. In PLS, variables that show a high correlation with the *response* variables are given extra weight because they will be more effective at prediction. In this way linear combinations of the predictor variables are chosen that are highly correlated with the response variables and also explain the variation in the predictor variables. As with PCR, it is hoped that only a few of the linear combinations of the predictor variables will be required to describe most of the variation.

Example 8.11.1

Carry out PLS regression on the data in Table 8.4 in order to obtain an equation for predicting c_1 .

PLS can be carried out using a number of computer packages (for example, Minitab® and The Unscrambler®). The following printout was obtained from Minitab®.

PLS Regression: c1 versus A1, A2, A3, A4, A5, A6

Number of components selected by cross-validation: 4

Number of observations left out per group: 1

Number of components cross-validated: 6

Analysis of Variance for c1

Source	DF	SS	MS	F	P
Regression	4	0.289476	0.0723690	333.84	0.000
Residual Error	5	0.001084	0.0002168		
Total	9	0.290560			

Model Selection and Validation for c1

Components	X Variance	Error SS	R-Sq	PRESS	R-Sq (pred)
1	0.457325	0.0287984	0.900887	0.0469069	0.838564
2	0.957200	0.0255230	0.912159	0.0511899	0.823823
3	0.988793	0.0021123	0.992730	0.0078758	0.972894
4	0.992990	0.0010839	0.996270	0.0052733	0.981851
5		0.0010724	0.996309	0.0186933	0.935664
6		0.0010681	0.996324	0.0274584	0.905498

	c1	c1 standardized
Constant	0.0426293	0.00000
A1	0.0039542	0.11981
A2	-0.0111737	-0.27695
A3	0.0038227	0.10753
A4	-0.0092380	-0.22261
A5	-0.0003408	-0.01425
A6	0.0176165	1.16114

The results have been evaluated using the 'leave-one-out' method (see Example 8.9.1). The first block in the printout shows that using this method of cross-validation the number of components required to model c_1 is 4. The third block in the table gives the reason for this choice: it shows that value of the PRESS is lowest for a 4-component model, taking the value 0.0052733. (This, incidentally, is higher than it was for the PCR model.) **Note that the predictive value of the model, as measured by the PRESS value, decreases if more components are added.** The first column of the last block in the table gives the coefficients in the equation for this model. So the regression equation is:

$$c_1 = 0.0426 + 0.0040A_1 - 0.0112A_2 + 0.0038A_3 - 0.0092A_4 - 0.0003A_5 + 0.0176A_6$$

Again an analysis of the residuals should be carried out.

Equations for predicting c_2 and c_3 can be found in a similar way.

It is interesting to compare the equations for c_1 obtained by MLR, PCR and PLS. These are


$$\text{MLR: } c_1 = 0.0501 + 0.00252A_1 - 0.00939A_2 + 0.00375A_3 - 0.00920A_4 - 0.00106A_5 + 0.0179A_6$$

$$\text{PCR: } c_1 = 0.06849 + 0.00037A_1 - 0.00317A_2 + 0.00014A_3 - 0.00792A_4 - 0.00343A_5 + 0.01909A_6$$

$$\text{PLS: } c_1 = 0.0426 + 0.0040A_1 - 0.0112A_2 + 0.0038A_3 - 0.0092A_4 - 0.0003A_5 + 0.0176A_6$$

Although the coefficients differ from one equation to another, they have the same sign in each equation and in all three equations the term in A_6 dominates.

In Example 8.11.1, each response variable has been treated separately. This is known as PLS1. The response variables can also be treated collectively. This is known as PLS2. It is usually used only when the response variables are correlated with each other. There is often little to choose between the two methods in terms of predictive ability.

Sections 8.9 to 8.11 have given a brief description of methods for making a regression model for multivariate calibration. To summarise, MLR would rarely be used because it cannot be carried out when the number of predictor variables is greater than the number of specimens. Rather than select a few of the predictor variables, it is better to reduce their number to just a few by using PCR or PLS. These methods give satisfactory results when there is correlation between the predictor variables. The preferred method in a given situation will depend on the precise nature of the data: an analysis can be carried out by each method and the results evaluated in order to find which method performs better. For example, for the data in Table 8.4 PCR performed better than PLS as measured by the PRESS statistic.



Many recent applications of PCR and PLS have arisen in molecular spectroscopy, where strongly overlapping absorption and emission spectra often arise, even in simple mixtures. For example a pesticide and its metabolites have been successfully analysed using Fourier transform infrared spectroscopy, and a mixture of very similar phenols was resolved by means of their fluorescence excitation spectra.

8.12 Natural computation methods: artificial neural networks

Recent years have seen a substantial growth in the application to chemical problems of computation methods that emulate natural, and especially biological, processes. As with the other methods outlined in this chapter these **natural computation** methods require computers for their implementation, but modern desktop computers are quite adequate in many cases. Simulated annealing, summarised in Chapter 7 as an approach to optimisation problems, can be regarded as a natural computation method based on a physical phenomenon. **Genetic algorithms** have a genuinely biological background. They are again used for optimisation and for other applications such as wavelength selection in spectroscopy. In general such an algorithm begins

with a random selection of the initial population (using this term biologically, not statistically), for example 10 wavelengths out of the hundreds possible in UV-visible spectroscopy. Subsequent generations are generated from this initial population by processes that mirror heredity in living beings, such as the selection of suitable parents, the crossover between the genes of two parents, and mutations. Each new generation is tested for its output (in our example this might be the accuracy of a multi-component spectroscopic analysis), and the quasi-genetic processes continue until an acceptable outcome is generated. The widely used MATLAB[®] technical software provides genetic algorithm facilities.

Artificial neural networks (ANN) are now finding many uses in analytical science. In a simple form ANNs attempt to imitate the operation of neurons in the brain. Brain neurons receive input signals via numerous filamentous extensions called dendrites, and send out signals through another very long, thin strand called an axon, which can transmit electrical signals. The axon also has many branches at the terminus distant from the cell nucleus. At the end of these branches synapses use neurotransmitter molecules to pass on signals to the dendrites of other neurons. Analogously, ANNs have a number of linked layers of artificial neurons, including an input and an output layer (Fig. 8.13). Measured variables are presented to the input layer and are processed by mathematical operations in one or more intermediate ('hidden') layers, to produce one or more outputs. (More details are given in the text by Otto, listed in the Bibliography.) In inverse calibration, the inputs could be the absorbances at various wavelengths and the output could be the concentrations of one or more analytes. The network is trained by an interactive procedure using a training set: in the calibration example the ANN would calculate concentrations for each member of the training set, and any discrepancies between the network's output and the known concentrations would be used to adjust internal parameters in

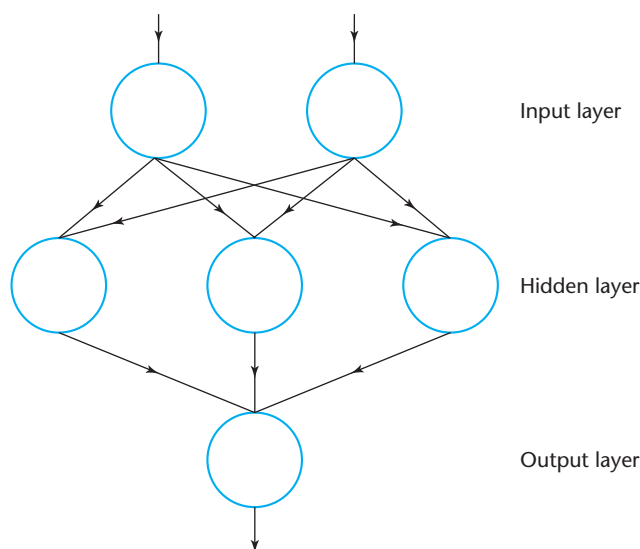


Figure 8.13 An example of a neural network.

the network. These prediction and adjustment steps are repeated until the required degree of accuracy, evaluated with a test set, is achieved. Since the training and test sets are bound to differ to some extent, it is important not to over-fit the training set, otherwise the network may perform less well with the test set, and subsequently with 'unknown' samples.

Unlike MLR, PCR and PLS methods, ANNs do not assume any initial mathematical relationship between the input and output variables, so they are particularly useful when the underlying mathematical model is unknown or uncertain. For example, they are appropriate in multivariate calibration when the analytes interfere with each other strongly. However, the lack of an assumed mathematical model has its disadvantages. Larger training sets than those used in the other techniques described in this chapter may be necessary, and there is no direct way to extract information about a suitable mathematical model, or to estimate confidence intervals.

Neural networks are versatile and flexible tools for modelling complex relationships between variables. They have been widely used in pattern recognition and calibration studies in conjunction with many spectroscopic, chromatographic and electrochemical methods. Many neural network designs have been studied, and their performances have been compared with the other multivariate approaches described in this chapter. It seems that in at least some cases, ANNs achieve results of high quality that are not obtainable with the other methods. Again MATLAB[®] supplies facilities for ANNs, with a range of options in terms of the number of layers available and the data propagation methods used.

8.13 Conclusions

The aim of this chapter has been to give an introduction to the methods of multivariate analysis which are most commonly used in analytical chemistry. In most cases there is a choice of several different multivariate methods which could be applied to the same set of data. For example, in cluster analysis there is a choice between a hierarchical and a non-hierarchical approach, and each of these approaches offers a choice of several different methods. In multivariate calibration there is a choice between multiple regression, PCR and PLS regression. In addition, several approaches might be tried in the initial analysis. For example cluster analysis and principal components analysis might be used prior to linear discriminant analysis, in order to see whether the objects being analysed fall naturally into groups.

This chapter has been able to introduce only a small number of the numerous multivariate analysis methods that have become relatively commonplace in the analytical sciences in recent years. This is still a rapidly developing field, with further new methods becoming ever more widely available as the power and speed of desktop computers grow. Similarly there is now a great range of websites, downloads and software packages to encourage their use (see below), which will doubtless continue to increase.

Bibliography and resources

Books

- Brereton, R.G., 2003, *Chemometrics: Data Analysis for the Laboratory and Chemical Plant*, John Wiley, Chichester. Detailed coverage of pattern recognition and related methods, with many chemical examples.
- Flury, B. and Riedwyl, H., 1988, *Multivariate Statistics: A Practical Approach*, Chapman & Hall, London. Introduces selected methods of multivariate analysis at a non-technical level, with an emphasis on their basic principles.
- Manly, B.F.J., 2004, *Multivariate Statistical Methods: A Primer*, 3rd edn, Chapman & Hall, London. A general introduction to multivariate analysis at a non-technical level.
- Martens, H. and Naes, T., 1989, *Multivariate Calibration*, John Wiley, Chichester. The book is structured so as to give a tutorial on the practical use of multivariate calibration techniques. It compares several calibration models, validation approaches, and ways to optimise models.
- Otto, M., 2007, *Chemometrics: Statistics and Computer Application in Analytical Chemistry*, 2nd edn, Wiley-VCH, Weinheim. Gives a detailed treatment of the topics introduced in this chapter.
- Vandeginste, B.G.M., Massart, D.L., Buydens, L.M.C., De Jong, S., Lewi, P.L. and Smeyers-Verbeke, J., 1998, *Handbook of Chemometrics and Qualimetrics: Part B*, Elsevier, Amsterdam. A detailed and comprehensive account of the application of multivariate techniques in analytical chemistry.

Software and internet resources

The advanced technical computing program Matlab® (the name derives from its ability to handle matrix-based problems) is available from www.mathworks.com. It operates from a variety of computing platforms, and provides extensive statistical and chemometric facilities. Downloads are available, and the large number of worldwide users ensures that there is much advice and tutorial material available. There is a chemometrics toolbox with over 40 specialised functions.

Most of the methods described in this chapter are also available through the software package Pirouette®, using the Windows environment. It is available from Infometrix at www.infometrix.com. Smaller ('Pirouette Lite®') packages covering exploratory data analysis, regression and classification are available separately.

Exercises

- 1 For the data in Example 8.5.1 carry out a linear discriminant analysis working with the standardised variables. Hence identify the two variables which are most effective at discriminating between the two groups. Repeat the discriminant analysis with these two variables. Use the cross-classification success rate to compare the performance using two variables with that using all four variables.
- 2 The data below give the concentration (in mg kg^{-1}) of four elements found in samples of rice. The rice was one of two types (polished (P) or unpolished (U)),

was one of two varieties (A or B), and was grown either in the wet season (W) or the dry season (D).

Variety	Type	Season	P	K	Ni	Mo
A	U	D	3555	2581	0.328	0.535
A	U	D	3535	2421	0.425	0.538
A	U	D	3294	2274	0.263	0.509
A	P	D	1682	1017	0.859	0.494
A	P	D	1593	1032	1.560	0.498
A	P	D	1554	984	1.013	0.478
B	U	D	3593	2791	0.301	0.771
B	U	D	3467	2833	0.384	0.407
B	P	D	2003	1690	0.216	0.728
B	P	D	1323	1327	0.924	0.393
A	U	W	3066	1961	0.256	0.481
A	P	W	1478	813	0.974	0.486
B	U	W	3629	2846	1.131	0.357
B	U	W	3256	2431	0.390	0.644
B	P	W	2041	1796	0.803	0.321
B	P	W	1745	1383	0.324	0.619

(Adapted from Phuong, T.D., Choung, P.V., Khiem, D.T., and Kokot, S., 1999, *Analyst*, 124, 553)

- Carry out a cluster analysis. Do the samples appear to fall into groups? What characteristic is important in determining group membership?
 - Calculate the correlation matrix. Which pairs of variables are strongly correlated? Which variable(s) show little correlation with the other variables?
 - Carry out a principal components analysis and obtain a score plot. Does it confirm your analysis in (a)?
 - Is it possible to identify the variety of a sample of rice by measuring the concentration of these four elements? Answer this question by carrying out a linear discriminant analysis. Investigate whether it is necessary to measure the concentration of all four elements in order to achieve satisfactory discrimination.
- 3 The table below shows the fluorescence intensity (arbitrary units) recorded at 10 different wavelengths, I_1 , I_2 , I_3 , etc., for nine specimens containing measured concentrations (nM), c_1 and c_2 , of two analytes.

c_1	c_2	I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9	I_{10}
0.38	0.10	33	31	28	26	24	21	19	17	14	12
0.60	0.90	62	64	62	69	72	74	77	79	82	84
0.88	0.96	89	89	83	91	91	92	92	93	94	94
0.01	0.41	5	8	11	14	18	21	24	28	31	34
0.86	0.14	74	68	61	56	50	44	38	32	26	20
0.25	0.05	21	19	17	16	14	13	11	9	8	6
0.03	0.16	4	5	6	7	9	10	11	12	13	14
0.22	0.02	18	17	15	13	12	10	8	7	5	3
0.29	0.34	27	28	28	29	29	30	30	31	31	32

- (a) Why is not possible to use MLR to carry out an inverse calibration for c_1 or c_2 for these data if the fluorescence intensities at all the different wavelengths are used?
- (b) Carry out a PCA of the fluorescence intensities and hence obtain a regression equation for predicting c_1 from the first two principal components of the fluorescence intensities. Check that the residuals are approximately randomly and normally distributed.
- (c) Use PLS1 to obtain a regression equation for predicting c_1 from the fluorescence intensities. Check that the residuals are approximately randomly and normally distributed.
- (d) Compare the predictive power of the models obtained by PCR and PLS1 using the PRESS statistic.
- (e) The fluorescence intensities, I_1 , I_2 , I_3 , etc., for a new specimen (measured in the same units and in the same conditions) are 51, 49, 46, 46, 44, 43, 41, 39, 38 and 36. Use the equations obtained in (b) and (c) to calculate the value of c_1 for this specimen.

Solutions to exercises

(NB. Outline solutions are provided here: fuller solutions with commentaries are included in the Instructors' Manual.)

Chapter 1

- 1 Mean results (g l^{-1}) for laboratories A–E are: 41.9, 41.9, 43.2, 39.1, 41.5. Hence A – precise, little bias, mean accurate; B – poor precision, little bias, mean accurate but not very reliable; C – precise but biased to high values, poor accuracy; D – poor precision, biased to low values, poor accuracy; E – similar to A, but the last result might be an 'outlier'.
- 2 Laboratory A still shows little bias, but precision is poorer, reflecting reproducibility (i.e. between-day precision) rather than repeatability (within-day precision).
- 3 Number of binding sites must be an integer, clearly 2 here, so results are precise, but biased to low values. The bias does not matter much, as two binding sites can be deduced.
- 4 (a) Blood lactate levels vary a lot in healthy patients, so great precision and accuracy are not needed. (b) Unbiased results could be crucial because of the great economic importance of U. (c) Speed of analysis is crucial here, so precision and accuracy are less important. (d) The aim is to detect even small changes over time, so precision is most important.
- 5 (a) Sample might not be representative, and/or reduction of Fe(III) to Fe(II) might be incomplete, giving biased results in each case. Completeness of reduction could be tested using a standard material. Random errors in each stage, including titrimetry, where they should be small. (b) Sampling problem as in (a), and also incomplete extraction, leading to bias (checked with standard). Random errors in spectrometry, which again should be relatively small. (c) Random errors in gravimetry should be very small: more significant will be chemical problems such as co-precipitation, giving biased results.

Chapter 2

- 1 Mean = $0.077 \mu\text{g ml}^{-1}$, SD = $0.007 \mu\text{g ml}^{-1}$, RSD = 9%.
- 2 19.01 ± 3.17 (95%); 19.01 ± 4.81 (99%).
- 3 Mean = 22.3 ng ml^{-1} , SD = 1.4 ng ml^{-1} , RSD = 6.2%, 99% CI = $22.3 \pm 1.4 \text{ ng ml}^{-1}$. Mean = 12.83 ng ml^{-1} , SD = 0.95 ng ml^{-1} , RSD = 7.4%, 99% CI = $12.8 \pm 1.6 \text{ ng ml}^{-1}$.
- 4 $10.12 \pm 0.18 \text{ ng ml}^{-1}$. Approximately 160.
- 5 $49.5 \pm 1.1 \text{ ng ml}^{-1}$. Yes.
- 6 $10.18 \pm 0.23 \text{ ml}$. No evidence for systematic error.
- 7 For weight of reagent: SD = 0.14 mg, RSD = 0.028% (0.029%).
For volume of solvent: RSD = 0.02%.
For molarity: RSD = 0.034% (0.020%).
Values for reagent with formula weight 392 are given in brackets.
- 8 s.d. = $0.44 \times 10^{-6} \text{ M}$.

Chapter 3

- 1 The points lie approximately on a straight line, indicating that the data are drawn from a normal distribution.
- 2 $t = 1.54, 1.60, 1.18, 1.60$. None of the means differs significantly from certified value.
- 3 (a) $Q = 0.565$ or $G = 1.97$. Not significant at $P = 0.05$. (b) $F = 34$. Significant at $P = 0.05$.
- 4 (a) $F = 1.70$. Not significant at $P = 0.05$. (b) $t = \pm 1.28$. Not significant at $P = 0.05$.
- 5 Between-sample mean square = 2121.9, within-sample mean square = 8.1. $F = 262$. Highly significant difference between depths. All pairs, except deepest pair, differ significantly from each other.
- 6 $t = \pm 1.20$. Sexes do not differ significantly.
- 7 $\chi^2 = 16.8$. No evidence that some digits are preferred to others.
- 8 Pine: $t = \pm 2.27$, not significant. Beech: $t = \pm 5.27$, significant at $P = 0.01$. Aquatic: $t = \pm 3.73$, significant at $P = 0.01$.
- 9 (a) $\chi^2 = 5.95$. The first worker differs significantly from the other three.
(b) $\chi^2 = 2.81$. The last three workers do not differ significantly from each other.
- 10 $t = \pm 1.02$. Methods do not differ significantly.
- 11 Between-samples mean square = 0.1144, within-samples mean square = 0.0445. $F = 2.57$. Just significant at $P = 0.05$. Least significant difference (0.25) indicates that A differs from B, D and E.

- 12 $t = \pm 2.2$. Men and women differ significantly.
- 13 $t = \pm 3.4$. Methods differ significantly.
- 14 Minimum size is 12.
- 15 The estimated mean and standard deviation from the data are 1.08 and 0.41 respectively. When the z -values (0.54, 1.02, etc.) are plotted the maximum difference is only 0.11 at $z = 0.54$. The critical value is 0.262 so the null hypothesis can be retained: the data fit this normal distribution very well.
- 16 (a) The posterior distribution is a normal distribution with mean equal to the mean of the four measurements, that is 7.5, and standard deviation $= 0.2/\sqrt{n}$ where $n = 4$, that is 0.1. The curve is truncated at 7 and 9 but this would not be apparent in a sketch of the curve since these values are well over 3 standard deviations from the mean. (b) Taking 2 standard deviations of either side of the mean gives 7.5 ± 0.2 or 7.3 to 7.5. This range includes approximately 95% of the area under the curve.

Chapter 4

- 1 For scheme 1, $\sigma^2 = (4/2) + (10/5) = 4$. For scheme 2, $\sigma^2 = 4/(2 \times 3) + 10/3 = 4$. If S is the cost of sampling and A the cost of the analysis, then (cost of scheme 1/cost of scheme 2) $= (5S + 2A)/(3S + 6A)$. This ratio is > 1 if $S/A > 2$.
- 2 ANOVA calculations show that the mean squares for the between-days and within-days variations are 111 and 3.25 respectively. Hence $F = 111/3.25 = 34$. The critical value of $F_{3,8}$ is 4.066 ($P = 0.05$), so the mean concentrations differ significantly. The sampling variance is given by $(111 - 3.25)/3 = 35.9$.
- 3 The mean squares for the between-sample and within-sample variations are 8.31×10^{-4} and 1.75×10^{-4} respectively, so $F = 8.31/1.75 = 4.746$. The critical value of $F_{3,8}$ is 4.066 ($P = 0.05$), so the between-sample mean square cannot be explained by measurement variation only. The latter variation, σ_0^2 , is estimated as 1.75×10^{-4} . The estimate of the sampling variance, σ_1^2 , is $[(8.31 - 1.75) \times 10^{-4}]/3 = 2.19 \times 10^{-4}$. Hence the variance of the mean for scheme 1 is $0.000175/4 + 0.000219/6 = 0.00008025$, and the variance of the mean for scheme 2 is $[0.000175/(2 \times 3)] + 0.000219/3 = 0.0001022$.
- 4 The six samples give six estimates of σ^2 , which have an average of 2.795. So $\sigma = 1.67$. Hence the action and warning lines are at $50 \pm (2 \times 1.67)/\sqrt{4}$ and $50 \pm (3 \times 1.67)/\sqrt{4}$ respectively, i.e. at 50 ± 1.67 and 50 ± 2.50 respectively.
- 5 Samples A and B give mean values of 7.01 and 7.75 ppm respectively. Using a table of D and T values (e.g. for laboratory 1 these are -1.2 and 18.8 respectively), we find that $s_R^2 = 11.027$ and $s_T^2 = 0.793$. So $F = 11.027/0.793 = 13.905$, far higher than the critical $F_{14,14}$ value of ca. 2.48 ($P = 0.05$), obtained from the table by interpolation. Systematic errors are thus significant, and s_T^2 is found to be 5.117.
- 6 For the Shewhart chart for the mean, the values of W and A are found from tables ($n = 5$) to be 0.3768 and 0.5942 respectively. Hence the warning lines are

at $120 \pm (7 \times 0.3768) = 120 \pm 2.64$, and the action lines are at $120 \pm (7 \times 0.5942) = 120 \pm 4.16$. For the range chart, the tables give w_1 , w_2 , a_1 and a_2 as 0.3653, 1.8045, 0.1580 and 2.3577 respectively so the lower warning line is at $7 \times 0.3653 = 2.56$, the upper warning line is at 12.63, and the lower and upper action lines are at 1.11 and 16.50 respectively.

- 7 Since $\sigma = 0.6$ and $n = 4$, the warning and action lines for the Shewhart chart for the mean are at 80 ± 0.6 and 80 ± 0.9 respectively. On this chart, the points for days 14–16 fall between the warning and action lines and point 17 is below the lower action line. So the chart suggests that the analytical process has gone out of control at about day 14. The CUSUM chart shows a steady negative trend from day 9 onwards, suggesting that the method was going out of control a good deal earlier.

Chapter 5

- Here $r = -0.8569$. This suggests a strong correlation; Eq. (5.3.2) gives $t = 3.33$, well above the critical value ($P = 0.05$) of 2.78. But (a) a non-linear relationship is more likely, and (b) correlation is not causation – the Hg contamination may arise elsewhere.
- In this case $r = 0.99982$. But the increase in the value of y (absorbance) with x is by a slightly decreasing amount at each point, i.e. this is really a curve, though little harm would come from treating it as a straight line.
- The usual equations give $a = 0.0021$, $b = 0.0252$ and $s_{y/x} = 0.00703$. We then obtain $s_a = 0.00479$ and $s_b = 0.000266$. To convert the two latter values into 95% confidence intervals we multiply by $t = 2.57$, giving intervals for the intercept and slope of 0.0021 ± 0.0123 and 0.0252 ± 0.0007 respectively.
- (a) A y -value of 0.456 corresponds to a concentration of 18.04 ng ml^{-1} . The s_{x_0} -value is 0.300 so the confidence limits are $18.04 \pm (2.57 \times 0.300) = 18.04 \pm 0.77 \text{ ng ml}^{-1}$. (b) The Q-test shows that the absorbance reading of 0.347 can be rejected as an outlier, the mean of the remaining three readings being 0.311, i.e. a concentration of 12.28 ng ml^{-1} . With $m = 3$ in this case, $s_{x_0} = 0.195$, giving confidence limits of $12.28 \pm 0.50 \text{ ng ml}^{-1}$.
- The absorbance at the limit of detection is given by $a + 3s_{y/x} = 0.0021 + (3 \times 0.00703) = 0.0232$. This corresponds to an x -value of 0.84 ng ml^{-1} , which is the limit of detection.
- Here $a = 0.2569$ and $b = 0.005349$, so the Au concentration is $0.2569/0.005349 = 48.0 \text{ ng ml}^{-1}$. The value of $s_{y/x}$ is 0.003693, so s_{x_E} is 0.9179. In this case $t = 2.45$, so the 95% confidence limits for the concentration are $48.0 \pm (2.45 \times 0.9179) = 48.0 \pm 2.2 \text{ ng ml}^{-1}$.
- The unweighted regression line has $b = 1.982$ and $a = 2.924$ respectively. Intensity values of 15 and 90 correspond to 6.09 and 43.9 ng ml^{-1} respectively. Then $s_{y/x} = 2.991$ and $s_{x_E} = 1.767$. So the confidence limits for the two concentrations are 6.09 ± 4.9 and $43.9 \pm 4.9 \text{ ng ml}^{-1}$. The weighted line is found

from the s -values for each point, in increasing order 0.71, 0.84, 0.89, 1.64, 2.24, 3.03. The corresponding weights are 2.23, 1.59, 1.42, 0.42, 0.22 and 0.12 (totalling 6 as expected). The weighted line then has $b = 1.964$ and $a = 3.483$, so the intensity values of 15 and 90 correspond to concentrations of 5.87 and 44.1 ng ml⁻¹ respectively. Estimated weights for these two points are 1.8 and 0.18 respectively, giving $s_{x_{\text{low}}}$ -values of 0.906 and 2.716, and confidence limits of 5.9 ± 2.5 and 44.1 ± 7.6 ng ml⁻¹.

- 8 If the ISE results are plotted as y and the gravimetric data are plotted as x the resulting line has $a = 4.48$ and $b = 0.963$. The r -value is 0.970. The confidence limits for a are 4.5 ± 20.1 , which includes zero, and the limits for b are 0.96 ± 0.20 , which includes 1, so there is no evidence of bias between the two methods.
- 9 Inspection suggests that the plot is linear up to $A = 0.7$ – 0.8 . The line through all six points gives $r = 0.9936$, and residuals of -0.07 , -0.02 , $+0.02$, $+0.06$, $+0.07$ and -0.07 . The trend suggests a curve. The SS for these values is 0.0191. If the last value is omitted, we find $r = 0.9972$, the residuals are -0.04 , 0 , $+0.02$, $+0.04$ and -0.02 (SS = 0.0040). Similar calculations show that the fifth point can be omitted also, at some cost in the range of the experiment.
- 10 The two straight line graphs are $y = 0.0014 + 0.0384x$, and $y = 0.1058 - 0.012x$. These intersect at an x -value of $(0.1058 - 0.0014)/(0.0384 - [-0.012]) = (0.1044/0.0504) = 2.07$, suggesting the formation of a 2 : 1 DPA : europium complex.
- 11 The best quadratic fit is $y = 0.0165 + 0.600x - 0.113x^2$. This gives $R^2 = 0.9991$ and $R'^2 = 0.9981$. The cubic fit is $y = -0.00552 + 0.764x - 0.383x^2 + 0.117x^3$. This gives $R^2 = 0.9999$ and $R'^2 = 0.9997$, so is a rather better fit.
- 12 For a straight line, a quadratic fit and a cubic fit, the R^2 values are 0.9238, 0.9786 and 0.9786 respectively, suggesting that a quadratic fit will be excellent. This is confirmed by the R'^2 values, which are 0.9085, 0.9679 and 0.9573 respectively, the quadratic fit giving the highest value.

Chapter 6

- 1 Mean = 9.96 ml, median = 9.90 ml. Q-test shows that the 10.20 value cannot quite be omitted ($P = 0.05$). If it is omitted, mean = 9.88, median = 9.89. The median is insensitive to outliers.
- 2 Sign test: compared with the median, the values give signs of $- + 0 + - + + +$. So eight signs, of which six are positive. Probability of this is 0.29, i.e. >0.05 , so null hypothesis is retained: median sulphur content could be 0.10%. In the signed rank test the zero is neglected, and the ranked differences are -0.01 , 0.01 , 0.01 , -0.02 , 0.02 , 0.02 , 0.04 , 0.07 . So signed ranks are -2 , 2 , 2 , -5 , 5 , 5 , 6 , 7 . Negative ranks total $(-)7$, but at $P = 0.05$, critical region is ≤ 3 . So the null hypothesis is again retained.
- 3 (RID–EID) results give signs of $+ - + + + + + 0 +$. So nine results, eight positive. $P = 0.04$ for this outcome, so the null hypothesis (that the methods

- give indistinguishable results) can be rejected. In the signed rank test, the negative ranks total $(-)2.5$, well below the critical level of 5, so again the null hypothesis can be rejected.
- 4 Arranging the results in order, the median is 23.5. So individual values have signs $+++- - - -++$. This sequence has three runs, but for $M = N = 3$, the critical value is 3, so the null hypothesis of a random sequence must be retained.
 - 5 Mann–Whitney U -test: ‘beer’ values are expected to be larger than ‘lager’ values. Number of lager values greater than the individual values = 4.5 (one tie). Critical value for a one-sided test is 5, so we can just reject the null hypothesis ($P = 0.05$). Tukey’s quick test: count is 5.5, just below the critical value of 6. So tests disagree: more data needed.
 - 6 For instruments A–G student rankings are 3, 1, 5, 4, 7, 6, 2, and staff rankings are 5, 3, 6, 2, 4, 7, 1. So the d -values are $-2, -2, -1, 2, 3, -1, 1$, and the d^2 values are 4, 4, 1, 4, 9, 1, 1, totalling 24. Hence $r_s = 1 - [(6 \times 24)/(7 \times 48)] = 0.571$. For $n = 7$ the critical value at $P = 0.05$ is 0.786: no evidence of correlation between student and staff opinions.
 - 7 If the x -values are the distances and the y -values the mercury levels, Theil’s method gives $a = 2.575$, $b = -0.125$. (The least-squares method gives $a = 2.573$, $b = -0.122$.)
 - 8 If the nickel levels are replaced by ranks (one tie occurs) the sums of the ranks for the three samples are 39, 52.5 and 79.5. (These add up to 171, as expected for 18 values, as $1/2 \times 18 \times 19 = 171$.) The corresponding value of $\chi^2 = 4.97$, below the critical value of 5.99 ($P = 0.05$, 2 degrees of freedom) so the null hypothesis of no significant difference in the nickel levels in the oils must be retained.

Chapter 7

- 1 This is two-way ANOVA without replication. The between-row (i.e. between-solution) mean square is 0.00370 (3 d.f.); the between-column (i.e. between-method) mean square is 0.00601 (2 d.f.); and the residual mean square is 0.00470 (6 d.f.). The between-solution mean square is less than the residual one, so is not significant. Comparison of the between-method and residual mean squares gives $F = 0.00601/0.00470 = 1.28$. The critical value of $F_{2,6}$ ($P = 0.05$) is 5.14, so the between-method variation is not significant.
- 2 Again, a two-way ANOVA experiment without replication. The between-soil, between-day and residual mean squares are respectively 4.67 (4 d.f.), 144.8 (2 d.f.) and 26.47 (8 d.f.). The between-soil mean square is less than the residual mean square, so there are no significant differences between soils. Comparing the between-day and residual mean squares gives $F = 144.8/26.47 = 5.47$. The critical value of $F_{2,8}$ is 4.46, so this source of variation is significant at $P = 0.05$. The actual probability (Excel) is 0.0318.
- 3 Another two-way ANOVA experiment without replication. (Replication would be needed to study possible interaction effects.) The between-compound,

between-molar ratio and residual mean squares are respectively 4204 (3 d.f.), 584 (2 d.f.) and 706 (6 d.f.). Thus molar ratios have no significant effect. Comparing the between-compound and residual mean squares gives $F = 4204/706 = 5.95$. The critical value of $F_{3,6}$ is 4.76 ($P = 0.05$), so this variation is significant. (P is given by Excel[®] as 0.0313.) Common sense should be applied to these and all other data – diphenylamine seems to behave differently from the other three compounds.

- 4 The single-factor effects are A: -0.0215 , C: 0.0005 , T: -0.0265 . The two-factor effects are AC: -0.0005 , CT: 0.0025 , AT: -0.0065 . The three-factor effect ACT is -0.0005 .
- 5 This is a two-way ANOVA experiment with replication. The mean squares for between-row, between-column, interaction and residual variations are respectively 2.53 (2 d.f.), 0.0939 (2 d.f.), 0.0256 (4 d.f.), and 0.0406 (9 d.f.). The interaction mean square is less than the residual mean square, so sample-laboratory interactions are not significant. Comparing the between-column (i.e. between-laboratory) and the residual mean squares gives $F = 0.0939/0.0406 = 2.31$. The critical value of $F_{2,9}$ is 4.256 ($P = 0.05$), so the between-laboratory variation is not significant.
- 6 (a) Using the Fibonacci approach to achieve a 40-fold reduction in the optimum range, we use the terms F_7 and F_9 (as F_9 is the first Fibonacci term above 40) to give the ratio 21/55. The starting pHs are then $5 + [(21 \times 4)/55] = 6.53$ and $9 - [(21 \times 4)/55] = 7.47$. (b) When six experiments are to be performed the Fibonacci method uses F_6 and F_4 to form the fraction 5/13, so the starting pHs are $5 + (20/13)$ and $9 - (20/13)$, i.e. 6.54 and 7.46 (similar values again). The degree of optimisation is $1/F_6$, i.e. $1/13$, so the optimum pH range will be defined within an envelope of $4/13 = 0.31$ pH units.
- 7 Vertex 1 should be rejected. The new vertex 8 should have co-ordinates 5.8, 9.4, 18.1, 9.2, 8.8 for factors A–E respectively, all values being given to one decimal place.

Chapter 8

- 1 The printout below was obtained using Minitab[®].

Linear Discriminant Function for Group			
	A	B	C
Constant	-14.538	-2.439	-8.782
Sucrose	15.039	-3.697	-11.342
Glucose	-1.829	2.931	-1.102
Fructose	-9.612	0.363	9.249
Sorbitol	-2.191	-0.229	2.421

This suggests that sucrose and fructose may be the variables which are most effective at discriminating between varieties.

The cross-classification success rate with just these two variables is:

Summary of Classification with Cross-validation

Put into	True Group		
Group	A	B	C
A	5	0	0
B	0	5	1
C	0	0	4
Total N	5	5	5
N Correct	5	5	4
Proportion	1.000	1.000	0.800

N = 15 N Correct = 14 Proportion Correct = 0.933

- 2 (a) A dendrogram shows two clear groups with group membership depending on whether the rice is polished or not.

(b)

	P	K	Ni
K	0.954		
Ni	-0.531	-0.528	
Mo	0.150	0.117	-0.527

Strong positive correlation between P and K. Little correlation between Mo and K and between Mo and P.

- (c) Carrying out PCA on the standardised values gives:

Eigenanalysis of the Correlation Matrix

Eigenvalue	2.4884	1.1201	0.3464	0.0451
Proportion	0.622	0.280	0.087	0.011
Cumulative	0.622	0.902	0.989	1.000

Variable	PC1	PC2
P	0.577	0.340
K	0.572	0.366
Ni	-0.509	0.357
Mo	0.283	-0.789

A score plot shows two fairly well-defined groups: one for polished and the other for unpolished samples.

- (d) The results of LDA using the standardised values are:

Summary of Classification with Cross-validation

Put into	True Group	
Group	A	B
A	7	1
B	1	7
Total N	8	8
N Correct	7	7
Proportion	0.875	0.875

N = 16 N Correct = 14 Proportion Correct = 0.875

Linear Discriminant Function for Group

	A	B
Constant	-2.608	-2.608
P	18.016	-18.016
K	-19.319	19.319
Ni	-0.051	0.051
Mo	-1.198	1.198

The discrimination between varieties is good (87.5% success). Results suggest that P and K are most effective at discriminating between varieties. Using just these two elements, a cross-classification rate of 15/16 is achieved.

- 3 (a) MLR cannot be used, as the number of specimens is not greater than the number of predictors.
- (b) The printout below was obtained from Minitab®. Between them the first two eigenvectors account for virtually all the variance.

Principal Component Analysis: I1, I2, I3, I4, I5, I6, I7, I8, I9, I10

Eigenanalysis of the Covariance Matrix

Eigenvalue	8537.3	659.1	0.5	0.2	0.1	0.1	0.0	0.0	-0.0
Proportion	0.928	0.072	0.000	0.000	0.000	0.000	0.000	0.000	-0.000
Cumulative	0.928	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Eigenvalue	-0.0								
Proportion	-0.000								
Cumulative	1.000								
Variable	PC1	PC2							
I1	-0.302	0.502							
I2	-0.303	0.398							
I3	-0.283	0.301							
I4	-0.314	0.181							
I5	-0.315	0.065							
I6	-0.320	-0.043							
I7	-0.324	-0.151							
I8	-0.327	-0.263							
I9	-0.334	-0.373							
I10	-0.336	-0.479							

The scores for the first two eigenvectors are

Z_1	Z_2
-70.155	24.3663
-230.122	-14.0245
-287.257	5.0160
-62.560	-27.0383
-145.815	60.5942
-41.701	16.5704
-29.195	-9.1504
-33.454	16.6590
-93.381	-0.6222

The regression equation of c_1 on Z_1 and Z_2 can be obtained from Minitab®. It is

$$c_1 = 0.0116 - 0.00294 Z_1 + 0.00686 Z_2$$

The residuals show no patterns and are approximately normally distributed.

(c) The following printout was obtained from Minitab®.

PLS Regression: c1 versus I1, I2, I3, I4, I5, I6, I7, I8, I9, I10

Number of components selected by cross-validation: 3

Number of observations left out per group: 1

Number of components cross-validated: 7

Model Selection and Validation for c1

Components	X Variance	Error SS	R-Sq	PRESS	R-Sq (pred)
1	0.929412	0.198592	0.76293	0.310446	0.629402
2	0.999900	0.000299	0.99964	0.001029	0.998772
3	0.999957	0.000137	0.99984	0.000906	0.998919
4		0.000053	0.99994	0.001248	0.998510
5		0.000030	0.99996	0.001756	0.997904
6		0.000006	0.99999	0.001987	0.997628
7		0.000000	1.00000	0.001976	0.997641

Regression Coefficients

	c1	c1 standardized
Constant	0.0048163	0.000000
I1	0.0025364	0.240705
I2	0.0024281	0.223514
I3	0.0097208	0.818996
I4	0.0009896	0.089914
I5	-0.0005505	-0.049677
I6	-0.0002864	-0.026187
I7	0.0007365	0.068765
I8	-0.0022379	-0.214303
I9	-0.0016805	-0.167711
I10	-0.0009030	-0.093197

The regression equation is

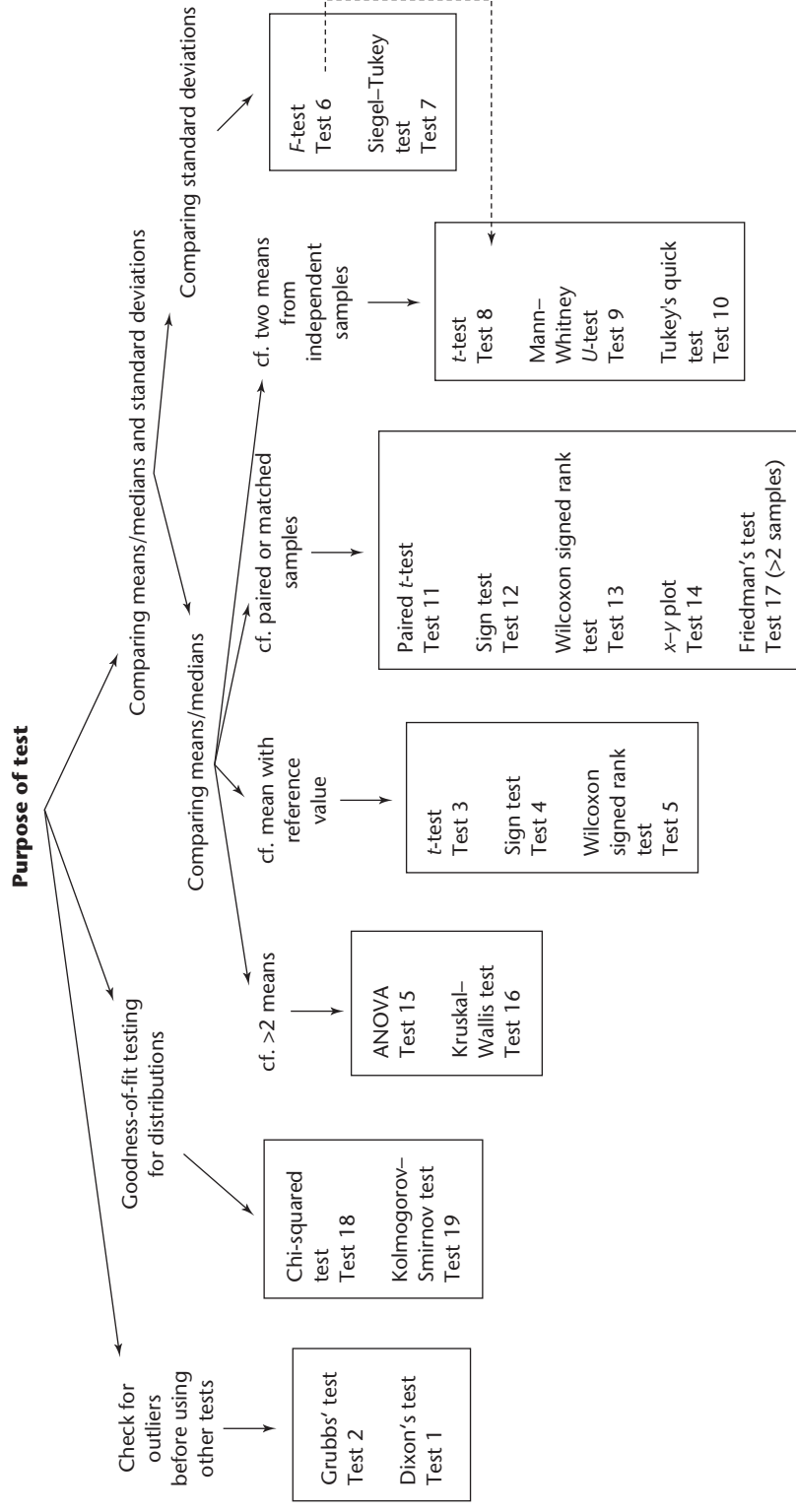
$$c_1 = 0.0048163 + 0.0025364I_1 + 0.0024281I_2 + 0.0097208I_3 + 0.0009896I_4 - 0.0005505I_5 - 0.0002864I_6 + 0.0007365I_7 - 0.0022379I_8 - 0.0016805I_9 - 0.0009030I_{10}$$

The residuals show no patterns and are approximately normally distributed.

- (d) The PRESS statistic is 0.00111959 for the model in (b) and 0.000906 for the model in (c) so PLS1 performs better than PCR by this measure.
- (e) PCR: The scores calculated for the new specimen are $Z_1 = -136.172$ and $Z_2 = 20.3898$. Substituting these values in the regression equation gives $c_1 = 0.552$. PLS: Substitution in the regression equation gives $c_1 = 0.556$. (Minitab® also gives confidence intervals: for PCR (0.544, 0.558), for PLS (0.549, 0.563).)

Appendix 1: Commonly used statistical significance tests

Problem	Tests available	See Section	Comments
Testing for outliers	1 Dixon's test 2 Grubbs' test	3.7 3.7	ISO recommended
Comparison of mean/ median with standard value	3 <i>t</i> -test 4 Sign test 5 Wilcoxon signed rank test	3.2 6.3 6.5	Non-parametric Non-parametric
Comparison of spreads of two data sets	6 <i>F</i> -test 7 Siegel–Tukey test	3.6 6.6	Precedes test 8 Non-parametric
Comparison of means or medians of two samples	8 <i>t</i> -test 9 Mann–Whitney <i>U</i> -test 10 Tukey's quick test	3.3 6.6 6.6	Non-parametric Non-parametric
Comparison of two sets of paired data	11 Paired <i>t</i> -test 12 Sign test 13 Wilcoxon signed rank test 14 <i>x</i> – <i>y</i> plot	3.4 6.3 6.5 5.9	Small range of values Non-parametric Non-parametric Large range of values
Comparison of means/ medians of >2 samples	15 ANOVA 16 Kruskal–Wallis test	3.9 6.7	See index Non-parametric
Comparison of >2 matched data sets	17 Friedman's test	6.7	Non-parametric
Testing for occurrence of a particular distribution	18 Chi-squared test 19 Kolmogorov– Smirnov tests	3.11 3.12	Small samples



Flow chart for statistical tests.

The flow chart

The flow chart is designed for use in conjunction with the table to aid the choice of appropriate significance test. It is intended only as a guide, and should not be used blindly. That is, once the chart has indicated which test or tests are most suitable to a given experimental situation, the analyst must become familiar with the principles of the selected test, the reasons for its selection, any limitations on its validity and so on. Only in this way will the results of the test be applied properly in all cases. For example, most non-parametric tests are not so powerful as parametric ones in conditions where the latter are appropriate, but may be more reliable where serious deviations from the normal distribution are known or suspected.

In the chart 'cf.' is used as an abbreviation for 'comparison of'. The test numbers refer to the table. Robust methods have not been included in either the table or the chart. Despite their growing importance they are still applied more usually by researchers and expert statisticians than by many laboratory workers, and the basic software packages referred to in Chapter 1 do not give a very comprehensive treatment of such methods. It is important to notice that ANOVA is a very widely used method, the exact form used depending on the problem to be solved: only the first reference to one-way ANOVA has been given in the table. The Cochran test (Section 4.12) and the least significant difference method (Section 3.9) used in conjunction with ANOVA, and the Wald–Wolfowitz test for runs (Section 6.4) have also been omitted for simplicity. The broken line linking Tests 6 and 8 is a reminder that, strictly speaking, the *F*-test should be applied to check whether the variances of the two samples under study are similar, before the *t*-test is applied. Some of the tests listed under 'Comparing means' actually compare medians; this has also been omitted in places in the interests of clarity.

Finally it is important to note that there are many tests in everyday use in addition to the ones listed above, as noted in the reference below.

Bibliography

Kanji, G.K., 1999, *100 Statistical Tests*, 2nd edn, Sage Publications, London.

Appendix 2: Statistical tables

The following tables are presented for the convenience of the reader, and for use with the simple statistical tests, examples and exercises in this book. They are presented in a format that is compatible with the needs of analytical chemists: the significance level $P = 0.05$ has been used in most cases, and it has been assumed that the number of measurements available is fairly small. Most of these abbreviated tables have been taken, with permission, from *Elementary Statistics Tables* by Henry R. Neave, published by Routledge (Tables A.2–A.4, A.7, A.8, A.12–A.14). The reader requiring statistical data corresponding to significance levels and/or numbers of measurements not covered in the tables is referred to these sources.

Table A.1 $F(z)$, the standard normal cumulative distribution function

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−3.4	0.0003	0.0003	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0005	0.0005
−3.3	0.0005	0.0005	0.0005	0.0005	0.0006	0.0006	0.0006	0.0006	0.0006	0.0007
−3.2	0.0007	0.0007	0.0007	0.0008	0.0008	0.0008	0.0008	0.0009	0.0009	0.0009
−3.1	0.0010	0.0010	0.0010	0.0011	0.0011	0.0011	0.0012	0.0012	0.0013	0.0013
−3.0	0.0013	0.0014	0.0014	0.0015	0.0015	0.0016	0.0016	0.0017	0.0018	0.0018
−2.9	0.0019	0.0019	0.0020	0.0021	0.0021	0.0022	0.0023	0.0023	0.0024	0.0025
−2.8	0.0026	0.0026	0.0027	0.0028	0.0029	0.0030	0.0031	0.0032	0.0033	0.0034
−2.7	0.0035	0.0036	0.0037	0.0038	0.0039	0.0040	0.0041	0.0043	0.0044	0.0045
−2.6	0.0047	0.0048	0.0049	0.0051	0.0052	0.0054	0.0055	0.0057	0.0059	0.0060
−2.5	0.0062	0.0064	0.0066	0.0068	0.0069	0.0071	0.0073	0.0075	0.0078	0.0080
−2.4	0.0082	0.0084	0.0087	0.0089	0.0091	0.0094	0.0096	0.0099	0.0102	0.0104
−2.3	0.0107	0.0110	0.0113	0.0116	0.0119	0.0122	0.0125	0.0129	0.0132	0.0136
−2.2	0.0139	0.0143	0.0146	0.0150	0.0154	0.0158	0.0162	0.0166	0.0170	0.0174
−2.1	0.0179	0.0183	0.0188	0.0192	0.0197	0.0202	0.0207	0.0212	0.0217	0.0222
−2.0	0.0228	0.0233	0.0239	0.0244	0.0250	0.0256	0.0262	0.0268	0.0274	0.0281
−1.9	0.0287	0.0294	0.0301	0.0307	0.0314	0.0322	0.0329	0.0336	0.0344	0.0351
−1.8	0.0359	0.0367	0.0375	0.0384	0.0392	0.0401	0.0409	0.0418	0.0427	0.0436
−1.7	0.0446	0.0455	0.0465	0.0475	0.0485	0.0495	0.0505	0.0516	0.0526	0.0537
−1.6	0.0548	0.0559	0.0571	0.0582	0.0594	0.0606	0.0618	0.0630	0.0643	0.0655
−1.5	0.0668	0.0681	0.0694	0.0708	0.0721	0.0735	0.0749	0.0764	0.0778	0.0793

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
−1.4	0.0808	0.0823	0.0838	0.0853	0.0869	0.0885	0.0901	0.0918	0.0934	0.0951
−1.3	0.0968	0.0985	0.1003	0.1020	0.1038	0.1056	0.1075	0.1093	0.1112	0.1131
−1.2	0.1151	0.1170	0.1190	0.1210	0.1230	0.1251	0.1271	0.1292	0.1314	0.1335
−1.1	0.1357	0.1379	0.1401	0.1423	0.1446	0.1469	0.1492	0.1515	0.1539	0.1562
−1.0	0.1587	0.1611	0.1635	0.1660	0.1685	0.1711	0.1736	0.1762	0.1788	0.1814
−0.9	0.1841	0.1867	0.1894	0.1922	0.1949	0.1977	0.2005	0.2033	0.2061	0.2090
−0.8	0.2119	0.2148	0.2177	0.2206	0.2236	0.2266	0.2296	0.2327	0.2358	0.2389
−0.7	0.2420	0.2451	0.2483	0.2514	0.2546	0.2578	0.2611	0.2643	0.2676	0.2709
−0.6	0.2743	0.2776	0.2810	0.2843	0.2877	0.2912	0.2946	0.2981	0.3015	0.3050
−0.5	0.3085	0.3121	0.3156	0.3192	0.3228	0.3264	0.3300	0.3336	0.3372	0.3409
−0.4	0.3446	0.3483	0.3520	0.3557	0.3594	0.3632	0.3669	0.3707	0.3745	0.3783
−0.3	0.3821	0.3859	0.3897	0.3936	0.3974	0.4013	0.4052	0.4090	0.4129	0.4168
−0.2	0.4207	0.4247	0.4286	0.4325	0.4364	0.4404	0.4443	0.4483	0.4522	0.4562
−0.1	0.4602	0.4641	0.4681	0.4721	0.4761	0.4801	0.4840	0.4880	0.4920	0.4960
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6579
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678				

Table A.2 The *t*-distribution

Value of <i>t</i> for a confidence interval of Critical value of $ t $ for <i>P</i> values of number of degrees of freedom	90% 0.10	95% 0.05	98% 0.02	99% 0.01
1	6.31	12.71	31.82	63.66
2	2.92	4.30	6.96	9.92
3	2.35	3.18	4.54	5.84
4	2.13	2.78	3.75	4.60
5	2.02	2.57	3.36	4.03
6	1.94	2.45	3.14	3.71
7	1.89	2.36	3.00	3.50
8	1.86	2.31	2.90	3.36
9	1.83	2.26	2.82	3.25
10	1.81	2.23	2.76	3.17
12	1.78	2.18	2.68	3.05
14	1.76	2.14	2.62	2.98
16	1.75	2.12	2.58	2.92
18	1.73	2.10	2.55	2.88
20	1.72	2.09	2.53	2.85
30	1.70	2.04	2.46	2.75
50	1.68	2.01	2.40	2.68
∞	1.64	1.96	2.33	2.58

The critical values of $|t|$ are appropriate for a two-tailed test. For a one-tailed test the value is taken from the column for *twice* the desired *P*-value, e.g. for a one-tailed test, $P = 0.05$, 5 degrees of freedom, the critical value is read from the $P = 0.10$ column and is equal to 2.02.

Table A.3 Critical values of *F* for a one-tailed test ($P = 0.05$)

v_2	v_1												
	1	2	3	4	5	6	7	8	9	10	12	15	20
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5	241.9	243.9	245.9	248.0
2	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37	19.38	19.40	19.41	19.43	19.45
3	10.13	9.552	9.277	9.117	9.013	8.941	8.887	8.845	8.812	8.786	8.745	8.703	8.660
4	7.709	6.944	6.591	6.388	6.256	6.163	6.094	6.041	5.999	5.964	5.912	5.858	5.803
5	6.608	5.786	5.409	5.192	5.050	4.950	4.876	4.818	4.772	4.735	4.678	4.619	4.558
6	5.987	5.143	4.757	4.534	4.387	4.284	4.207	4.147	4.099	4.060	4.000	3.938	3.874
7	5.591	4.737	4.347	4.120	3.972	3.866	3.787	3.726	3.677	3.637	3.575	3.511	3.445
8	5.318	4.459	4.066	3.838	3.687	3.581	3.500	3.438	3.388	3.347	3.284	3.218	3.150
9	5.117	4.256	3.863	3.633	3.482	3.374	3.293	3.230	3.179	3.137	3.073	3.006	2.936
10	4.965	4.103	3.708	3.478	3.326	3.217	3.135	3.072	3.020	2.978	2.913	2.845	2.774
11	4.844	3.982	3.587	3.357	3.204	3.095	3.012	2.948	2.896	2.854	2.788	2.719	2.646
12	4.747	3.885	3.490	3.259	3.106	2.996	2.913	2.849	2.796	2.753	2.687	2.617	2.544
13	4.667	3.806	3.411	3.179	3.025	2.915	2.832	2.767	2.714	2.671	2.604	2.533	2.459
14	4.600	3.739	3.344	3.112	2.958	2.848	2.764	2.699	2.646	2.602	2.534	2.463	2.388
15	4.543	3.682	3.287	3.056	2.901	2.790	2.707	2.641	2.588	2.544	2.475	2.403	2.328
16	4.494	3.634	3.239	3.007	2.852	2.741	2.657	2.591	2.538	2.494	2.425	2.352	2.276
17	4.451	3.592	3.197	2.965	2.810	2.699	2.614	2.548	2.494	2.450	2.381	2.308	2.230
18	4.414	3.555	3.160	2.928	2.773	2.661	2.577	2.510	2.456	2.412	2.342	2.269	2.191
19	4.381	3.522	3.127	2.895	2.740	2.628	2.544	2.477	2.423	2.378	2.308	2.234	2.155
20	4.351	3.493	3.098	2.866	2.711	2.599	2.514	2.447	2.393	2.348	2.278	2.203	2.124

v_1 = number of degrees of freedom of the numerator; v_2 = number of degrees of freedom of the denominator.

Table A.4 Critical values of F for a two-tailed test ($P = 0.05$)

v ₂	v ₁													
	1	2	3	4	5	6	7	8	9	10	12	15	20	
1	647.8	799.5	864.2	899.6	921.8	937.1	948.2	956.7	963.3	968.6	976.7	984.9	993.1	
2	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37	39.39	39.40	39.41	39.43	39.45	
3	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54	14.47	14.42	14.34	14.25	14.17	
4	12.22	10.65	9.979	9.605	9.364	9.197	9.074	8.980	8.905	8.844	8.751	8.657	8.560	
5	10.01	8.434	7.764	7.388	7.146	6.978	6.853	6.757	6.681	6.619	6.525	6.428	6.329	
6	8.813	7.260	6.599	6.227	5.988	5.820	5.695	5.600	5.523	5.461	5.366	5.269	5.168	
7	8.073	6.542	5.890	5.523	5.285	5.119	4.995	4.899	4.823	4.761	4.666	4.568	4.467	
8	7.571	6.059	5.416	5.053	4.817	4.652	4.529	4.433	4.357	4.295	4.200	4.101	3.999	
9	7.209	5.715	5.078	4.718	4.484	4.320	4.197	4.102	4.026	3.964	3.868	3.769	3.667	
10	6.937	5.456	4.826	4.468	4.236	4.072	3.950	3.855	3.779	3.717	3.621	3.522	3.419	
11	6.724	5.256	4.630	4.275	4.044	3.881	3.759	3.664	3.588	3.526	3.430	3.330	3.226	
12	6.554	5.096	4.474	4.121	3.891	3.728	3.607	3.512	3.436	3.374	3.277	3.177	3.073	
13	6.414	4.965	4.347	3.996	3.767	3.604	3.483	3.388	3.312	3.250	3.153	3.053	2.948	
14	6.298	4.857	4.242	3.892	3.663	3.501	3.380	3.285	3.209	3.147	3.050	2.949	2.844	
15	6.200	4.765	4.153	3.804	3.576	3.415	3.293	3.199	3.123	3.060	2.963	2.862	2.756	
16	6.115	4.687	4.077	3.729	3.502	3.341	3.219	3.125	3.049	2.986	2.889	2.788	2.681	
17	6.042	4.619	4.011	3.665	3.438	3.277	3.156	3.061	2.985	2.922	2.825	2.723	2.616	
18	5.978	4.560	3.954	3.608	3.382	3.221	3.100	3.005	2.929	2.866	2.769	2.667	2.559	
19	5.922	4.508	3.903	3.559	3.333	3.172	3.051	2.956	2.880	2.817	2.720	2.617	2.509	
20	5.871	4.461	3.859	3.515	3.289	3.128	3.007	2.913	2.837	2.774	2.676	2.573	2.464	

v_1 = number of degrees of freedom of the numerator; v_2 = number of degrees of freedom of the denominator.

Table A.5 Critical values of G ($P = 0.05$) for a two-sided test

Sample size	Critical value
3	1.155
4	1.481
5	1.715
6	1.887
7	2.020
8	2.126
9	2.215
10	2.290

Taken from Barnett, V. and Lewis, T., 1984, *Outliers in Statistical Data*, 2nd edn, John Wiley & Sons Limited.

Table A.6 Critical values of Q ($P = 0.05$) for a two-sided test

Sample size	Critical value
4	0.829
5	0.710
6	0.625
7	0.568

Adapted with permission from Statistical treatment for rejection of deviant values: critical values of Dixon's "Q" parameter and related subrange ratios at the 95% confidence level, *Analytical Chemistry*, **63**(2), pp. 139–46 (Rorabacher, D.B. 1991), American Chemical Society. Copyright 1991 American Chemical Society.

Table A.7 Critical values of χ^2 ($P = 0.05$)

Number of degrees of freedom	Critical value
1	3.84
2	5.99
3	7.81
4	9.49
5	11.07
6	12.59
7	14.07
8	15.51
9	16.92
10	18.31

Table A.8 Random numbers

02484	88139	31788	35873	63259	99886	20644	41853	41915	02944
83680	56131	12238	68291	95093	07362	74354	13071	77901	63058
37336	63266	18632	79781	09184	83909	77232	57571	25413	82680
04060	46030	23751	61880	40119	88098	75956	85250	05015	99184
62040	01812	46847	79352	42478	71784	65864	84904	48901	17115
96417	63336	88491	73259	21086	51932	32304	45021	61697	73953
42293	29755	24119	62125	33717	20284	55606	33308	51007	68272
31378	35714	00941	53042	99174	30596	67769	59343	53193	19203
27098	38959	49721	69341	40475	55998	87510	55523	15549	32402
66527	73898	66912	76300	52782	29356	35332	52387	29194	21591
61621	52967	40644	91293	80576	67485	88715	45293	59454	76218
18798	99633	32948	49802	40261	35555	76229	00486	64236	74782
36864	66460	87303	13788	04806	31140	75253	79692	47618	20024
10346	28822	51891	04097	98009	58042	67833	23539	37668	16324
20582	49576	91822	63807	99450	18240	70002	75386	26035	21459
12023	82328	54810	64766	58954	76201	78456	98467	34166	84186
48255	20815	51322	04936	33413	43128	21643	90674	98858	26060
92956	09401	58892	59686	10899	89780	57080	82799	70178	40399
87300	04729	57966	95672	49036	24993	69827	67637	09472	63356
69101	21192	00256	81645	48500	73237	95420	98974	36036	21781
22084	03117	96937	86176	80102	48211	61149	71246	19993	79708
28000	44301	40028	88132	07083	50818	09104	92449	27860	90196
41662	20930	32856	91566	64917	18709	79884	44742	18010	11599
91398	16841	51399	82654	00857	21068	94121	39197	27752	67308
46560	00597	84561	42334	06695	26306	16832	63140	13762	15598

Table A.9 The sign test

n	$r = 0$	1	2	3	4	5	6	7
4	0.063	0.313	0.688					
5	0.031	0.188	0.500					
6	0.016	0.109	0.344	0.656				
7	0.008	0.063	0.227	0.500				
8	0.004	0.035	0.144	0.363	0.637			
9	0.002	0.020	0.090	0.254	0.500			
10	0.001	0.011	0.055	0.172	0.377	0.623		
11	0.001	0.006	0.033	0.113	0.274	0.500		
12	0.000	0.003	0.019	0.073	0.194	0.387	0.613	
13	0.000	0.002	0.011	0.046	0.133	0.290	0.500	
14	0.000	0.001	0.006	0.029	0.090	0.212	0.395	0.605
15	0.000	0.000	0.004	0.018	0.059	0.151	0.304	0.500

The table uses the binomial distribution with $P = 0.5$ to give the probabilities of r or fewer successes for $n = 4 - 15$. These values correspond to a one-tailed sign test and should be doubled for a two-tailed test.

Table A.10 The Wald-Wolfowitz runs test

N	M	At $P = 0.05$, the number of runs is significant if it is:	
		Less than	Greater than
2	12–20	3	NA
3	6–14	3	NA
3	15–20	4	NA
4	5–6	3	8
4	7	3	NA
4	8–15	4	NA
4	16–20	5	NA
5	5	3	9
5	6	4	9
5	7–8	4	10
5	9–10	4	NA
5	11–17	5	NA
6	6	4	10
6	7–8	4	11
6	9–12	5	12
6	13–18	6	NA
7	7	4	12
7	8	5	12
7	9	5	13
7	10–12	6	13
8	8	5	13
8	9	6	13
8	10–11	6	14
8	12–15	7	15

Adapted from Swed, F.S. and Eisenhart, C., 1943, *Ann. Math. Statist.*, 14: 66.

The test cannot be applied to data with N , M smaller than the given numbers, or to cases marked NA.

Table A.11 Wilcoxon signed rank test. Critical values for the test statistic at $P = 0.05$

n	One-tailed test	Two-tailed test
5	0	NA
6	2	0
7	3	2
8	5	3
9	8	5
10	10	8
11	13	10
12	17	13
13	21	17
14	25	21
15	30	25

The null hypothesis can be rejected when the test statistic is less than or equal to the tabulated value. NA indicates that the test cannot be applied.

Table A.12 Mann–Whitney U -test. Critical values for U or the lower of T_1 and T_2 at $P = 0.05$

n_1	n_2	One-tailed test	Two-tailed test
3	3	0	NA
3	4	0	NA
3	5	1	0
3	6	2	1
4	4	1	0
4	5	2	1
4	6	3	2
4	7	4	3
5	5	4	2
5	6	5	3
5	7	6	5
6	6	7	5
6	7	8	6
7	7	11	8

The null hypothesis can be rejected when U or the lower T value is less than or equal to the tabulated value. NA indicates that the test cannot be applied.

Table A.13 The Spearman rank correlation coefficient. Critical values for ρ at $P = 0.05$

n	One-tailed test	Two-tailed test
5	0.900	1.000
6	0.829	0.886
7	0.714	0.786
8	0.643	0.738
9	0.600	0.700
10	0.564	0.649
11	0.536	0.618
12	0.504	0.587
13	0.483	0.560
14	0.464	0.538
15	0.446	0.521
16	0.429	0.503
17	0.414	0.488
18	0.401	0.472
19	0.391	0.460
20	0.380	0.447

Table A.14 The Kolmogorov test for normality. Critical two-tailed values at $P = 0.05$

n	Critical values
3	0.376
4	0.375
5	0.343
6	0.323
7	0.304
8	0.288
9	0.274
10	0.262
11	0.251
12	0.242
13	0.234
14	0.226
15	0.219
16	0.213
17	0.207
18	0.202
19	0.197
20	0.192

The appropriate value is compared with the maximum difference between the hypothetical and sample functions as described in the text.

Table A.15 Critical values for C ($P = 0.05$) for $n = 2$

k	Critical value
3	0.967
4	0.906
5	0.841
6	0.781
7	0.727
8	0.680
9	0.638
10	0.602

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